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Technical Specification

O-RAN Open Xhaul Transport Working Group 9

Xhaul Transport Testing

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1 Revision History

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07/09/2021	V01.00	R Vaez-Ghaemi	Version 01.00
07/22/2022	V02.00	Satheesh Kumar, S, Jiang, J Zou, R Vaez-Ghaemi	Addition of Synchronization, latency and OAM Test Cases
07/06/2023	V03.00	Satheesh Kumar, M Zuo	Addition of WDM OAM (7.1.6), and Synchronization Test cases (9.3)
01/20/2024	V04.00	Satheesh Kumar	Addition of Sync Test Cases in section 9.4 (9.4.1-9.4.7)
10/03/2024	V05.00	Satheesh Kumar, Jiang Sun	Added WDM test cases, updated Inclusive Terminology in synchronization section

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3 Scope

This document defines the testing requirements for an Open Xhaul transport infrastructure. It is based on Xhaul Transport Requirement document, Packet Switched Architectures and Solutions document, the WDM-based Fronthaul Transport, and Synchronization Architecture and Solution documents. It defines test cases and provides dedicated chapters for different solutions.

This document uses information published by O-RAN, 3GPP, IEEE, ITU-T, IETF, CableLabs, NGMN and several other relevant standard bodies and industry associations. It contains educational, informative, and normative content.

Version 02.00 of this document adds new test cases for synchronization (9.2) and WDM (7.1.5 & 7.1.6). Version 3.0 adds new test cases for Synchronization (9.3) and WDM (7.1.6).

What is not covered in this document:

- The document shall minimize any unnecessary overlap with content presented in other O-RAN testing specifications such as WG4 Conformance Test (S-Plane) specification. Whereas the latter is about testing an O-RU or O-DU, this document verifies the transport networks placed between those O-RAN components. Some of the presented synchronization test cases may be moved from this document to other documents such as the WG4 Conformance Test specification in agreement with respective authors of the latter document.

4 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document in Release 15.

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24	[82]	ITU-T G.8275.1, 03/2020, "Precision time protocol telecom profile for phase/time synchronization with full timing support from the network "
26	[83]	ITU-T G.8273.2 Amendment, 03/2020, "Timing characteristics of telecom boundary clocks and telecom timeReceiver clocks"
28	[84]	ITU-T G.8272/Y.1367, 11/2018. "Timing characteristics of primary reference time clocks"
29	[85]	3GPP TS38.801, version 14.0.0, 03/2017, "Study on new radio access technology: Radio access architecture and interfaces"
31	[86]	NGMN "Overview on 5G RAN function decomposition".
32	[87]	3GPP TR22.804 v16.2.0 (2018-12), "Study on communications for automation in vertical domains"
35	[88]	ITU-T G.8271.1, 03/2020, "Network limits for time synchronization in packet networks with full timing support from the network", Amendment 1, 10/2020.

5 Definitions and abbreviations

5.1 Definitions

The key words "**SHALL**", "**SHALL NOT**", "**SHOULD**", "**SHOULD NOT**", "**MAY**", and "**OPTIONAL**" in this document are to be interpreted as described in IETF RFC 2119 [25]. All key words must be in upper case, bold text.

Items that are **REQUIRED** (contain the words **SHALL** or **SHALL NOT**) will be labelled as **[Rx]** for required. Items that are **RECOMMENDED** (contain the words **SHOULD** or **SHOULD NOT**) will be labelled as **[Dx]** for desirable. Items that are **OPTIONAL** (contain the words **MAY** or **OPTIONAL**) will be labelled as **[Ox]** for optional.

Items, if supported, are not meant to be active at all times, but should be available for use. Their state (active or not active) should be based on configuration.

5.2 Abbreviations

Abbreviations defined in this document take precedence over the definition of 3GPP.

AF	Application Function	ITU-T	International Telecom Union-Telecom
AMF	Access and Mobility Management Function	LAN	Local Area Network
AN	Access Node	LLS	Lower Layer Split
ARP	Address Resolution Protocol	LTE	Long Term Evolution
ABW	Antenna Bandwidth	MAC	Medium Access Layer
BBU	Baseband Unit	MBSFN	Multimedia Broadcast multicast service Single Frequency Network
BER	Bit Error Ratio	MCU	Microprogrammed Control Unit
BERT	Bit Error Rate Tester	MPLS	Multi-Protocol Label Switching
BH	Backhaul	MIMO	Multiple Inputs Multiple Outputs
BiDi	Bidirectional	MNO	Mobile Network Operator
BTCA	Best TimeTransmitter Clock Algorithm		
BS	Base Station	MRTD	Maximum Receive Time Difference
BW	Bandwidth	NGMN	Next Generation Mobile Network
CAPEX	Capital Expense	MUX/DEMUX	Multiplexer/Demultiplexer
CBS	Committed Burst Size	NR	New Radio
CFP	Common Form factor Pluggable	NSA	Non-Stand Alone
CIR	Committed Information Rate	NSSI	Subnet Networking Slices Instance
CN	Core Network	OAM	Operation Administration Maintenance
CoMP	Coordinated Multipoint	O-CU	O-RAN Central Unit
CP	Control Plane	O-DU	O-RAN Distributed Unit
CPRI	Common Public Radio Interface	OPEX	Operation Expense
CSR	Cell Site Router	O-RU	O-RAN Radio Unit
CU	Central Unit	PCF	Policy Control Function
DC	Data Center	PDCP	Packet Data Convergence Protocol
DL	Downlink	ppb	Parts per billion
DN	Data Network	PRB	Physical Resource Block
DHCP	Dynamic Host Configuration Protocol	PRTC	Primary Reference Telecom Clock
DSCP	Differentiated Services Codepoint	PTP	Precision Time Protocol
dTE _H	Dynamic Time Error High	OFDM	Orthogonal Frequency Division Multiplexing
dTE _L	Dynamic Time Error Low	QAM	Quadrature Amplitude Modulation
DU	Distributed Unit	QoS	Quality of Service
eCPRI	evolved Common Public Radio Interface	QSFP	Quad SFP
eMBB	enhanced Mobile Broadband	RB	Resource Block
eNB	Evolved NodeB	RRH	Remote Radio Head

EP	Ethernet Private	RU	Radio Unit
EPC	Evolved Packet Core	SCS	Sub Carrier Spacing
EPL	Ethernet Private Line	SCTP	Stream Control Transmission Protocol
ePRC	Enhanced Primary Reference Clock	SFF	Small Form Factor
ePRTC	Enhanced Primary Reference Telecom Clock	SFP	Small Form factor Pluggable
E-UTRA	evolved UMTS Terrestrial Radio Access	SLA	Service Level Agreement
EVP	Ethernet Virtual Private	SOH	Shared O-RU Host
EVPL	Ethernet Virtual Private Line	SRO	Shared Resource Operator
FDD	Frequency Division Duplex	TAE	Time Alignment Error
FFO	Fractional Frequency Offset	T-BC	Telecom Boundary Clock
FFS	For Further Study	TDD	Time Division Duplex
FH	Fronthaul	TE	Time Error (in the context of synchronization)
FLR	Frame Loss Ratio	T-GM	Telecom Grand Master
		TT	TimeTransmitter
		TR	TimeReceiver
FR1	Frequency Range 1	TN	Transport Node
FR2	Frequency Range 2	TNE	Transport Network Equipment
FTTH	Fiber To The Home	T-TSC	Telecom Time Synchronous Clock
gNB	gNodeB	TX	Transmit
GNSS	Global Navigation Satellite System	UDP	User Datagram Protocol
GPRS	General Packet Radio Service	UE	User equipment
GTP	GPRS Tunnelling Protocol	UL	Uplink
HSR	Hub Site Router	UNI	Universal Network Interface
ICMP	Internet Control Message Protocol	UNI-C	UNI-Customer edge
IoT	Internet of Things	UPF	User Plane Function
IP	Internet Protocol	URLLC	Ultra-Reliable Low Latency Communication
IQ	In phase Quadrature	VPN	Virtual Private Network
		WDM	Wavelength Division Multiplexing

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6 Test approach and methodology

This section provides a definition of test use cases in Open Xhaul transport networks.

6.1 System under test

This specification covers cases that test:

1. Xhaul networks from a UNI interface to another UNI interface
2. Xhaul network segments that carry traffic for an overlay network.
3. Individual TNEs or their components

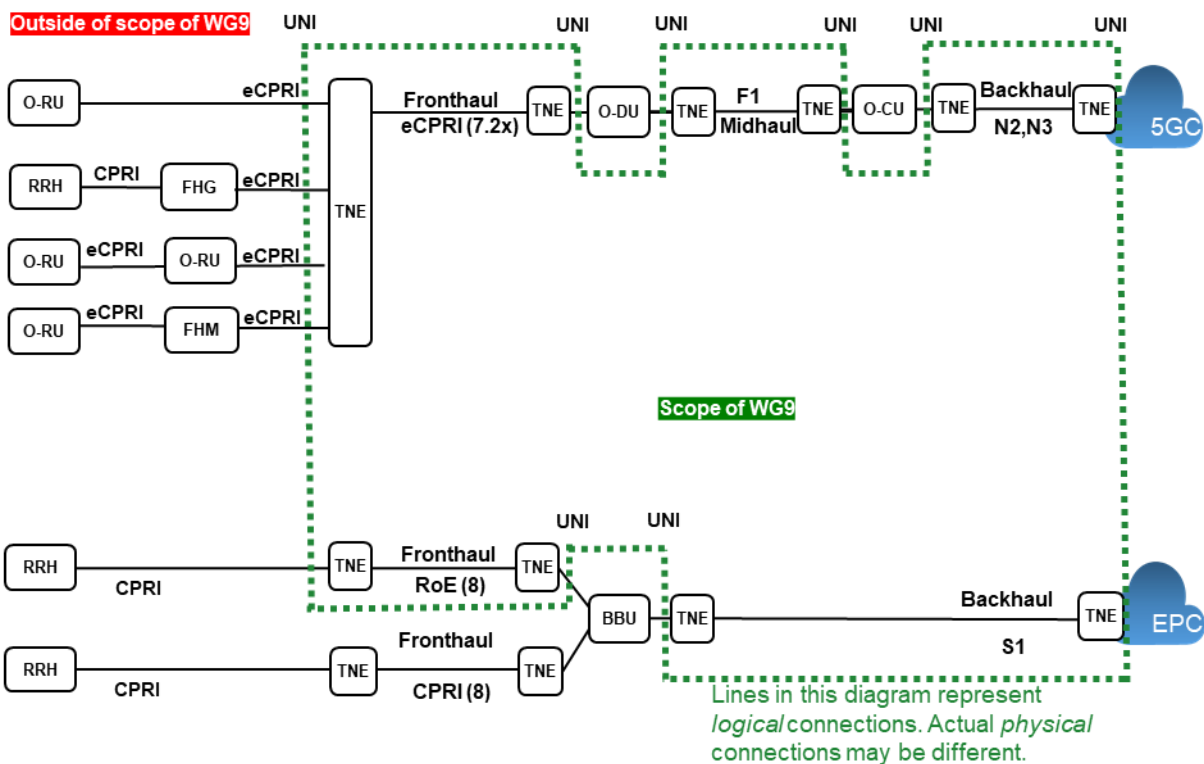


Figure 6-1: Xhaul Transport Networks

Figure 6-1 depicts a simplified overview of Xhaul transport networks that are composed of Fronthaul, Midhaul and Backhaul. The first category of test cases covers the network segments between UNI interfaces of Fronthaul, Midhaul or Backhaul. These test cases do not cross the O-DU or O-CU, and remain within the same network (Fronthaul, Midhaul, or Backhaul).

Beyond these UNI interfaces, there may be other UNI interfaces within the individual Xhaul networks. Figure 6-2 illustrates one use case that includes UNI interfaces within a Fronthaul Network. An example is the use of a WDM physical interface (layer below) between packet-based solutions (layer above). Other use cases include -but are not limited to- use of wireless, PON, or DOCSIS infrastructure in Backhaul, Midhaul, or Fronthaul if the underlying technologies meet the necessary SLAs.

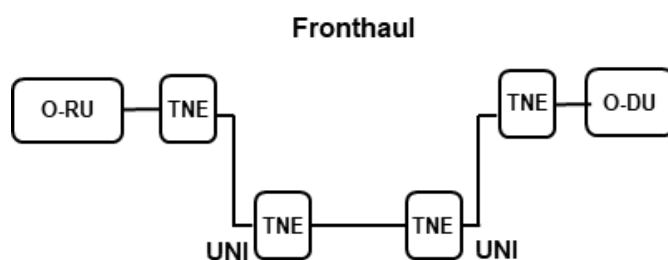


Figure 6-2: UNI interfaces within Xhaul Networks

6.2 Testing Parameters

The test cases in this specification relate to the metrics listed in existing WG9 requirement and solution documents. Examples include throughput, delay, time error or optical power level.

6.3 Testing Methodologies

Test methodologies can be categorized as follows:

- Equipment level vs. network level test
- Technology specific test methodologies

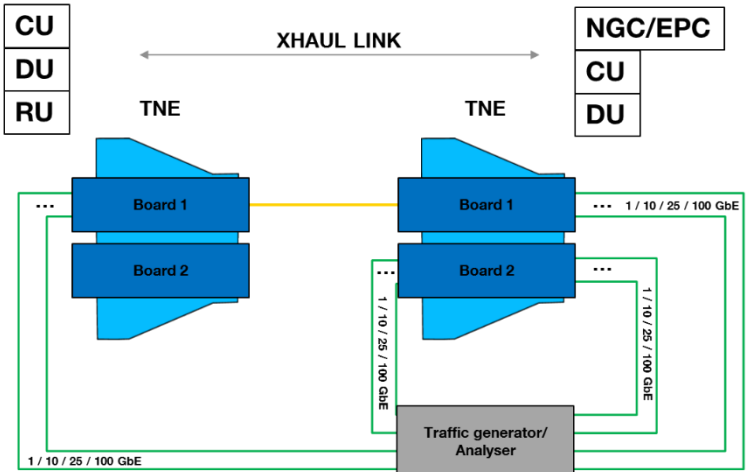
The equipment level tests verify the functionality or performance of an individual equipment or its components and are covered in the next section. The network level tests perform tests from a network UNI to another UNI across one or multiple equipment and are described in remaining chapters of this document.

There are technology specification test cases. Optical layer tests perform tests at optical layer, they include power level or wavelength measurements. Packet layer tests use packet-based methods that generate and analyse Ethernet/eCPRI packets. Synchronization tests include measurements of time or frequency error that measure synchronizations signal or PTP timestamps against a reference signal that may be derived from a GNSS receiver. They may need to emulate a PTP timeTransmitter or timeReceiver function to be able to perform these measurements.

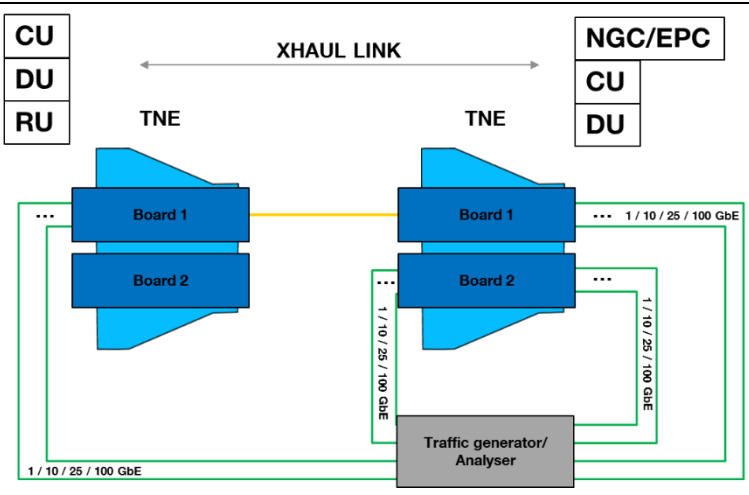
6.4 General (solution/technology agnostic) test use cases

6.4.1 Equipment Testing

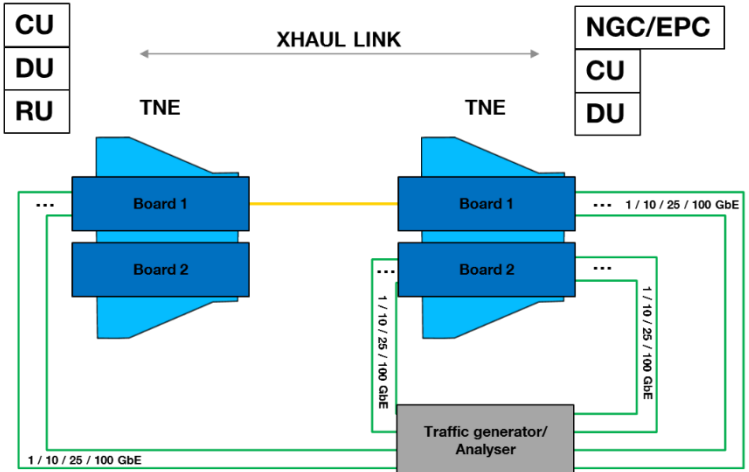
6.4.1.1 TNE board extraction – insertion

Test ID	Equipment.01	Classification	Functionality
Test Title	TNE board extraction – insertion		
Test Objective:	The aim of the test is to check the system behavior in case of line board is physically removed from the TME shelf		
Pre-Req	Running service will be interrupted		
Test Topology:			
Procedure:	<u>Detailed Test procedure:</u> <u>Test Procedure:</u> 1. TNE is connected to another TNE or O-RU/DU/CU & NGC/EPC, all services are running as defined in Traffic and services configuration. 2. Make a line board extraction. 3. Wait 2 minutes. 4. Make a line board insertion. <u>Verifications needed:</u> 1. Verify that traffic and services emulation is running correctly after operation. 2. Check services recovery time after line board extraction and insertion. 3. Check generation and clearance of proper alarms.		
Test Configuration	Xhaul traffic and RAN services configuration		
Pass/Fail Criteria:	All services MUST recover after extraction and insertion of line board		
Results	[] Pass [] Pass w/X [] Dropped [] Blocked [] Failed		
Test Output	First service recovery time in seconds: Last service recovery time in seconds:		

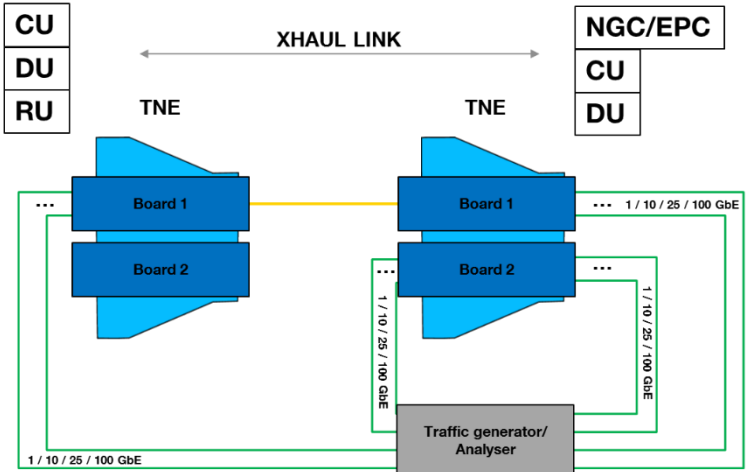
1 6.4.1.2 Fiber network disconnection-connection

Test ID	Equipment.02	Classification	Functionality
Test Title	Fiber network disconnection-connection		
Test Objective:	The aim of the test is to check the system behavior after xhaul link connection/connection		
Pre-Req	Running service will be interrupted		
Test Topology:			
Procedure:	<u>Detailed Test procedure:</u> <u>Test Procedure:</u> 1. Make a network link disconnection/connection. <u>Verifications needed:</u> 1. Verify that traffic is running correctly after operation. 2. Check services recovery time after the network link disconnection/connection. 3. Check generation and clearance of proper alarms		
Test Configuration	Xhaul traffic and RAN services configuration		
Pass/Fail Criteria:	1- All services MUST recover after network link disconnection/connection 2- There is no impact for other links		
Results	☐ Pass ☐ Pass w/X ☐ Dropped ☐ Blocked ☐ Failed		
Test Output	First service recovery time in seconds: Last service recovery time in seconds:		

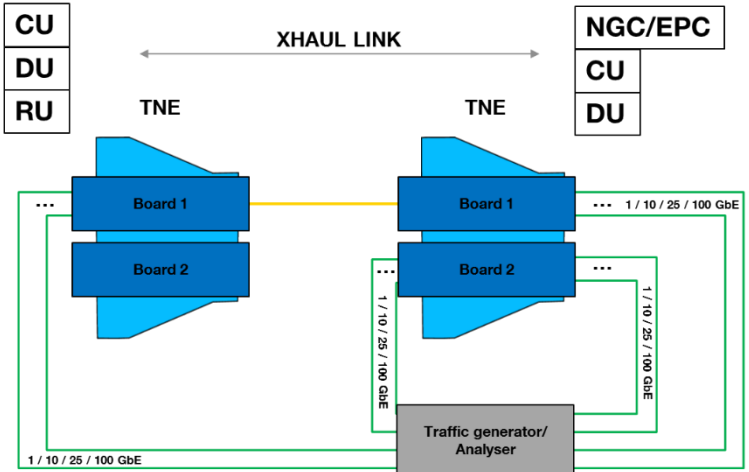
1 6.4.1.3 Transceivers (SFP, QSFP, ...) extraction – insertion

Test ID	Equipment.03	Classification	Functionality
Test Title	Transceivers (SFP, QSFP,...) extraction – insertion		
Test Objective:	The aim of the test is to check if transceivers can be easily extracted/inserted		
Pre-Req	Running service will be interrupted		
Test Topology:			
Procedure:	<u>Detailed Test procedure:</u> <u>Test Procedure:</u> 1. Check if network side SFPs are easy-going to extracted/inserted. 2. Make tests for all SFPs types used with devices. <u>Verifications needed:</u> 1. Verify that traffic is running correctly after operation. 2. Check if the fibres are not impacted by cabinet door close		
Test Configuration	Xhaul traffic and RAN services configuration		
Pass/Fail Criteria:	1- Transceivers MUST be easy extracted and inserted 2- The closed door of the shelf/cabinet MUST NOT impact fibres by too much bending or other ways		
Results	[] Pass [] Pass w/X [] Dropped [] Blocked [] Failed		
Test Output			

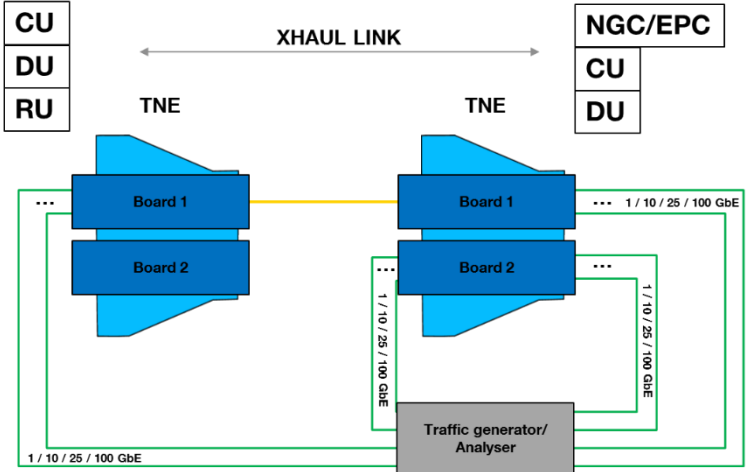
1 6.4.1.4 TNE board swap

Test ID	Equipment.04	Classification	Functionality
Test Title	TNE board swap		
Test Objective:	The aim of the test is to check TME behavior when Board is swapped		
Pre-Req	Running service will be interrupted		
Test Topology:			
Procedure:	<u>Detailed Test procedure:</u> <u>Test Procedure:</u> 1. TNE is connected to another TNE or O-RU/DU/CU & NGC/EPC, start traffic over the services as defined in Traffic and services configuration 2. Disconnect customers fiber and extract the TNE board 3. Insert new TNE board in the same slot and connect the fiber to new TNE board (ex. Board 10GEth to 25GEth, 10GETh w/o sync. features to 10GEth with sync. feature...) <u>Verifications needed:</u> 1. Check services impact and alarms		
Test Configuration	Xhaul traffic and RAN services configuration		
Pass/Fail Criteria:	1. The services MUST recover on the new line board on step 3 2. 2-LT board is operational without alarm		
Results	[] Pass [] Pass w/X [] Dropped [] Blocked [] Failed		
Test Output			
Comment	Swap could be done between - two identical boards (same type and hardware version) - two boards with same type but with different hardware version - two boards with different type and hardware version (for example 1GETh board replaced by 10GETh board)		

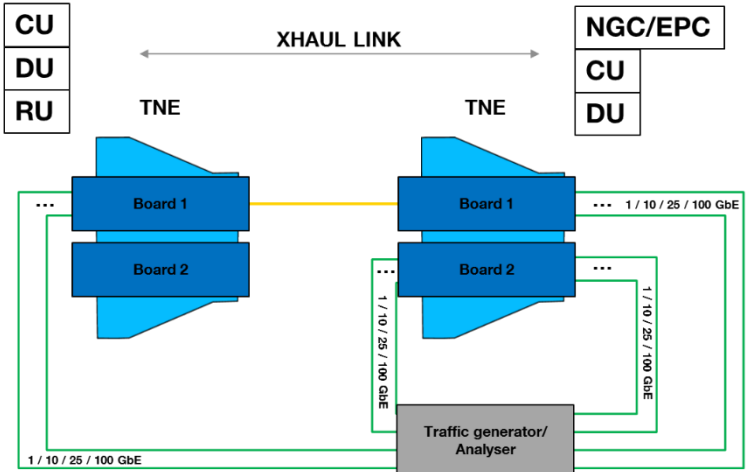
6.4.1.5 Restarts – TNE electric power off-on

Test ID	Equipment.05	Classification	Functionality
Test Title	Restarts – TNE electric power off-on		
Test Objective:	The aim of the test is to check if the system behavior after electrical power off/on		
Pre-Req	Running service will be interrupted		
Test Topology:			
Procedure:	Detailed Test procedure: Test Procedure: 1. Connect TNE to another TNE or O-RU/DU/CU & NGC/EPC, start traffic over the services as defined in Traffic and services configuration chapter 2. Save the configuration 3. Make an TNE electrical off/on Verifications needed: 1. Check the relevant table (MAC, vMAC, DHCP, ARP) persistency/recovery. 2. Verify that traffic is running correctly after operation. 3. Check services recovery time after operation. 4. Note time recover		
Test Configuration	Xhaul traffic and RAN services configuration		
Pass/Fail Criteria:	All services MUST recover after electrical TNE power off/on		
Results	[] Pass [] Pass w/X [] Dropped [] Blocked [] Failed		
Test Output	First service recovery time in seconds: Last service recovery time in seconds:		

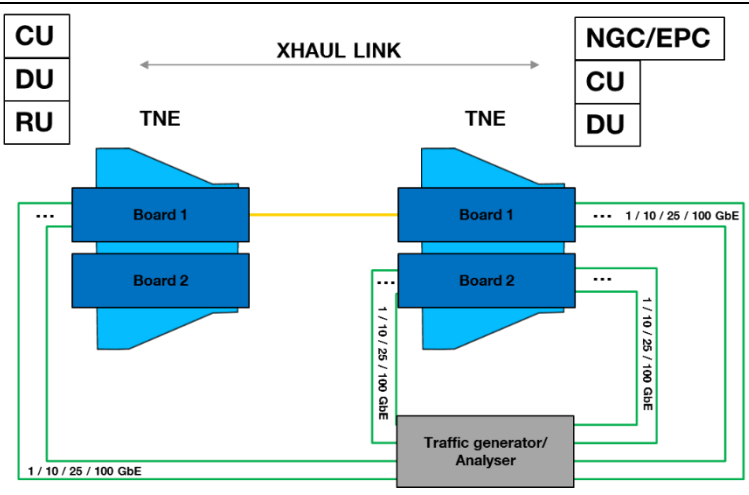
1 Restarts – TNE board software reset

Test ID	Equipment.06	Classification	Functionality
Test Title	Restarts – TNE board software reset		
Test Objective:	The aim of the test is to check if the system behavior after board software reset		
Pre-Req	Running service will be interrupted		
Test Topology:			
Procedure:	<u>Detailed Test procedure:</u> <u>Test Procedure:</u> 1. Connect TNE to another TNE or O-RU/DU/CU & NGC/EPC, start traffic over the services as defined in Traffic and services configuration chapter 2. Save the configuration 3. Make an TNE board software reset <u>Verifications needed:</u> 1. Check the relevant table (MAC, vMAC, DHCP, ARP) persistency/recovery. 2. Verify that traffic is running correctly after operation. 3. Check services recovery time after operation. 4. Note time recover		
Test Configuration	Xhaul traffic and RAN services configuration		
Pass/Fail Criteria:	All services MUST recover after TNE board software reset		
Results	[] Pass [] Pass w/X [] Dropped [] Blocked [] Failed		
Test Output	First service recovery time in seconds: Last service recovery time in seconds:		

1 6.4.1.6 Mirror port

Test ID	Equipment.07	Classification	Functionality
Test Title	Mirror port		
Test Objective:	The aim of the test is to check if TNE supports port monitoring		
Pre-Req	Running service will be interrupted		
Test Topology:			
Procedure:	<u>Detailed Test procedure:</u> <u>Test Procedure:</u> 1. Configure TNE port. 2. Configure TNE mirror port to other port. 3. Send bidirectional traffic through TNE. 4. Capture traffic on mirror port <u>Verifications needed:</u> 1. Verify that the traffic was captured on mirror port.		
Test Configuration	Xhaul traffic and RAN services configuration		
Pass/Fail Criteria:	All traffic on source port MUST be visible on mirror port		
Results	[] Pass [] Pass w/X [] Dropped [] Blocked [] Failed		
Test Output			

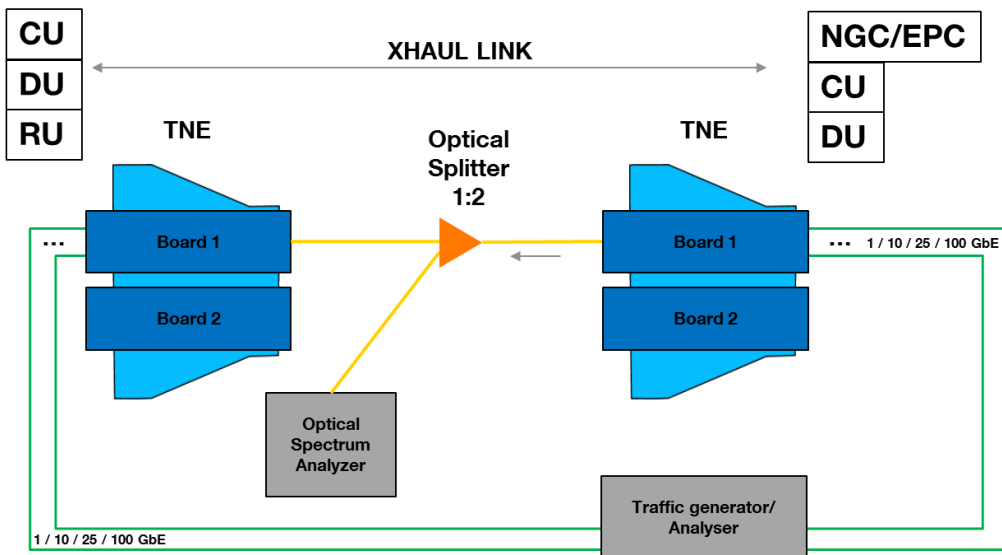
6.4.1.7 Board provisioning

Test ID	Equipment.08	Classification	Functionality
Test Title	Board provisioning		
Test Objective:	The aim of the test is to check no problem is observed when a new board is installed and configured in the TNE shelf		
Pre-Req	Running service will be interrupted		
Test Topology:			
Procedure:	<u>Detailed Test procedure:</u> <u>Test Procedure:</u> 1. Insert board in an empty slot. 2. Make relevant action if necessary (Plan board, acknowledgment, etc.). 3. List alarm(s) reported during operation. 4. Connect and provision traffic. 5. Send traffic. <u>Verifications needed:</u> 1. Board is detected in step 1. 2. Board is recognized in step 2. 3. Connection is created in step 4. 4. Traffic is running without error in step 5		
Test Configuration	Xhaul traffic and RAN services configuration		
Pass/Fail Criteria:			
Results	[] Pass [] Pass w/X [] Dropped [] Blocked [] Failed		
Test Output			

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6.4.2 PHY layer test

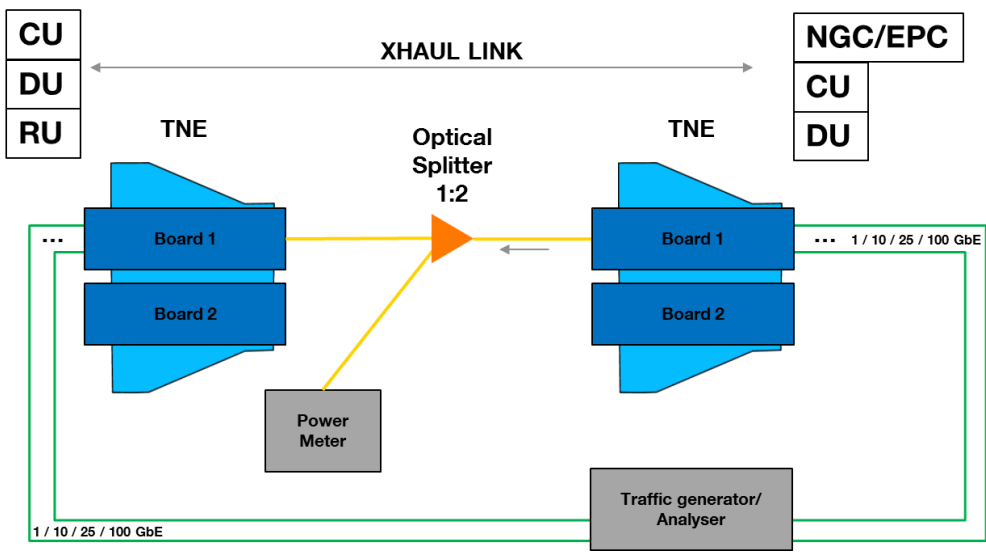
6.4.2.1 Transceiver operating wavelength

Test ID	PHY.01	Classification	Functionality
Test Title	Transceiver operating wavelength		
Test Objective:	The aim of the test is to check the conformity of operating wavelength of the transceiver with adequate specification (IEEE, MSA, ITU,...)		
Pre-Req	Running service will be interrupted		
Test Topology:			
Procedure:	<u>Detailed Test procedure:</u> <u>Test Procedure:</u> 1. Insert an optical spectrum analyser and 1 to 2 splitter as described on relevant test setup a. For bidirectional link (half duplex) b. For full duplex link: insert splitter on the required transmission link and connect the other link directly. 2. Start traffic as defined in Ethernet tests case. 3. Measure the wavelength of generated signal with optical spectrum analyzer. 4. Check the wavelength value of the transceiver remotely based on digital diagnostic. 5. Note the result. <u>Verifications needed:</u> 1. Traffic correctly forwarded on step 2 and step 3 2. TNE operational without alarms on step 2 and step 3 3. Wavelength of generated signal in accordance with the specification		
Test Configuration	Xhaul traffic and RAN services configuration		

Pass/Fail Criteria:	1. Wavelength of generated signal MUST be in accordance with the specification. 2. Digital diagnostic wavelength value in accordance with measure value														
Results	☐ Pass ☐ Pass w/X ☐ Dropped ☐ Blocked ☐ Failed														
Test Output	Comments: The results are presented in the table below: <table><thead><tr><th>Transceiver</th><th>Recommendation</th><th>Measured values</th></tr></thead><tbody><tr><td>Vendor:</td><td></td><td></td></tr><tr><td>Model:</td><td>xxxx +/- yy nm</td><td></td></tr><tr><td>SN:</td><td></td><td></td></tr></tbody></table>			Transceiver	Recommendation	Measured values	Vendor:			Model:	xxxx +/- yy nm		SN:		
Transceiver	Recommendation	Measured values													
Vendor:															
Model:	xxxx +/- yy nm														
SN:															

6.4.2.2 Mean launched optical power of transceiver.

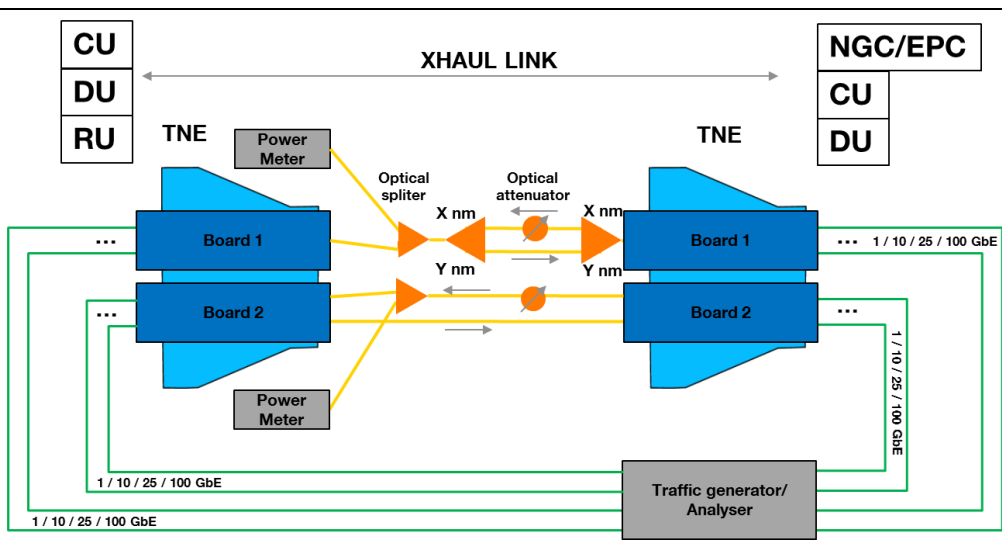
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Test ID	PHY.02	Classification	Functionality
Test Title	Mean launched optical power of transceiver		
Test Objective:	The aim of the test is to check the conformity of mean launched optical power of the transceiver with adequate specification (IEEE, MSA, ITU,...)		
Pre-Req	Running service will be interrupted		
Test Topology:			
Procedure:	<u>Detailed Test procedure:</u> <u>Test Procedure:</u> 1. Insert an optical power meter and 1 to 2 splitter as described on relevant test setup a. For bidirectional link (half duplex) b. For full duplex link: insert splitter on the required transmission link and connect the other link directly. 2. Start traffic as defined in Ethernet tests case. 3. Measure and report the optical power by power meter at the adequate wavelength displayed by power meter (1 measurement) [calibration is required including optical splitter loss] 4. Check the received optical output power of the transceiver remotely based on digital diagnostic. 5. Note the result. 6. Measure and report the repeatability of the optical power by power meter and remotely by digital diagnostic after fiber disconnection / connection on the transceiver (3 additional measurements) <u>Verifications needed:</u> 1. Traffic correctly forwarded on step 2, 3 and 4 2. TNE operational without alarms on step 2 and step 3 3. Optical power of generated signal in accordance with the specification		

	4. Digital diagnostic received optical value in accordance with measure value					
Test Configuration	Xhaul traffic and RAN services configuration					
Pass/Fail Criteria:	1. All optical power measurements of generated signal MUST be in accordance with the specification. 2. Digital diagnostic transmitter optical output power MUST be in accordance with measured values					
Results	☐ Pass ☐ Pass w/X ☐ Dropped ☐ Blocked ☐ Failed					
Test Output	Comments: The results are presented in the table below:					
	Transceiver	Recommendation	Measured values (after fibre disconnection / connection) by power meter and by digital diagnostic			
	Vendor Model SN	(xx, yy) dBm				

6.4.2.3 Optical sensitivity of transceiver

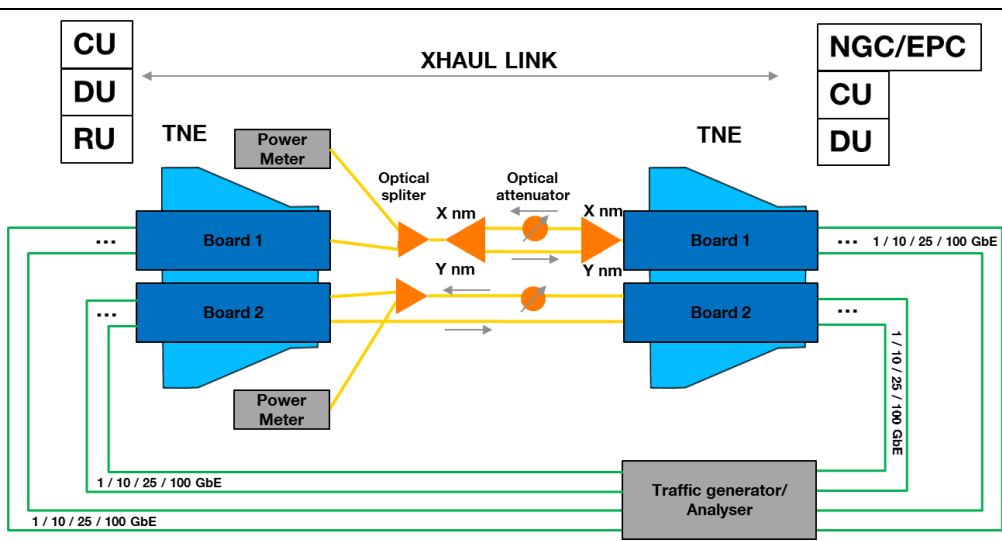
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Test ID	PHY.03	Classification	Functionality
Test Title	Optical sensitivity of transceiver		
Test Objective:	The aim of the test is to check the conformity of optical sensitivity of the transceiver with adequate specification (IEEE, MSA, ITU,...)		
Pre-Req	Running service will be interrupted		
Test Topology:			
Procedure:	Detailed Test procedure: Test Procedure: 1. Insert an optical power meter with 1 to 2 splitter and a variable optical attenuator with optical multiplexor (or circulator) at adequate wavelength and link as described on relevant test setup. a. For bidirectional link (half duplex) : top part of the set up b. For full duplex link: bottom part of the set up. 2. Configure if possible FEC off or FEC on or type of FEC on 3. Start traffic as defined in Ethernet tests case. 4. Increase the optical attenuation gradually on attenuator [dB] till the moment when dropping of Ethernet frames / bit errors are observed. 5. Measure and report the optical power at the adequate wavelength displayed by optical power meter [calibration is required including optical splitter loss] 6. Check the received optical power of the transceiver remotely based on digital diagnostic. 7. Note the result. Verifications needed: 1. Traffic correctly forwarded on step 3 and 4 (before drop frame) 2. TNE operational without alarms on step 3 and 4 (before drop frame) 3. Optical sensitivity of transceiver in accordance with the specification 4. Digital diagnostic received optical value in accordance with measure value		

Test Configuration	Xhaul traffic and RAN services configuration		
Pass/Fail Criteria:	1. Optical sensitivity of transceiver MUST be in accordance with the specification. 2. Digital diagnostic received optical value MUST be in accordance with measured value		
Results	[] Pass [] Pass w/X [] Dropped [] Blocked [] Failed		
Test Output	Comments: The results are presented in the table below: Forward Error Correction (FEC): state/type		
	Transceiver	Recommendation	Measured values (FEC state)
	Vendor Model SN	xx dBm	

6.4.2.4 Minimum optical overload of transceiver

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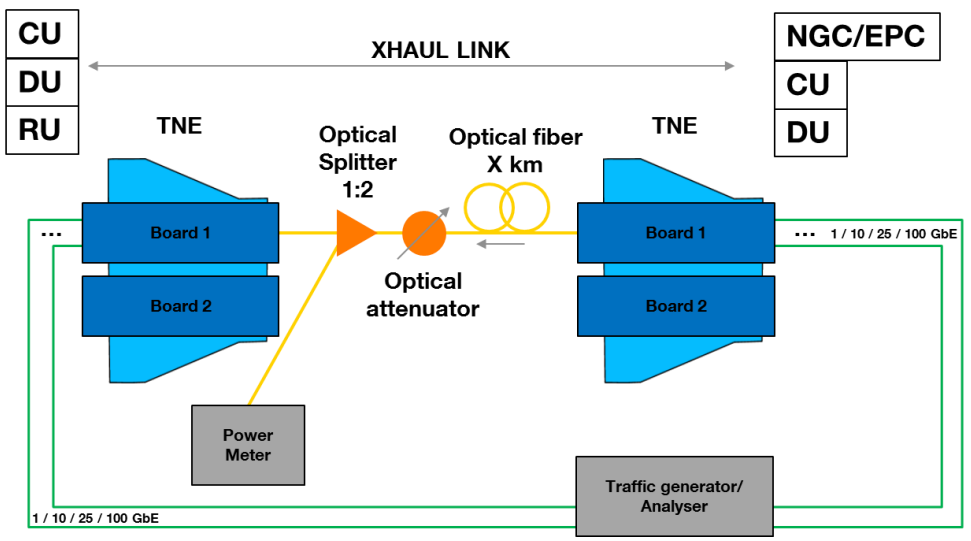
Test ID	PHY.04	Classification	Functionality
Test Title	Minimum optical overload of transceiver		
Test Objective:	The aim of the test is to check the conformity of minimum optical overload of the transceiver with adequate specification (IEEE, MSA, ITU,...)		
Pre-Req	Running service will be interrupted		
Test Topology:			
Procedure:	Detailed Test procedure: Test Procedure: 1. Insert an optical power meter with 1 to 2 splitter and a variable optical attenuator with optical multiplexor (or circulator) at adequate wavelength and link as described on relevant test setup. a. For bidirectional link (half duplex): top part of the set up. b. For full duplex link: bottom part of the set up. 2. Start traffic as defined in Ethernet tests case. 3. Decrease the optical attenuation gradually on attenuator [dB] till the moment when dropping of Ethernet frames / bit errors are observed. 4. Measure and report the optical power at the adequate wavelength displayed by optical power meter [calibration is required including optical splitter loss] 5. Check the received optical power of the transceiver remotely based on digital diagnostic. 6. Note the result. Verifications needed: 1. Traffic correctly forwarded on step 2 and 3 (before drop frame) 2. TNE operational without alarms on step 2 and 3 (before drop frame) 3. Minimum optical overload of transceiver in accordance with the specification 4. Digital diagnostic received optical value in accordance with measure value		
Test Configuration	Xhaul traffic and RAN services configuration		

Pass/Fail Criteria:	1. Minimum optical overload of transceiver MUST be in accordance with the specification. 2. Digital diagnostic received optical value MUST be in accordance with measured value			
Results	☐ Pass ☐ Pass w/X ☐ Dropped ☐ Blocked ☐ Failed			
Test Output	Comments: The results are presented in the table below: Forward Error Correction (FEC): state/type			
	Transceiver	Recommendation	Measured value	Digital diagnostic value
	Vendor Model SN	xx dBm		

1

1 6.4.2.5 Transceiver at minimum and maximum distance (km)

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Test ID	PHY.05	Classification	Functionality
Test Title	Transceiver at minimum and maximum distance (km)		
Test Objective:	The aim of the test is to check the conformity to distance parameter with adequate specification (IEEE, MSA, ITU,...)		
Pre-Req	Running service will be interrupted		
Test Topology:			
Procedure:	Detailed Test procedure: Test Procedure: 1. Insert an optical fiber loop of maximum km length, a variable optical attenuator, and an optical power meter connected to a 1 to 2 splitter as described on relevant test setup. a. For bidirectional link (half duplex) b. For full duplex link: insert test devices on the required transmission link and connect the other link directly. 2. Configure if possible FEC off or FEC on or type of FEC on 3. Set an attenuation of optical attenuator to the value relating to the optical sensitivity value. 4. Start traffic as defined in Ethernet tests case. 5. Fifteen minutes later, disconnect / connect the transceivers. 6. Control the traffic. 7. Insert minimum km of optical. 8. Set an attenuation of optical attenuator to the value relating to the optical sensitivity value. 9. Control the traffic. 10. Fifteen minutes later, disconnect / connect the transceivers. 11. Control the traffic.		

	12. Set an attenuation of optical attenuator to the value relating to the optical overload value. 13. Control the traffic. 14. Fifteen minutes later, disconnect / connect the transceivers. 15. Control the traffic. 16. Check the link length supported pf the transceiver remotely based on digital diagnostic. 17. Note the result. Verifications needed: 1. Traffic correctly forwarded on step 4, 6, 9, 11, 13 and 15. 2. TNE operational without alarms on step 4, 6, 9, 11, 13 and 15. 3. Digital diagnostic link length supported in accordance with maximum km of fiber												
Test Configuration	Xhaul traffic and RAN services configuration												
Pass/Fail Criteria:	1. Traffic is correctly forwarded without frame loss on step 4, 6, 9, 11, 13 and 15. 2. Maximum km of fiber must be equal or greater than the specified value. 3. Minimum km of fiber must be equal or less than the specified value. 4. Digital diagnostic link length supported must be in accordance with maximum km of fiber.												
Results	[] Pass [] Pass w/X [] Dropped [] Blocked [] Failed												
Test Output	Comments: The results are presented in the table below: Forward Error Correction (FEC): state/type <table><tr><td>Transceiver</td><td>Recommendation</td><td>Measured values</td></tr><tr><td>Vendor</td><td>Minimum km</td><td></td></tr><tr><td>Model</td><td></td><td></td></tr><tr><td>SN</td><td>Maximum km</td><td></td></tr></table>	Transceiver	Recommendation	Measured values	Vendor	Minimum km		Model			SN	Maximum km	
Transceiver	Recommendation	Measured values											
Vendor	Minimum km												
Model													
SN	Maximum km												

6.4.2.6 Transceiver extinction ratio and eye diagram

This test case is FFS.

3

Test ID	PHY.06	Classification	Functionality												
Test Title	Transceiver extinction ratio and eye diagram														
Test Objective:	The aim of the test is to check the conformity														
Pre-Req	Running service will be interrupted														
Test Topology:															
Procedure:	<u>Detailed Test procedure:</u> <u>Test Procedure:</u> 18. x <u>Verifications needed:</u> 1. x 2.														
Test Configuration	Xhaul traffic and RAN services configuration														
Pass/Fail Criteria:	5. x														
Results	[] Pass [] Pass w/X [] Dropped [] Blocked [] Failed														
Test Output	Comments: The results are presented in the table below: <table><tr><td>Transceiver</td><td>Recommendation</td><td>Measured values</td></tr><tr><td>Vendor</td><td></td><td></td></tr><tr><td>Model</td><td></td><td></td></tr><tr><td>SN</td><td></td><td></td></tr></table>			Transceiver	Recommendation	Measured values	Vendor			Model			SN		
Transceiver	Recommendation	Measured values													
Vendor															
Model															
SN															

4

1 6.4.2.7 Transceiver information display

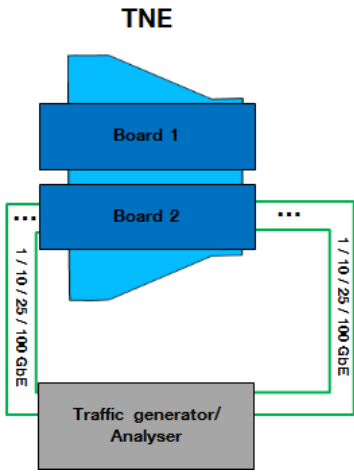
2

Test ID	PHY.07	Classification	Functionality
Test Title	Transceiver information display		
Test Objective:	The aim of the test is to check information of optical interface remotely		
Pre-Req			
Test Topology:			
Procedure:	<u>Detailed Test procedure:</u> <u>Test Procedure and Verification:</u> 1. Insert the transceiver. 2. Check the type of optical transceiver remotely and all available digital diagnostic values. 3. Note the result.		
Test Configuration	Xhaul traffic and RAN services configuration		
Pass/Fail Criteria:	The system MUST show: • Type/class • SFP vendor • Part number • S/N • Manufacturing date • Link length supported. • Temperature • Supply voltage • Tx bias current • Tx output power • Rx optical power		
Results	[] Pass [] Pass w/X [] Dropped [] Blocked [] Failed		
Test Output	Comments:		

3

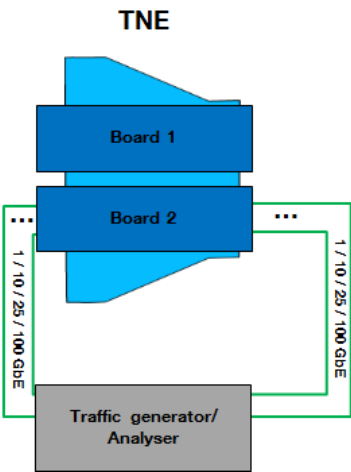
6.4.3 Alarm and Monitoring Test Cases

6.4.3.1 Alarm on board reset and plug out.

Test ID	Alarm & Monitoring.01	Classification	Functionality
Test Title	Alarm on board reset and plug out		
Test Objective:	The aim of the test is to check if TNE send and cleared alarm when board is rebooted or plugout/plugin		
Pre-Req	Running service will be interrupted		
Test Topology:			
Procedure:	<u>Detailed Test procedure:</u> <u>Test Procedure & Verifications needed:</u> 1. Perform a board reset/reboot. 2. Measure and report the duration to raise the alarm from the line/interface disconnection. 3. After board recover 4. Measure and report the duration to clear the alarm from the line/interface disconnection. 5. Perform a board plug-out. 6. Measure and report the duration to raise the alarm from the line/interface disconnection. 7. Perform a board plugin. 8. Measure and report the duration to clear the alarm from the line/interface disconnection		
Test Configuration	Xhaul traffic and RAN services configuration		
Pass/Fail Criteria:	1- Alarm reported within 1 minute on step 2 and 6 2- Alarm cleared within 1 minute on step 4 and 8		

Results	☐ Pass ☐ Pass w/X ☐ Dropped ☐ Blocked ☐ Failed
Test Output	Alarms is reported xx second. Alarms is cleared xx second

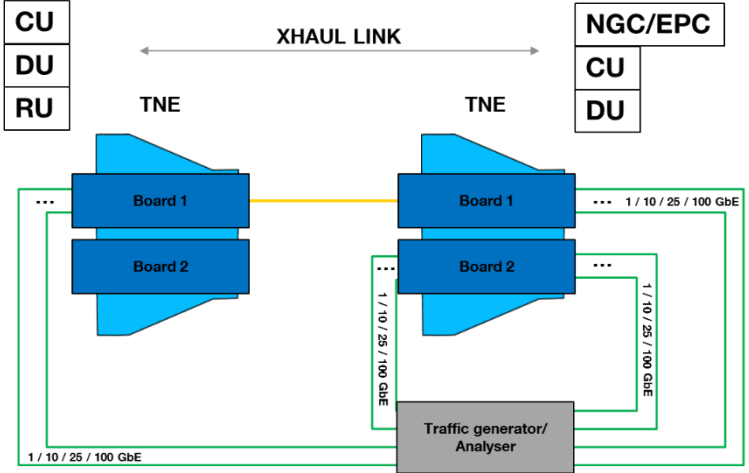
1 6.4.3.2 Alarm time stamping

Test ID	Alarm & Monitoring.02	Classification	Functionality
Test Title	Alarm time stamping		
Test Objective:	The aim of the test is to check if for TNE is possible to determine the exact time when the alarm was raised		
Pre-Req	Running service will be interrupted		
Test Topology:			
Procedure:	<u>Detailed Test procedure:</u> <u>Test Procedure:</u> 1. Simulate few alarms. 2. Show alarms history. <u>Verifications needed:</u> 1. Check if time stamping (Date and Time) is available for all alarms and events/warnings. 2. Check if is possible to retrieve the exact time when the alarm was raised from the alarm timestamp		
Test Configuration	Xhaul traffic and RAN services configuration		
Pass/Fail Criteria:	1- There MUST be possible to check time stamping on TNE 2- There MUST be possible to determine the exact time when the alarm was raised		
Results	[] Pass [] Pass w/X [] Dropped [] Blocked [] Failed		
Test Output			

6.4.3.3 Ethernet link down alarm

This test is FFS.

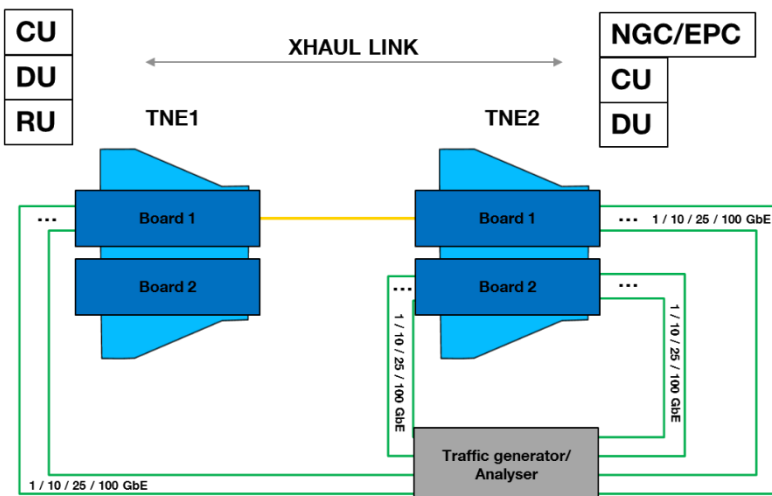
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Test ID	Alarm & Monitoring.03	Classification	Functionality
Test Title	Ethernet link down alarm		
Test Objective:	The aim of the test is to generate and clear an Ethernet link down alarm xhaul link		
Pre-Req	Running service will be interrupted		
Test Topology:			
Procedure:	<u>Detailed Test procedure:</u> <u>Test Procedure and Verifications needed:</u> 1. Perform optical Point to Point port fibre disconnection. 2. Measure and report the duration to raise the alarm. 3. Perform optical Point to Point port fibre connection. 4. Measure and report the duration to clear the alarm.		
Test Configuration	Xhaul traffic and RAN services configuration		
Pass/Fail Criteria:	1. "Ethernet link down" for optical Point to Point reported within 1 minute on step. 2. "Ethernet link down" for optical Point to Point cleared within 1 minute on step 4		
Results	[] Pass [] Pass w/X [] Dropped [] Blocked [] Failed		
Test Output	"Ethernet link down" alarm is reported xx minutes yy seconds. "Ethernet link down" alarm is cleared yy seconds		

4

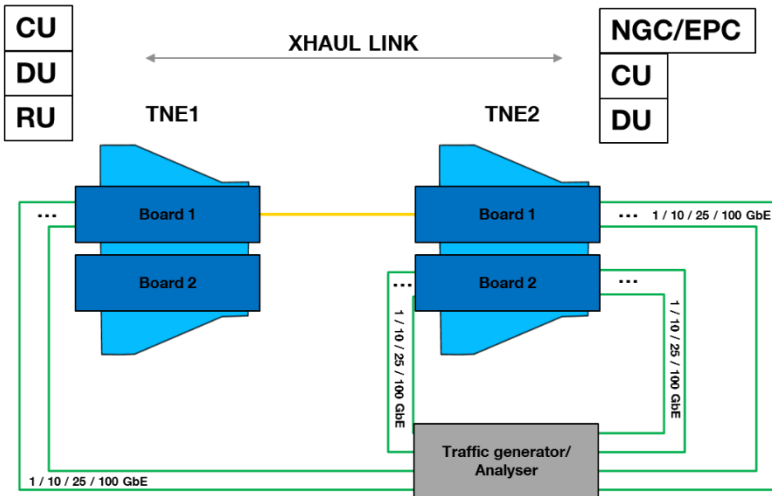
6.4.3.4 Dying Gasp (DG)

This test is FFS.

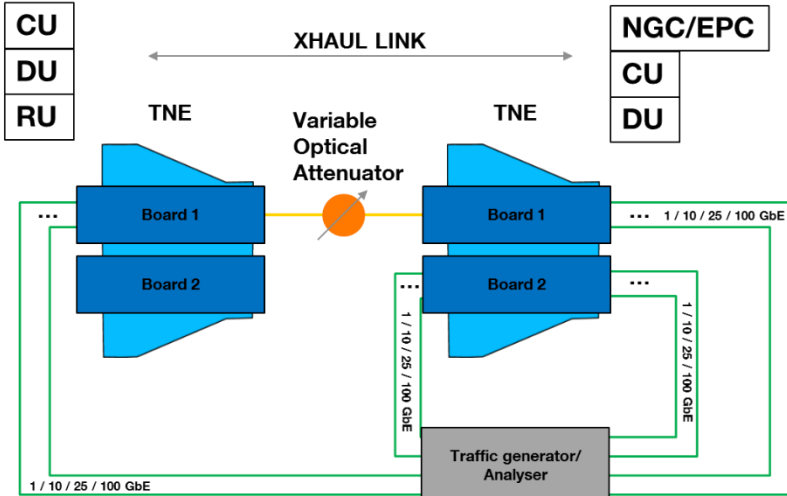
Test ID	Alarm & Monitoring.04	Classification	Functionality
Test Title	Dying Gasp (DG)		
Test Objective:	The aim of the test is to generate and clear a Dying Gasp (DG) of TNE alarm		
Pre-Req	Running service will be interrupted		
Test Topology:			
Procedure:	<u>Detailed Test procedure:</u> <u>Test Procedure and Verifications needed:</u> 1. Perform the TNE1 or O-RU/DU/CU electrical power off. 2. Measure and report the duration to raise the alarm on the TNE2 when the second TNE1 electrical power off 3. Perform an TNE1 electrical power on 4. Measure and report the duration to clear the alarm on the TNE2 when the TNE1 electrical power on 5. Repeat the test few times.		
Test Configuration	Xhaul traffic and RAN services configuration		
Pass/Fail Criteria:	1- DG alarm reported within 1 minute on step 2 2- DG alarm cleared within 2 minutes on step 4		
Results	[] Pass [] Pass w/X [] Dropped [] Blocked [] Failed		
Test Output	DG alarm is reported xx minutes yy seconds. DG alarm is cleared xx minutes yy seconds		

6.4.3.5 No Loss Of Signal alarm for Dying Gasp (DG) alarm

This test is FFS.

Test ID	Alarm & Monitoring.05	Classification	Functionality
Test Title	No Loss Of Signal alarm for Dying Gasp (DG) alarm		
Test Objective:	The aim of the test is to check alarms reported when all/many links on one TNE report DG (Dying Gasp) alarm		
Pre-Req	Running service will be interrupted		
Test Topology:			
Procedure:	<u>Detailed Test procedure:</u> <u>Test Procedure:</u> 1. Perform all/many TNE1 or O-RU/DU/CU electrical power off. <u>Verifications needed:</u> 1. Check if last link reports DG then no LOS alarm is reported (Check if this behaviour is enabled) 2. Wait and check if no LOS alarm was reported		
Test Configuration	Xhaul traffic and RAN services configuration		
Pass/Fail Criteria:	1- No LOS alarm is reported		
Results	[] Pass [] Pass w/X [] Dropped [] Blocked [] Failed		
Test Output	DG alarm is reported xx minutes yy seconds DG alarm is cleared xx minutes yy seconds		

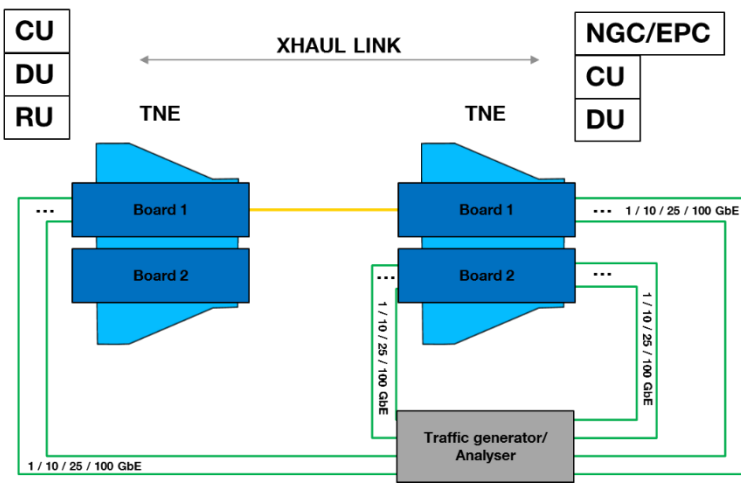
1 6.4.3.6 Transceiver (SFP, QSFP,...) Rx optical power alarm threshold

Test ID	Alarm & Monitoring.06	Classification	Functionality
Test Title	Transceiver (SFP, QSFP,...) Rx optical power alarm threshold		
Test Objective:	The aim of the test is to check if the transceiver received optical power parameters exceed warning high and low threshold then alarm is reported. When Rx signal is lower and higher than threshold defined then alarm is cleared properly.		
Pre-Req	Running service will be interrupted		
Test Topology:			
Procedure:	<u>Detailed Test procedure:</u> <u>Test Procedure and Verifications needed:</u> 1. Configure Rx power parameter threshold high and low. 2. Connect variable optical attenuator and change optical power. Rx signal must be higher than the threshold range. 3. Measure and report the duration to raise the alarm. 4. Check that alarm is cleared if Rx power is in the threshold range. 5. Measure and report the duration to clear the alarm. 6. Connect variable optical attenuator and change optical power. Rx signal must be lower than the threshold range. 7. Measure and report the duration to raise the alarm. 8. Check that alarm is cleared if Rx power is in the threshold range. 9. Measure and report the duration to clear the alarm.		
Test Configuration	Xhaul traffic and RAN services configuration		
Pass/Fail Criteria:	1- Alarm "RX high power" reported within 1 minute on step 2 2- Alarm "RX high power" cleared within 1 minute on step 4 3- Alarm "RX low power" reported within 1 minute on step 6 4- Alarm "RX low power" cleared within 1 minute on step 8		
Results	[] Pass [] Pass w/X [] Dropped [] Blocked [] Failed		
Test Output	Alarm "RX high power" is reported xx second. Alarm "RX high power" is cleared xx second.		

Alarm "RX low power" is reported xx second.
Alarm "RX low power" is cleared xx second

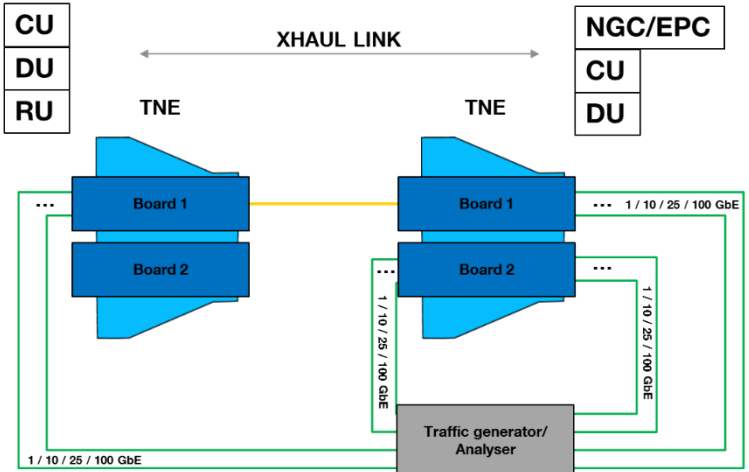
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6.4.3.7 Transceiver (SFP, QSFP,...) Tx optical power alarm threshold

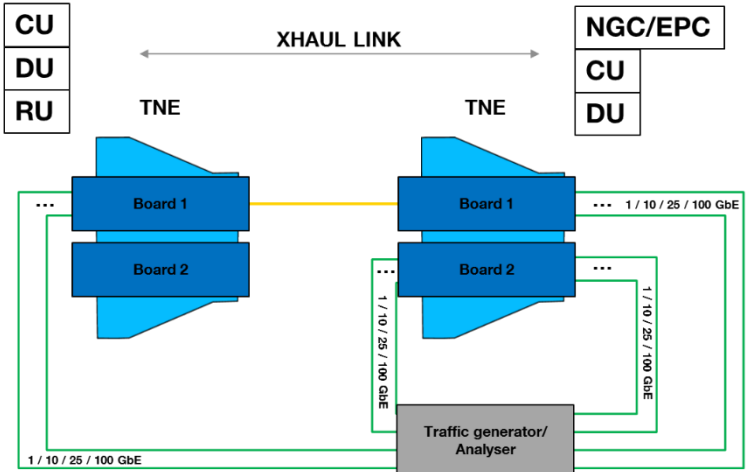
Test ID	Alarm & Monitoring.07	Classification	Functionality
Test Title	Transceiver (SFP, QSFP,...) Tx optical power alarm threshold		
Test Objective:	The aim of the test is to check if the Ethernet transceiver emitted optical power parameters exceed warning high and low threshold then alarm is reported. When Tx signal is lower and higher than threshold defined then alarm is cleared properly.		
Pre-Req	Running service will be interrupted		
Test Topology:			
Procedure:	Detailed Test procedure: Test Procedure and Verifications needed: 1. Configure Tx power parameter threshold high, Tx signal must be higher than the threshold range. 2. Measure and report the duration to raise the alarm. 3. Configure Tx power parameter threshold high, Tx signal must be in the threshold range. 4. Check that alarm is cleared. 5. Measure and report the duration to clear the alarm. 6. Configure Tx power parameter threshold low, Tx signal must be lower than the threshold range. 7. Measure and report the duration to raise the alarm. 8. Configure Tx power parameter threshold low, Tx signal must be in the threshold range. 9. Check that alarm is cleared. 10. Measure and report the duration to clear the alarm.		
Test Configuration	Xhaul traffic and RAN services configuration		

Pass/Fail Criteria:	1- Alarm "TX high power" reported within 1 minute on step 2 2- Alarm "TX high power" cleared within 1 minute on step 4 3- Alarm "TX low power" reported within 1 minute on step 7 4- Alarm "TX low power" cleared within 1 minute on step 9
Results	☐ Pass ☐ Pass w/X ☐ Dropped ☐ Blocked ☐ Failed
Test Output	Alarm "TX high power" is reported xx second. Alarm "TX high power" is cleared xx second. Alarm "TX low power" is reported xx second. Alarm "TX low power" is cleared xx second. xx is to be defined by the vendor.

1 6.4.3.8 Severity handling of alarms

Test ID	Alarm & Monitoring.08	Classification	Functionality
Test Title	Severity handling of alarms		
Test Objective:	The aim of the test is to verify the severity handling of alarms		
Pre-Req	Running service will be interrupted		
Test Topology:			
Procedure:	<u>Detailed Test procedure:</u> <u>Test Procedure and Verifications needed:</u> 1. Change the severity of an alarm. 2. Generate the alarm. 3. Check that the alarm is reported with the configured severity level. 4. Change the severity of the selected alarm with the default value. 5. Generate the alarm. 6. Check that the alarm is reported with the configured severity level.		
Test Configuration	Xhaul traffic and RAN services configuration		
Pass/Fail Criteria:	Alarm must be reported with the configured severity level on step 3 and step 6		
Results	[] Pass [] Pass w/X [] Dropped [] Blocked [] Failed		
Test Output	“		

6.4.3.9 Port-ID (IDentification) maintenance

Test ID	Alarm & Monitoring.08	Classification	Functionality
Test Title	Port-ID (IDentification) maintenance		
Test Objective:	The aim of the test is to check the ID feature		
Pre-Req	Running service will be interrupted		
Test Topology:			
Procedure:	<u>Detailed Test procedure:</u> <u>Test Procedure and Verifications needed:</u> 1. Configure Port-ID at TNE 2. Check if Port-ID modification can be done with TNE in administrative state up. 3. Connect an Ethernet teste on the link.		
Test Configuration	Xhaul traffic and RAN services configuration		
Pass/Fail Criteria:	1. Port-ID creation and modification should be possible without service impact for customer on step 2 2. Port-ID should be retrieved on step 3		
Results	[] Pass [] Pass w/X [] Dropped [] Blocked [] Failed		
Test Output			

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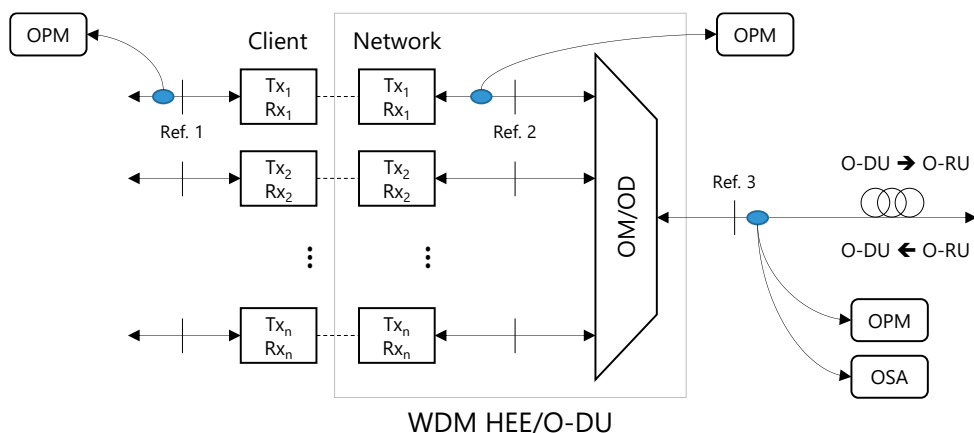
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7 WDM-based Fronthaul Transport Testing

Fronthaul in O-RAN is defined as the connectivity in the RAN infrastructure between the Distributed Unit (O-DU) and Radio Unit (O-RU).

7.1 Fronthaul Testing use cases.

7.1.1 Transmit Power

Test ID	DWDM.HEE.01	Classification	Functionality
Test Title	Transmitting power at the DWDM head end equipment (HEE)		
Test Objective:	To test transmitting power levels at the HEE		
Pre-Req	Running service will be interrupted		
Test Topology:	Refer to O-RAN.WG9.WDM-v01.00, Section 2.3 Semi-Active WDM system  The diagram illustrates the test topology for DWDM.HEE.01. It shows a 'Client' section with multiple transmitters (Tx₁, Tx₂, ..., Tx_n) and receivers (Rx₁, Rx₂, ..., Rx_n). A reference point 'Ref. 1' is located at the output of the Client Tx section. The 'Network' section contains corresponding receivers (Rx₁, Rx₂, ..., Rx_n) and transmitters (Tx₁, Tx₂, ..., Tx_n). A reference point 'Ref. 2' is located at the output of the Network Tx section. The 'WDM HEE/O-DU' section is represented by a large trapezoidal block labeled 'OM/OD'. A reference point 'Ref. 3' is located at the input of the OM/OD block. The output of the OM/OD block is connected to an 'O-DU → O-RU' link, which is represented by two overlapping circles. This link is connected to an 'O-DU ← O-RU' link, which is also represented by two overlapping circles. The output of the O-DU ← O-RU link is connected to an 'OPM' (Optical Power Meter) and an 'OSA' (Optical Spectrum Analyzer).		
Required tester	Optical power meter (OPM), optical spectrum analyzer (OSA)		
Test Configuration	1. Disable all the other transmitters (e.g. Tx _n) which are NOT under test;		
Procedure:	Detailed Test procedure: 1. Connect the OPM and OSA to the reference points shown in the above test configuration. 2. Read the power level of Ref. 1 (Client Tx). 3. Read the power level of Ref. 2 (Network Tx). 4. Read the power level of Ref. 3 (Network Port of OM/OD) 5. Measure the center frequency and the optical spectrum at Ref. 3 (Network Port of OM/OD) 6. To carry out the procedure for all the other transmitters.		

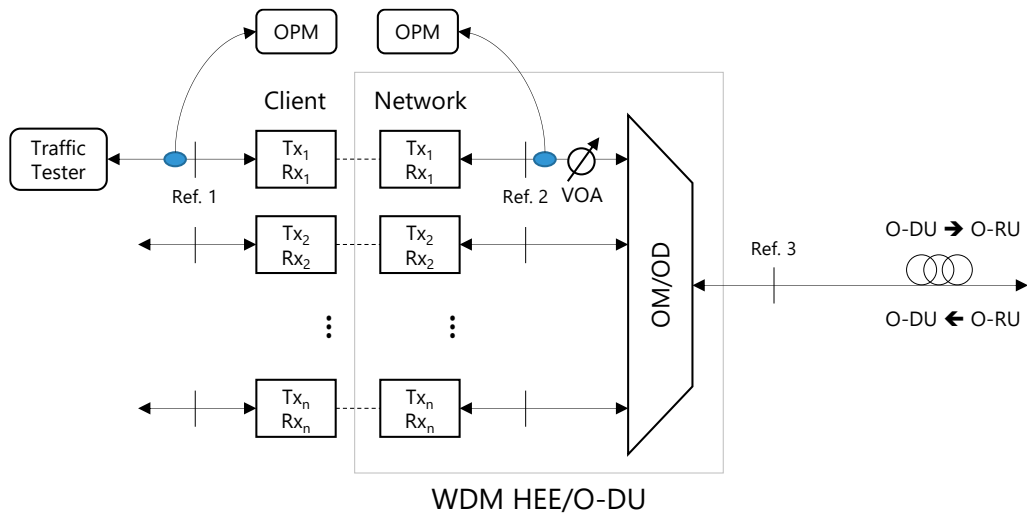
	<u>Verifications needed:</u> 1. Record Tx power readings from the host network management.														
Pass/Fail Criteria:	1. Tx power at Ref. 1 should be -8 to 0 dBm. 2. Tx power at Ref. 2 complies with O-RAN.WG9.WDM-v01.00, Table 6-10, depending on the pluggable variants. 3. Tx power at Ref. 3 should be within the range of Ref. 2 deducting maximum insertion loss of 4.6 dB. 4. Optical spectrum complies with O-RAN.WG9.WDM-v01.00, Table 6-10, depending on the pluggable variants.														
Test Output	<table><tr><td>Interface</td><td>Network Mgmt. Tx power (dBm)</td><td>OPM Tx power (dBm)</td></tr><tr><td>Client</td><td></td><td></td></tr><tr><td>Network</td><td></td><td></td></tr><tr><td>OM/OD</td><td></td><td></td></tr></table>			Interface	Network Mgmt. Tx power (dBm)	OPM Tx power (dBm)	Client			Network			OM/OD		
Interface	Network Mgmt. Tx power (dBm)	OPM Tx power (dBm)													
Client															
Network															
OM/OD															

1

2

7.1.2 Receive Sensitivity

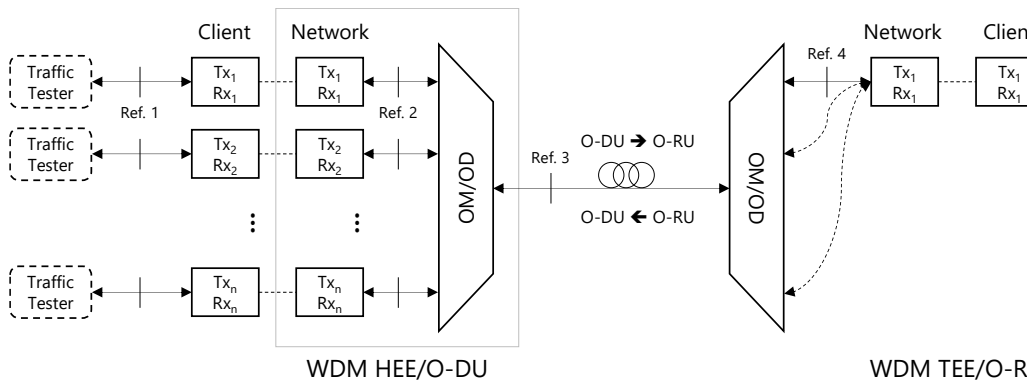
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Test ID	DWDM.HEE.02	Classification	Functionality
Test Title	Receiving sensitivity at the DWDM head end equipment (HEE)		
Test Objective:	To test received power level and receiving sensitivity at the HEE		
Pre-Req	Running service will be interrupted		
Test Topology:	Refer to O-RAN.WG9.WDM-v01.00, Section 2.3 Semi-Active WDM system 		
Required tester	Optical power meter (OPM), variable optical attenuator (VOA), traffic tester (eCPRI)		
Test Configuration			
Procedure:	Detailed Test procedure: 1. Connect the OPM, VOA, and traffic tester to the reference points shown in the above test configuration. 2. Disable all the other transmitters (e.g. Tx_n) on both O-DU and O-RU ends which are NOT under test. 3. Configure the traffic tester accordingly to terminate the traffic flow from the O-RU. 4. Gradually increase the attenuation of VOA, until the BER is stable at the threshold of 1e-12 for 10 minutes. 5. Read the Tx/Rx power level of Ref. 1 (Client Tx/Rx); 6. Read the Rx power level of Ref. 2 (Network Rx). Verifications needed: 1. Verify the Tx/Rx power levels between the traffic tester and the HEE Client are in the dynamic range of client transceivers. 2. Record Rx power readings from the host network management.		

Pass/Fail Criteria:	1. Rx power at Ref. 2 complies with O-RAN.WG9.WDM-v01.00, Table 6-10, depending on the pluggable variants.		
Test Output	Interface	Network Mgmt. Rx power (dBm)	OPM Rx power (dBm)
	Network		

1 7.1.3 Auto-tuning test

2

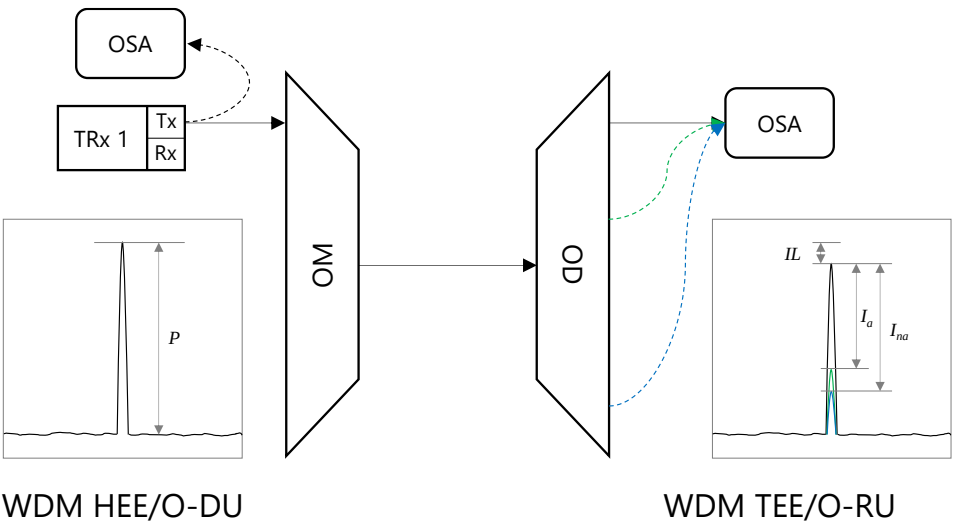
Test ID	WDM.auto-tuning.01	Classification	Functionality
Test Title	Paired frequency tuning at tail end (TE)		
Test Objective:	To verify if the tail-end transceiver can be auto-tuned to the connected filter port		
Pre-Req	Optical transceivers supporting paired auto-tuning		
Test Topology:	Applicable to O-RAN.WG9.WDM-v01.00, Passive, Semi-Active, and Active WDM fronthaul system  WDM HEE/O-DU WDM TEE/O-R		
Required tester	Optional traffic tester (eCPRI)		
Test Configuration			
Procedure:	Detailed Test procedure: 1. Connect the client and network transceivers at the WDM HEE/O-DU according to the test topology above. 2. At the WDM HEE/O-DU, provision the frequency of all the populated HE Network transceiver (Tx₁/Rx₁ ... Tx_n/Rx_n), as well as the paired frequency for the TE transceivers.		

	3. Enable the channel ports (laser on) and optionally testing traffic. 4. Plug and enable the TE transceiver under test into the O-RU (passive WDM) or remote WDM equipment. 5. Connect fibers between the TE transceiver and one of the provisioned OM/OD ports. 6. Monitor and read the Rx signal at the corresponding HE transceiver. 7. Repeat Step 5 and 6 for the other OM/OD ports. Verifications needed: 1. Verify whether the TE transceiver can be automatically tuned to the correct uplink frequency. 2. Verify whether the channel status at the WDM HEE/O-DU is “link up” and read the received optical power. 3. Record the time duration from the moment connecting the OM/OD port to the TE transceiver being auto tuned. 4. Optionally verify whether the traffic is resumed. The traffic needs to be looped back from the WDM TEE/O-RU.								
Pass/Fail Criteria:	1. After power cycle or being reconnected to a new OM/OD port, the TE transceiver is automatically tuned to the correct uplink frequency. 2. Rx power at Ref. 2 complies with O-RAN.WG9.WDM-v01.00, Table 6-10, depending on the pluggable variants. 3. Optionally the traffic can be resumed.								
Test Output	<table><tr><td>TE Channel Frequency</td><td>Tuning Time</td><td>Link Status</td><td>Rx Power at HE</td></tr><tr><td></td><td></td><td></td><td></td></tr></table>	TE Channel Frequency	Tuning Time	Link Status	Rx Power at HE				
TE Channel Frequency	Tuning Time	Link Status	Rx Power at HE						

1

2

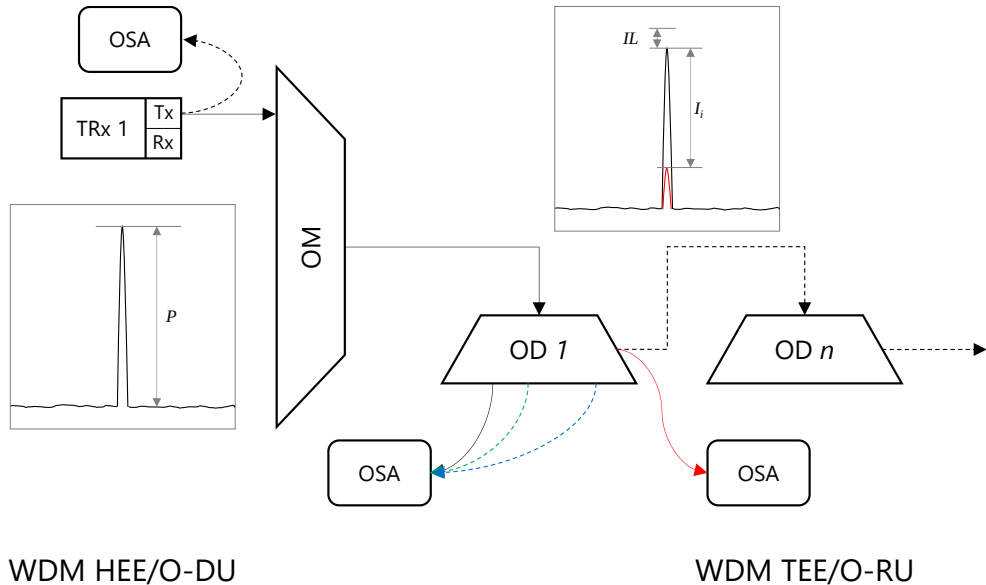
1 7.1.4 Crosstalk testing

Test ID	WDM.crosstalk.01	Classification	Functionality
Test Title	WDM Xhaul transport inter-channel crosstalk		
Test Objective:	To verify if the OM/OD meets the isolation specification of the WDM transport link		
Pre-Req			
Test Topology:	Applicable to O-RAN.WG9.WDM-v01.00, Passive, Semi-Active, and Active WDM fronthaul system  WDM HEE/O-DU WDM TEE/O-RU		
Required tester	Optical spectrum analyzer (OSA)		
Test Configuration	- Definitions of IL, I_a, and I_{na} are referred to ITU-T G.671 - OSA shall be capable of a sensitivity of at least -65 dBm		
Procedure:	Detailed Test procedure: - Provision and enable a WDM transceiver for the respective WDM solution (e.g., DWDM, MWDM); - Use the OSA to measure the spectrum peak power at Tx (P in dBm); - Connect the transceiver to the OM, and only a short jumper cable is needed between the OM and the OD. - Connect to the OSA to the OD output port of the wanted channel, and measure the spectrum peak power to calculate the insertion loss (IL in dB); - Connect to the OSA to one of the adjacent channel ports of the wanted one, and measure the spectrum peak power to calculate the adjacent channel isolation (I_a in dB); - Connect to the OSA to any of the other OD output ports and measure the spectrum peak power to calculate the non-adjacent channel isolation (I_{na} in dB).		

	<u>Verifications needed:</u> 1. OSA settings shall be always kept the same. 2. For the measurement of each wanted channel, at least one I_a and one I_{na} shall be measured. 3. If more than one I_a or I_{na} are measured, always take the smallest (worst) value from the result set.												
Pass/Fail Criteria:	1. Depending on the type of WDM OM/OD, IL, I_a, and I_{na} shall be compliant with O-RAN.WG9.WDM-v01.00, Table 9-10. 2. The actual tolerable inter-channel crosstalk (C_c in dB) for 100GHz-spaced DWDM is specified in G.698.1, Tables 8-1 to 8-4. C_c can be calculated following: $C_c = d + 10 \log \left[2 \times 10^{\frac{-I_a}{10}} + (k - 3) 10^{\frac{-I_{na}}{10}} \right]$where d is the maximum allowable power difference of a particular transceiver type specified in O-RAN.WG9.WDM-v01.00												
Test Output	<table><tr><td>WDM Type</td><td>Transceiver</td><td>Channel</td><td>IL</td><td>I_a</td><td>I_{na}</td></tr><tr><td>e.g. DWDM or MWDM</td><td>e.g. 10G, 25G, BiDi</td><td></td><td></td><td></td><td></td></tr></table>	WDM Type	Transceiver	Channel	IL	I_a	I_{na}	e.g. DWDM or MWDM	e.g. 10G, 25G, BiDi				
WDM Type	Transceiver	Channel	IL	I_a	I_{na}								
e.g. DWDM or MWDM	e.g. 10G, 25G, BiDi												

1

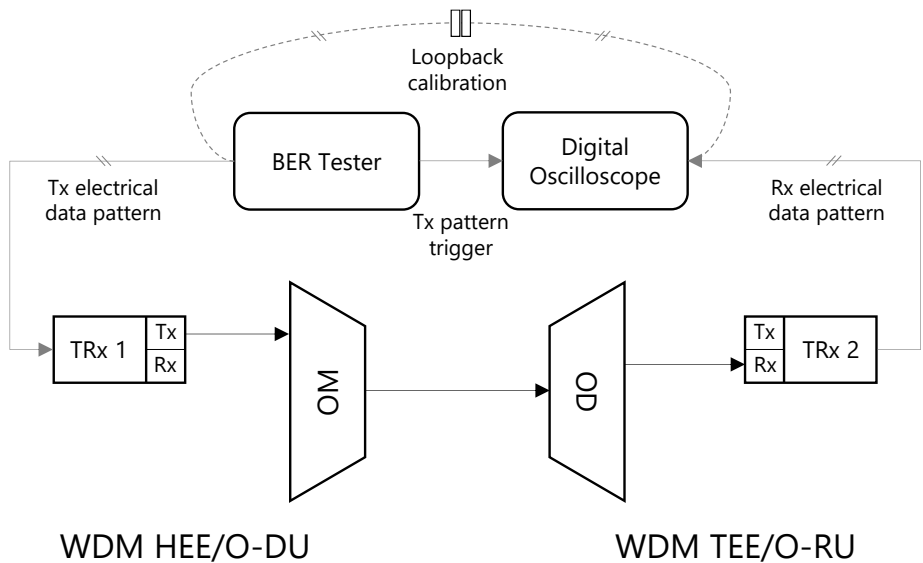
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Test ID	WDM.crosstalk.02	Classification	Functionality
Test Title	WDM Xhaul transport interferometric crosstalk		
Test Objective:	To verify if the OM/OD meets the isolation specification of a WDM transport link with add/drop nodes		
Pre-Req			
Test Topology:	Applicable to O-RAN.WG9.WDM-v01.00, Passive, Semi-Active, and Active WDM fronthaul system  WDM HEE/O-DU WDM TEE/O-RU		
Required tester	Optical spectrum analyzer (OSA)		
Test Configuration	OSA shall be capable of a sensitivity of at least -65 dBm		
Procedure:	Detailed Test procedure: - Provision and enable a WDM transceiver for the respective WDM solution (e.g., DWDM, MWDM), of which the channel is dropped at the measured OD (i.e. OD 1 in the test example); - Follow the procedure specified in WDM.crosstalk.01 (Step 2 to 6). - Measure the spectrum peak power at the OD network port to calculate the insulation (I_i) for interferometric crosstalk. Note: The signal from the first source will be extinguished by the OADMs where the channel is dropped, and the interferometric crosstalk will be dominated by the performance of the last OADM that dropped and added that channel. This interferometric crosstalk, therefore, does not constrain the number of OADMs in a link. [Ref to G.698.1] Verifications needed: - OSA settings shall be always kept the same. - For the measurement of each wanted channel, at least one I_a and one I_{na} shall be measured.		

	3. If more than one I_a or I_{na} are measured, always take the smallest (worst) value from the result set.														
Pass/Fail Criteria:	1. Depending on the type of WDM OM/OD, IL, I_a, and I_{na} shall be compliant with O-RAN.WG9.WDM-v01.00, Table 9-10. 2. The actual tolerable interferometric crosstalk for 100GHz-spaced DWDM is specified in G.698.1, Tables 8-1 to 8-4.														
Test Output	<table><tr><th>WDM Type</th><th>Transceiver</th><th>Dropped Channel</th><th>IL</th><th>I_a</th><th>I_{na}</th><th>I_i</th></tr><tr><td>e.g. DWDM or MWDM</td><td>e.g. 10G, 25G, BiDi</td><td></td><td></td><td></td><td></td><td></td></tr></table>	WDM Type	Transceiver	Dropped Channel	IL	I_a	I_{na}	I_i	e.g. DWDM or MWDM	e.g. 10G, 25G, BiDi					
WDM Type	Transceiver	Dropped Channel	IL	I_a	I_{na}	I_i									
e.g. DWDM or MWDM	e.g. 10G, 25G, BiDi														

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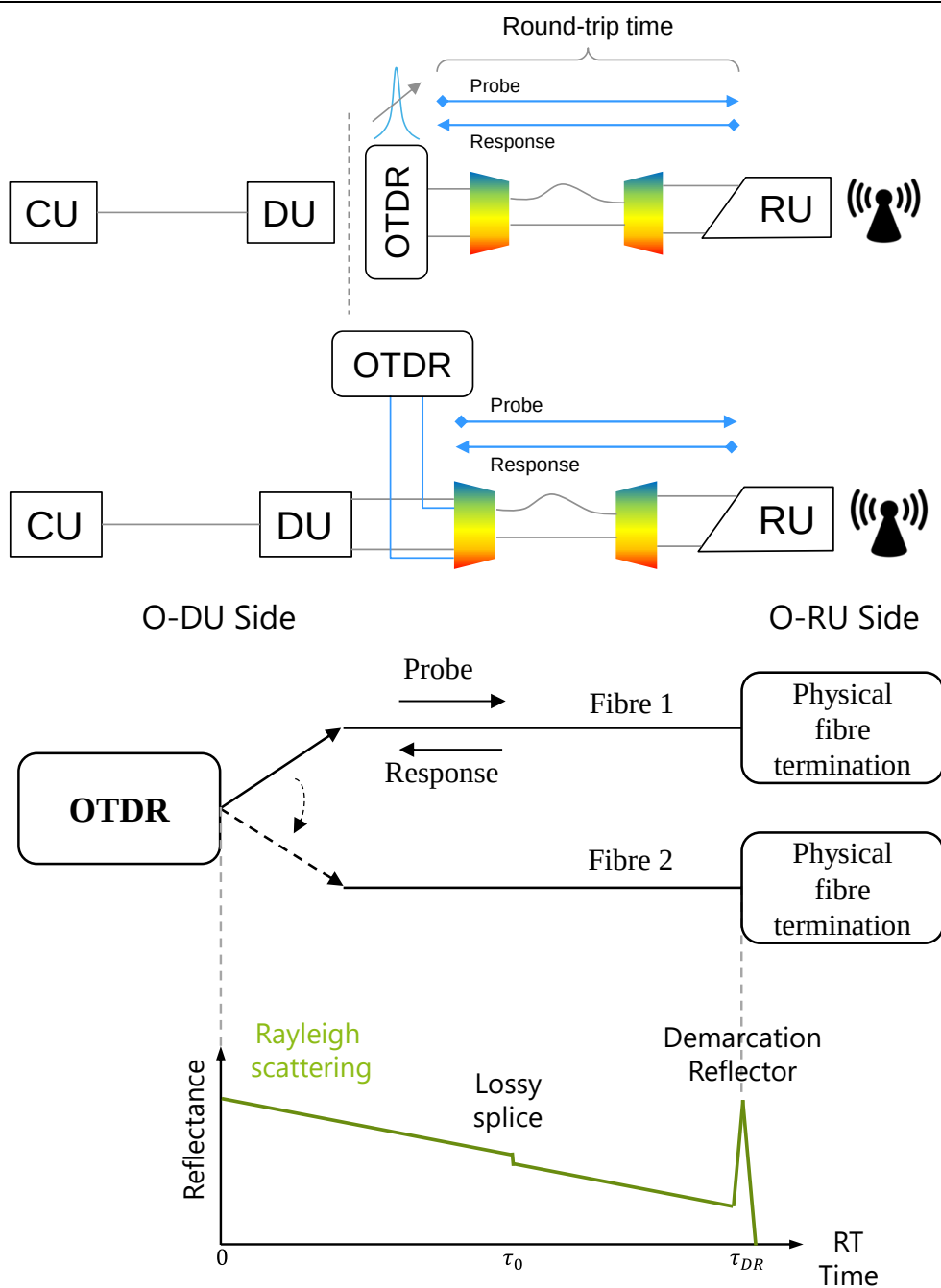
7.1.5 Latency Test

Test ID	WDM.latency.01	Classification	Characterization
Test Title	WDM Xhaul transport latency measurement		
Test Objective:	To measure the PHY latency from an optical transceiver to the other of the WDM transport link		
Pre-Req			
Test Topology:	Applicable to O-RAN.WG9.WDM-v01.00, Passive, Semi-Active, and Active WDM fronthaul system 		
Required tester	General purpose bit error rate tester (BERT); highspped digital oscilloscope (OSC)		

Test Configuration	<
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Test ID	WDM.latency.02	Classification	Characterization
Test Title	Fiber latency and asymmetry measurement		
Test Objective:	To measure the fiber latency and asymmetry for a particular x-haul PHY link		
Pre-Req			
Test Topology:	Applicable to O-RAN.WG9.WDM-v01.00, Passive, Semi-Active, and Active WDM fronthaul system		



Required tester

General purpose optical time-domain reflectometer (OTDR)

Test Configuration

1. The accuracy of the measurement depends on the length of the OTDR pulse, usually in a range of few nanoseconds.
2. To achieve a sufficient signal-to-noise ratio, longer probe pulses may be required for longer fiber links, because of a higher loss.
3. In case of using a correlated OTDR, a bit sequence at a rate of a few Gbit/s can be used as the probe signal. The reflected signals are then cross-correlated with the transmitted bit pattern, improving the resolution of this correlated OTDR to picoseconds.

	4. For measuring any wavelength asymmetry, CWDM, DWDM or tunable OTDR source is required. Note 1: Link asymmetry compensation is elaborated in Appendix IV, ITU-T G.8271/Y.1366 (03/2020). Note 2: OTDR is specified as a fiber link monitoring for WDM systems in ITU-T G.697.										
Procedure:	Detailed Test procedure: 1. Insert the OTDR at the optical transmitter interface (e.g.O-DU/CSR output, WDM HEE Tx, etc.); 2. If possible, insert also a demarcation reflector in front of the actual fiber termination; 3. Configure the pulse length, wavelength, measurement duration etc. of the OTDR, according to the expected link range; 4. The round-trip latency can be calculated between the start event of pulse launch and the end/demarcation reflective event, and the one-way latency of the measured link is exactly half of it; 5. In case of measuring the latency of multiple wavelengths, change the OTDR wavelength accordingly and repeat Step 3-4; 6. In case of measuring the latency of multiple links from the same transmitter side, switch the OTDR connection to another link and repeat Step 3-5.										
Pass/Fail Criteria:	1. Informative result only										
Test Output	<table><tr><td>Link</td><td>Transceiver</td><td>Channel</td><td>RTL</td><td>Asymmetry</td></tr><tr><td></td><td>e.g. 10G, 25G, BiDi</td><td></td><td></td><td></td></tr></table>	Link	Transceiver	Channel	RTL	Asymmetry		e.g. 10G, 25G, BiDi			
Link	Transceiver	Channel	RTL	Asymmetry							
	e.g. 10G, 25G, BiDi										

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7.1.6 OAM Test

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Applicable to O-RAN.WG9.WDM-v01.00, in semi-active or active fronthaul WDM systems, active

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WDM equipment on the O-DU side is configured to transmit management requests to O-RUs and

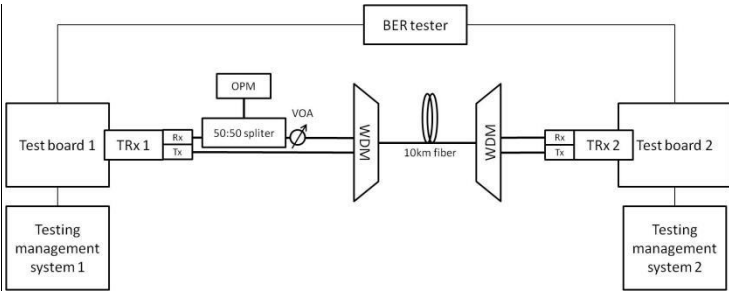
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manage the colored optical modules within the O-RUs. The optical modules in O-RUs are

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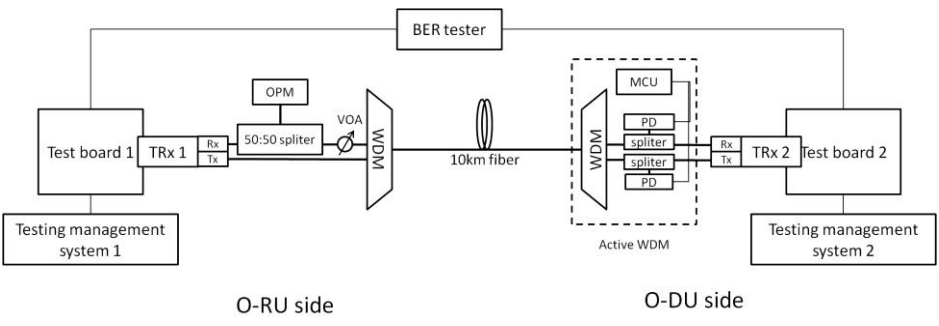
configured to send back management request responses to the active WDM equipment. The

management request responses carry the status information of the colored optical modules in the O-RUs in the table of Key Performance Indicators in section 6.1 of O-RAN.WG9.WDM-v01.00, including the wavelength, driving voltage, driving current, the launched optical power of the transmitter, the received optical power of the receiver, etc. In this way, OAM tests are crucial for the interoperability between optical modules and active WDM equipment from different vendors.

Test ID	WDM.OAM.01	Classification	Characterization
Test Title	OAM test for optical module OAM data link layer		
Test Objective:	To test the OAM data link communication of the optical modules		
Pre-Req			
Test Topology:	Applicable to O-RAN.WG9.WDM-v01.00, optical modules.  An optical link consists of WDM components, like MUX/DEMUX and 10km optical fiber, which are used to connect optical modules at two ends to simulate the Fronthaul optical transport channel. Each optical module is installed on a test board like a transponder, on which a management system installs. The information of optical modules can be configured and read by the test boards through the management system. To test the active function specified in section 6.1 of O-RAN.WG9.WDM-v01.00, the optical module on one side can receive and verify the OAM information from the optical module configured by the corresponding management system on the other side, and send back the verification result to the optical module on the other side. To test the configuration and inquiry functions specified in section 6.1 of O-RAN.WG9.WDM-v01.00, the optical module on one side can also be configured by the corresponding testing management system to send the testing OAM configuration or inquiry messages to the optical module on the other side. Then, the optical module on one side can receive and verify the corresponding OAM responses from the optical module on the other side. If the corresponding OAM responses are correct, the optical modules can send the results to the optical module on the other side.		
Required tester	General purpose bit error rate tester (BERT); digital oscilloscope; Optical power meter (OPM); variable optical attenuator (VOA); test board;		
Test Configuration	1. Connect the OPM, VOA, splitters and BER tester to the reference points shown in the above test configuration.		

	2. Configure the BERT to simulate service information transmission. 3. Adjust the VOA to make the attenuation of the link meet the requirements of the power budget. 4. Configure OAM connections between the testing management systems and TRxs.
Procedure:	1. Verify that the service channel is working properly through the BERT. 2. Start OAM frame statistics at the testing management system and test for 10 minutes to check the number of bits sent and received, and the number of error frames in the testing management systems.
Pass/Fail Criteria:	No alarms, no error frames, and the number of sent and received frames is the same.
Test Output	The OAM data link communication of the optical modules is OK.

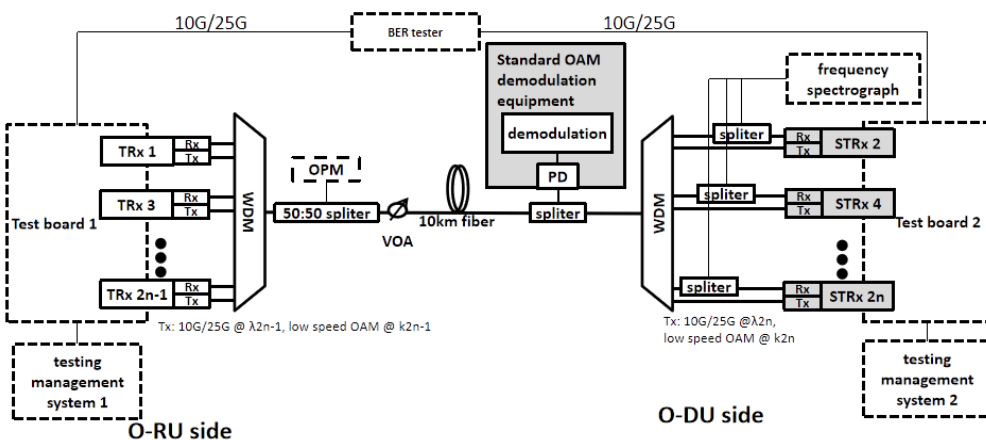
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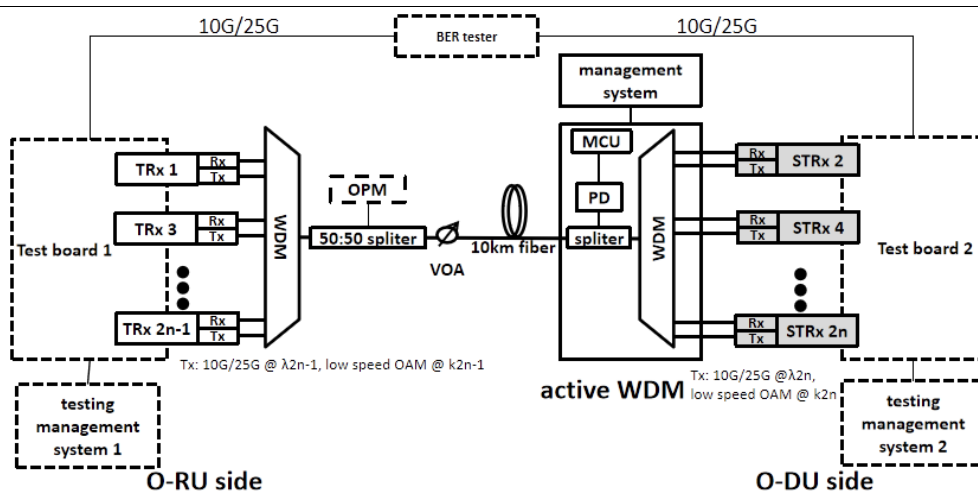
Test ID	WDM.OAM.02	Classification	Characterization
Test Title	OAM test for active WDM equipment OAM data link layer		
Test Objective:	To test the OAM data link communication of active WDM equipment and optical modules.		
Pre-Req			
Test Topology:	Applicable to O-RAN.WG9.WDM-v01.00, active WDM equipment for semi-active WDM system and active WDM system. 		
Required tester	General purpose bit error rate tester (BERT); digital oscilloscope; Optical power meter (OPM); variable optical attenuator (VOA); test board;		
Test Configuration	1. Connect the OPM, VOA, splitters and BERT tester to the reference points shown in the above test configuration. 2. Configure the BERT to simulate service information transmission. 3. Adjust the VOA to make the attenuation of the link meet the requirements of the power budget. 4. Configure OAM connections between testing management system and TRx.		
Procedure:	1. Verify that the service channel is working properly through the BERT. 2. Verify that the OAM data link communication of optical modules is OK through the testing management systems.		

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	status inquiry and configuration requests to optical modules at the O-RU side, and check whether the corresponding responses are received.
Pass/Fail Criteria:	1. The active WDM equipment can get the correct information of optical module at O-RU side. 2. For semi-active WDM type I system and active WDM system, the active WDM equipment can send inquiry and configuration messages to optical modules at the O-RU side and receive the corresponding response messages.
Test Output	The OAM service layer function of active WDM equipment is OK, including inquiry, configuration and active.

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2

Test ID	WDM.OAM.04	Classification	Characterization
Test Title	OAM test for technical characteristics of multi-carrier OAM modulation		
Test Objective:	To test the technical characteristics of multi-carrier OAM modulation		
Pre-Req			
Test Topology:	Applicable to O-RAN.WG9.WDM-v01.00, active WDM equipment for semi-active WDM system.  The diagram illustrates the test setup topology for OAM modulation. It shows two sides: the O-RU side and the O-DU side. On the O-RU side, there is a 'Test board 1' containing multiple transceiver pairs (TRx 1, TRx 3, ..., TRx 2n-1). These are connected to a WDM multiplexer. The output of the WDM is connected to a 50:50 splitter, which then connects to an OPM (Optical Power Meter) and a VOA (Variable Optical Attenuator). The signal then travels through a 10km fiber to the O-DU side. On the O-DU side, the signal passes through a PD (Photodetector) and a splitter. It then enters a 'Standard OAM demodulation equipment' block, which includes a 'demodulation' section. The output of the demodulation equipment is connected to a frequency spectrograph. Additionally, there is a 'testing management system 2' connected to the O-DU side. The O-DU side also features a 'Test board 2' with multiple transceiver pairs (STRx 2, STRx 4, ..., STRx 2n). The diagram also shows a 'testing management system 1' connected to the O-RU side. The entire setup is labeled 'Test setup topology 1'.		



Test setup topology 2

An optical link consists of WDM components, like MUX/DEMUX and 10km optical fiber, which are used to connect optical modules at two ends to simulate the fronthaul optical transport channel.

Colored optical modules are inserted on a test board like a transponder, on which a management system installs. The information of colored optical modules can be configured and read by the test boards through the management system.

The standardized colored optical modules (STRxs) and standardized OAM demodulation equipment based on multi-carrier OAM modulation are used for the verification of the interoperation of optical module and active WDM equipment of different vendors.

At the STRxs, the transmitting and receiving frequency of service signal is 10G or 25G, the carrier modulation frequencies of in-band OAM signals are unique for each WDM optical signals with different wavelengths ($\lambda_2 \sim \lambda_{2n}$) and are between 1k to 100M.

At the standardized OAM demodulation equipment, the optical signal with different wavelengths ($\lambda_2 \sim \lambda_{2n}$) is converted into the electrical signal with different carrier modulation frequencies ($k_1 \sim k_{2n-1}$) by a photoelectric detector (PD), and all in-band OAM messages with different carrier modulation frequencies ($k_1 \sim k_{2n-1}$) can be demodulated together correctly from the electrical signal by a demodulation.

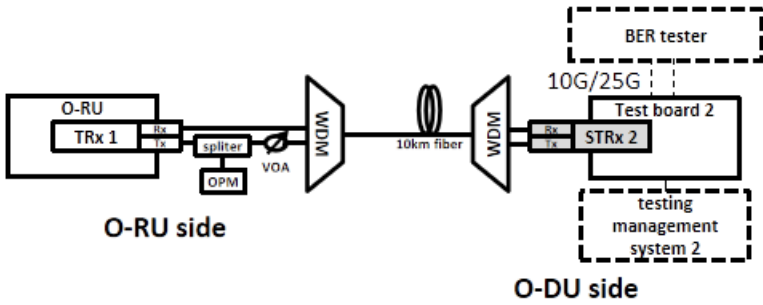
Required tester	General purpose bit error rate tester (BERT); frequency spectrograph; Optical power meter (OPM); variable optical attenuator (VOA); test board;
Test Configuration	1. Connect the OPM, VOA, splitters, frequency spectrograph and BER tester to the reference points shown in the test setup topology 1. 2. Configure the BERT to simulate service information transmission. 3. Adjust the VOA to make the attenuation of the link meet the requirements of the power budget. 4. Configure OAM connections between testing management systems and TRxs.
Procedure:	1. Check whether the service signal link communication between optical modules at different wavelength ($\lambda_1 \sim \lambda_{2n}$) are working correctly through the BERT. 2. Check that the OAM data link communication of optical modules is OK through the testing management systems. 3. Measure the carrier modulation frequencies ($k_1 \sim k_{2n}$) of in-band OAM signals at different wavelength channels through the frequency spectrograph, check

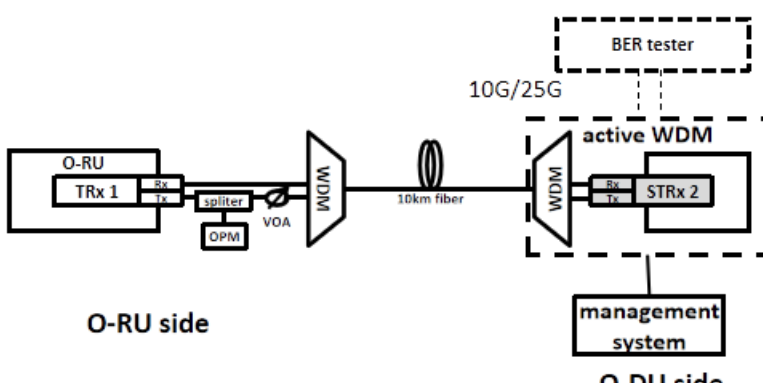
	whether they are unique for each WDM optical signals with different wavelengths ($\lambda_1 \sim \lambda_{2n}$). 4. Check whether the OAM messages from different OAM carriers are extracted correctly by the standardized OAM demodulation equipment. 5. Connect the test setup as the test setup topology 2. 6. Check whether the active WDM equipment could extract the OAM messages from different OAM carriers.
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Pass/Fail Criteria:	the OAM data link communication of optical modules based on multi-carrier OAM modulation is OK. The carrier modulation frequencies ($k_1 \sim k_{2n}$) of in-band OAM signals are unique for each WDM optical signals with different wavelengths ($\lambda_1 \sim \lambda_{2n}$). The OAM messages from different OAM carriers are extracted correctly by the standardized OAM demodulation equipment. The active WDM equipment can receive the OAM messages from optical modules at O-RU side.
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Test Output	optical module serial number	Optical transmission wavelength	low-speed carrier modulation frequency
	1	λ_1	k_1
	2	λ_2	k_2
	2n	λ_{2n}	k_{2n}

1

Test ID	WDM.OAM.05	Classification	Characterization
Test Title	OAM test for the loopback configuration function of semi-active WDM system		
Test Objective:	To test the loopback configuration function of semi-active WDM system		
Pre-Req			
Test Topology:	Applicable to O-RAN.WG9.WDM-v01.00, active WDM equipment for semi-active WDM system. 		

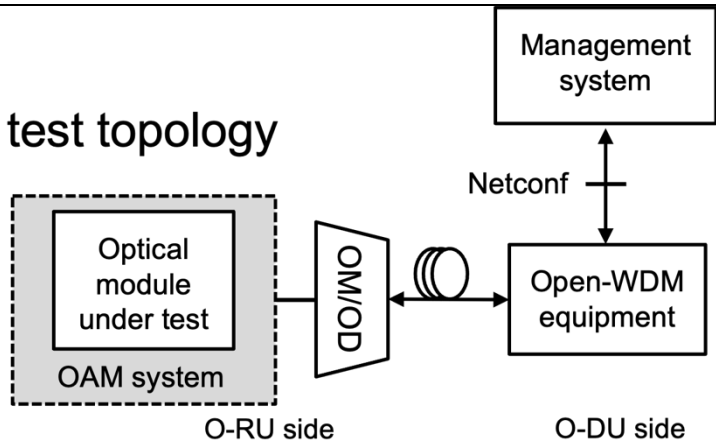
	Test setup topology 1  Test setup topology 2 An optical link consists of WDM components, like MUX/DEMUX and 10km optical fiber, which are used to connect optical modules at two ends to simulate the fronthaul optical transport channel. Optical modules are inserted on a test board like a transponder, on which a management system installs. The information of optical modules can be configured and read by the test boards through the management system. The standardized colored optical modules (STRx) are used for the verification of the optical module interoperation of different vendors. The STRxs can send the loopback configuration message and receive the response message of the loopback configuration success from the tested optical modules through the in-band OAM channel. The STRxs also can receive the loopback configuration message from the tested optical modules, configure itself on the loopback mode and send the response message of the loopback configuration success to the tested optical modules through the in-band OAM channel. If the response message of the loopback configuration success from the tested optical modules is received by STRxs, STRxs can send the 10G/25G PRBS test data stream and receive the 10G/25G data stream from the tested optical modules. It can be estimated by STRx whether there is a fault of the tested optical module: If the standardized response message of loopback configuration success isn't responded or the received 10G/25G data stream from the tested optical modules has error, the tested optical module is faulty. If standardized response message of the loopback configuration success is responded and the received 10G/25G data stream from the tested optical modules is error-free, the tested optical module is normal.
Required tester	General purpose bit error rate tester (BERT); digital oscilloscope; Optical power meter (OPM); variable optical attenuator (VOA); test board;
Test Configuration	1. Connect the OPM, VOA, splitters and BER tester to the reference points shown in the test setup topology 1. 2. Configure the BERT to simulate service information transmission. 3. Adjust the VOA to make the attenuation of the link meet the requirements of the power budget. 4. Configure OAM connections between testing management systems and TRxs.
Procedure:	1. Verify that the service channel isn't working through the BERT. 2. Verify that the OAM data link communication of optical modules is OK.

	3. Configure the STRxs to send the loopback configuration OAM message to optical module at the O-RU by the test management system 2. 4. Check whether the service channel is working properly through the BERT. 5. Check whether the standardized response OAM message from optical module at the O-RU could be extracted by the STRxs. 6. Connect the test setup as the test setup topology 2. 7. Configure the STRxs to send the loopback configuration OAM message to optical module at the O-RU by the active WDM equipment. 8. Check whether the service channel is working properly through the BERT. 9. Check whether the standardized response OAM message from optical module at the O-RU could be extracted by active WDM equipment.
Pass/Fail Criteria:	If the service channel is working properly through the BERT and the standardized response OAM message could be extracted, the optical module satisfies the loopback configuration requirement. If the service channel is working properly through the BERT and the standardized response OAM message could be extracted by the active WDM equipment, the active WDM equipment satisfies the loopback configuration requirement.
Test Output	The loopback configuration function of semi-active WDM system is OK.

7.1.7 Management Interface Test

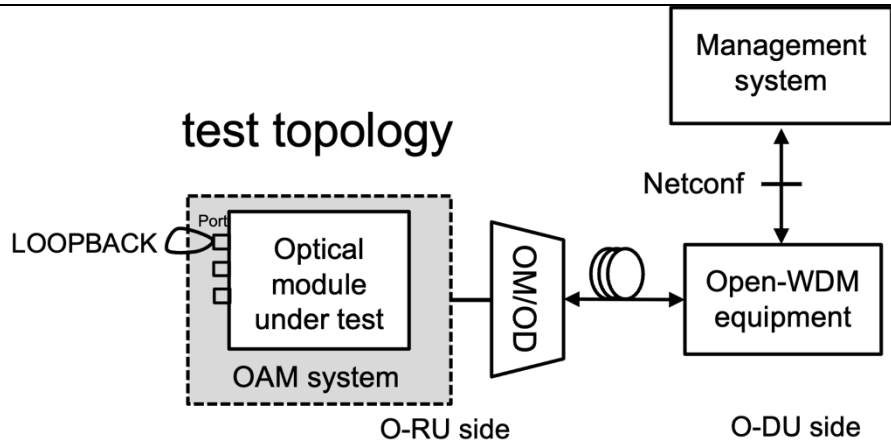
Functional requirements for Semi-active Front-haul WDM systems have been defined in the current version of Management Interfaces for Transport Network Elements (O-RAN.WG9.XTRP-MGT.0-v04.0). The centralized management system could send management requests to the active WDM equipment and control the semi-active WDM front-haul systems of multiple vendors through the unified Netconf interface. To fully verify the functional requirements of the management system and interface, typical test cases shall be included.

Test ID	WDM.management.01	Classification	Management System Functionality
Test Title	Network element information query		
Test Objective:	To test if the WDM management southbound interface support network element information query function		
Test Topology:	Refer to O-RAN.WG9.WDM-v01.00, Section 2.3 Semi-Active WDM system		

	 The diagram illustrates the test topology. At the top, a 'Management system' box is connected via a bidirectional arrow labeled 'Netconf' to an 'Open-WDM equipment' box. Below the 'Open-WDM equipment' box is the label 'O-DU side'. To the left of the 'Open-WDM equipment' box is a trapezoidal box labeled 'O/OD'. A bidirectional arrow connects the 'Open-WDM equipment' box and the 'O/OD' box. To the left of the 'O/OD' box is a dashed rectangular box labeled 'OAM system'. Inside this dashed box is a smaller box labeled 'Optical module under test'. Below the 'OAM system' box is the label 'O-RU side'. The entire setup is labeled 'test topology' at the top left.		
Required tester	Southbound interface test tool		
Pre-Req	1. Connect the test topology according to the test configuration;		
Procedure:	<u>Detailed Test procedure:</u> - Query the network element information through the 'get' command on the management and control system. Verify that some typical network element information can be obtained. - Verify whether the queried information is consistent with the actual situation of the device. <u>Verifications needed:</u> - Record the actual situation of the device.		
Pass/Fail Criteria:	1. Step 2 can query the device network element information, and the information is consistent with the actual situation of the device.		
Test Output	Typical network element information: - Network element information list; - Network element ID; - Network element name; - The manufacturer name of the network element; - Product name; - Software version; - Additional attributes; - Network element classification: semi-active, etc.; - Hardware version; - Network element IP; - Board list; - Layer protocol; - Gateway.		

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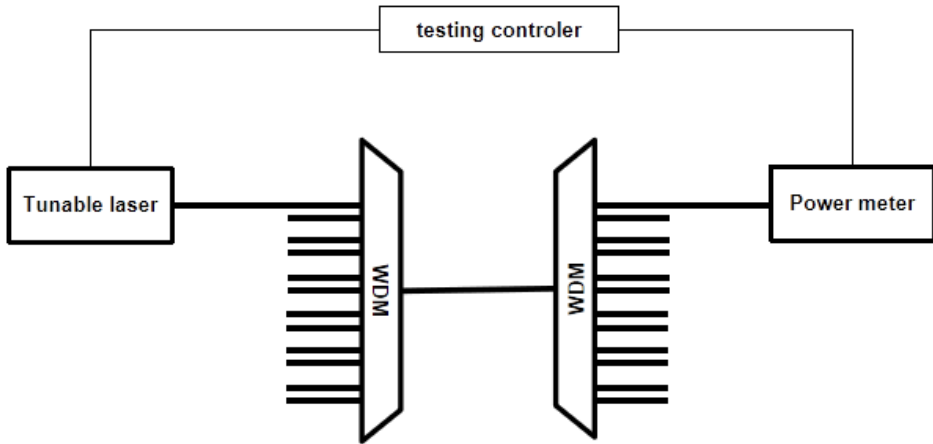
Test ID	WDM.management.02	Classification	Management System Functionality
Test Title	Loopback configuration and query		
Test Objective:	To test if the WDM management southbound interface support loopback configuration and query function		
Test Topology:	Refer to O-RAN.WG9.WDM-v01.00, Section 2.3 Semi-Active WDM system		

	 The diagram illustrates the test topology. On the O-RU side, there is an 'Optical module under test' within an 'OAM system'. A 'LOOPBACK' is indicated on a 'Port' of the optical module. This is connected to an 'OM/OD' (Optical Module/Optical Device) interface. On the O-DU side, there is 'Open-WDM equipment'. A bidirectional connection exists between the 'OM/OD' and the 'Open-WDM equipment'. A 'Management system' is connected to the 'Open-WDM equipment' via 'Netconf'.
Required tester	Southbound interface test tool
Pre-Req	1. Connect the test topology according to the test configuration;
Procedure:	Detailed Test procedure: - Loopback setting and query: - Query the WDM device port information on the management and control system. - Use the '<i>edit-config</i>' command to configure several ports loopback into the following types respectively, and verify that the loopback setting is successful: – FACILITY_LOOPBACK – TERMINAL_LOOPBACK – NON_LOOPBACK - Use the '<i>get</i>' command on the management system to query the loopback types of all the network element ports and the loopback type of the specified port, and verify that the query is successful.
Pass/Fail Criteria:	- The loopback configuration should be able to return the operation result (success/failure). - The loopback query results in step 3 should be consistent with the actual configuration in step 2. - Supports two query granularities: network element and port.
Test Output	- The loopback query result must include the port loopback status; - The query result value without loopback type is empty.

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7.1.8 Performance of a pair of MUX/DEMUXs

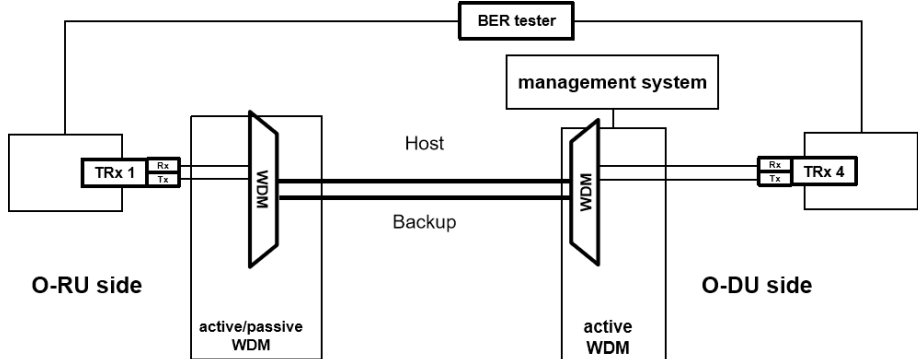
Test ID	WDM.MUX-DEMUX	Classification	Functionality
Test Title	Performance of a pair of MUX/DEMUXs		
Test Objective:	The aim of the test is to measure the performance of a pair of MUX/DEMUXs, including the max shift of central wavelength, max 1 dB bandwidth and insertion loss of a pair of MUX/DeMUXs.		
Pre-Req	Connect the test topology according to the test configuration		

Test Topology:	Applicable to O-RAN.WG9.WDM-v01.00, Passive, Semi-Active, and Active WDM fronthaul system 																																																																														
Required tester	Tunable Laser, Power meter																																																																														
Test Configuration	1. Connect the tunable Laser and power meter directly, and measure and store the output power curves '$P_{btb}(\lambda)$'(dBm) as the reference while the tunable laser wavelength is swept within the wavelength range of WDM system requirement. 2. Connect the tunable Laser and power meter to the reference points shown in the above test configuration.																																																																														
Procedure:	3. Detailed Test procedure: 4. The tunable laser is connected to the left port 'i' ($i=1,....n$), and the power meter is connected to the right port 'i' ($i=1,....n$) of the paired device; 5. The wavelength of the tunable laser is swept within the working wavelength range and the power curve '$P_i(\lambda)$'(dBm) is measured and stored; 6. The Transmission power spectrum curve '$T_i(\lambda)$' of pair of MUX/DEMUXs is got by the formula: $T_i(\lambda)=P_{btb}(\lambda)-P_i(\lambda)$; 7. 1 dB bandwidth of wavelength channel 'i': Find the wavelength bandwidth that the difference between '$T_i(\lambda)$' and max value of '$T_i(\lambda)$' is no more than 1dB; 8. Insertion loss of a pair of MUX/DeMUXs of wavelength channel 'i': find the min value of '$T_i(\lambda)$' at 1 dB wavelength bandwidth; 9. Shift of central wavelength of wavelength channel 'i': the difference between the central wavelength of 1 dB bandwidth wavelength range and the standardized central wavelength.																																																																														
Pass/Fail Criteria:	If the value of performance is in the standardized range, pass; if not, fail. <table><tr><th>Parameters</th><th>Unit</th><th colspan="14">Proposed values</th></tr><tr><td>Central wavelength</td><td>nm</td><td>1267.5</td><td>1274.5</td><td>1287.5</td><td>1294.5</td><td>1307.5</td><td>1314.5</td><td>1327.5</td><td>1334.5</td><td>1347.5</td><td>1354.5</td><td>1367.5</td><td>1374.5</td><td></td></tr><tr><td>Max shift of central wavelength</td><td>nm</td><td colspan="14">± 1.2</td></tr><tr><td>1 dB bandwidth</td><td>nm</td><td colspan="14">≥ 5.0</td></tr><tr><td>Max Insertion loss @ O-RU+O-DU pairs, including the IL of an optical splitter</td><td>dB</td><td>6.1</td><td>6.1</td><td>6.1</td><td>6.1</td><td>6.1</td><td>6.1</td><td>4.5</td><td>4.5</td><td>3.0</td><td>3.0</td><td>3.0</td><td>3.0</td><td></td></tr></table>	Parameters	Unit	Proposed values														Central wavelength	nm	1267.5	1274.5	1287.5	1294.5	1307.5	1314.5	1327.5	1334.5	1347.5	1354.5	1367.5	1374.5		Max shift of central wavelength	nm	± 1.2														1 dB bandwidth	nm	≥ 5.0														Max Insertion loss @ O-RU+O-DU pairs, including the IL of an optical splitter	dB	6.1	6.1	6.1	6.1	6.1	6.1	4.5	4.5	3.0	3.0	3.0	3.0	
Parameters	Unit	Proposed values																																																																													
Central wavelength	nm	1267.5	1274.5	1287.5	1294.5	1307.5	1314.5	1327.5	1334.5	1347.5	1354.5	1367.5	1374.5																																																																		
Max shift of central wavelength	nm	± 1.2																																																																													
1 dB bandwidth	nm	≥ 5.0																																																																													
Max Insertion loss @ O-RU+O-DU pairs, including the IL of an optical splitter	dB	6.1	6.1	6.1	6.1	6.1	6.1	4.5	4.5	3.0	3.0	3.0	3.0																																																																		

Test Output	
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1

7.1.9 Protection function for WDM system

Test ID	WDM.protection	Classification	Functionality
Test Title	protection function for WDM system		
Test Objective:	The aim of the test is to check protection function for WDM system defined in “WDM-based Fronthaul Transport”		
Pre-Req	Running service will be interrupted		
Test Topology:	Applicable to O-RAN.WG9.WDM-v01.00, Semi-Active and Active WDM fronthaul system 		
Required tester	bit error rate tester (BERT)		
Test Configuration	1 . Connect the BER tester to the reference points shown in the above test configuration. 2. Configure the BER tester to simulate service information transmission.		
Procedure:	1. On the management and control system, create or modify protection function through “edit-config” command. 2. On the management and control system, execute protection switching through RPC commands, and the commands include the following input content: a) Protection ID b) Switching command, such as manual switching, forced switching, c) protection Direction, such as Host and Backup 3. Check the service connection status through the data network analyzer and record the service protection switching downtime. 4. On the management and control system, query the protection information through “get-config” command to verify whether the switching has been successful. 5. Create a link break fault and repeat step 3 and 4.		
Pass/Fail Criteria:	Verify whether protection functions can be queried and configured through management system for active and semi-active WDM system. Verify whether the performance of the protection function meets the requirements.		
Test Output	Protection functions can be queried and configured through management system for active and semi-active WDM system. The performance of the protection function meets the requirements.		

2

8 Packet-Switched Architectures and Solutions Testing

This chapter is composed of four sections:

- Generalized basic verification test.
- Service level tests
- Network element test
- Additional network tests

The first section covers the most common and basic packet tests of an Xhaul network (from a UNI interface to another UNI interface). The second section delivers test cases for service level testing at UNI interfaces.

The third section is limited to testing the functions of a packet-switched Transport Network Element (TNE).

The last section provides other test cases outside of the above the three categories.

8.1 Generalized basic verification methods for Xhaul Transport packet switched Network.

This section covers the most common and basic verification methods using Traffic Generator and Analyzer tool.

Table 8.1-1 summarizes the encapsulation of traffic streams combination of which shall be used in generalized verification methods that must be executed to validate any transport network.

Transport Network Type	Traffic stream	Packet Encapsulation	VLAN PCP bits	DSCP bits	Frame Size (min, max)
Fronthaul	Split option 7-2x eCPRI U-plane and C-plane traffic	Option 1 –Ethernet + eCPRI			As per network
		Option 2 – Ethernet + VLAN + eCPRI	7 (default)		As per network
		Option 3 – Ethernet + IPv4 + UDP + eCPRI		EF	As per network
		Option 4 - Ethernet + VLAN + IPv4 + UDP + eCPRI	7 (default)	EF	As per network
		Option 5 – Ethernet + IPv6 + UDP + eCPRI		EF	As per network
		Option 6 - Ethernet + VLAN + IPv6 + UDP + eCPRI	7 (default)	EF	As per network
	M-plane Traffic	Option 1 –Ethernet + IPv4 + Netconf		AF2x	As per network
		Option 2 – Ethernet + VLAN + IPv4 + Netconf	2 (default)	AF2x	As per network
		Option 3 – Ethernet + IPv6 + Netconf		AF2x	As per network
		Option 4 - Ethernet + VLAN + IPv6 + Netconf	2 (default)	AF2x	As per network
	RoE Traffic	Option 1 - Ethernet + RoE			As per network

		Option 2 - Ethernet + VLAN + RoE	7 (default)		As per network
Midhaul	User plane and control plane traffic	Option 1 – Ethernet + IPv4		As per QoS scheme of the network	As per network
		Option 2 – Ethernet + VLAN + IPv4	As per QoS scheme of the network	As per QoS scheme of the network	As per network
		Option 3 – Ethernet + IPv6		As per QoS scheme of the network	As per network
		Option 4 – Ethernet + VLAN + IPv6	As per QoS scheme of the network	As per QoS scheme of the network	As per network
	M-plane traffic	Option 1 – Ethernet + IPv4		As per QoS scheme of the network	As per network
		Option 2 – Ethernet + VLAN + IPv4	As per QoS scheme of the network	As per QoS scheme of the network	As per network
		Option 3 – Ethernet + IPv6		As per QoS scheme of the network	As per network
		Option 4 – Ethernet + VLAN + IPv6	As per QoS scheme of the network	As per QoS scheme of the network	As per network
Backhaul	User plane and control plane traffic	Option 1 – Ethernet + IPv4		As per QoS scheme of the network	As per network
		Option 2 – Ethernet + VLAN + IPv4	As per QoS scheme of the network	As per QoS scheme of the network	As per network
		Option 3 – Ethernet + IPv6		As per QoS scheme of the network	As per network
		Option 4 – Ethernet + VLAN + IPv6	As per QoS scheme of the network	As per QoS scheme of the network	As per network
	M-plane traffic	Option 1 – Ethernet + IPv4		As per QoS scheme of the network	As per network
		Option 2 – Ethernet + VLAN + IPv4	As per QoS scheme of the network	As per QoS scheme of the network	As per network
		Option 3 – Ethernet + IPv6		As per QoS scheme of the network	As per network
		Option 4 – Ethernet + VLAN + IPv6	As per QoS scheme of the network	As per QoS scheme of the network	As per network

Table 8.1-1

Peak fronthaul bandwidth required for 7-2x split based eCPRI traffic can be estimated by the following formula as specified in section 7.2 of Xhaul Transport Requirements doc.

$$FH\ BW = 2 \times 10^{-9} (1 + c) \frac{v_{layer} N_{PRB} (12 N_{mantissa} + N_{exponent})}{T_s^{\mu}} \text{ (Gbps)}$$

v_{layer} is the maximum number of supported layers.

N_{PRB} is the maximum RB allocation for a given channel bandwidth and numerology μ

$N_{mantissa}$ is the number of mantissa bits.

$N_{exponent}$ is the number of the exponent bits.

T_s^{μ} is the average OFDM symbol duration in a subframe for numerology, i.e., $T_s^{\mu} = \frac{10^{-3}}{14 \times 2^{\mu}}$

c is C-plane overhead. For downlink $c \approx 10\%$ and for uplink $c \approx 0$

Table 8.1-2 provides some examples for peak fronthaul bandwidth calculated for different scenarios assuming $N_{mantissa} = 9$ and $N_{exponent} = 4$

Number of Sectors	CBW (MHz)	μ	MIMO Layers	ABW (MHz)	FH BW (Gbps)
1	100	1	4	400	7.7
1	400	3	2	800	14.57

Table 8.1-2

The above estimate does not include protocol encapsulation overhead which should be added based upon encapsulation type and average packet size.

Table 8.1-3 provides traffic profiles to be used to simulate eCPRI traffic for Fronthaul network testing

Traffic Profile	Packet Encapsulation and QoS marking	Stream Pattern	Stream Property	Traffic Load percentage
FH eCPRI traffic profile 1	As per Table 8.1-1	Constant Bit Rate	Frame size = maximum allowed frame size Frame rate = as calculated using the formula for peak fronthaul bandwidth * traffic load percentage	20%,50%,100%
FH eCPRI traffic profile 2	As per Table 8.1-1	Constant Bit Rate	Frame size = minimum allowed frame size Frame rate = as calculated using the formula for peak fronthaul bandwidth * traffic load percentage	20%,50%,100%

FH eCPRI traffic profile 3	As per Table 8.1-1	Constant Bit Rate	Frame size = Random or any other distribution such as custom, IMIX, Gaussian etc. Frame rate = as calculated using the formula for peak fronthaul bandwidth * traffic load percentage	20%,50%,100%
FH eCPRI traffic profile 4	As per Table 8.1-1	Burst	Frame size = maximum allowed frame size Gap between two consecutive bursts = 1 ms or less Frame Count in burst is such that average frame rate $\approx$ calculated peak fronthaul bandwidth * traffic load percentage	20%,50%,100%
FH eCPRI traffic profile 5	As per Table 8.1-1	Burst	Frame size = minimum allowed frame size Gap between two consecutive bursts = 1 ms or less Frame Count in burst is such that average frame rate $\approx$ calculated peak fronthaul bandwidth * traffic load percentage	20%,50%,100%

Table 8.1-3

Table 8.1-4 provides traffic profiles to be used to simulate M-plane and background traffic for Fronthaul network testing. It has been assumed that M-plane traffic would be bursty in nature and may occupy the rest of the available bandwidth after bandwidth utilization by eCPRI traffic.

Traffic Profile	Packet Encapsulation and QoS marking	Stream Pattern	Stream Property
FH M-plane traffic profile 1	As per Table 8.1-1	Burst	Frame size = maximum allowed frame size Gap between two consecutive bursts = 1 ms or less Frame Count in burst is such that average frame rate $\approx$ interface rate - calculated peak fronthaul bandwidth
FH M-plane traffic profile 2	As per Table 8.1-1	Burst	Frame size = minimum allowed frame size Gap between two consecutive bursts = 1 ms or less Frame Count in burst is such that average frame rate $\approx$ interface rate - calculated peak fronthaul bandwidth

Table 8.1-4

Figure 8.1-1 depicts a generic test set up for network under test with Traffic Generator and Analyzer tool.

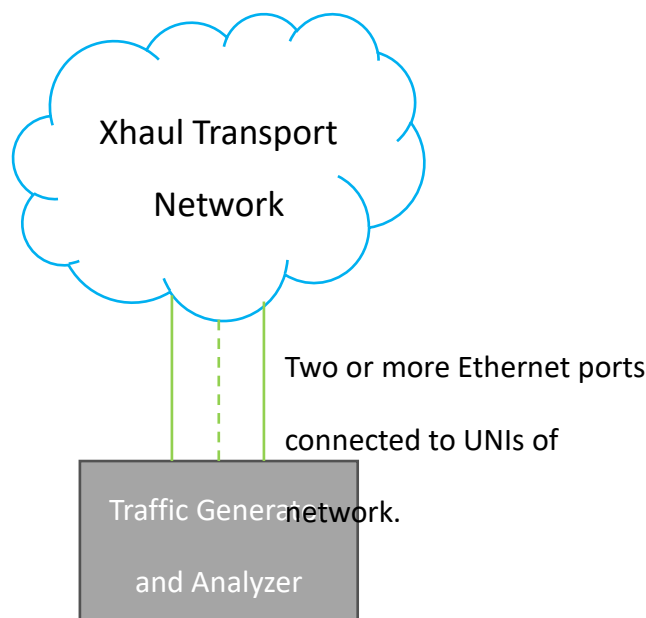


Figure 8.1-1

8.1.1 Forwarding Verification

Once the required combination of traffic streams is set up in Traffic Generator and Analyzer Tool with desired scale of source and destination endpoints in transport layer (for example thousands of Ethernet/IPv4/IPv6 endpoints), steps from table 8.1.1-1 shall be executed. Since this step is to just ensure basic forwarding capability, the configured rate of the traffic streams could be low.

Test Steps	Expected result
Send traffic streams with minimum and maximum frame size and verify that no frames are dropped from the counters of Traffic Generator and Analyzer tool	No frames shall be dropped by network

Table 8.1.1-1

8.1.2 Latency Measurement and Verification

Depending upon network type and application type, latency requirement could be different in network. For fronthaul network, maximum tolerable latency for different use cases is described in Table 3 of Xhaul Transport Requirements document. For midhaul and backhaul networks latency limits are described in Table 5 and Table 6 of the same document respectively.

Latency in packet switched network can be measured in four different ways –

3. Store and Forward Latency (The time interval starting when the last bit of the frames leaves sending port and ending when the first bit of the frame is seen on the receiving port or LIFO)
4. Cut Through Latency (The time interval starting when the first bit of the frames leaves sending port and ending when the first bit of the frame is seen on the receiving port or FIFO)
5. MEF Frame Delay (The time interval starting when the first bit of the frames leaves sending port and ending when the last bit of the frame is seen on the receiving port or as FILO)
6. Forwarding Delay (The time interval starting when the last bit of the frames leaves sending port and ending when the last bit of the frame is seen on the receiving port or as LILO)

Anyone of the four methods could be used. However, the last method is recommended as it captures the delay caused by only the network under test.

Step from table 8.1.2-1 below shall be executed for measuring latency

Test Steps	Expected result
While executing step from Table 8.1.1-1, measure the latency for both minimum and maximum sized frames	Measured latency of individual traffic streams shall not be more than applicable latency limit depending upon the targeted use cases and network type.

Table 8.1.2-1

8.1.3 RFC 2544 Benchmarking test for frame loss rate, latency, throughput

Once results from 8.1.1 and 8.1.2 are satisfactory, RFC 2544 benchmarking tests shall be executed for frame loss, latency, and throughput as mentioned in table 8.1.3-1.

Test Steps	Expected result
<u>RFC 2544 frame loss test</u> Execute RFC 2544 frame loss test starting with 100% interface speed and different frame size (as an example 70, 100, 250, 300, 400, 500, 600, 700, 800, 900, 1000, 1100, 1200, 1300, 1400, 1500, 1522 bytes or up to size of jumbo frames if they are supported) for each class of traffic stream identified. Test duration for each iteration should be 120 seconds minimum. Observe the maximum rate achieved for each class of stream without any frame loss.	Maximum rate achieved for each class of traffic stream shall be equal to more than the demand of respective application.

<u>RFC 2544 latency test</u> Execute RFC 2544 latency test with maximum rate achieved in previous step without any frame loss and different frame size (as an example 70, 100, 250, 300, 400, 500, 600, 700, 800, 900, 1000, 1100, 1200, 1300, 1400, 1500, 1522 bytes) for each class of traffic stream identified. Test duration for each iteration should be 120 seconds minimum. Observe the minimum, maximum, average latency, and delay variation for each class of stream without any frame loss.	Maximum latency measured for each traffic stream shall be within tolerable latency limit of respective application.
<u>RFC 2544 throughput test</u> While executing previous step also measure the throughput achieved without any loss	Throughput achieved without any loss and within tolerable latency limit shall meet the application demand.

Table 8.1.3-1

8.2 Service Level Test Cases

This section covers the baseline test case scenarios to be validated for service and underlying transport network setup. These scenarios utilize general basic verification mentioned in Section 8.1 and can also be used in various other Network Testing use cases mentioned in section 8.4.

Although the test cases and their respective topologies shown in this section shows a single interface simulating the O-RU and O-DU connecting to the CSR and HSR, it is recommended that basic functionality for Fronthaul scenarios involving multiple O-RU's connected to the CSR should also be verified as part of these test cases. This verification will help in ensuring real-life deployments, where a lot of time multiple O-RU's (or its simulated interfaces) are connected to the same CSR are covered. Some of these "typical" scenarios are covered in WG9 - Xhaul Transport Requirements specification, specifically Table 4. A sample scenario with 1 and 3 O-RU's, each representing a sector, along with the bandwidth requirements for FH is shown below. Please note that this is just an example and actual test and deployment scenario may vary, some of which are documented in the Xhaul Transport Requirements specification

Cell Size	Sectors	Total CBW (MHz)	MIMO Layers	ABW (MHz)	FH BW (Gbps)
Small	1		-		
Medium	3	100	4	1200	23
Large	3	100	16	4800	90

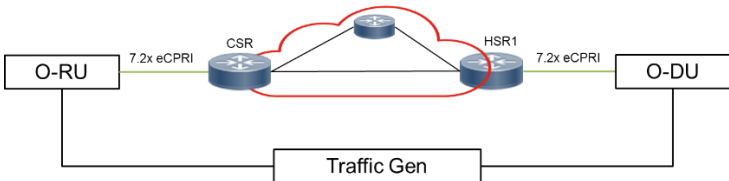
Table 8.2-1

It is also recommended to use "background traffic" alongside the Split 7-2x eCPRI traffic on the CSR and HSR as well. Another interface for this background traffic will be used from test equipment to the O-RU and O-DU. This ethernet interface will carry "other" traffic such as Wireline or Enterprise traffic, primarily in the

form on MPLS L3VPN. Each of the testcases in this section reflect the topology with additional interface for background traffic.

In the following test cases a Traffic Gen (also referred to as an O-DU or O-RU simulator) is used to generate and analyze test traffic towards O-RU or O-DU. It is connected directly to a CSR or an HSR port that usually connects to an O-RU or an O-DU. For illustration purposes the O-RU or O-DU are left in the test setup diagrams, although they are not needed for the test application.

8.2.1 Baseline 7-2x eCPRI transport over Packetized Fronthaul

Test ID	Ethernet.FH.01	Classification	Fronthaul Services
Test Title	Baseline 7-2x eCPRI transport over Packetized Fronthaul		
Test Objective:	1. Verify O-RU – O-DU communication over packetized Fronthaul network using SR-MPLS and/or SRv6 based transport		
Pre-Req	SR-MPLS and/or SRv6 baseline configuration on CSR, HSR and any other Fronthaul routers Optional Timing and syn configuration through FH network for RAN Nodes if actual O-RU and O-DU are used instead of O-RU and O-DU simulators		
Test Topology:			
Test Configuration	1. Connect O-RU or O-RU simulator to the UNI of the CSR and O-DU or O-DU simulator to the HSR. 2. Although not required for this baseline test, 1 or more routers should be used in the Fronthaul test network for future test cases (SRTE, Redundancy, etc) 3. Configure L2 transport service between CSR and HSR from O-RU Uni to O-DU Uni using L2 VPN or EVPN VPWS or both 4. Connect a tester to O-RU and O-DU to send simulated UE traffic 5. If actual O-RU and O-DU are used, connect a tester to O-RU and O-DU to send simulated UE traffic. If not, send eCPRI 7-2x traffic directly from simulators.		
Procedure:	Detailed Test procedure: 1. Verify O-RU and O-DU are up and communicating if actual O-RU and O-DU are used instead of simulators. 2. Start bi-directional eCPRI traffic through CSR and HSR between the RAN node, or their simulators, verify traffic is being received on either ends. 3. Simultaneously send background L3VPN traffic using the additional ethernet interface on CSR and HSR. Ensure that combined eCPRI and L3VPN traffic do not exceed CSR and HSR's uplink capacity. 4. Execute Test cases documented in Sections 8.1 and 8.4 using both SR-MPLS and/or SRv6 based transport		

Pass/Fail Criteria:	100% successful 7-2x eCPRI traffic transport between O-RU and O-DU, or their simulators 100% successful background L3VPN traffic transport through the packetized Fronthaul - Verify traffic stats on: - Test equipment - O-RU and O-DU, if possible - CSR and HSR interface and L2VPN/EVPN VPWS service counters - The test topology and setup could be used for more granular test cases and measurements (various traffic mix, Latency, etc) documented in Section 8.4
Test Output	Capture configuration used for reference. Capture verification commands (show commands) as appropriate.

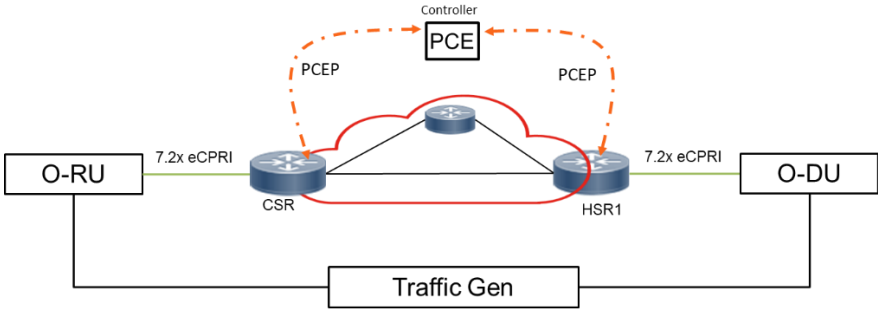
8.2.2 Constraints based 7-2x eCPRI transport over Packetized Fronthaul using Headend Path calculation

Test ID	Ethernet.FH.02	Classification	Fronthaul Services
Test Title	Constraints based 7-2x eCPRI transport over Packetized Fronthaul using Headend Path calculation		
Test Objective:	Verify O-RU – O-DU communication over packetized Fronthaul network using SR-MPLS and/or SRv6 based transport		
Pre-Req	SR-MPLS and/or SRv6 baseline configuration on CSR, HSR and any other Fronthaul routers Optional Timing and syn configuration through FH network for RAN Nodes, if actual O-RU and O-DU are used instead of O-RU and O-DU simulators		
Test Topology:	Same as described in Ethernet.FH.01		
Test Configuration	- Connect O-RU or the O-RU simulator to the UNI of the CSR, and O-DU or O-DU simulator to the HSR. 1 or more routers should be used in the Fronthaul test network to ensure constrains based path calculation. - Configure L2 transport service between CSR and HSR from O-RU Uni to O-DU Uni using L2 VPN or EVPN VPWS or both - Connect an additional interface from traffic generator to the CSR and the HSR to carry the background L3VPN traffic. The CSR and HSR should be pre-configured with L3VPN services to carry this background traffic. - If actual O-RU and O-DU are used, connect a tester to O-RU and O-DU to send simulated UE traffic. If not, send eCPRI 7-2x traffic directly from simulators.		
Procedure:	<u>Detailed Test procedure:</u> Verify O-RU and O-DU are up and communicating if actual O-RU and O-DU are used instead of simulators.		

	- Start bi-directional eCPRI traffic through CSR and HSR between the RAN nodes, or their simulators - Verify traffic is being received on either ends using shortest IGP path between CSR and HSR. - Simultaneously send background L3VPN traffic using the additional ethernet interface on CSR and HSR. Ensure that combined eCPRI and L3VPN traffic do not exceed CSR and HSR's uplink capacity. - Configure an SRTE policy defining a constrain based path, different than the IGP shortest path. SRTE policy should use Headend (CSR or HSR) based path calculation. - Constraints may include: - Include or exclude certain links. - Lowest latency path (latency may be increased on the IGP shortest path to force a longer IGP path but with lower latency) - Use of Flex-Algo to enforce path selection. - Apply the policy to the L2VPN/EVPN VPWS service as well as L3VPN services using static configuration or route-coloring. - Execute Test cases documented in Sections 8.1 and 8.4 using both SR-MPLS and/or SRv6 based transport
Pass/Fail Criteria:	- Verify traffic used IGP best path before applying SRTE Policy. - Verify route coloring and the use of SRTE policy for EVPN VPWS and L3VPN service between HSR and CSR once SRTE is configured. - Verify traffic used updated path based on constraints defined by SRTE policy. - Verify SID path list accurately reflects the path used by EVPN VPWS and L3VPN background traffic. - Ensure 100% traffic send/received success. - The test topology and setup could be used for more granular test cases and measurements (various traffic mix, Latency, etc) documented in 8. and 8.4.
Test Output	Capture configuration used for reference. Capture verification commands (show commands) as appropriate for: - Route coloring and SRTE policy use with EVPN VPWS and L3VPN. - SID list/SRTE Path calculation performed by headend

8.2.3 Constraints based 7-2x eCPRI transport over Packetized Fronthaul using Controller based Path calculation.

Test ID	Ethernet.FH.03	Classification	Fronthaul Services
Test Title	Constraints based 7-2x eCPRI transport over Packetized Fronthaul using Controller based Path calculation		
Test Objective:	Verify O-RU – O-DU communication over packetized Fronthaul network using SR-MPLS and/or SRv6 based transport and controller-based Path Calculation		

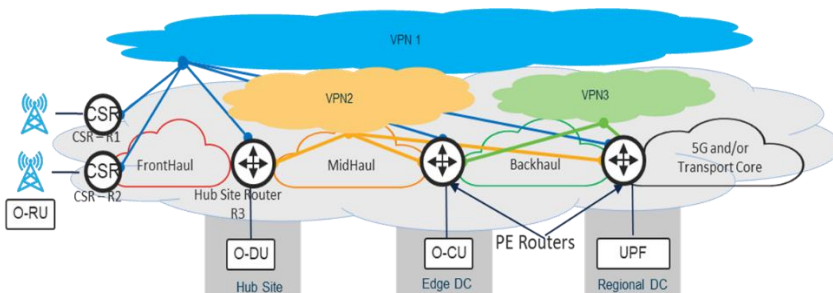
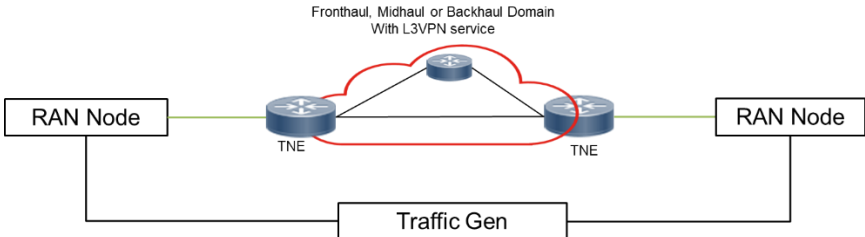
Pre-Req	SR-MPLS and/or SRv6 baseline configuration on CSR, HSR and any other Fronthaul routers Optional Timing and syn configuration through FH network for RAN Nodes to sync if actual O-RU and O-DU are used instead of O-Ru and O-DU simulators SR PCE controller setup and communication with CSR/HSR through PCEP as shown in topology
Test Topology:	
Test Configuration	1. Connect O-RU or O-RU simulator to the UNI of the CSR and O-DU or O-DU simulator to the HSR. 2. 1 or more routers should be used in the Fronthaul test network to ensure constrains based path calculation. 3. Configure L2 transport service between CSR and HSR from O-RU Uni to O-DU Uni using L2 VPN or EVPN VPWS or both 4. Connect an additional interface from traffic generator to the CSR and the HSR to carry the background L3VPN traffic. The CSR and HSR should be pre-configured with L3VPN services to carry this background traffic 5. If actual O-RU and O-DU are used, connect a tester to O-RU and O-DU to send simulated UE traffic. If not, send eCPRI 7-2x traffic directly from simulators.
Procedure:	<u>Detailed Test procedure:</u> 1. Verify O-RU and O-DU are up and communicating, if actual O-RU and O-DU are used instead of simulators. 2. Start bi-directional traffic between the RAN nodes, verify traffic is being received on either ends using shortest IGP path between CSR and HSR. 3. Start bi-directional eCPRI traffic through CSR and HSR between the RAN nodes, or their simulators. 4. Verify traffic is being received on either ends using shortest IGP path between CSR and HSR. 5. Simultaneously send background L3VPN traffic using the additional ethernet interface on CSR and HSR. Ensure that combined eCPRI and L3VPN traffic do not exceed CSR and HSR's uplink capacity 6. Configure an SRTE policy defining a constrain based path, different than the IGP shortest path. 7. SRTE policy should use PCE controller-based path calculation. CSR and HSR should request this path calculation from PCE based on the constraints defined. 8. Constraints may include: a. Include or exclude certain links. b. Lowest latency path (latency may be increased on the IGP shortest path to force a longer IGP path but with lower latency c. Use of Flex-Algo to enforce path selection.

	- Apply the policy to the EVPN VPWS service as well as L3VPN services, using static configuration or route-coloring. - Execute Test cases documented in 8.1 and 8.4 using both SR-MPLS and/or SRv6 based transport
Pass/Fail Criteria:	- Verify traffic used IGP best path before applying SRTE Policy. - Verify route coloring and the use of SRTE policy for EVPN VPWS and L3VPN service between HSR and CSR once SRTE is configured. - Verify updated path is calculated on the PCE and sent to the HSR and CSR to use with SRTE policy. - Verify CSR and HSR install/use the path received from PCE for the SRTE policy - Verify traffic uses updated path based on constraints defined by SRTE policy and computed by PCE controller. - Verify SID path list accurately reflects the path used by EVPN VPWS and L3VPN traffic. - Ensure 100% success for traffic send and received. - The test topology and setup could be used for more granular test cases and measurements (various traffic mix, Latency, etc) documented in Sections 8.1 and 8.4.
Test Output	Capture configuration used for reference. Capture verification commands (show commands) as appropriate for: - Route coloring and SRTE policy use with EVPN VPWS. - PCEP communication between CSR/HSR and PCE - SRTE Path calculation performed by PCE. - SRTE Path installation and use by CSR/HSR

8.2.4 L3VPN based Services Verification for Xhaul network

This section covers the test case scenarios to be validated for L3 Service across various domains. L3VPN is a mature technology, hence this case should act only as a guidance and included for completeness.

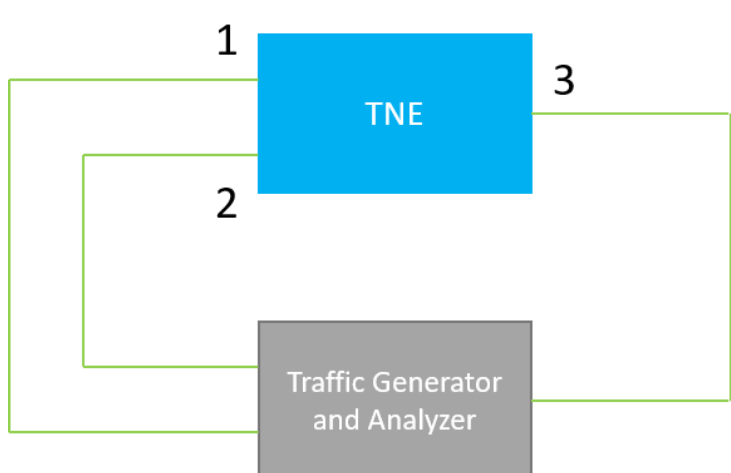
Test ID	Ethernet.L3VPN.01	Classification	L3VPN Services
Test Title	L3VPN Services verification		
Test Objective:	- Verify L3VPN services for FH (M-Plane only), MH and BH over SR-MPLS and SRv6 - Test could be repeated for constraints-based Path using SR-TE with Headend and/or controller-based path computation		
Pre-Req	SR-MPLS and/or SRv6 baseline configuration on FH, MH and BH routers		

	Timing and syn configuration through FH network for RAN Nodes to sync if actual O-RU, O-DU and O-CU are used. L3VPN services configuration as shown in the test topology
Test Topology:	Either of the two test topologies shown could be used. First topology is for L3VPN Service across all the domains, second topology is for 1 domain if testing if cross domains is not preferred. Topology Choice1:  Topology Choice2: 
Test Configuration	- As shown in the topology - Configure L3VPN services as desired
Procedure:	Detailed Test procedure: - Verify O-RU, O-DU and O-CU are up and communicating if actual devices are used. Else connect traffic simulator to appropriate RAN nodes. - Start bi-directional traffic between the RAN nodes, verify traffic is being received on all ends. - Execute Test cases documented in Section 0 and 8.4 using both SR-MPLS and/or SRv6 based transport. - Execute the test case with Constraints based path calculation for: - Headend based Path Computation - Controller based path computation. - Constraints may include, but not limited: - Include or exclude certain links. - Lowest latency path (latency may be increased on the IGP shortest path to force a longer IGP path but with lower latency) - Use of Flex-Algo to enforce path selection.
Pass/Fail Criteria:	100% successful traffic transport between O-RU, O-DU and O-CU - Verify traffic stats on: - Test equipment

	b. O-RU, O-DU and O-CU, if used c. TNE devices in the CSR and HSR interface and L2VPN/EVPN VPWS service counters 3. The test topology and setup could be used for more granular test cases and measurements (various traffic mix, Latency, etc) documented in Section 8.4.
Test Output	Capture configuration used for reference. Capture verification commands (show commands) as appropriate. For SRTE constraints based testcases verify: 1. Route coloring and SRTE policy use with L3VPN services. 2. L3VPN Path calculation at the headend (With headend based path calculation) 3. PCEP communication between various TNE and PCE (for controller-based Path computation) 4. SRTE Path calculation performed by PCE (for controller-based Path computation) 5. SRTE Path installation and use by TNE's (for controller-based Path computation)

1 8.3 Network element testing

2 8.3.1 Frame Pre-emption Testing

Test ID	ETH.PREEMPT.01	Classification	Ethernet, Frame Preemption
Test Title	Fragmentation of latency sensitive Fronthaul traffic by other background traffic by TNE supporting TSN profile B		
Test Objective:	Verify fragmentation capability of TNE supporting TSN profile B while queuing delay for latency sensitive Fronthaul traffic is bounded by 155 bytes transmission time at egress port		
Pre-Req	TNE supporting TSN profile B. This test needs to be done only at TNE interface speed of 1 Gbps/10 Gbps/25 Gbps.		
Test Topology:	 Interface speed: 1Gbps/10Gbps/25Gbps		

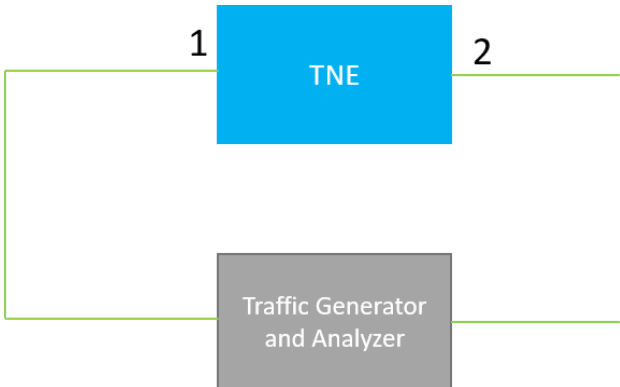
Express frame size (in bytes)	Express frame count in burst	Preemptible Frame Size (in bytes)	Preemptible frame count in burst
64	1000	123	10000
64	1000	124	10000
64	1000	1000	10000
64	1000	4500	10000
64	1000	9000	10000
500	1000	4500	10000
500	1000	9000	10000
2000	1000	4500	10000
2000	1000	9000	10000

	1. Verify that at least some preemptible frames are fragmented (except when preemptible frame size is 123 bytes) 2. Verify that difference between worst case latency of express frames and latency measured in step 5 is no more than 155 bytes transmission time. In the table below, 155 bytes transmission time on different line speed is mentioned. <table border="1"> <thead> <tr> <th>Port speed</th><th>Transmission time of 155 bytes</th></tr> </thead> <tbody> <tr> <td>1 Gbps</td><td>1.24 μs</td></tr> <tr> <td>10 Gbps</td><td>0.124 μs</td></tr> <tr> <td>25 Gbps</td><td>0.05 μs</td></tr> </tbody> </table>	Port speed	Transmission time of 155 bytes	1 Gbps	1.24 μ s	10 Gbps	0.124 μ s	25 Gbps	0.05 μ s
Port speed	Transmission time of 155 bytes								
1 Gbps	1.24 μ s								
10 Gbps	0.124 μ s								
25 Gbps	0.05 μ s								
Pass/Fail Criteria:	Test shall pass if some of preemptible traffic is fragmented and queuing delay of fronthaul traffic is bounded by 155 bytes transmission time on egress port. 155 bytes transmission time at individual line speed is mentioned in the table above.								
Test Output	This section is the placeholder that will capture the output/configurations after test is executed, typically done after test execution								

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Test ID	ETH.PREEMPT.02	Classification	Ethernet, Frame Preemption
Test Title	Reassembly of preemptible traffic by TNE supporting TSN profile B		
Test Objective:	Verify reassembly capability of TNE supporting TSN profile B		
Pre-Req	TNE supporting TSN profile B. This test needs to be done only at TNE interface speed of 1 Gbps/10 Gbps/25 Gbps.		
Test Topology:	 Interface speed: 1Gbps/10Gbps/25Gbps		
Test Configuration	This section to be filled out when putting the test report (may include device and/or tester configuration)		

Procedure:	<u>Detailed Test procedure:</u> 1. Enable TSN profile B and Frame Preemption on TNE port 1 and disable frame preemption on TNE port 2 2. From a tester port send interspersed fragments from same preemptible frame and express frames of various size according to the table below. Measure latency of express frames in case. <table border="1"> <thead> <tr> <th>Preemptible Frame Size (in bytes)</th><th>Fragment size (in bytes)</th><th>Fragment count interspersed with express frame size</th><th>Express Frame size interspersed with fragments (in bytes)</th></tr> </thead> <tbody> <tr> <td>256</td><td>64</td><td>4</td><td>64 to 2000</td></tr> <tr> <td>2048</td><td>128</td><td>8</td><td>64 to 2000</td></tr> <tr> <td>5000</td><td>256</td><td>8</td><td>64 to 2000</td></tr> </tbody> </table> 3. For each express frame size, also configure and send express traffic without any interspersed fragments or preemptible traffic. Measure latency of express traffic. 4. Observe latency experienced by reassembled frames by considering receive time of last bit of last fragment in the DUT and the last bit of reassembled frame in traffic generator port. <u>Verifications needed:</u> 1. Verify that TNE reassembles all the fragments and sends out a preemptible frame successfully on egress port for each combination. 2. Verify that express frame experiences almost same latency in step 2 as in step 3 since reassembly must not affect express frame latency. 3. Recorded latency in step 4 is only for information since there is no limit imposed on latency of fragmented frames.			Preemptible Frame Size (in bytes)	Fragment size (in bytes)	Fragment count interspersed with express frame size	Express Frame size interspersed with fragments (in bytes)	256	64	4	64 to 2000	2048	128	8	64 to 2000	5000	256	8	64 to 2000
Preemptible Frame Size (in bytes)	Fragment size (in bytes)	Fragment count interspersed with express frame size	Express Frame size interspersed with fragments (in bytes)																
256	64	4	64 to 2000																
2048	128	8	64 to 2000																
5000	256	8	64 to 2000																
Pass/Fail Criteria:	Test shall pass if above verification steps are successful																		
Test Output	This section is the placeholder that will capture the output/configurations after test is executed, typically done after test execution																		

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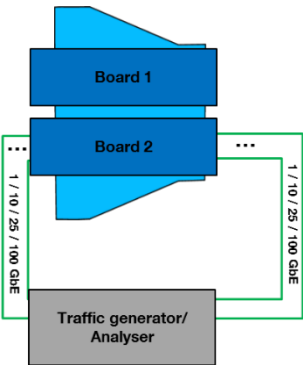
2

8.3.1 Auto-negotiation

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Test ID	Ethernet.AUTONEG.01	Classification	Functionality
Test Title	Auto-negotiation		
Test Objective:	The aim of the test is to check the system behavior after fiber network link disconnection/connection		

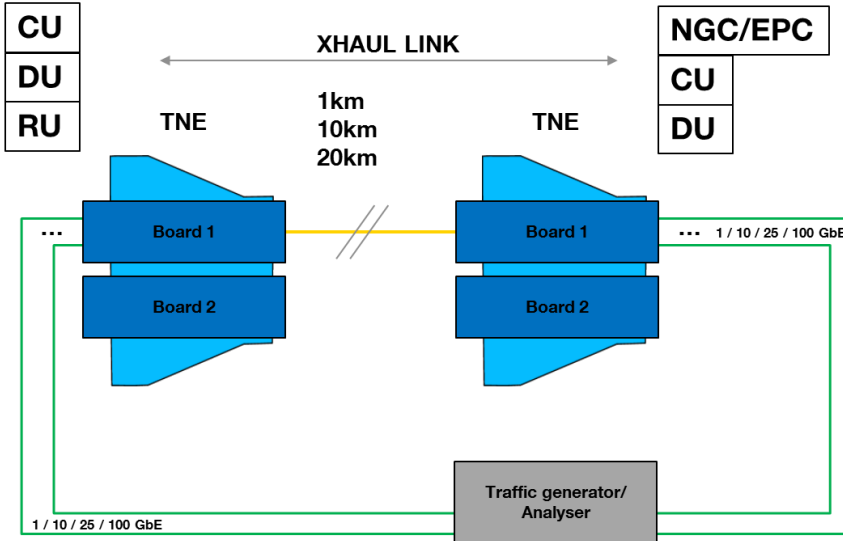
Pre-Req	Running service will be interrupted
Test Topology:	TNE 
Procedure:	<u>Detailed Test procedure:</u> <u>Test Procedure:</u> 1. Check the configuration, devices should support auto-negotiation mode. 2. Enable the auto-negotiation of the port. 3. Connect various link (port rate, duplex mode) of an Ethernet port. <u>Verifications needed:</u> 1. Verify if the port automatically negotiates the force mode
Test Configuration	IEEE 802.3
Pass/Fail Criteria:	1. Devices must support auto-negotiation mode. 2. Port must automatically negotiate the force mode
Results	[] Pass [] Pass w/X [] Dropped [] Blocked [] Failed
Test Output	

8.4 Additional Network testing

- Traffic configuration
 - Backhaul and midhaul traffic profile.
 - O-RAN fronthaul traffic profile
- Jumbo frame
- Auto-negotiation
- Frame Loss
- Latency TNE at minimum distance (0km)
- Throughput
- Latency TNE at 1, 10 and 20 km distance

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8.4.1 Latency TNE at 1, 10 and 20 km distance

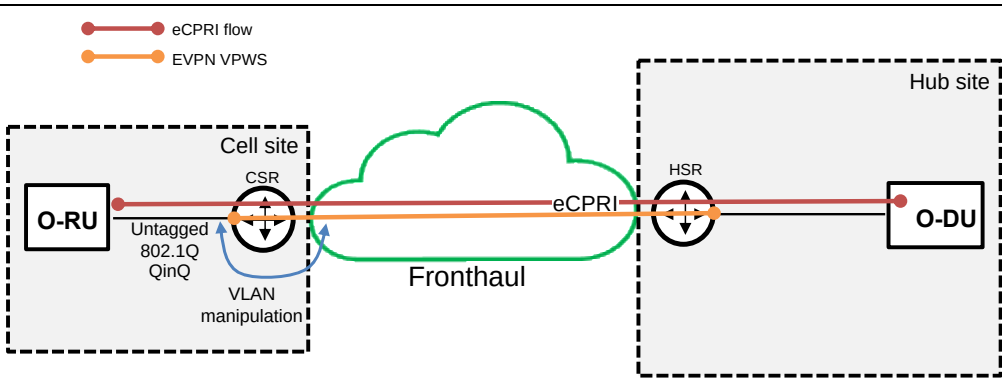
Test ID	Ethernet.LAT.01	Classification	Functionality
Test Title	Latency TNE with Xhaul link		
Test Objective:	The aim of the test is to check the maximum latency at which the TNE receives and forwards frames with no frame loss per packet size with Xhaul link (fiber or Microwave link)		
Pre-Req	Running service will be interrupted		
Test Topology:			
Procedure:	<u>Detailed Test procedure:</u> <u>Test Procedure:</u> 1. TNE is connected to another TNE or O-RU/DU/CU & NGC/EPC by 1, 10 and 20 km using fibre loop or microwave link, all services are running as defined in Traffic and services configuration. 2. Send traffic for size of frames from 70 to 1522 bytes (70, 100, 250, 300, 400, 500, 600, 700, 800, 900, 1000, 1100, 1200, 1300, 1400, 1500, 1522) frame size. Traffic should be bidirectional. 3. The test should be at least 120 seconds in duration. <u>Verifications needed:</u> 1. Make measure and report Maximum Latency Rx and Tx rate for each size of frames in included table. 2. Make measure and report Minimum Latency Rx and Tx rate for each size of frames in included table. 3. Make measure and report Average Latency Rx and Tx rate for each size of frames in included table		
Test Configuration	RFC2544		
Pass/Fail Criteria:	1- Latency MUST be below 1ms 2- Latency SHOULD be in terms of tens of microseconds		
Results	[] Pass [] Pass w/X [] Dropped [] Blocked [] Failed		

Test Output	frame size	Latency					
		Upstream			Downstream		
		Average (us)	Minimum (us)	Maximum (us)	Average (us)	Minimum (us)	Maximum (us)
	1522						
	1400						
	1300						
	1200						
	1100						
	1000						
	900						
	800						
	700						
	600						
	500						
	400						
	300						
	200						
	100						
	70						

8.4.2 VLAN Manipulation Tests

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Test ID	Ethernet.VLAN.01	Classification	Functional
Test Title	VLAN manipulation for eCPRI EVPN VPWS service		
Test Objective:	Validate VLAN manipulation (pop 1, pop 2, push 1, push2, translate 1 to 1, 1 to 2, 2 to 1, 1 to 2)		
Pre-Req	eCPRI EVPN VPWS service between CSR and HSR up and running		
Test Topology:			
Procedure:	Detailed Test procedure: Please note the following test steps partially use commands that are specific to a particular vendor. They can certainly be replaced by the commands of equipment deployed by the users of this document. Perform the following VLAN translations. - pop 1 - pop 2 - push 1 (VLAN 1100) - push 2 (VLAN 1101.2101) - translate 1 to 1 (VLAN 1102) - translate 1 to 2 (VLAN 1103.2103) - translate 2 to 1 (VLAN 1104) - translate 2 to 2 (VLAN 1105) This test shall be performed on the eCPRI EVPN VPWS service between CSR and HSR. Verifications needed: Confirm via traffic generator that there is connectivity (i.e. eCPRI traffic is flowing bidirectionally end-to-end across EVPN VPWS service between emulated O-RU and emulated O-DU).		
Test Configuration	pop 1 <pre>kszarkowicz@AN17-re0# show interfaces ae1 unit 1600</pre>		

	<pre>W50: Thu 2014-12-11T16:29:48 WET (UTC+0000) encapsulation vlan-ccc; vlan-tags outer 0x8100:1681 inner 0x8100:1600; input-vlan-map pop; output-vlan-map { push; tag-protocol-id 0x8100; inner-tag-protocol-id 0x8100; } pop 2 kszarkowicz@AN17-re0# show interfaces ae1.1603 W50: Thu 2014-12-11T16:22:36 WET (UTC+0000) encapsulation vlan-ccc; vlan-tags outer 0x8100:1681 inner 0x8100:1603; input-vlan-map pop-pop; output-vlan-map { push-push; tag-protocol-id 0x8100; inner-tag-protocol-id 0x8100; }</pre>																																																																								
Pass/Fail Criteria:	Traffic flows via EVPN VPWS service with all VLAN manipulation combinations.																																																																								
Results	[X] Pass [] Pass w/X [] Dropped [] Blocked [] Failed																																																																								
Test Output	<table><tr><th></th><th>Tx Port</th><th>Rx Port</th><th>Traffic Item</th><th>Tx Frames</th><th>Rx Frames</th><th>Frames Delta</th><th>Loss %</th></tr><tr><td>1</td><td>100GE - Broadband MX240</td><td>100GE - Broadband ACX710</td><td>Broadband</td><td>1,492,928...</td><td>1,492,928...</td><td>0</td><td></td></tr><tr><td>2</td><td>100GE - Broadband ACX710</td><td>100GE - Broadband MX240</td><td>Broadband</td><td>1,492,928...</td><td>1,492,928...</td><td>0</td><td></td></tr><tr><td>3</td><td>10GE - LTE MX240</td><td>10GE - LTE ACX710</td><td>LTE</td><td>191,948,290</td><td>191,948,290</td><td>0</td><td></td></tr><tr><td>4</td><td>10GE - LTE ACX710</td><td>10GE - LTE MX240</td><td>LTE</td><td>191,948,290</td><td>191,948,290</td><td>0</td><td></td></tr><tr><td>▶ 5</td><td>10GE - eCPRI ACX710</td><td>10GE - eCPRI ACX5448-16</td><td>eCPRI 1-2</td><td>0</td><td>0</td><td>0</td><td></td></tr><tr><td>6</td><td>10GE - eCPRI ACX5448-16</td><td>10GE - eCPRI ACX710</td><td>eCPRI 1-2</td><td>0</td><td>0</td><td>0</td><td></td></tr><tr><td>7</td><td>10GE - eCPRI ACX710</td><td>10GE - eCPRI ACX5448-24</td><td>eCPRI 1-3</td><td>71,855,403</td><td>71,855,403</td><td>0</td><td></td></tr><tr><td>8</td><td>10GE - eCPRI ACX5448-24</td><td>10GE - eCPRI ACX710</td><td>eCPRI 1-3</td><td>71,959,914</td><td>71,959,914</td><td>0</td><td></td></tr></table>		Tx Port	Rx Port	Traffic Item	Tx Frames	Rx Frames	Frames Delta	Loss %	1	100GE - Broadband MX240	100GE - Broadband ACX710	Broadband	1,492,928...	1,492,928...	0		2	100GE - Broadband ACX710	100GE - Broadband MX240	Broadband	1,492,928...	1,492,928...	0		3	10GE - LTE MX240	10GE - LTE ACX710	LTE	191,948,290	191,948,290	0		4	10GE - LTE ACX710	10GE - LTE MX240	LTE	191,948,290	191,948,290	0		▶ 5	10GE - eCPRI ACX710	10GE - eCPRI ACX5448-16	eCPRI 1-2	0	0	0		6	10GE - eCPRI ACX5448-16	10GE - eCPRI ACX710	eCPRI 1-2	0	0	0		7	10GE - eCPRI ACX710	10GE - eCPRI ACX5448-24	eCPRI 1-3	71,855,403	71,855,403	0		8	10GE - eCPRI ACX5448-24	10GE - eCPRI ACX710	eCPRI 1-3	71,959,914	71,959,914	0	
	Tx Port	Rx Port	Traffic Item	Tx Frames	Rx Frames	Frames Delta	Loss %																																																																		
1	100GE - Broadband MX240	100GE - Broadband ACX710	Broadband	1,492,928...	1,492,928...	0																																																																			
2	100GE - Broadband ACX710	100GE - Broadband MX240	Broadband	1,492,928...	1,492,928...	0																																																																			
3	10GE - LTE MX240	10GE - LTE ACX710	LTE	191,948,290	191,948,290	0																																																																			
4	10GE - LTE ACX710	10GE - LTE MX240	LTE	191,948,290	191,948,290	0																																																																			
▶ 5	10GE - eCPRI ACX710	10GE - eCPRI ACX5448-16	eCPRI 1-2	0	0	0																																																																			
6	10GE - eCPRI ACX5448-16	10GE - eCPRI ACX710	eCPRI 1-2	0	0	0																																																																			
7	10GE - eCPRI ACX710	10GE - eCPRI ACX5448-24	eCPRI 1-3	71,855,403	71,855,403	0																																																																			
8	10GE - eCPRI ACX5448-24	10GE - eCPRI ACX710	eCPRI 1-3	71,959,914	71,959,914	0																																																																			

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8.4.3 QoS testing

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Test ID	Ethernet.QOS.01	Classification	QoS
Test Title	End to end latency, loss of classified traffic		
Test Objective:	Verify end to end latency, loss of classified traffic		
Pre-Req	Each TNE supporting QoS in Xhaul Transport Network		

Test Topology:	Ports carrying Ingress traffic towards transport network Ports carrying egress traffic outwards transport network Xhaul Transport Network Traffic Generator and Analyzer			
Test Configuration	This section to be filled out when putting the test report (may include device and/or tester configuration)			
Procedure:	Detailed Test procedure: 1. Enable QoS in each TNE (core and edge devices) with operator specific scheme and connect traffic generator ports at the edge of the transport network. 2. Configure continuous traffic flows in traffic generator which would be mapped to strict priority queue(s) with a rate equal to the configured rate in policing/shaping at edge TNE(s). In case of Ethernet traffic, VLAN PCP bits/ VLAN ID/ingress port ID could be used for traffic classification in edge TNE. In case of IPv4 traffic. DSCP bits/ingress port could be used for traffic classification in edge TNE. Rate and size of these flows would be operator's choice based upon deployment scenario. 3. Configure continuous traffic flows in traffic generator which would be mapped to WRR/WFQ/MDRR queue(s) with rate more than configured rate of each of these queue(s) so that traffic congestion occurs in core. In case of Ethernet traffic, VLAN PCP bits/ VLAN ID/ingress port ID could be used for traffic classification in edge TNE. In case of IPv4 traffic. DSCP bits/ingress port could be used for traffic classification in edge TNE. Rate and size of these flows would be operator's choice based upon deployment scenario. 4. Create latency bins in traffic generator rx port for each traffic flows based on expected latency range and traffic marking such as PCP bits, IPv4 DSCP bits etc. Latency ranges are decided by operator based upon deployment scenario. For example, if latency class for fronthaul traffic is "High25", then a latency bin must be created with 0-25 us range. Other possible two bins could be 25 – 500 us and 500-max us for traffic flows other than latency sensitive ones. Table 3 of Xhaul Transport Requirement Doc is repeated here that helps in identifying latency bins in case of fronthaul traffic. <table><tr><td>Latency Class</td><td>Max. One-way Frame Delay Performance</td><td>Use case</td></tr></table>	Latency Class	Max. One-way Frame Delay Performance	Use case
Latency Class	Max. One-way Frame Delay Performance	Use case		

	High25	25 us	Ultra-low latency performance
	High75	75 us	For full NR performance with fiber lengths in 10km range
	High100	100 us	For standard NR performance with fiber lengths in 10km range
	High200	200 us	For installations with fiber lengths in 30km range
	High500	500 us	Large latency installations > 30 km
	Medium	User Plane (slow) & C&M Plane (fast)	1 ms
	Low	C&M Plane	100 ms
5. Send all the traffic flows simultaneously. Observe packets/frames received in each latency bins. Also observe loss in each traffic flows <u>Verifications needed:</u> 1. Latency sensitive traffics which are mapped to strict priority queue(s) must not suffer any frame loss. 2. All Latency sensitive traffic frames which are mapped to strict priority queue(s) must be received in respective configured latency bins in traffic generator. 3. Traffic frames mapped to WRR/WFQ/MDRR queue(s) must maintain a long term receive rate equal to the configured rate in those queues. Due to congestion excess traffic would be dropped and loss is reported in traffic generator for these traffic flows. 4. Latency reported for traffic flows mapped to WRR/WFQ/MDRR queues in general must be higher than that of traffic flows mapped to strict priority queue(s)			
Pass/Fail Criteria:		Test shall pass if all verifications steps pass	
Test Output		This section is the placeholder that will capture the output/configurations after test is executed, typically done after test execution	

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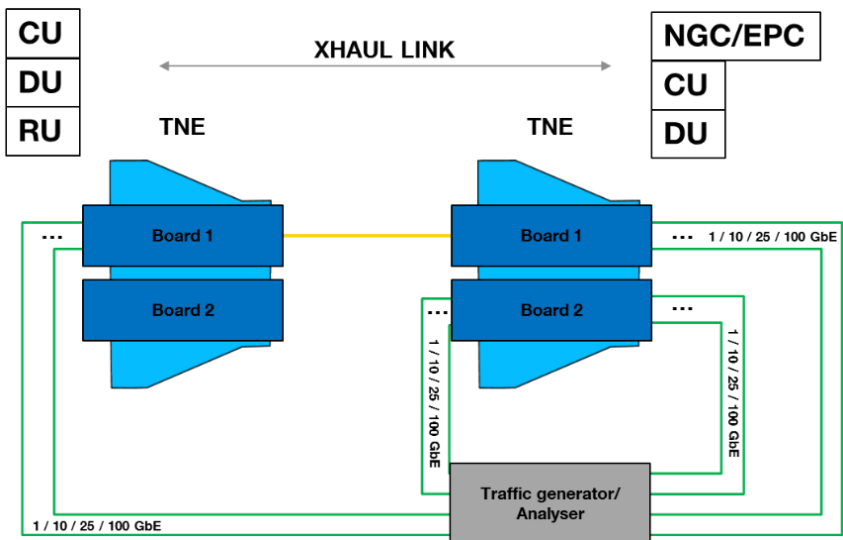
8.4.4 Jitter Testing

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This test case is FFS as it needs inputs from WG4 CUS-Specification team on the maximum jitter (packet delay variation) acceptable at an O-RU Fronthaul UNI. Future versions of this document may update this section as inputs from WG4 become available,

Test ID	Ethernet.JIT.01	Classification	Functionality				
Test Title	Delay variation						
Test Objective:	The aim of the test is to check the delay variation at which the TNE receives frames with no frame loss per packet size						
Pre-Req	N/A						
Test Topology:							
Procedure:	<u>Detailed Test procedure:</u> <u>Test Procedure:</u> - TNE is connected to another TNE. Two services are defined, one representing the Xhaul traffic, the other one representing background traffic. - Xhaul and background traffic are marked with different QoS values, being the Xhaul traffic of higher priority vs the background traffic. - Send traffic for size of frames from 70 to 1522 bytes (70, 100, 250, 300, 400, 500, 600, 700, 800, 900, 1000, 1100, 1200, 1300, 1400, 1500, 1522) frame size. Traffic should be bidirectional. - The test should be at least 120 seconds in duration. <u>Verifications needed:</u> - Make measure and report Maximum Jitter Rx and Tx rate for each size of frames in included table						
Test Configuration	RFC3393						
Pass/Fail Criteria:	1- Jitter SHOULD be below TBD microseconds						
Results	[X] Pass [] Pass w/X [] Dropped [] Blocked [] Failed						
Test Output	<table><tr><td>Frame size</td><td>Observed Jitter (µs)</td></tr><tr><td>1522</td><td></td></tr></table>			Frame size	Observed Jitter (µs)	1522	
Frame size	Observed Jitter (µs)						
1522							

		1400	
		1300	
		1200	
		1100	
		1000	
		900	
		800	
		700	
		600	
		500	
		400	
		300	
		200	
		100	
		70	

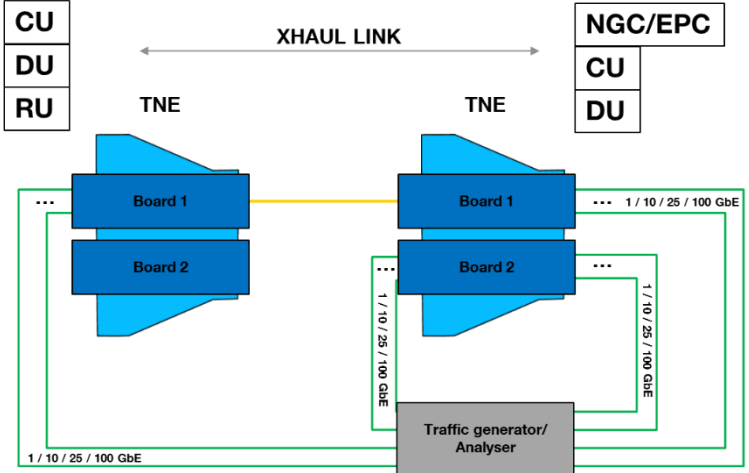
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8.4.5 Test for frame fragmentation

Test ID	Ethernet.FRAG.01	Classification	Functionality
Test Title	Impact of fragmentation		
Test Objective:	The aim of the test is to check the impacts of frame fragmentation in throughput, latency and delay variation with no frame loss per packet size (frame fragmentation occurs in the link between TNEs)		
Pre-Req	N/A		

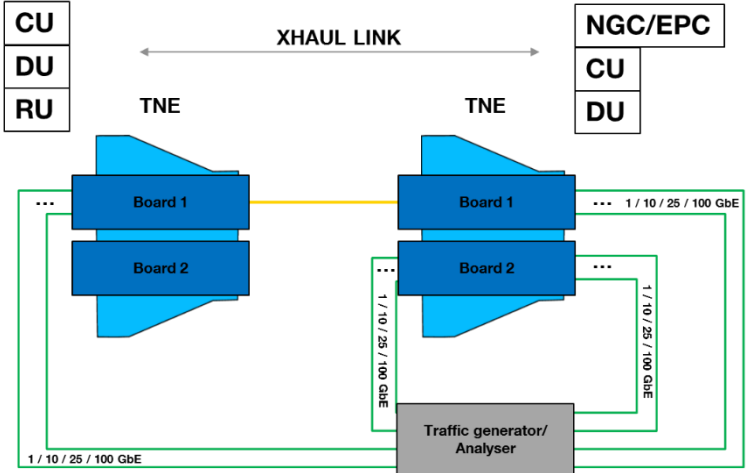
Test Topology:	
Procedure:	<u>Detailed Test procedure:</u> <u>Test Procedure:</u> 1. TNE is connected to another TNE, and the MTU among them is lower than the MTU allowed at client side (i.e., O-RU / O-DU / O-CU). 2. Two frame sizes are established, one for client side and another one between TNEs. 3. Send traffic for size higher MTU at sending client port and verify permitted MTU at receiver port. 4. The test should be at least 120 seconds in duration. <u>Verifications needed:</u> 1. Make measure and report throughput, latency and jitter with and without fragmentation
Test Configuration	
Pass/Fail Criteria:	1-
Results	[] Pass [] Pass w/X [] Dropped [] Blocked [] Failed
Test Output	

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8.4.6 Test for encapsulations and protocols

Test ID	Ethernet.ENC.01	Classification	Functionality
Test Title	Impacts of encapsulations and protocols used		
Test Objective:	The aim of the test is to check the impacts due to the usage of different encapsulations and protocols for the establishing the communication between O-RU, O-DU and O-CU. This test applies as pre-condition to the tests on latency, jitter and throughput.		
Pre-Req	N/A		
Test Topology:			
Procedure:	<u>Detailed Test procedure:</u> <u>Test Procedure:</u> - Define a proper technology selection for both O-RU <-> O-DU and O-DU <-> O-CU communication. - Perform the tests for latency, jitter and throughput, collecting the information indicated for their respective tests. - The test should be at least 15 min in duration. <u>Verifications needed:</u> - Validate the configured solution. - Make measure and report traffic passing rate for each size of frames in included table		
Test Configuration	Potential protocols and encapsulations are listed here below. - Ethernet traffic (e.g., eCPRI between O-RU and O-DU) (UDP/) IP traffic (e.g., eCPRI between O-RU and O-DU) - TSN capabilities between TNEs - FlexE connectivity between TNEs - SR-MPLS encapsulation between TNEs - SRv6 encapsulation between TNEs - L2 VPN between TNEs (e.g., for midhaul connection) - L3 VPN between TNEs (e.g., for midhaul connection)		

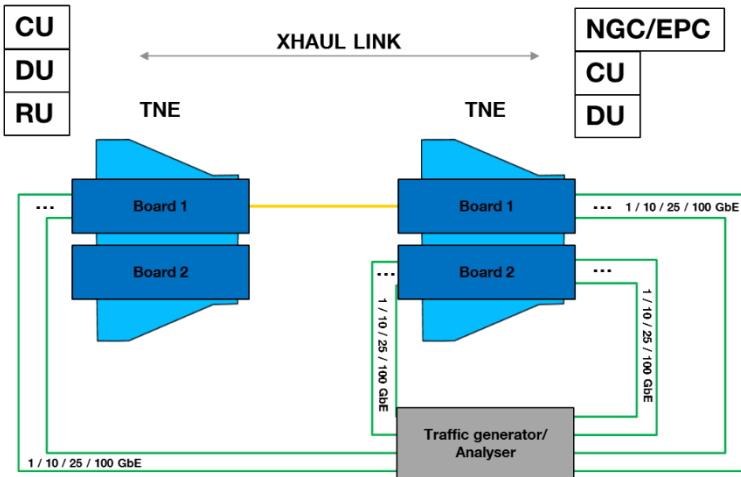
	• EVPN between TNEs (e.g., for eCPRI connection)
Pass/Fail Criteria:	Compare the results obtained for latency, jitter and throughput for the different selected combinations. Particular criteria defined for latency, jitter and throughput apply.
Results	[] Pass [] Pass w/X [] Dropped [] Blocked [] Failed
Test Output	The output of the tests will be the set of tables defined for latency, jitter and throughput, for each of the selected combinations of protocols and encapsulations.

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1 8.4.7 Addition of background traffic

2 To check if specific tests is needed or simply to amend existing ones.

Test ID	Ethernet.BAC.01	Classification	Functionality
Test Title	System behavior with the presence of background traffic		
Test Objective:	The aim of the test is to check the impacts due to the introduction of background traffic in the system. This background traffic can represent other kinds of traffic present on the TNEs (e.g., LTE traffic, enterprise traffic, fixed service traffic, etc). This test applies as complements the tests on latency, jitter and throughput.		
Pre-Req	Identify the proportion and the marking of the background traffic under consideration. Each network can have different scopes, thus influencing on the kind of background traffic under consideration.		
Test Topology:			
Procedure:	<u>Detailed Test procedure:</u> <u>Test Procedure:</u> 1. Define the kind of background traffic under consideration (e.g., LTE, enterprise, fixed, etc). Identify peak/average traffic volumes as well as applicable marking. 2. Perform the tests for latency, jitter and throughput adding this background traffic collecting the information indicated for their respective tests. 3. The test should be at least 15 min in duration. Alternative test – to fill the link between TNEs. <u>Verifications needed:</u> 1. Validate the kind of background traffic introduced in the test. 2. Make measure and report traffic passing rate for each size of frames in included table, comparing with the scenario where no background traffic is considered.		
Test Configuration			
Pass/Fail Criteria:	Compare the results obtained for latency, jitter and throughput for the scenarios with and without background traffic.		
Results	☐ Pass ☐ Pass w/X ☐ Dropped ☐ Blocked ☐ Failed		

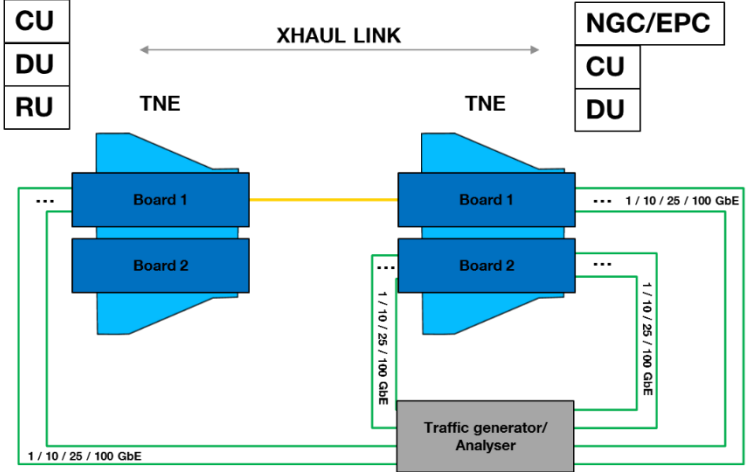
Test Output

The output of the tests will be the set of tables defined for latency, jitter and throughput, not repeated here for simplicity.

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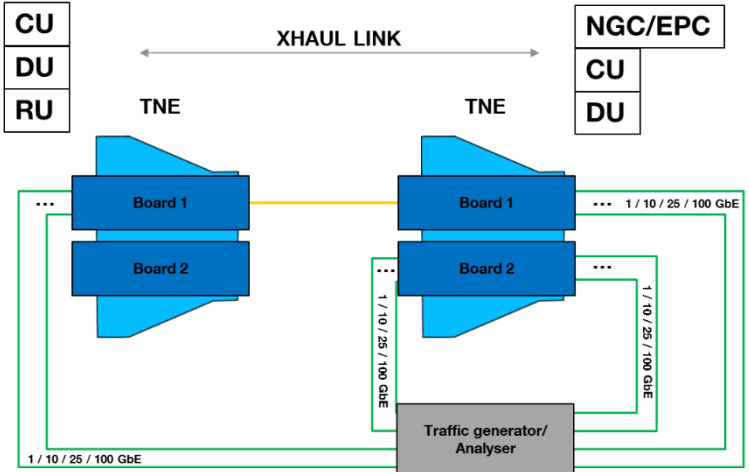
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8.4.8 Support of native IPv6 connectivity

Test ID	Ethernet.IPv6.01	Classification	Functionality
Test Title	Native support of IPv6 connectivity		
Test Objective:	The aim of the test is to check the impacts due to the usage of IPv6 in O-RU, O-DU and O-CU. IPv6 support is optional in ORAN-WG4.CUS. This is a functional test but can also have performance implications to be validated by collecting statistics of latency, jitter and throughput.		
Pre-Req	Support of native IPv6 in TNEs		
Test Topology:			
Procedure:	Detailed Test procedure: Test Procedure: 1. Define a native IPv6 connectivity in TNEs. 2. Perform connectivity between O-RU, O-DU and O-CU natively in IPv6 (if eCPRI is directly encapsulated in Eth, then this would apply solely to midhaul traffic). 3. The test should be at least 120 secs in duration. Verifications needed: 1. Validate IPv6 connectivity 2. Make measure and report traffic passing rate for each size of frames in included table, comparing with the scenario of native IPv4 connectivity.		
Test Configuration			
Pass/Fail Criteria:	Compare the results obtained for latency, jitter and throughput for the scenarios with and without background traffic.		
Results	[] Pass [] Pass w/X [] Dropped [] Blocked [] Failed		
Test Output	The output of the tests will be the set of tables defined for latency, jitter and throughput, not repeated here for simplicity.		

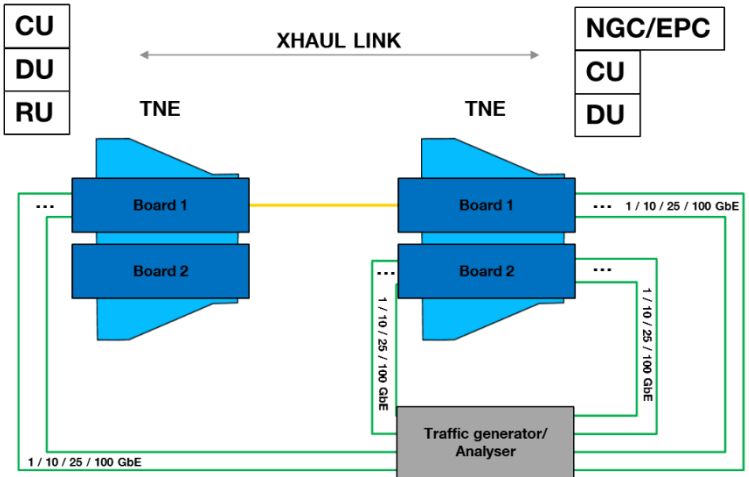
1 8.4.9 Support of IPv6 connectivity in an IPv4 network (dual stack)

2 To discuss if tunneling or translation methods should be detailed as separated tests or included here

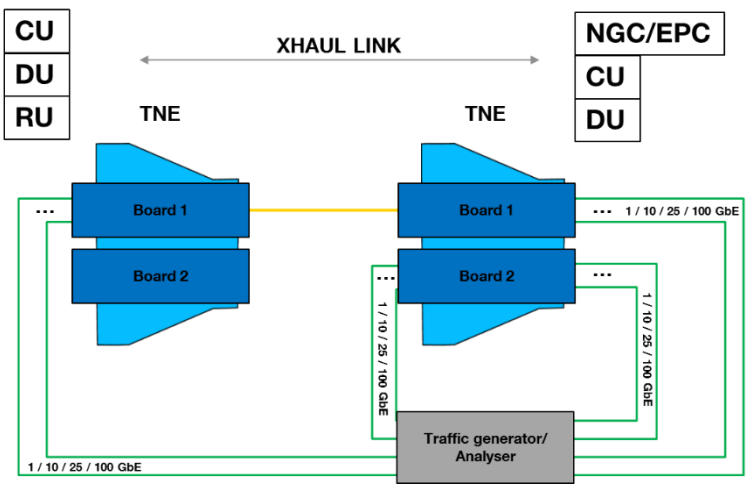
Test ID	Ethernet.DUAL.01	Classification	Functionality
Test Title	Support of IPv6 connectivity in an IPv4 network		
Test Objective:	The aim of the test is to check the impacts due to the usage of IPv6 in O-RU, O-DU and O-CU when the underlying network is IPv4-based. IPv6 support is optional in ORAN-WG4.CUS. This is a functional test but can also have performance implications to be validates by collecting statistics of latency, jitter and throughput.		
Pre-Req	Proper dual stack support (IPv4 and IPv6) by the TNEs		
Test Topology:			
Procedure:	<u>Detailed Test procedure:</u> <u>Test Procedure:</u> 1. Define a native IPv4 connectivity in TNEs. 2. Perform connectivity between O-RU, O-DU and O-CU natively in IPv6 (if eCPRI is directly encapsulated in Eth, then this would apply solely to midhaul traffic). 3. The test should be at least 120 secs in duration. <u>Verifications needed:</u> 1. Validate IPv6 connectivity with dual stack capabilities in the TNEs. 2. Make measure and report traffic passing rate for each size of frames in included table, comparing with the scenario of native IPv4 connectivity.		
Test Configuration	To be defined the mechanisms for IPv6 to IPv4 consideration		
Pass/Fail Criteria:	Compare the results obtained for latency, jitter and throughput for the scenarios with and without background traffic.		
Results	[X] Pass [] Pass w/X [] Dropped [] Blocked [] Failed		
Test Output	The output of the tests will be the set of tables defined for latency, jitter and throughput, not repeated here for simplicity.		

8.4.10 Support of IPv4 connectivity in an IPv6 network (dual stack)

To discuss if tunneling or translation methods should be detailed as separated tests or included here.

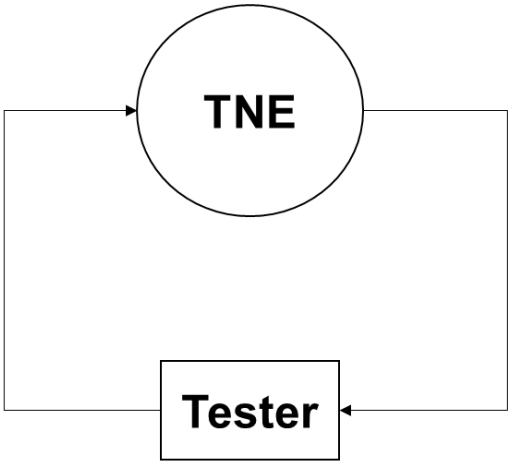
Test ID	Ethernet.DUAL.02	Classification	Functionality
Test Title	Support of IPv4 connectivity in an IPv6 network		
Test Objective:	The aim of the test is to check the impacts due to the usage of IPv4 in O-RU, O-DU and O-CU when the underlying network is IPv6-based. This is a functional test but can have also performance implications to be validates by collecting statistics of latency, jitter and throughput.		
Pre-Req	Proper dual stack support (IPv4 and IPv6) by the TNEs		
Test Topology:			
Procedure:	<u>Detailed Test procedure:</u> <u>Test Procedure:</u> - Define a native IPv4 connectivity in TNEs. - Perform connectivity between O-RU, O-DU and O-CU natively in IPv6 (if eCPRI is directly encapsulated in Eth, then this would apply solely to midhaul traffic). - The test should be at least 120 secs in duration. <u>Verifications needed:</u> - Validate IPv6 connectivity with dual stack capabilities in the TNEs. - Make measure and report traffic passing rate for each size of frames in included table, comparing with the scenario of native IPv4 connectivity.		
Test Configuration	To be defined the mechanisms for IPv6 to IPv4 consideration		
Pass/Fail Criteria:	Compare the results obtained for latency, jitter and throughput for the scenarios with and without background traffic.		
Results	[] Pass [] Pass w/X [] Dropped [] Blocked [] Failed		
Test Output	The output of the tests will be the set of tables defined for latency, jitter and throughput, not repeated here for simplicity.		

8.4.11 Test for in-band / out-band management

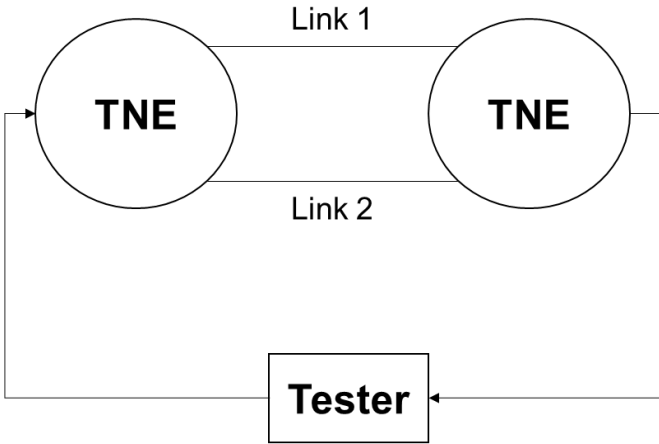
Test ID	Ethernet.IOUT.01	Classification	Functionality
Test Title	Validation of in-band /out-band management of TNEs		
Test Objective:	The aim of the test is to validate the in-band management of the device. TNEs addressing fronthaul / midhaul connectivity will usually lack of dedicated out-band management network.		
Pre-Req	Configure in-band management plane connection. According to O-RAN.WG4.CUS specification, the management plane Cos is 2, and the DSCP code is AF2x. Introduce background traffic of higher and lower priority and validate management actions on the device.		
Test Topology:			
Procedure:	<u>Detailed Test procedure:</u> <u>Test Procedure:</u> - Defined a proper reachability of management IP address in-band, in such a way that the management address of one TNE can be accessed through the other TNE. - Configure background traffic (with higher and lower priorities) to compete for the network resources. Proper dimensioning can follow the scenarios described in O-RAN-WG9.Transport requirement document. - The test should be at least 5 min in duration. <u>Verifications needed:</u> - Validate the possibility of managing the device in-band. - Compare the responses to configuration actions between out-band and in-band management of the device.		
Test Configuration			
Pass/Fail Criteria:	Compare the results obtained with out-band management.		
Results	[] Pass [] Pass w/X [] Dropped [] Blocked [] Failed		

Test Output	The output of the tests will be the set of tables defined for latency, jitter and throughput, for each of the selected combinations of protocols and encapsulations.
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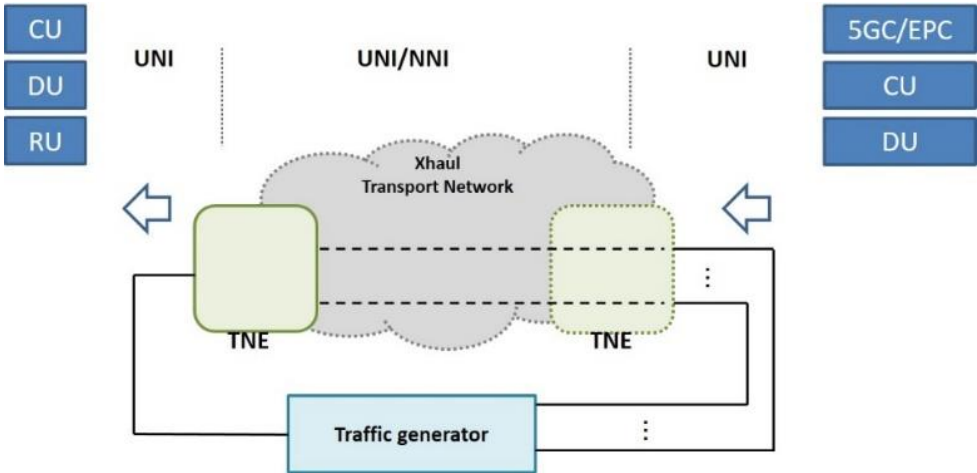
8.4.12 Transparency of eCPRI Ethertype

Test ID	Ethernet.ETHTYPE.01	Classification	Functionality
Test Title	Transparency of eCPRI ethertype		
Test Objective:	The aim of the test is to validate that eCPRI Ethertype is passed transparently and frames with such Ethertype are not discarded		
Pre-Req			
Test Topology:			
Procedure:	<u>Detailed Test procedure:</u> To check if different encapsulations can have any impact. To check if the support of IEEE 1914 ethertype is also required. <u>Test Procedure:</u> 1. The tester will inject eCPRI frames with the corresponding eCPRI Ethertype. 2. The test should be at least 120 secs in duration. <u>Verifications needed:</u> 1. Validate that eCPRI Ethertype is transparently supported		
Test Configuration			
Pass/Fail Criteria:	No loss frame		
Results	[] Pass [] Pass w/X [] Dropped [] Blocked [] Failed		
Test Output	No losses.		

8.4.13 Recovery times

Test ID	Ethernet.REC.01	Classification	Functionality
Test Title	Service restoration		
Test Objective:	The aim of the test is to measure the time for service restoration in the event of link failure		
Pre-Req	Configure to links between TNEs. Configure the proper encapsulations / protocol support for the service.		
Test Topology:			
Procedure:	<u>Detailed Test procedure:</u> <u>Test Procedure:</u> 1. Configure the service emulating fronthaul / midhaul / backhaul connection. 2. Inject traffic. 3. Shut down the link from where the traffic is passing. 4. Measure the frame losses and the time to recover. <u>Verifications needed:</u> 1. Validate the time for service recovery		
Test Configuration			
Pass/Fail Criteria:	Recovery time		
Results	☐ Pass ☐ Pass w/X ☐ Dropped ☐ Blocked ☐ Failed		
Test Output	Recovery time		

8.4.14 Test of Buffer Sizing

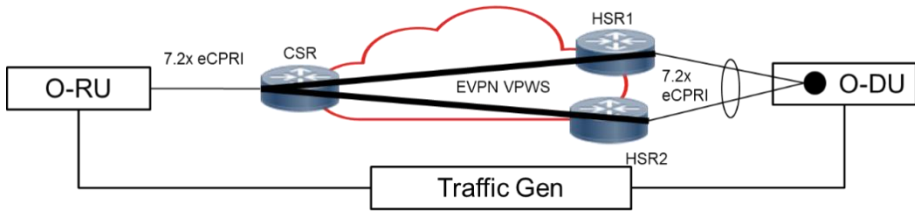
Test ID	Ethernet.BUF.01	Classification	Functionality
Test Title	Buffer Sizing		
Test Objective:	This test is For Further Study. It is meant for midhaul and backhaul cases. The aim of the test is to check the buffering capacity at which the TNE receives and forwards the traffic which the amount of the ingress traffic is greater than the throughput of the egress interface with no frame loss with Xhaul (Midhaul/ Backhaul) link.		
Pre-Req	N/A		
Test Topology:			
Test Configuration	- There could be one TNE or two or more TNEs. Service is configured in one TNE /among TNEs to make traffic be switched. - The buffer size parameter of shaping function is not defined but no frame loss is desired.		
Pass/Fail Criteria:			
Results	☒ Pass ☐ Pass w/X ☐ Dropped ☐ Blocked ☐ Failed		

8.5 Fronthaul Services Redundancy and High Availability

This section covers the Fronthaul redundancy scenarios where the CSR (on the O-RU side) is connected to 2 (or more) HSR's that provide connectivity to the O-DU. EVPN VPWS is used between the two sites in a redundant configuration as defined in the O-RAN Xhaul Packet switched Network Architecture. All test cases in this section correspond to section 13 of the Packet Switched Specification.

In the following test cases a Traffic Gen (also referred to as an O-DU or O-RU simulator) is used to generate and analyze test traffic towards O-RU or O-DU. It is connected directly to a CSR or an HSR port that usually connects to an O-RU or an O-DU. For illustration purposes the O-RU or O-DU are left in the test setup diagrams, although they are not needed for the test application.

8.5.1 Multi-Home Active/Active Fronthaul Redundancy

Test ID	Ethernet.HA.01	Classification	Redundancy & High Availability
Test Title	Active-active Fronthaul service redundancy		
Test Objective:	Verify O-RU – O-DU redundancy using Active-Active EVPN VPWS services		
Pre-Req	SR-MPLS and/or SRv6 baseline configuration on CSR, HSR and any other Fronthaul routers Timing and syn configuration through FH network for RAN Nodes to sync		
Test Topology:			
Test Configuration	1. Connect O-RU to the UNI of the CSR and dual-home O-DU to multiple HSRs. 2. Configure EVPN VPWS Active -Active between CSR and HSR as shown in test topology. 3. O-DU should support terminating both port from HSRs on a single logical BD		
Procedure:	Detailed Test procedure: 1. Verify O-RU and O-DU are up and communicating. 2. Start bi-directional traffic between the RAN nodes, verify traffic is being received on either ends. 3. Fail 1 of the two Active-Active EVPN VPWS services by either: a. Powering off HSR-1, or b. Disconnect the UNI from HSR-1 to O-DU 4. Verify all traffic is converged on the other EVPN VPWS circuit. 5. Wait for traffic to regain pre-failure rates, then recover the failure, by powering on the HSR-1 or reconnecting UNI on HSR-1.		

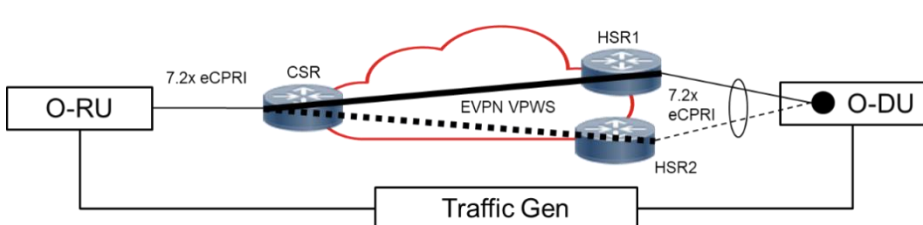
	6. Wait for EVPN services to recover to pre-failure state with Active-Active configuration
Pass/Fail Criteria:	During Failure: 1. No application layer connectivity loss between O-RU and O-DU (O-RU should not reboot) 2. Document Failure convergence time. Ensure the convergence time is within Operator defined upper limits. 3. All traffic converges on the Active VPWS link with 0 Packet loss post convergence. During Recovery 4. No application layer connectivity loss between O-RU and O-DU (O-RU should not reboot) 5. Document recovery convergence time. Ensure the convergence time is within Operator defined upper limits. 6. All traffic reverts to using Active-Active VPWS configuration with 0 Packet loss post convergence.
Test Output	Capture syslogs and any error messages received during failure and recovery. Document failure and recovery convergence times

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8.5.2 Multi-Home Single-Active Bundle Fronthaul Redundancy

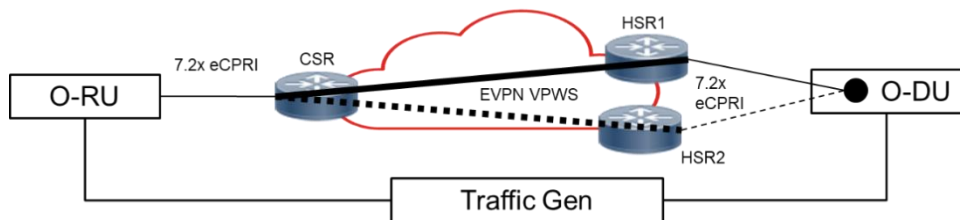
Test ID	Ethernet.HA.02	Classification	Redundancy & High Availability
Test Title	Active-active Fronthaul service redundancy		
Test Objective:	1. Verify O-RU – O-DU redundancy using Single-Active EVPN VPWS services		
Pre-Req	SR-MPLS and/or SRv6 baseline configuration on CSR, HSR and any other Fronthaul routers Timing and syn configuration through FH network for RAN Nodes to sync		
Test Topology:			
Test Configuration	1. Connect O-RU to the UNI of the CSR and dual-home O-DU to multiple HSRs. 2. Configure EVPN VPWS Single-Active between CSR and HSR as shown in test topology. 3. O-DU should support terminating both port from HSRs on a single logical BD		
Procedure:	Detailed Test procedure: 1. Verify O-RU and O-DU are up and communicating.		

	- Start bi-directional traffic between the RAN nodes, verify traffic is being received on either ends using active EVPN VPWS only, in this case through HSR1. - Fail the Active EVPN VPWS services by either: - Powering off HSR-1, or - Disconnect the UNI from HSR-1 to O-DU - Verify all traffic is converged on the other EVPN VPWS circuit through HSR2. - After full traffic convergence, recover the failure, by powering on the HSR-1 or reconnecting UNI on HSR-1. - Wait for EVPN services to recover to pre-failure state with Single-Active configuration
Pass/Fail Criteria:	During Failure: - No application layer connectivity loss between O-RU and O-DU (O-RU should not reboot) - Document Failure convergence time. Ensure the convergence time is within Operator defined upper limits. - All traffic converges on the backup VPWS link, that now becomes active, with 0 Packet loss post convergence. During Recovery - No application layer connectivity loss between O-RU and O-DU (O-RU should not reboot) - Document recovery convergence time, if any. Ensure the convergence time is within Operator defined upper limits. - All traffic revert to pre-failure state, Single-Active VPWS configuration with 0 Packet loss post convergence.
Test Output	Capture syslogs and any error messages received during failure and recover. Document failure and recovery convergence times

1

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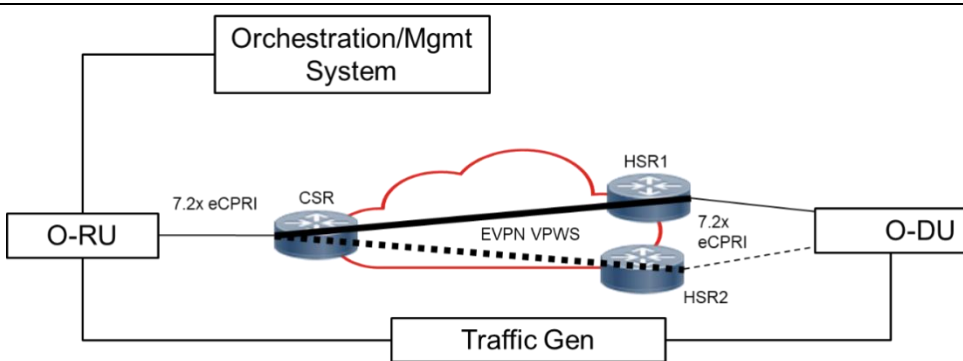
8.5.3 Multi-Home Single-Active Standalone Fronthaul Redundancy

Test ID	Ethernet.HA.03	Classification	Redundancy & High Availability
Test Title	Active-active Fronthaul service redundancy		
Test Objective:	2. Verify O-RU – O-DU redundancy using Single-Active EVPN VPWS services		
Pre-Req	SR-MPLS and/or SRv6 baseline configuration on CSR, HSR and any other Fronthaul routers Timing and syn configuration through FH network for RAN Nodes to sync		
Test Topology:			

Test Configuration	1. Connect O-RU to the UNI of the CSR and dual-home O-DU to multiple HSRs. 2. Configure EVPN VPWS Single-Active between CSR and HSR as shown in test topology. 3. O-DU should support terminating both port from HSRs on a single logical BD. 4. UNI's from HSR pair are not part of the port channel in this scenario
Procedure:	<u>Detailed Test procedure:</u> 1. Verify O-RU and O-DU are up and communicating. 2. Start bi-directional traffic between the RAN nodes, verify traffic is being received on either ends using active EVPN VPWS only, in this case through HSR1. 3. Fail the Active EVPN VPWS services by either: a. Powering off HSR-1, or b. Disconnect the UNI from HSR-1 to O-DU 4. Verify all traffic is converged on the other EVPN VPWS circuit through HSR2. 5. After full traffic convergence, recover the failure, by powering on the HSR-1 or reconnecting UNI on HSR-1. 6. Wait for EVPN services to recover to pre-failure state with Single-Active configuration
Pass/Fail Criteria:	During Failure: 1. No application layer connectivity loss between O-RU and O-DU (O-RU should not reboot) 2. Document Failure convergence time. Ensure the convergence time is within Operator defined upper limits. 3. All traffic converges on the backup VPWS link, that now becomes active, with 0 Packet loss post convergence. During Recovery 4. No application layer connectivity loss between O-RU and O-DU (O-RU should not reboot) 5. Document recovery convergence time, if any. Ensure the convergence time is within Operator defined upper limits. 6. All traffic revert to pre-failure state, Single-Active VPWS configuration with 0 Packet loss post convergence.
Test Output	Capture syslogs and any error messages received during failure and recover. Document failure and recovery convergence times

8.5.4 Multi-Home Single-Active Standalone, Fronthaul Redundancy, no BD on O-DU

Test ID	Ethernet.HA.04	Classification	Redundancy & High Availability
Test Title	Active-active Fronthaul service redundancy		
Test Objective:	1. Verify O-RU – O-DU redundancy using Single-Active EVPN VPWS services		

Pre-Req	SR-MPLS and/or SRv6 baseline configuration on CSR, HSR and any other Fronthaul routers Timing and syn configuration through FH network for RAN Nodes to sync
Test Topology:	
Test Configuration	1. Connect O-RU to the UNI of the CSR and dual-home O-DU to multiple HSRs. 2. Configure EVPN VPWS Single-Active between CSR and HSR as shown in test topology. 3. In this case, O-DU terminates both port from HSRs on separate interface each with its own MAC address, instead of a single MAC address using logical BD. 4. UNI's from HSR pair are not part of the port channel in this scenario
Procedure:	<u>Detailed Test procedure:</u> 1. Verify O-RU and O-DU are up and communicating. 2. Start bi-directional traffic between the RAN nodes, verify traffic is being received on either end using active EVPN VPWS only, in this case through HSR1. 3. Fail the Active EVPN VPWS service by either: a. Powering off HSR-1, or b. Disconnect the UNI from HSR-1 to O-DU 4. Upon detection of failure, the orchestration system should change the O-RU configuration to reflect the new MAC address to allow O-RU to communicate with the DU using new MAC. 5. Verify all traffic is converged on the other EVPN VPWS circuit through HSR2. 6. After full traffic convergence, recover the failure, by powering on the HSR-1 or reconnecting UNI on HSR-1. 7. Wait for EVPN services to recover to pre-failure state with Single-Active configuration
Pass/Fail Criteria:	<u>During Failure:</u> 1. Successful O-RU re-configuration. O-RU behavior on reconfiguration will depend upon RU vendor implementation. 2. Document Failure convergence time. Evaluate whether the convergence time is in-line with Operator expectations. The convergence will be higher as O-RU reconfiguration will have to take place. 3. All traffic converges on the backup VPWS link, that now becomes active, with 0 Packet loss post convergence. <u>During Recovery</u> 4. No application layer connectivity loss between O-RU and O-DU (O-RU should not reboot) 5. Document recovery convergence time, if any

	6. All traffic reverts to pre-failure state, Single-Active VPWS configuration with 0 Packet loss post convergence.
Test Output	Capture syslogs and any error messages received during failure and recover. Document failure and recovery convergence times

1

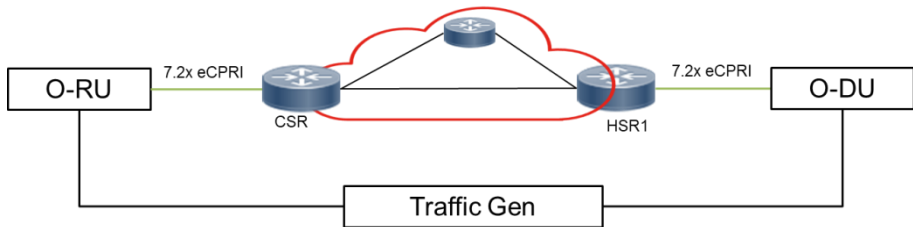
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8.5.5 Transport network Redundancy

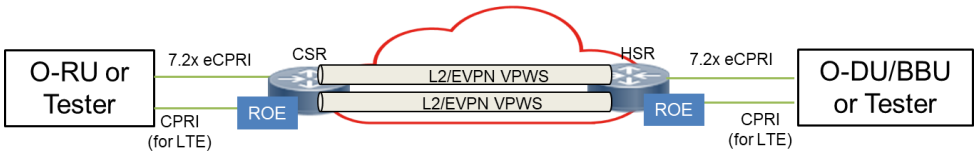
Test ID	Ethernet.HA.05	Classification	Redundancy & High Availability
Test Title	Transport network path redundancy		
Test Objective:	1. Verify traffic convergence following a link/node failure in FH Network comprising redundant paths(s)		
Pre-Req	SR-MPLS and/or SRv6 baseline configuration on CSR, HSR and any other Fronthaul routers Optional Timing and sync configuration through FH network for RAN Nodes if actual O-RU and O-DU are used instead of O-RU and O-DU simulators Configure Topology Independent Loop Free Alternate (TI-LFA) in the FH network		
Test Topology:			
Test Configuration	1. Connect O-RU or O-RU simulator to the UNI of the CSR and O-DU or O-DU simulator to the HSR. 2. Configure L2VPN or EVPN VPWS Service between CSR and HSR as shown in test topology. 3. Configure L3VPN services between CSR and HSR for M-Plane 4. Verify TI-LFA backup path is programmed in the forwarding table for all services		
Procedure:	1. Verify O-RU and O-DU are up and communicating. If actual O-RU and O-DU are used 2. Start bi-directional traffic between the RAN nodes for both L2VPN/EVPN and L3VPN services, verify traffic is being received on either end . 3. Fail the IGP best path link from CSR to HSR, through which the traffic is being forwarded. Types of failure may include, but not limited: a. Physical link failure on the CSR or HSR' directly connected interface, b. A link failure on intermediary nodes along the traffic path, or c. A node failure on intermediary node along the traffic path		

	4. Verify Upon detection of failure, traffic convergences over the backup path for all services. 5. After full traffic convergence, gather data and recover the failure, by reconnecting the failed link. 6. Ensure all services recover to pre-failure state with Single-Active configuration
Pass/Fail Criteria:	During Failure: 1. Document the failure impact on O-RU and O-DU, including whether there is any application layer connectivity loss between O-RU and O-DU or if O-RU reboots during the process. 2. Successful traffic convergence over backup path for all services. 3. Document Failure convergence time. Evaluate whether the convergence time is in-line with Operator expectations. 4. No packet loss for any service once all traffic is converged. During Recovery 5. Document the recovery impact on O-RU and O-DU, including whether there is any application layer connectivity loss between O-RU and O-DU or if O-RU reboots during the process. 6. Document recovery convergence time if any 7. All traffic reverts to pre-failure state with 0 Packet loss post convergence.
Test Output	Capture syslogs and any error messages received during failure and recover. Document failure and recovery convergence times

8.6 Legacy Fronthaul and C-RAN Coexistence Verification

This section covers the scenarios where an operator may have both 4G and 5G Fronthaul networks using C-RAN. In those cases operator may need to have traditional CPRI and 7-2x traffic coexisting on the same Packetized fronthaul requiring the CPRI traffic converted to packet using IEEE 1914-3 ROE specification as described in O-RAN Xhaul Packet Switched network architecture. Following test cases could be used to support and validate this scenario.

8.6.1 Legacy CPRI with RoE Type 0 Coexistence with 7-2x eCPRI

Test ID	Ethernet.CPRI.01	Classification	Functionality
Test Title	Coexistence and transport of Legacy CPRI over Packetized Fronthaul using IEEE 1914.3 ROE Structure agnostic Type 0 encapsulation		
Test Objective:	- Verify that Legacy CPRI and O-RAN 7-2x traffic can coexist on Packet Fronthaul. - Verify that CSR and HSR can use ROE mapper/demapper functions to convert CPRI into Packet and vice versa		
Pre-Req	SR-MPLS and/or SRv6 baseline configuration on CSR, HSR and any other Fronthaul routers Timing and syn configuration through FH network if RAN Nodes are used, not required if test equipment or CPRI/7-2x simulator is used		
Test Topology:			
Test Configuration	- O-RU and O-DU receives CPRI and 7-2x traffic on its UNI through tester or actual RAN Nodes. - Interface speed on the uplink must be capable of transporting combine CPRI and 7-2X traffic. - IEEE 1914.3 ROE Structure Agnostic Type 0 configuration on UNI at CSR and HSR		
Procedure:	Detailed Test procedure: - Connect 5G O-RU and O-DU to appropriate ethernet ports on CSR and HSR. - Simultaneously, connect 4G RU and corresponding BBU to CPRI port on CSR and HSR. - Configure EVPN VPWS circuits for each of the two connections between CSR and HSR. - Bi-directional traffic should be sent on both the Ethernet (7-2x connected) port and ROE-enabled (CPRI facing) ports. - Test should be repeated with SR-MPLS and/or SRv6 control planes depending on FH transport technology used, in accordance with the O-RAN Xhaul packet switched document. - Test could be repeated for various CPRI rate options (typical are option 3 – 8, but others could be tested as well)		
Pass/Fail Criteria:	Successful Transport of both traffic types with 0 packet loss, within FH latency budget: Correlation of counters on both CSR and HSR for: 7-2x Traffic: Interface counters, EVPN VPWS counters, Bridge-domain or others as implemented by the TNE vendor CPRI Traffic: Two set of measurements/correlation should be used for CPRI traffic: - Hyperframe counters on Test equipment should match on both sides. - ROE Mapper/Demapper Counters on CSR/HSR and CPRI controller should match on both side.		

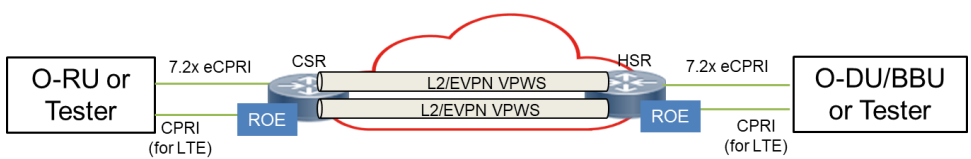
	3. EVPN VPWS services counters, others as implemented by the TNE vendor
Test Output	Capture any data w.r.t: Transport underlay: SRv6 or SR-MPLS control plane, routing (ISIS/OSPF) adjacencies, EVPN VPWS service verification (PW is up on both sides), CPRI Transport: CPRI controller information, ROE Mapper/Demapper function verification, Traffic Forwarding: Interface forwarding on CPRI and 7-2x UNI, EVPN service and CPRI controller. Document data rates comparing CPRI with packet data after ROE Structure agnostic Type 0 mapping for legacy traffic.

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8.6.2 Legacy CPRI with RoE Type 1 Coexistence with 7-2x eCPRI

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Test ID	Ethernet.CPRI.02	Classification	Functionality
Test Title	Coexistence and transport of Legacy CPRI over Packetized Fronthaul using IEEE 1914.3 ROE Structure agnostic Type 1 encapsulation		
Test Objective:	1. Verify that Legacy CPRI and O-RAN 7-2x traffic can coexist on Packet Fronthaul. 2. Verify that CSR and HSR can use ROE Mapper/Demapper functions to convert CPRI into Packet using Structure agnostic Type 1 function and vice versa		
Pre-Req	SR-MPLS and/or SRv6 baseline configuration on CSR, HSR and any other Fronthaul routers Timing and syn configuration through FH network if RAN Nodes are used, not required if test equipment or CPRI/7-2x simulator is used		
Test Topology:			
Test Configuration	1. O-RU and O-DU receives CPRI and 7-2x traffic on its UNI through tester or actual RAN Nodes. 2. Interface speed on the uplink must be capable of transporting combine CPRI and 7-2X traffic. 3. IEEE 1914.3 ROE Structure Agnostic Type 1 configuration on UNI at CSR and HSR		
Procedure:	Detailed Test procedure: 1. Connect 5G O-RU and O-DU to appropriate ethernet ports on CSR and HSR. 2. Simultaneously, connect 4G RU and corresponding BBU to CPRI port on CSR and HSR.		

	3. Configure EVPN VPWS circuits for each of the two connections between CSR and HSR. 4. Bi-directional traffic should be sent on both the Ethernet (7-2x connected) port and ROE-enabled (CPRI facing) ports. 5. Test should be repeated with SR-MPLS and/or SRv6 control planes depending on FH transport technology used, in accordance with the O-RAN Xhaul packet switched document. 6. Test could be repeated for various CPRI rate options (typical are option 3 – 7, but others could be tested as well)
Pass/Fail Criteria:	Successful Transport of both traffic types with 0 packet loss Correlation of counters on both CSR and HSR for: 7-2x Traffic: Interface counters, EVPN VPWS counters, Bridge-domain or others as implemented by the TNE vendor CPRI Traffic: ROE Mapper/Demapper Counters, CPRI controller counters, Interface, EVPN VPWS services counters, others as implemented by the TNE vendor. Expected ~20% bandwidth savings from CPRI and ROE based Packet transport for CPRI Rate Options 3-7 due to 8B10B line coding removal
Test Output	Capture any data w.r.t: Transport underlay: SRv6 or SR-MPLS control plane, routing (ISIS/OSPF) adjacencies, EVPN VPWS service verification (PW is up on both sides), CPRI Transport: CPRI controller information, ROE Mapper/Demapper function verification, Traffic Forwarding: Interface forwarding on CPRI and 7-2x UNI, EVPN service and CPRI controller. Document data rates comparing CPRI with packet data after ROE Structure agnostic Type 1 mapping for legacy traffic.

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9 Xhaul Synchronization Testing

9.1 Synchronization test use cases, parameters and methodologies.

This chapter delivers the testing methodologies, parameters and limits for PTP-based network technologies. The chapter is organized in different sections each representing a typical synchronization configuration mode. For each mode, all related metrics are identified, and limits are provided to deliver pass/fail verdicts.

Various synchronization options have been defined in the O-RAN WG4 CUS-Plane Specification [2] (LLS-C1, LLS-C2, LLS-C3 and LLS-C4). Depending on the specific O-RAN deployment being considered, not all of them might be relevant. When testing the S-Plane, the System Integrator shall identify which of the test cases are relevant depending on the specific deployment scenarios addressed. As a general guideline the following applies:

1. Direct connection between O-DU and O-RU:

LLS-C1 is generally the main sync option to be validated.

LLS-C4 may be considered as an alternative or as a complement to LLS-C1.

2. Bridged network between O-DU and O-RU

LLS-C2 for cases where the synchronization is delivered to the O-RU via the O-DU and over the bridged network. In this case the PRTC/PRC may be embedded in the O-DU or may be located anywhere in the network (connected via backhaul or FH transport).

LLS-C3 for cases when the synchronization is distributed to the O-RU without involving the O-DU In this case the PRTC/PRC may be located anywhere in the network (connected via backhaul or FH transport) and may also be co-located with the O-DU.

LLS-C4 may be considered as alternative or as a complement to LLS-C2/LLS-C3. LLS-C4 is considered for future versions of the specification.

The Fronthaul focused tests for S-Plane for the current version of this specification covers LLS-C1, LLS-C2 and LLS-C3 using the ITU-T G.8275.1 [5] profile (Full Timing Support). Fronthaul focused tests for S-Plane for LLS-C1, LLS-C2 and LLS-C3 using the ITU-T G.8275.2 [6] profile (Partial Timing Support), and LLS-C4 are for future study.

The test cases apply to both lab and field deployment cases. For lab cases, the type and number of inserted Transport Node Elements (TNE) will be determined by operators (or any other applicable end customer). The tests are performed at Xhaul UNI interfaces, and not between individual TNEs. The Xhaul UNI interfaces include O-RU Fronthaul UNI, O-DU Fronthaul or Midhaul UNI.

The Test Topologies in all the test cases uses GNSS simulator to generate the GNSS signal to the network PRTC/T-GM. This high-quality reference from the simulator is used as the external reference for clock testers as well. The resultant accuracy of this method is a few nanoseconds. However, one can also use a GNSS signal feeding both the initial PRTC/T-GM in the network and the reference clock generating a 1pps/10MHz used to synchronize the clock testers for the measurements. Operator can make an allowance in this case, for all of the measurements for the accuracy of the reference. The magnitude of this allowance will depend on the specific GNSS reference in use.

For relative time error measurements at UNI (e.g., O-RU Fronthaul UNI) this document uses the metric $\max|TE_{RL}|$ (1,2) as defined in [88], Appendix XI. Please note this metric represents an estimation and the worst-case scenario. Tests that meet (are below) the limits are seen as PASS. Tests not meeting this metric may still be acceptable. The decision is for operators to make. However, if the measurement equipment support relative time error measurement, operator can use such equipment. Refer ITU-T G.8271.1 Appendix XI for more details.

For Frequency Limit at UNI (e.g., O-RU Fronthaul UNI) this document uses the metric defined as per WG4 section H.2 based on O-DU class for LLS-C1 and LLS-C2 configuration and based on PRTC class for LLS-C3 configuration. Applicable MTIE limit shall apply for Frequency as per WG4 specification.

ITU-T G.8273 Annex B mentions the Phase/time clock equipment specification related measurement methods. One can refer to this section for any additional information and guideline for such measurements.

The acceptable limits in all the below tests shall meet CUS specification worst case limits based on the maximum network size, but operator “should” apply the limits more aligned with the real tested configuration based on the network size.

9.1.1 Network Time Synchronization-LLS-C1 (Option-A) configuration

Test ID	SYNC.TE.01	Classification	Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Measurement of Frequency Error, Time Error, Relative Time Error at O-RU UNI. Configuration-LLS-C1(Option-A) (T-GM embedded in O-DU)		
Test Objective:	1. Verify the Maximum Absolute Packet Time Error ($\max TE_L$) at O-RU UNI. 2. Verify the Maximum Relative Packet Time Error ($\max TE_{RL}$) between two O-RU UNI ports. 3. Verify the Maximum Absolute 1PPS Time Error ($\max TE_L$) at O-RU UNI. 4. Verify the Time Wander at the O-RU UNI.		
Pre-Req	ODU (Integrated T-GM) connected to PRTC/GNSS and locked to PRTC. O-RU UNI is the measurement Plane		
Test Topology:	For Enhanced O-RU (FR2-FR1) $\max TE_{RL} =60-190\text{ns}$ at O-RU UNI $TAE=130-260\text{ns}$ For Regular O-RU (FR1) $\max TE_{RL} =100\text{ns}$ at O-RU UNI $TAE=260\text{ns}$ $TE_L \leq 1420\text{ns}$ $\pm 30-95\text{ns} \dots \pm 50\text{ns}$ (FR2-FR1) ---- (FR1)		

	The test topology above illustrates the placement of clock tester at the measurement plane with 1PPS and 10MHz reference from GNSS simulator connected to clock tester. The tester measurement ports at O-RU UNI connected to O-DU.
Test Configuration	1. Clock Tester at O-RU UNI receives timing input on its measurement port. 2. Provide 1PPS and 10MHz reference to Clock Tester from GNSS simulator. 3. G.8275.1 Full Timing Support profile.
Procedure:	<u>Detailed Test procedure:</u> 1. Connect T-GM (O-DU) to GNSS Simulator. 2. Configure O-DU as T-GM with G.8275.1 profile. 3. Configure SyncE on O-DU with ESMC/eESMC as applicable. 4. Connect Clock Tester measurement ports to O-DU TimeTransmitter ports to monitor PTP at the O-RU UNI. 5. Establish a PTP and SyncE session to the fronthaul network. <u>Verifications for Maximum Absolute Packet Time Error</u> 1. Verify that PTP and SyncE session is up and running. 2. Perform time error measurements on Clock Tester at O-RU1 and O-RU2 UNI independently. a. $\max TE_L$ measured on Clock Tester at the O-RU UNI should be $\leq 1420ns$ (For Regular RU) b. $\max TE_L$ measured on Clock Tester at the O-RU UNI should be $\leq 1465ns$ (For Enhanced RU) <u>Verifications for Maximum Relative Packet Time Error (NOTE1)</u> 1. Perform the time error measurements on Clock Tester at O-RU1 and O-RU2 UNI independently. Collect the data ($\max TE1, \min TE1$) and ($\max TE2, \min TE2$) for O-RU1 UNI and O-RU2 UNI respectively at least for a period of 1000sec. 2. Estimate the Maximum Relative Time Error measurements between the two O-RU UNI ports. a. $\max TE_{RL} (1,2) \leq 60 (FR2)$ (For Enhanced RU) b. $\max TE_{RL} (1,2) \leq 190ns (FR1)$ (For Enhanced RU) c. $\max TE_{RL} (1,2) \leq 100ns (FR1)$ (For Regular RU) As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TE_{RL}(t)$, but to measure the maximum possible value it can take. $\max TE_{RL} (1,2) \text{ the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$ <u>Verifications for Maximum Absolute 1PPS Time Error</u> 1. Verify that PTP and SyncE session is up and running. 2. Perform 1PPS time error measurements on Clock Tester at O-RU1 and O-RU2 UNI independently. a. $\max TE_L$ measured on Clock Tester at the O-RU UNI $\leq 1420ns$ (For Regular RU) b. $\max TE_L$ measured on Clock Tester at the O-RU UNI $\leq 1465ns$ (For Enhanced RU) <u>Verifications for Time Wander</u>

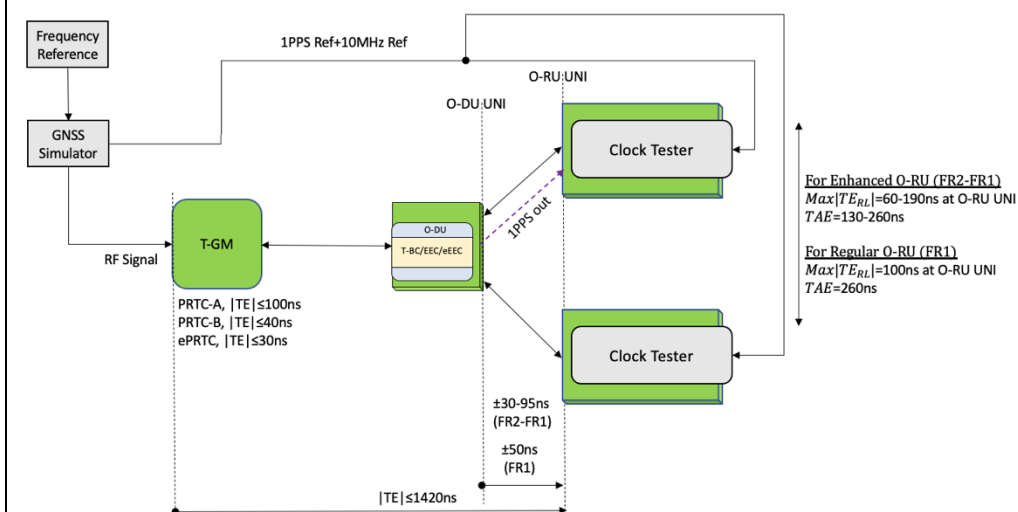
	1. Verify the Time Wander using Clock tester at the O-RU UNI. a. Frequency Limit is $\leq 15ppb$ (For O-DU Class-A) b. Frequency Limit is $\leq 5ppb$ (For O-DU Class-B) c. MTIE mask as per the limit specified. NOTE1 (As per G.8271.1): To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.			
Pass/Fail Criteria:	Test shall pass if the Maximum Absolute TE, Relative TE and Time Wander Limit result is within the specified limits.			
Test Output	End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value
		(For enhanced RU)	(For regular RU)	
	Packet $\max TE_L $	$\leq 1465ns$	$\leq 1420ns$	
	Packet $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
		$\leq 190ns(FR1)$		
	1PPS $\max TE_L $	$\leq 1465ns$	$\leq 1420ns$	
	Frequency Limit (For O-DU Class-A)	$\leq 15ppb$	$\leq 15ppb$	
Frequency Limit (For O-DU Class-B)	$\leq 5ppb$	$\leq 5ppb$		

1 9.1.2 Network Time Synchronization-LLS-C1 (Option-B) configuration:

2 Scenario-1

Test ID	SYNC.TE.02	Classification	Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Measurement of Frequency Error, Absolute Time Error, Relative Time Error at O-RU UNI. Configuration-LLS-C1(Option-B)		
Test Objective:	- Verify the Maximum Absolute Packet Time Error ($\max TE_L$) at O-RU UNI. - Verify the Maximum Relative Packet Time Error ($\max TE_{RL}$) between two O-RU UNI ports. - Verify the Maximum Absolute 1PPS Time Error ($\max TE_L$) at O-RU UNI. - Verify the Time Wander at the O-RU UNI.		
Pre-Req	T-GM is directly connected to O-DU. O-RU UNI is the measurement plane.		

Test Topology:



The test topology above illustrates the placement of clock tester at measurement plane with 1PPS and 10MHz reference connected to clock tester. The tester measurement ports at O-RU UNI connected to O-DU.

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- c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TE_{RL}(t)$, but to measure the maximum possible value it can take.

$$\max|TE_{RL}|(1,2) \text{ the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$$

Verifications for Maximum Absolute 1PPS Time Error

11. Verify that PTP and SyncE session is up and running.
12. Perform 1PPS time error measurements on Clock Tester at O-RU1 and O-RU2 UNI independently.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1420ns$ (For Regular RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1465ns$ (For Enhanced RU)

Verifications for Time Wander

13. Verify the Time Wander using Clock tester at the O-RU UNI.
 - a. Frequency Limit is $\leq 36ppb$ (For O-DU Class-A)
 - b. Frequency Limit is $\leq 32ppb$ (For O-DU Class-B)
 - c. MTIE mask as per the limit specified.

NOTE1 (As per G.8271.1):

To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.

Pass/Fail Criteria:

Test shall pass if the Maximum Absolute TE, Relative TE and Time Wander Limit result is within the specified limits.

Test Output

End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value
	(For Enhanced RU)	(For Regular RU)	
Packet $\max TE_L $	$\leq 1465ns$	$\leq 1420ns$	
Packet $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
	$\leq 190ns(FR1)$		
1PPS $\max TE_L $	$\leq 1465ns$	$\leq 1420ns$	
Frequency Limit (For O-DU Class-A)	$\leq 15ppb$	$\leq 15ppb$	
Frequency Limit (For O-DU Class-B)	$\leq 5ppb$	$\leq 5ppb$	

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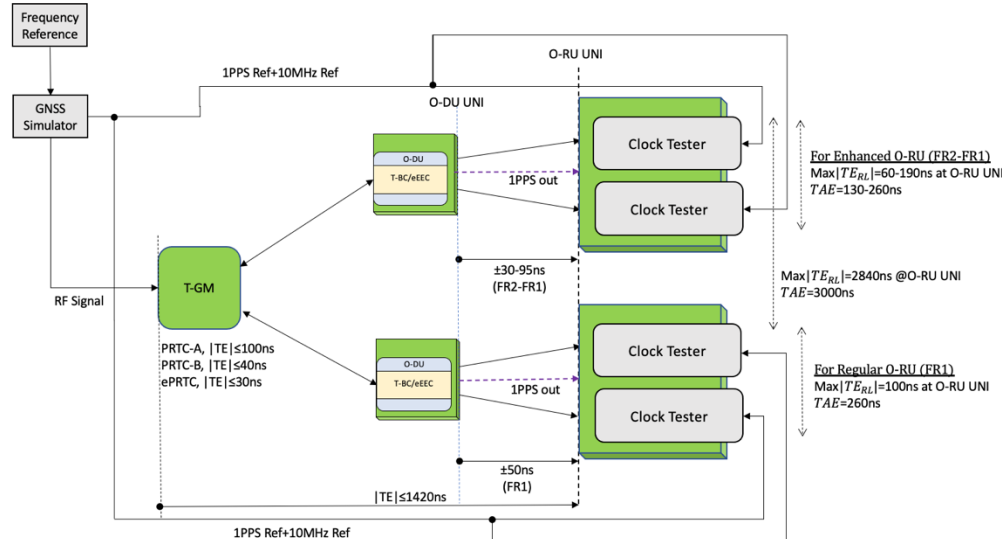
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9.1.3 Network Time Synchronization-LLS-C1 (Option-B) configuration:

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Scenario-2.

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Test ID	SYNC.TE.03	Classification	Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Measurement of Frequency Error, Absolute Time Error, Relative Time Error at O-RU UNI. LLS-C1(Optional-B)		
Test Objective:	1. Verify the Maximum Absolute Packet Time Error ($Max TE_L$) at O-RU UNI. 2. Verify the Maximum Relative Packet Time Error ($Max TE_{RL}$) between two O-RU UNI ports. 3. Verify the Maximum Absolute 1PPS Time Error ($Max TE_L$) at O-RU UNI. 4. Verify the Maximum Relative 1PPS Time Error ($Max TE_{RL}$) at O-RU UNI. 5. Verify the Time Wander at the O-RU UNI.		
Pre-Req	T-GM is dual-homed to ODUs. O-RU UNI is the measurement plane.		
Test Topology:	 The test topology above illustrates the placement of clock tester at the measurement plane with 1PPS and 10MHz reference connected to clock tester. The tester measurement ports at O-RU UNI connected to O-DU.		
Test Configuration	1. Clock Tester at O-RU UNI receives timing input on its measurement port. 2. Provide 1PPS and 10MHz reference to Clock Tester from GNSS simulator. 3. G.8275.1 Full Timing Support profile as applicable.		
Procedure:	Detailed Test procedure: 1. Connect T-GM/PRTC to both O-DUs directly. 2. Configure O-DU as T-BC. 3. Configure G.8275.1 profile on O-DU and T-GM. 4. Configure End-to-End SyncE with ESMC/eESMC as applicable. 5. Connect tester measurement ports to O-DU TimeTransmitter ports to monitor PTP at the O-RU1 and O-RU2 UNI. 6. Establish a PTP and SyncE session to the fronthaul network. Verifications for Maximum Absolute Packet Time Error 1. Verify that PTP and SyncE session is up and running. 2. Verify time error measurements using Clock Tester at O-RU UNI.		

- a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1420ns$ (For Regular RU)
- b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1465ns$ (For Enhanced RU)

Verifications for Maximum Relative Packet Time Error (NOTE1)

1. Perform time error measurements on Clock Tester at O-RU1 and O-RU2 plane independently. Collect the data ($\max TE1, \min TE1$) and ($\max TE2, \min TE2$) for O-RU UNI at least for a period of 1000sec.
2. Estimate the Maximum Relative Time Error measurements between the two O-RU UNI ports connected to O-DU1.
 - a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)
3. Estimate the Maximum Relative Time Error measurements between the two O-RU UNI ports connected to O-DU2.
 - a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)
4. Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports connected to O-DU1 and O-DU2.
 - a. $\max|TE_{RL}|(1,2) \leq 2930ns$ (For Enhanced RU)
 - b. $\max|TE_{RL}|(1,2) \leq 2840ns$ (For Regular RU)

As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TERL(t)$, but to measure the maximum possible value it can take.

$$\max|TE_{RL}|(1,2) \text{ the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$$

Verifications for Maximum Absolute 1PPS Time Error

1. Verify that PTP and SyncE session is up and running.
2. Perform 1PPS time error measurements on Clock Tester at O-RU1 and O-RU2 UNI independently.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1420ns$ (For Regular RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1465ns$ (For Enhanced RU)

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and SyncE session is up and running.
2. Estimate the Maximum Relative 1PPS Time Error measurements between the TimeTransmitter ports of O-DU1 and O-DU2 at O-RU UNI.
 - a. $\max|TE_{RL}|(1,2) \leq 2930ns$ (For Enhanced RU)
 - b. $\max|TE_{RL}|(1,2) \leq 2840ns$ (For Regular RU)

Verifications for Time Wander

1. Verify the Time Wander using Clock tester at the O-RU UNI.

- a. Frequency Limit is $\leq 15ppb$ (For O-DU Class-A)
- b. Frequency Limit is $\leq 5ppb$ (For O-DU Class-B)
- c. MTIE mask as per the limit specified.

NOTE1 (As per G.8271.1):

To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.

Pass/Fail Criteria:

Test shall pass if the Max Absolute TE, Maximum Relative TE and Time Wander Limit measurement result is below the specified limits.

Test Output

End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value
	(For Enhanced RU)	(For Regular RU)	
Packet $\max TE_L $	$\leq 1465ns$	$\leq 1420ns$	
Packet $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
	$\leq 190ns(FR1)$		
1PPS $\max TE_L $	$\leq 1465ns$	$\leq 1420ns$	
Frequency Limit (For O-DU Class-A)	$\leq 15ppb$	$\leq 15ppb$	
Frequency Limit (For O-DU Class-B)	$\leq 5ppb$	$\leq 5ppb$	
Relative Time Error between the O-RU UNI ports connected to O-DU1 and O-DU2			
Packet $\max TE_{RL} $	$\leq 2930ns$	$\leq 2840ns$	
1PPS $\max TE_{RL} $	$\leq 2930ns$	$\leq 2840ns$	

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9.1.4 Network Time Synchronization-LLS-C1 (Option-C) configuration

Test ID

SYNC.TE.04

Classification

Synchronization, Performance, Absolute Time Error, Relative Time Error

Test Title

Measurement of Frequency Error, Maximum Absolute Time Error, Maximum Relative Time Error at O-RU UNI. LLS-C1 (Option-C)

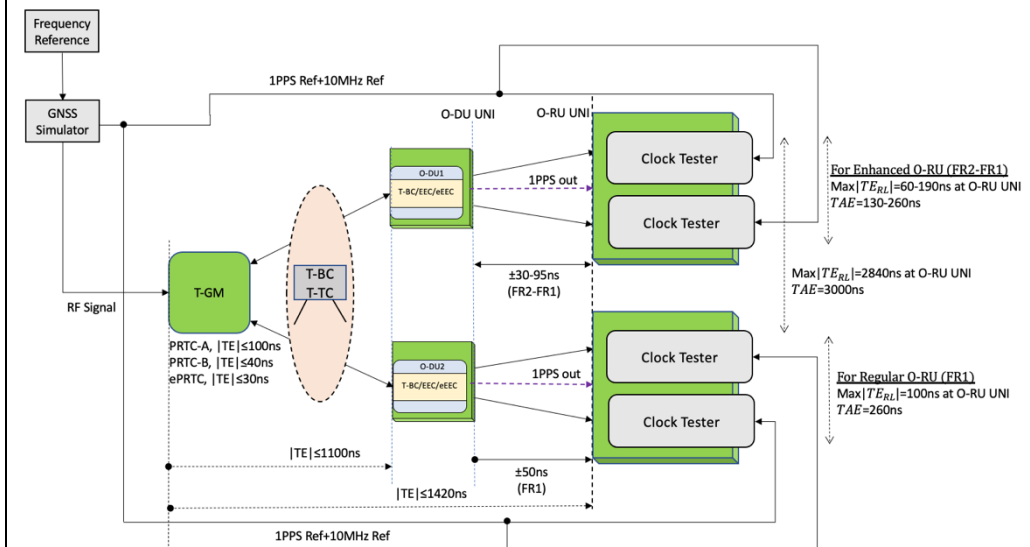
Test Objective:

1. Verify the Maximum Absolute Packet Time Error ($\max|TE_L|$) at O-RU UNI.
2. Verify the Maximum Relative Packet Time Error ($\max|TE_{RL}|$) between two O-RU UNI ports.
3. Verify the Maximum Absolute 1PPS Time Error ($\max|TE_L|$) at O-RU UNI.
4. Verify the Maximum Relative 1PPS Time Error ($\max|TE_{RL}|$) at O-RU UNI.
5. Verify the Time Wander at O-RU UNI.

Pre-Req

T-GM connected to O-DU via T-BC/T-TC chain.
O-RU UNI is the measurement plane.

Test Topology:



The test topology above illustrates the placement of clock tester at measurement plane with 1PPS and 10MHz reference connected to clock tester. The tester measurement ports at O-RU UNI connected to the O-DUs.

Test Configuration	1. Clock Tester at O-RU UNI receives timing input on its measurement port. 2. Provide 1PPS and 10MHz reference to Clock Tester from GNSS simulator. 3. G.8275.1 Full Timing Support profile as applicable.
Procedure:	<u>Detailed Test procedure:</u> 1. Connect T-GM/PRTC to O-DU via a chain of 1 or more T-BC/T-TCs. 2. Configure O-DU as T-BC. 3. Configure G.8275.1 profile on T-GM and O-DU. 4. Configure End-to-End SyncE with ESMC/eESMC as applicable. 7. Connect tester measurement ports to O-DU TimeTransmitter ports to monitor PTP at the O-RU1 and O-RU2 UNI. 5. Establish a PTP and SyncE session to the fronthaul network. <u>Verifications for Maximum Absolute Packet Time Error</u> 1. Verify that PTP and SyncE session is up and running. 2. Verify time error measurements using Clock Tester at O-RU UNI. a. $\max TE_L$ measured on Clock Tester at the O-RU UNI $\leq 1420ns$ (For Regular RU) b. $\max TE_L$ measured on Clock Tester at the O-RU UNI $\leq 1465ns$ (For Enhanced RU) <u>Verifications for Maximum Relative Packet Time Error (NOTE1)</u> 1. Perform time error measurements on Clock Tester at O-RU1 and O-RU2 plane independently. Collect the data ($\max TE1, \min TE1$) and ($\max TE2, \min TE2$) for O-RU UNI at least for a period of 1000sec. 2. Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports connected to O-DU1. a. $\max TE_{RL} (1,2) \leq 60ns$ (FR2) (For Enhanced RU) b. $\max TE_{RL} (1,2) \leq 190ns$ (FR1) (For Enhanced RU) c. $\max TE_{RL} (1,2) \leq 100ns$ (FR1) (For Regular RU)

3. Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports connected to O-DU2.
 - a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)
4. Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports connected to O-DU1 and O-DU2.
 - a. $\max|TE_{RL}|(1,2) \leq 2930ns$ (For Enhanced RU)
 - b. $\max|TE_{RL}|(1,2) \leq 2840ns$ (For Regular RU)

As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TE_{RL}(t)$, but to measure the maximum possible value it can take.

$$\max|TE_{RL}|(1,2) \text{ the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$$

Verifications for Maximum Absolute 1PPS Time Error

1. Verify that PTP and SyncE session is up and running.
2. Perform 1PPS time error measurements on Clock Tester at O-RU1 and O-RU2 UNI independently.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1420ns$ (For Regular RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1465ns$ (For Enhanced RU)

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and SyncE session is up and running.
2. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RU UNI ports connected to O-DU1 and O-DU2.
 - a. $\max|TE_{RL}|(1,2) \leq 2930ns$ (For Enhanced RU)
 - b. $\max|TE_{RL}|(1,2) \leq 2840ns$ (For Regular RU)

Verifications for Time Wander

1. Verify the Time Wander using Clock tester at the O-RU UNI.
 - a. Frequency Limit is $\leq 15ppb$ (For O-DU Class-A)
 - b. Frequency Limit is $\leq 5ppb$ (For O-DU Class-B)
 - c. MTIE mask as per the limit specified.

NOTE1 (As per G.8271.1):

To measure the maximum relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.

Pass/Fail Criteria:	Test shall pass if the Max Absolute TE, Maximum Relative TE and Time Wander Limit measurement result is below the specified limits.			
Test Output	End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value
		(For Enhanced RU)	(For Regular RU)	

	Packet $\max TE_L $	$\leq 1465ns$	$\leq 1420ns$	
	Packet $\max TE_{RL} $	$\leq 60ns(FR2)$ $\leq 190ns(FR1)$	$\leq 100ns(FR1)$	
	1PPS $\max TE_L $	$\leq 1465ns$	$\leq 1420ns$	
	Frequency Limit (For O-DU Class-A)	$\leq 15ppb$	$\leq 15ppb$	
	Frequency Limit (For O-DU Class-B)	$\leq 5ppb$	$\leq 5ppb$	
	Relative Time Error between the O-RU UNI ports connected to O-DU1 and O-DU2			
	Packet $\max TE_{RL} $	$\leq 2930ns$	$\leq 2840ns$	
	1PPS $\max TE_{RL} $	$\leq 2930ns$	$\leq 2840ns$	

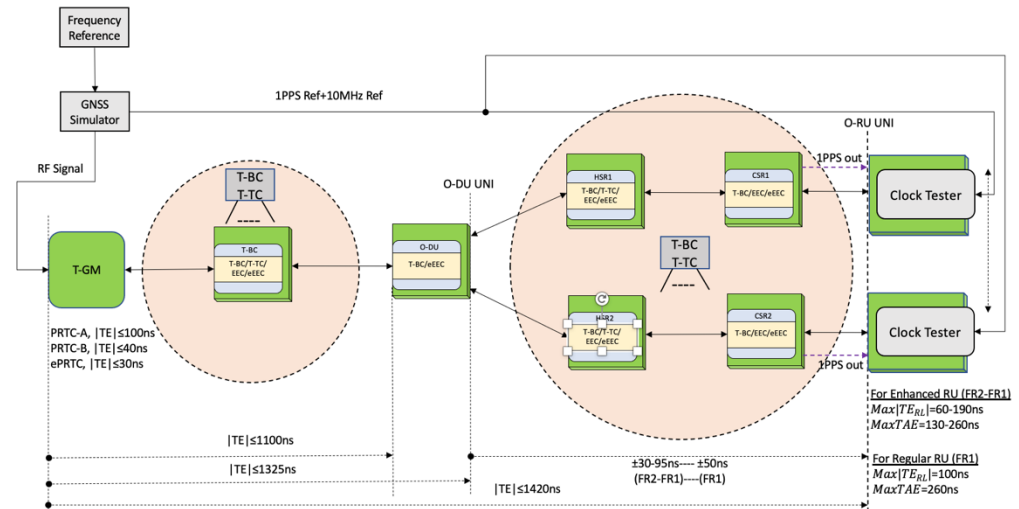
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9.1.5 Network Time Synchronization-LLS-C2 (Option-A) configuration

Test ID	SYNC.TE.05	Classification	Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Measurement of Frequency Error, Maximum Absolute Time Error, Maximum Relative Time Error at O-RU UNI. LLS-C2 (Option-A) configuration		
Test Objective:	1. Verify the Maximum Absolute Packet Time Error ($\max TE_L$) at O-RU UNI. 2. Verify the Maximum Relative Packet Time Error ($\max TE_{RL}$) between two O-RU UNI ports. 3. Verify the Maximum Absolute 1PPS Time Error ($\max TE_L$) at O-RU UNI. 4. Verify the Maximum Relative 1PPS Time Error ($\max TE_{RL}$) at O-RU UNI. 5. Verify the Time Wander at O-RU UNI.		
Pre-Req	T-GM/PRTC connected to O-DU via a chain of T-BC and O-DU locked. O-RU UNI is the measurement plane.		

Test Topology:



The test topology above illustrates the placement of clock tester at measurement plane with 1PPS and 10MHz reference connected to clock tester. The tester measurement ports at O-RU UNI connected to the CSRs.

Test Configuration	1. Clock Tester at O-RU UNI receives timing input on its measurement port. 2. Provide 1PPS and 10MHz reference to Clock Tester from GNSS simulator. 3. G.8275.1 Full Timing Support profile as applicable.
Procedure:	<u>Detailed Test procedure:</u> 1. Connect T-GM to O-DU via a chain of T-BC. 2. Configure O-DU as T-BC and HSR as either T-BC/T-TC. 3. Enable G.8275.1 profile on HSR and T-GM. 4. Connect Clock tester measurement ports to CSR1 and CSR2 TimeTransmitter ports to monitor the PTP at O-RU UNI. 5. Establish a PTP and SyncE session to the fronthaul network. <u>Verifications for Maximum Absolute Packet Time Error</u> 1. Verify that PTP and SyncE session is up and running. 2. Verify time error measurements using Clock Tester at O-RU UNI. a. $\max TE_L$ measured on Clock Tester at the O-RU UNI $\leq 1465ns$ (For Enhanced RU) b. $\max TE_L$ measured on Clock Tester at the O-RU UNI $\leq 1420ns$ (For Regular RU) <u>Verifications for Maximum Relative Packet Time Error (NOTE1)</u> 1. Perform time error measurements on Clock Tester at O-RU1 and O-RU2 UNI. Collect the data ($\max TE1, \min TE1$) and ($\max TE2, \min TE2$) for both the O-RU UNI at least for a period of 1000sec. 2. Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports. a. $\max TE_{RL} (1,2) \leq 60ns$ (FR2) (For Enhanced RU) b. $\max TE_{RL} (1,2) \leq 190ns$ (FR1) (For Enhanced RU) c. $\max TE_{RL} (1,2) \leq 100ns$ (FR1) (For Regular RU) As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TE_{RL}(t)$, but to measure the maximum possible value it can take. $\max TE_{RL} (1,2) \text{ the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$ <u>Verifications for Maximum 1PPS Absolute Time Error</u> 1. Verify that PTP and SyncE session is up and running. 2. Verify 1PPS time error measurements using Clock Tester at O-RU UNI. a. $\max TE_L$ measured on Clock Tester at the O-RU UNI $\leq 1465ns$ (For Enhanced RU) b. $\max TE_L$ measured on Clock Tester at the O-RU UNI $\leq 1420ns$ (For Regular RU) <u>Verifications for Maximum Relative 1PPS Time Error</u> 1. Verify that PTP and SyncE session is up and running. 2. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RU UNI ports. a. $\max TE_{RL} (1,2) \leq 60ns$ (FR2) (For Enhanced RU)

b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)

c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

Verifications for Time Wander

1. Verify the Time Wander using Clock tester at the O-RU UNI.

a. Frequency Limit is $\leq 36ppb$ (For O-DU Class-A)

b. Frequency Limit is $\leq 32ppb$ (For O-DU Class-B)

c. MTIE mask as per the limit specified.

NOTE1 (As per G.8271.1):

To measure the maximum relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.

Pass/Fail Criteria:

Test shall pass if the Maximum absolute TE, Maximum Relative TE and Time Wander Limit measurement result is below the limit specified.

Test Output

End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value
	(For Enhanced RU)	(For Regular RU)	
Packet $\max TE_L $	$\leq 1465ns$	$\leq 1420ns$	
Packet $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
	$\leq 190ns(FR1)$		
1PPS $\max TE_L $	$\leq 1465ns$	$\leq 1420ns$	
1PPS $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
	$\leq 190ns(FR1)$		
Frequency Limit (For O-DU Class-A)	$\leq 36ppb$	$\leq 36ppb$	
Frequency Limit (For O-DU Class-B)	$\leq 32ppb$	$\leq 32ppb$	

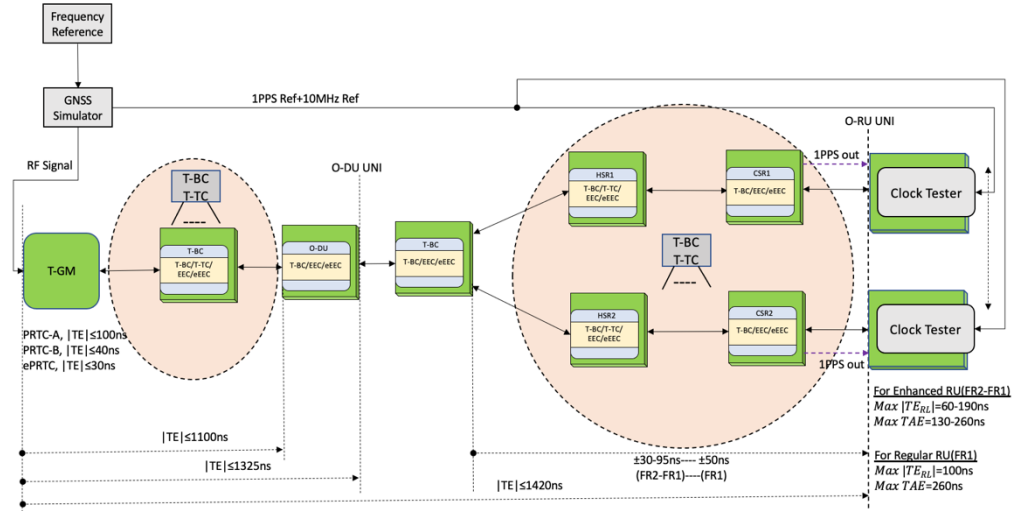
1 9.1.6 Network Time Synchronization-LLS-C2 (Option-B) configuration:

2 Scenario-1

Test ID	SYNC.TE.06	Classification	Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Measurement of Frequency Error, Maximum Absolute Time Error, Maximum Relative Error at O-RU UNI. LLS-C2 (Option-B)		
Test Objective:	1. Verify the Maximum Absolute Packet Time Error ($\max TE_L $) at O-RU UNI. 2. Verify the Maximum Relative Packet Time Error ($\max TE_{RL} $) between two O-RU UNI ports. 3. Verify the Maximum Absolute 1PPS Time Error ($\max TE_L $) at O-RU UNI. 4. Verify the Maximum Relative 1PPS Time Error ($\max TE_{RL} $) at O-RU UNI. 5. Verify the Time Wander at O-RU UNI.		
Pre-Req	Nearest common node for O-RU UNI is T-BC and not O-DU. T-GM or PRTC connected to O-DU via chain of T-BC/T-TCs.		

O-RU UNI is the measurement plane.

Test Topology:



The test topology above illustrates the placement of clock tester at measurement plane with 1PPS and 10MHz reference connected to clock tester. The tester measurement ports at O-RU UNI connected to the CSRs.

Test Configuration

1. Clock Tester at O-RU UNI receives timing input on its measurement port.
2. Provide 1PPS and 10MHz reference to Clock Tester from GNSS simulator.
3. G.8275.1 Full Timing Support profile as applicable.

Procedure:

Detailed Test procedure:

1. Connect T-GM to ODU via a chain of T-BC/T-TCs.
2. Configure O-DU and CSR as T-BCs and HSR is configured as either T-BC or T-TC.
3. Enable G.8275.1 profile on T-GM, O-DU, HSR (when acting as T-BC) and CSR.
4. Connect Clock tester measurement ports to CSR1 and CSR2 TimeTransmitter ports to monitor the PTP at O-RU UNI.
5. Establish PTP and SyncE session to the fronthaul network.

Verifications for Maximum Absolute Packet Time Error

1. Verify that PTP and SyncE session is up and running.
2. Verify time error measurements using Clock Tester at O-RU UNI.
 - a. $\max |TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1465ns$ (For Enhanced RU)
 - b. $\max |TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1420ns$ (For Regular RU)

Verifications for Maximum Relative Packet Time Error (NOTE1)

1. Perform time error measurements on Clock Tester at O-RU1 and O-RU2 UNI. Collect the data ($\max TE1, \min TE1$) and ($\max TE2, \min TE2$) for both the O-RU UNI at least for a period of 1000sec.
2. Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports.
 - a. $\max |TE_{RL}| (1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max |TE_{RL}| (1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max |TE_{RL}| (1,2) \leq 100ns$ (FR1) (For Regular RU)

As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TE_{RL}(t)$, but to measure the maximum possible value it can take.

$$\max|TE_{RL}|(1,2) \text{ the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$$

Verifications for Maximum 1PPS Absolute Time Error

1. Verify that PTP and SyncE session is up and running.
2. Verify 1PPS time error measurements using Clock Tester at O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1465ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1420ns$ (For Regular RU)

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and SyncE session is up and running.
2. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

Verifications for Time Wander

1. Verify the Time Wander using Clock tester at the O-RU UNI.
 - a. Frequency Limit is $\leq 36ppb$ (For O-DU Class-A)
 - b. Frequency Limit is $\leq 32ppb$ (For O-DU Class-B)
 - c. MTIE mask as per the limit specified.

NOTE1:

To measure the maximum relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.

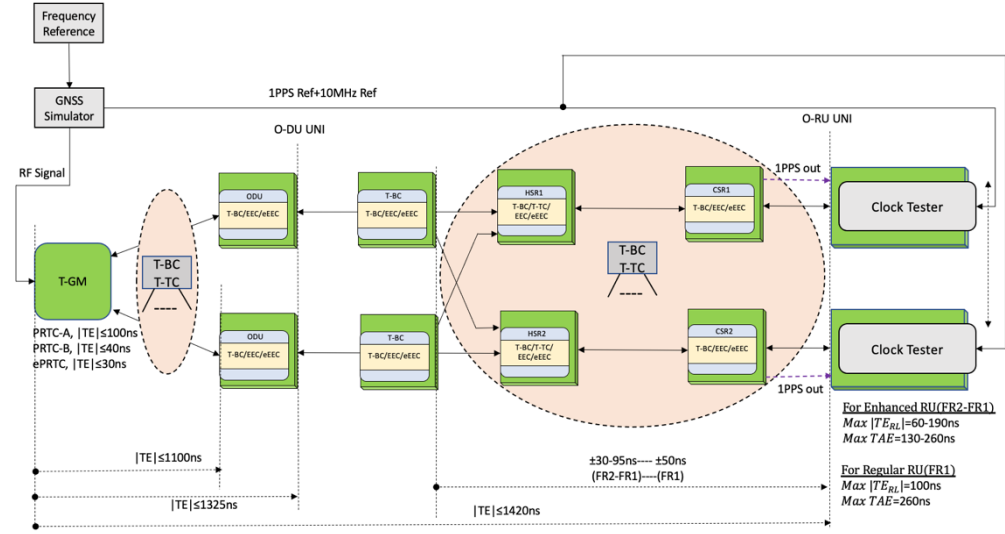
Pass/Fail Criteria:

Test shall pass if the Maximum absolute TE, Maximum Relative TE and Time Wander Limit measurement result is below the limit specified.

Test Output

End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value
	(For Enhanced RU)	(For Regular RU)	
Packet $\max TE_L $	$\leq 1465ns$	$\leq 1420ns$	
Packet $\max TE_{RL} $	$\leq 60ns$ (FR2)	$\leq 100ns$ (FR1)	
	$\leq 190ns$ (FR1)		
1PPS $\max TE_L $	$\leq 1465ns$	$\leq 1420ns$	
1PPS $\max TE_{RL} $	$\leq 60ns$ (FR2)	$\leq 100ns$ (FR1)	
	$\leq 190ns$ (FR1)		
Frequency Limit (For O-DU Class-A)	$\leq 36ppb$	$\leq 36ppb$	
Frequency Limit (For O-DU Class-B)	$\leq 32ppb$	$\leq 32ppb$	

9.1.7 Network Time Synchronization-LLS-C2 (Option-B) configuration: Scenario-2

Test ID	SYNC.TE.07	Classification	Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Measurement of Frequency Error, Maximum Absolute Time Error, Maximum Relative Time Error at O-RU UNI. LLS-C2 (Option-B) configuration		
Test Objective:	1. Verify the Maximum Absolute Packet Time Error ($Max TE_L$) at O-RU UNI. 2. Verify the Maximum Relative Packet Time Error ($Max TE_{RL}$) between two O-RU UNI ports. 3. Verify the Maximum Absolute 1PPS Time Error ($Max TE_L$) at O-RU UNI. 4. Verify the Maximum Relative 1PPS Time Error ($Max TE_{RL}$) at O-RU UNI. 5. Verify the Time Wander at O-RU UNI.		
Pre-Req	Nearest Common node for O-RU UNI is T-BC and not O-DU. T-GM/PRTC connected to O-DU via chain of T-BC/T-TCs. O-DU is locked. O-RU UNI is the measurement plane.		
Test Topology:	 The test topology above illustrates the placement of clock tester at the measurement plane with 1PPS and 10MHz reference connected to clock tester. The tester measurement ports at O-RU UNI connected to the CSRs.		
Test Configuration	1. Clock Tester at O-RU UNI receives timing input on its measurement port. 2. Provide 1PPS and 10MHz reference to Clock Tester from GNSS simulator. 3. G.8275.1 Full Timing Support profile as applicable.		
Procedure:	Detailed Test procedure: 1. Connect T-GM to O-DU via chain of T-BC/T-TCs. 2. Configure O-DU and CSR as T-BC and HSR as either T-BC or T-TC. 3. Enable G.8275.1 profile on HSR (when acting as T-BC), O-DU, CSR and T-GM.		

4. Connect Clock tester measurement ports to CSR1 and CSR2 TimeTransmitter ports to monitor the PTP at O-RU UNI.
5. Establish a PTP and SyncE session to the fronthaul network.

Verifications for Maximum Absolute Packet Time Error

1. Verify that PTP and SyncE session is up and running.
2. Perform time error measurements on Clock tester at O-RU UNI.
3. Verify time error measurements using Clock Tester at O-RU UNI.
 - a. $\text{Max } |TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1465ns$ (For Enhanced RU)
 - b. $\text{Max } |TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1420ns$ (For Regular RU)

Verifications for Maximum Relative Packet Time Error (NOTE1)

1. Perform time error measurements on Clock Tester at O-RU1 and O-RU2 UNI. Collect the data ($\text{maxTE1}, \text{minTE1}$) and ($\text{maxTE2}, \text{minTE2}$) for both the O-RU UNI at least for a period of 1000sec.
2. Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports.
 - a. $\text{max}|TE_{RL}| (1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\text{max}|TE_{RL}| (1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\text{max}|TE_{RL}| (1,2) \leq 100ns$ (FR1) (For Regular RU)

As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TE_{RL}(t)$, but to measure the maximum possible value it can take.

$$\text{max}|TE_{RL}| (1,2) \text{ the greatest of } \begin{cases} \text{maxTE}_L(1) - \text{minTE}_L(2) \\ \text{maxTE}_L(2) - \text{minTE}_L(1) \end{cases}$$

Verifications for Maximum 1PPS Absolute Time Error

1. Verify that PTP and SyncE session is up and running.
2. Verify 1PPS time error measurements using Clock Tester at O-RU UNI.
 - a. $\text{max}|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1465ns$ (For Enhanced RU)
 - b. $\text{max}|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1420ns$ (For Regular RU)

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and SyncE session is up and running.
2. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RU UNI ports.
 - a. $\text{max}|TE_{RL}| (1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\text{max}|TE_{RL}| (1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\text{max}|TE_{RL}| (1,2) \leq 100ns$ (FR1) (For Regular RU)

Verifications for Time Wander

1. Verify the Time Wander using Clock tester at the O-RU UNI.
 - a. Frequency Limit is $\leq 36ppb$ (For O-DU Class-A)
 - b. Frequency Limit is $\leq 32ppb$ (For O-DU Class-B)

c. MTIE mask as per the limit specified.

NOTE1:

To measure the maximum relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.

Pass/Fail Criteria:

Test shall pass if the Maximum absolute TE, Maximum Relative TE and Time Wander Limit measurement result is below the limit specified.

Test Output

End-to-End Network Limit at O-RU UNI.	Expected Value		Observed Value
	(For Enhanced RU)	(For Regular RU)	
Packet $\max TE_L $	$\leq 1465ns$	$\leq 1420ns$	
Packet $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
	$\leq 190ns(FR1)$		
1PPS $\max TE_L $	$\leq 1465ns$	$\leq 1420ns$	
1PPS $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
	$\leq 190ns(FR1)$		
Frequency Limit (For O-DU Class-A)	$\leq 36ppb$	$\leq 36ppb$	
Frequency Limit (For O-DU Class-B)	$\leq 32ppb$	$\leq 32ppb$	

1

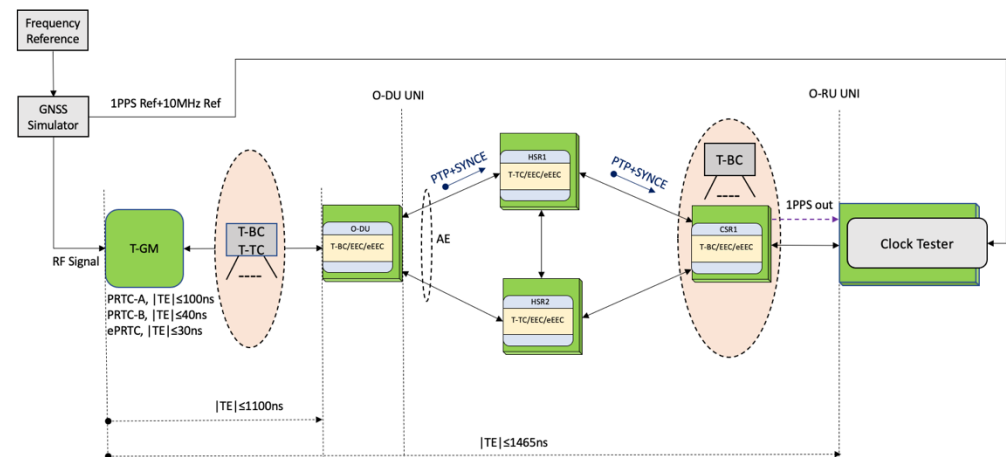
2

9.1.8 Network Time Synchronization-LLS-C2 (Option-B) configuration

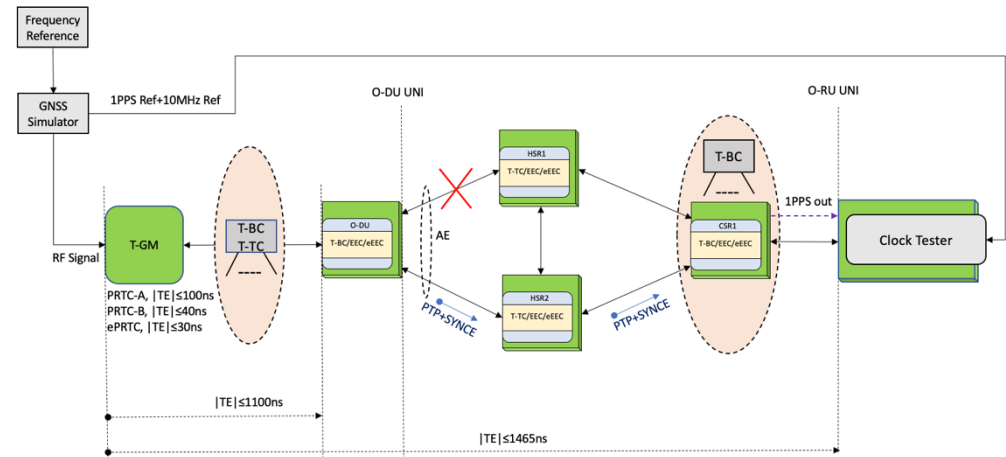
3

(Fail-over: Scenario-1)

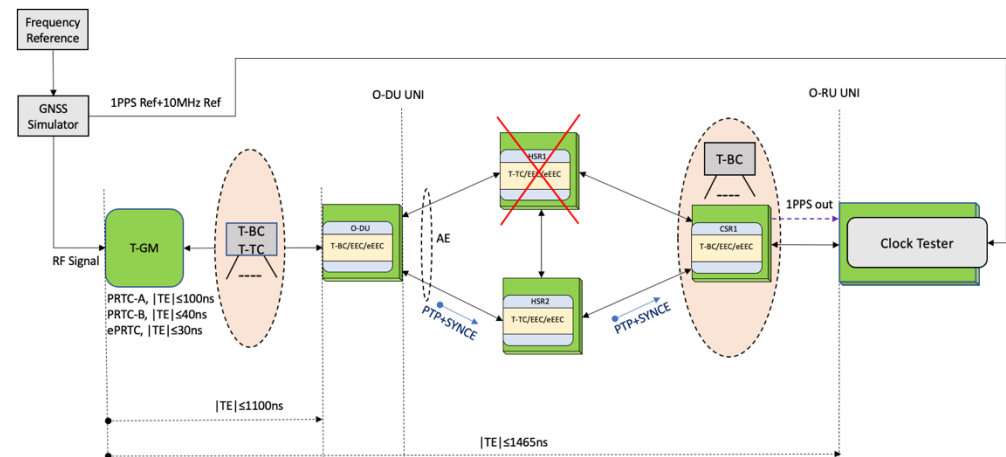
Test ID	SYNC.TE.08	Classification	Synchronization, Performance, Absolute Time Error
Test Title	Measurement of Frequency Error, Maximum Absolute Time Error at O-RU UNI LLS-C2 (Option-B): Dual-Homing with LAG at O-DU side		
Test Objective:	1. Verify the Maximum Absolute Packet Time Error ($\max TE_L $) at O-RU UNI. 2. Verify the Maximum Absolute 1PPS Time Error ($\max TE_L $) at O-RU UNI. 3. Verify the Time Wander at O-RU UNI.		
Pre-Req	O-DU is the nearest common T-BC to O-RU UNI. T-GM or PRTC connected to ODU via a chain of T-BC/T-TCs. A pair of HSR nodes as T-TCs, with CSR dual-homed to the HSR's on one side and a pair of O-DU uplinks are bundled to LAG and connected to HSR node on the other side. O-RU UNI is the measurement plane.		
Test Topology:	Normal Scenario		



Link Failure between O-DU and HSR1



HSR Node Failure



The test topology above illustrates the placement of clock tester at the measurement plane with 1PPS and 10MHz reference connected to clock tester. The tester measurement ports at O-RU UNI connected to the CSR.

Test Configuration

1. Clock Tester at O-RU UNI receives timing input on its measurement port.
2. Provide 1PPS and 10MHz reference to Clock Tester from GNSS simulator.
3. G.8275.1 Full Timing Support profile as applicable.

Procedure:

Case-1: Normal Scenario

Detailed Test procedure

1. Connect T-GM to O-DU via a chain of T-BC/T-TCs.
2. Configure O-DU and CSR as T-BC and HSR as T-TC.
3. Enable G.8275.1 profile on CSR and T-GM.
4. Connect Clock tester measurement ports to CSR TimeTransmitter port to monitor the PTP at O-RU UNI.
5. Establish PTP and SyncE session to the fronthaul network.

Verifications for Maximum Absolute Packet Time Error

1. Verify that PTP and SyncE session is up and running.
2. Verify time error measurements using Clock Tester at O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1465ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1420ns$ (For Regular RU)

Verifications for Maximum 1PPS Absolute Time Error

1. Verify that PTP and SyncE session is up and running.
2. Verify 1PPS time error measurements using Clock Tester at O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1465ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1420ns$ (For Regular RU)

Verifications for Time Wander

1. Verify the Time Wander using Clock tester at the O-RU UNI.
 - a. Frequency Limit is $\leq 36ppb$ (For O-DU Class-A)
 - b. Frequency Limit is $\leq 32ppb$ (For O-DU Class-B)
 - c. MTIE mask as per the limit specified.

Case-2: Link Fail between O-DU and HSR1

Detailed Test procedure

1. Fail the link between O-DU and HSR1.

Verifications for Maximum Absolute Packet Time Error

1. Verify that PTP session is established via the path O-DU-HSR2-CSR1 and CSR1 lock to O-DU via HSR-2.
2. Verify time error measurements using Clock tester on O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1465ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1420ns$ (For Regular RU)

Verifications for Maximum 1PPS Absolute Time Error

1. Verify that PTP and SyncE session is up and running.
2. Verify 1PPS time error measurements using Clock Tester at O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1465ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1420ns$ (For Regular RU)

Verifications for Time Wander

1. Verify the Time Wander using Clock tester at the O-RU UNI.
 - a. Frequency Limit is $\leq 36ppb$ (For O-DU Class-A)
 - b. Frequency Limit is $\leq 32ppb$ (For O-DU Class-B)
 - c. MTIE mask as per the limit specified.

Case-3: HSR1 Node Fail

Detailed Test procedure

1. Fail the node HSR1 by power off.

Verifications for Maximum Absolute Packet Time Error

1. Verify that PTP session is established via the path O-DU-HSR2-CSR1 and CSR1 lock to O-DU via HSR2.
2. Verify the time error measurements using Clock Tester on O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1465ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1420ns$ (For Regular RU)

Verifications for Maximum 1PPS Absolute Time Error

1. Verify that PTP and SyncE session is up and running.
2. Verify 1PPS time error measurements using Clock Tester at O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1465ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1420ns$ (For Regular RU)

Verifications for Time Wander

1. Verify the Time Wander using Clock tester at the O-RU UNI.
 - a. Frequency Limit is $\leq 36ppb$ (For O-DU Class-A)
 - b. Frequency Limit is $\leq 32ppb$ (For O-DU Class-B)
 - c. MTIE mask as per the limit specified.

Pass/Fail Criteria:

Test shall pass if the Maximum absolute TE and Time Wander Limit measurement result is below the limit specified.

Test Output

End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value
	(For Enhanced RU)	(For Regular RU)	
Packet $\max TE_L $	$\leq 1465ns$	$\leq 1420ns$	
1PPS $\max TE_L $	$\leq 1465ns$	$\leq 1420ns$	
Frequency Limit (For O-DU Class-A)	$\leq 36ppb$	$\leq 36ppb$	
Frequency Limit (For O-DU Class-B)	$\leq 32ppb$	$\leq 32ppb$	

1

9.1.9 Network Time Synchronization-LLS-C2 (Option-B) configuration

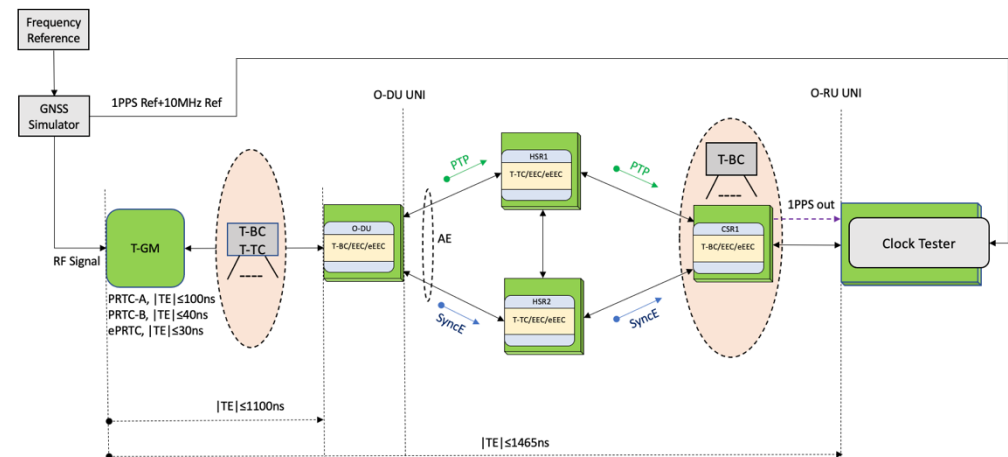
2

(Fail-over: Scenario-2)

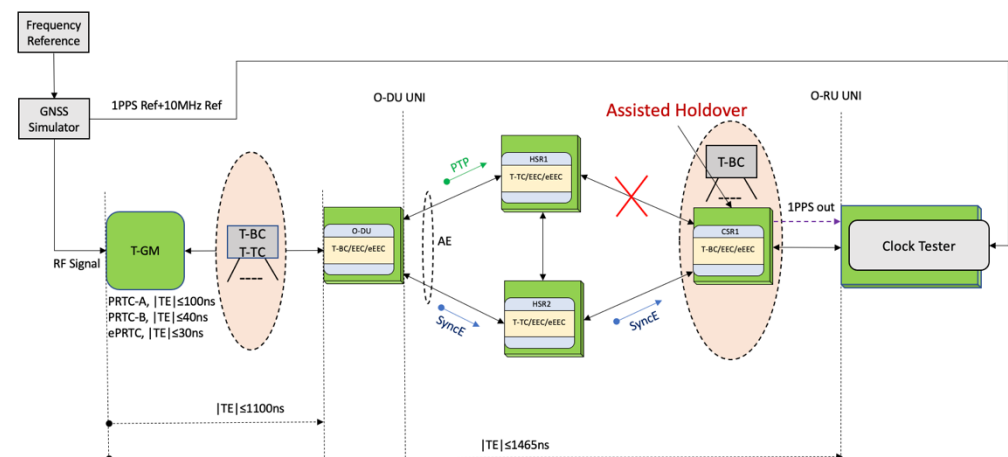
Test ID	SYNC.TE.09	Classification	Synchronization, Performance, Absolute Time Error
Test Title	Measurement of Frequency Error, Maximum Absolute Time Error at O-RU UNI LLS-C2 (Option-B): Dual-Homing with AE at O-DU side		
Test Objective:	1. Verify the Maximum Absolute Packet Time Error ($Max TE_L$) at O-RU UNI. 2. Verify the Maximum Absolute 1PPS Time Error ($Max TE_L$) at O-RU UNI. 3. Verify the Time Wander at O-RU UNI.		
Pre-Req	Nearest common node to O-RU UNI is O-DU. T-GM or PRTC connected to ODU via a chain of T-BC/T-TCs. A pair of HSR nodes as T-TCs, with CSR dual-homed to the HSR's on one side and a pair of O-DU uplinks are bundled to AE and connected to HSR node on the other side. O-RU UNI is the measurement plane.		

Test Topology:

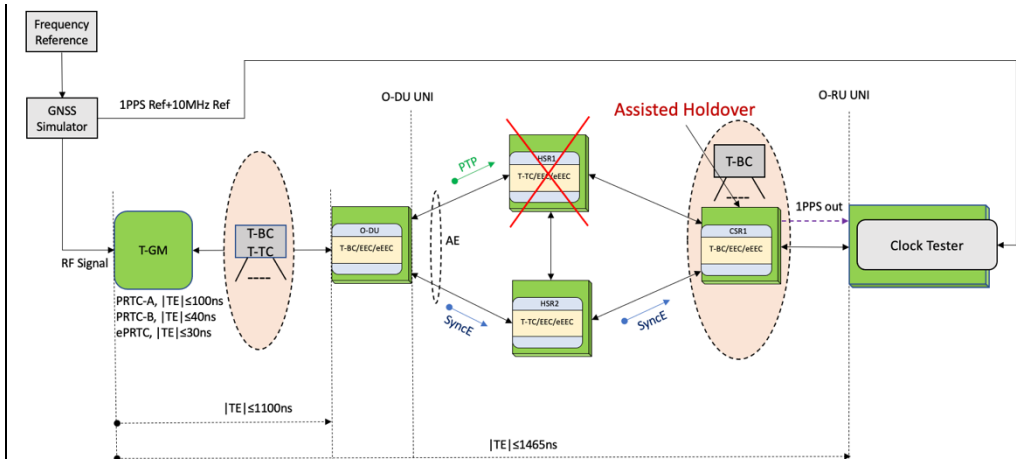
Normal Scenario



Link Fail between HSR1 and CSR1



HSR Node Failure



The test topology above illustrates the placement of clock tester at measurement plane with 1PPS and 10MHz reference connected to clock tester. The tester measurement ports at O-RU UNI connected to CSR.

Test Configuration

1. Clock Tester at O-RU UNI receives timing input on its measurement port.
2. Provide 1PPS and 10MHz reference to Clock Tester from GNSS simulator.
3. G.8275.1 Full Timing Support profile as applicable.

Procedure:

Case-1: Normal Scenario

Detailed Test procedure

1. Connect T-GM to O-DU via a chain of T-BC/T-TCs.
2. Configure O-DU and CSR as T-BC and HSR as T-TC.
3. Enable G.8275.1 profile on CSR and T-GM.
4. Connect Clock tester measurement ports to CSR TimeTransmitter port to monitor the PTP at O-RU UNI.
5. Establish PTP and SyncE session to the fronthaul network.

Verifications for Maximum Absolute Packet Time Error

1. Verify that PTP and SyncE session is up and running.
2. Configure PTP on the primary path and SyncE on the standby path.
3. Verify time error measurements using Clock Tester at O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1465ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1420ns$ (For Regular RU)

Verifications for Maximum 1PPS Absolute Time Error

1. Verify that PTP and SyncE session is up and running.
2. Verify 1PPS time error measurements using Clock Tester at O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1465ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1420ns$ (For Regular RU)

Verifications for Time Wander

1. Verify the Time Wander using Clock tester at the O-RU UNI.
 - a. Frequency Limit is $\leq 36ppb$ (For O-DU Class-A)

- b. Frequency Limit is $\leq 32ppb$ (For O-DU Class-B)
- c. MTIE mask as per the limit specified.

Case-2: Link Fail between HSR1 and CSR

Detailed Test procedure

1. Fail the link between HSR1 and CSR.

Verifications for Maximum Absolute Packet Time Error

1. Verify that the PTP session on CSR moves to Assisted Holdover with Physical layer SyncE via the standby path.
2. Perform time error measurements on Clock tester on O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1465ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1420ns$ (For Regular RU)

Verifications for Maximum 1PPS Absolute Time Error

1. Verify that PTP and SyncE session is up and running.
2. Verify 1PPS time error measurements using Clock Tester at O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1465ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1420ns$ (For Regular RU)

Verifications for Time Wander

1. Verify the Time Wander using Clock tester at the O-RU UNI.
 - a. Frequency Limit is $\leq 36ppb$ (For O-DU Class-A)
 - b. Frequency Limit is $\leq 32ppb$ (For O-DU Class-B)
 - c. MTIE mask as per the limit specified.

Case-3: HSR1 Node Fail

Detailed Test procedure

1. Fail the node HSR1 by power off.

Verifications for Maximum Absolute Packet Time Error

1. Verify that PTP session on CSR moves to Assisted Holdover with SyncE via the standby path.
2. Perform time error measurements on Tester on O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1465ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1420ns$ (For Regular RU)

Verifications for Maximum 1PPS Absolute Time Error

1. Verify that PTP and SyncE session is up and running.
2. Verify 1PPS time error measurements using Clock Tester at O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1465ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1420ns$ (For Regular RU)

Verifications for Time Wander

	1. Verify the Time Wander using Clock tester at the O-RU UNI. a. Frequency Limit is $\leq 36ppb$ (For O-DU Class-A) b. Frequency Limit is $\leq 32ppb$ (For O-DU Class-B) c. MTIE mask as per the limit specified.			
Pass/Fail Criteria:	Test shall pass if the Maximum absolute TE and Time Wander Limit measurement result is below the limit specified.			
Test Output	End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value
		(For Enhanced RU)	(For Regular RU)	
	Packet $\max TE_L $	$\leq 1465ns$	$\leq 1420ns$	
	1PPS $\max TE_L $	$\leq 1465ns$	$\leq 1420ns$	
	Frequency Limit (For O-DU Class-A)	$\leq 36ppb$	$\leq 36ppb$	
	Frequency Limit (For O-DU Class-B)	$\leq 32ppb$	$\leq 32ppb$	

1

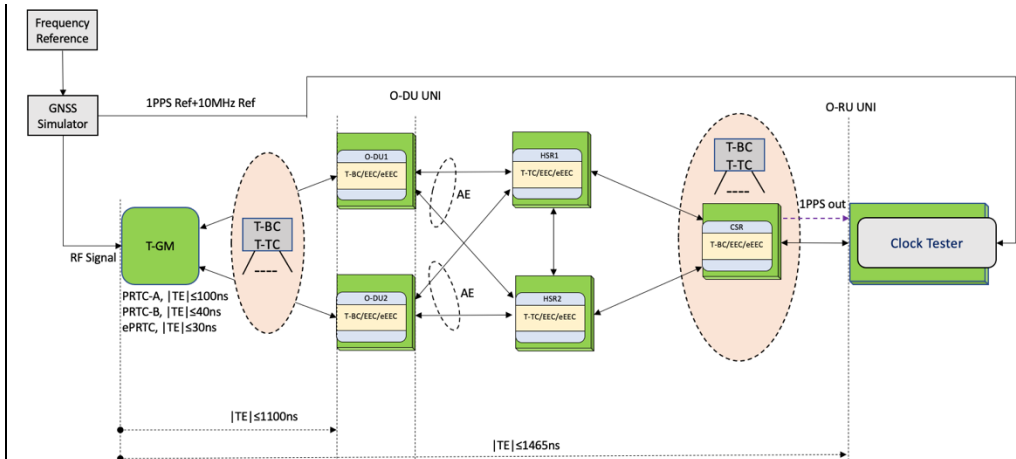
2

9.1.10 Network Time Synchronization-LLS-C2 (Option-B) configuration

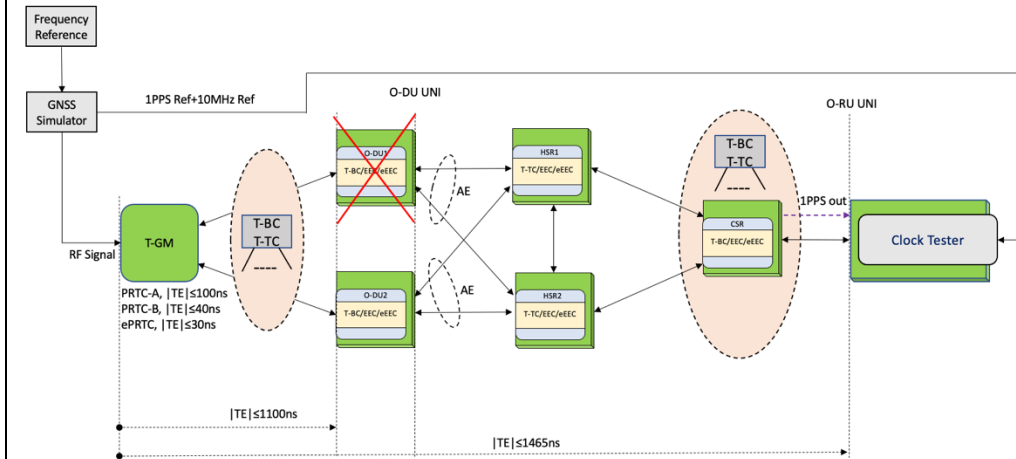
3

(Fail-over: Scenario-3)

Test ID	SYNC.TE.10	Classification	Synchronization, Performance, Absolute Time Error
Test Title	Measurement of Frequency Error, Maximum Absolute Time Error at O-RU UNI LLS-C2 (Option-B): Dual-Homing with dual/redundant O-DU		
Test Objective:	1. Verify the Maximum Absolute Packet Time Error ($\max TE_L$) at O-RU UNI. 2. Verify the Maximum Absolute 1PPS Time Error ($\max TE_L$) at O-RU UNI. 3. Verify the Time Wander at O-RU UNI.		
Pre-Req	T-GM or PRTC connected to O-DUs via T-BC chain. A pair of HSR nodes, with CSR dual-homed to the HSR's on one side and a pair of O-DU uplinks are bundled to AE and connected to HSR node on the other side. O-RU UNI is the measurement plane.		
Test Topology:	Normal Scenario		



O-DU Failure Scenario



The test topology above illustrates the placement of clock tester at the measurement plane with 1PPS and 10MHz reference connected to clock tester. The tester measurement ports at O-RU UNI connected to the CSR.

Test Configuration	1. Clock Tester at O-RU UNI receives timing input on its measurement port. 2. Provide 1PPS and 10MHz reference to Clock Tester from GNSS simulator. 3. G.8275.1 Full Timing Support profile as applicable.
Procedure:	Case-1: Normal Scenario Detailed Test procedure: 1. Connect T-GM to O-DU via a chain of T-BC/T-TCs. 2. Configure O-DU1, O-DU2 and CSR as T-BC. HSR1 and HSR2 as T-TC. 3. Enable G.8275.1 profile on CSR and T-GM. 4. Connect Clock tester measurement ports to CSR TimeTransmitter port to monitor the PTP at O-RU UNI. 5. Establish PTP and SyncE session to the fronthaul network. Verifications for Maximum Absolute Packet Time Error 1. Verify that PTP session is up and running. 2. Verify that PTP session is established on the Primary path T-GM-O-DU1-HSR1-CSR and CSR lock to O-DU1 via HSR1. 3. Perform time error measurements on Tester on O-RU UNI.

- a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1465ns$ (For Enhanced RU)
- b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1420ns$ (For Regular RU)

Verifications for Maximum 1PPS Absolute Time Error

1. Verify that PTP and SyncE session is up and running.
2. Verify the 1PPS time error measurements using Clock Tester at O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1465ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1420ns$ (For Regular RU)

Verifications for Time Wander

1. Verify the Time Wander using Clock tester at the O-RU UNI.
 - a. Frequency Limit is $\leq 36ppb$ (For O-DU Class-A)
 - b. Frequency Limit is $\leq 32ppb$ (For O-DU Class-B)
 - c. MTIE mask as per the limit specified.

Case-2: O-DU1 Fail

Detailed Test procedure

1. Fail the node O-DU1 by power off.

Verifications for Maximum Absolute Time Error

1. Verify that PTP session is established via the path O-DU2-HSR1-CSR and CSR lock to HSR1.
2. Perform time error measurements on Clock Tester at the O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1465ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1420ns$ (For Regular RU)

Verifications for Maximum 1PPS Absolute Time Error

1. Verify that PTP and SyncE session is up and running.
2. Verify 1PPS time error measurements using Clock Tester at O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1465ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1420ns$ (For Regular RU)

Verifications for Time Wander

1. Verify the Time Wander using Clock tester at the O-RU UNI.
 - a. Frequency Limit is $\leq 36ppb$ (For O-DU Class-A)
 - b. Frequency Limit is $\leq 32ppb$ (For O-DU Class-B)
 - c. MTIE mask as per the limit specified.

Pass/Fail Criteria:

Test shall pass if the Maximum absolute TE and Time Wander Limit measurement result is below the limit specified.

Test Output

End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value
	(For Enhanced RU)	(For Regular RU)	
Packet $\max TE_L $	$\leq 1465ns$	$\leq 1420ns$	

	1PPS max TE _L	≤ 1465ns	≤ 1420ns	
	Frequency Limit (For O-DU Class-A)	≤ 36ppb	≤ 36ppb	
	Frequency Limit (For O-DU Class-B)	≤ 32ppb	≤ 32ppb	

- 1
- 2
- 3

- 1
- 2
- 3

	The test topology above illustrates the placement of clock tester at the measurement plane with 1PPS and 10MHz reference connected to clock tester. The tester measurement ports at O-RU UNI connected to the CSRs.
Test Configuration	- Multiple Transport TNEs (HSR1, HSR2, CSR1 and CSR2) connected and configured as shown in the Test Topology. - Clock Tester at O-RU UNI receives timing input on its measurement port. - Provide 1PPS and 10MHz reference to Clock Tester from GNSS simulator. - Traffic Tester for emulating Background Traffic for the desired packet sizes and rate. - Proper QoS configuration on TNEs to make sure PTP packets not congested in the network. - G.8275.1 Full Timing Support profile.
Procedure:	<u>Case-1: Normal Scenario (with no background traffic)</u> <u>Detailed Test procedure:</u> - Connect T-GM to GNSS simulator and ensure that T-GM is locked. - Connect O-DU directly to T-GM. - Configure O-DU, CSR1 and CSR2 as T-BCs. HSR1 and HSR2 as either T-BCs or T-TCs. - Enable G.8275.1 profile on all transport TNEs (HSR and CSR). - Connect Clock tester measurement ports to CSRs TimeTransmitter port to monitor the PTP at O-RU UNI. - Establish PTP and SyncE session to the fronthaul network. <u>Verifications for Maximum Absolute Packet Time Error</u> - Verify that PTP session is up and running between T-GM and CSR. - Verify time error measurements using the Clock Tester for the PTP flows from CSR at the O-RU UNI. - Max TE measured on Clock Tester at the O-RU UNI $\leq 1465ns$ (For Enhanced RU) - Max TE measured on Clock Tester at the O-RU UNI $\leq 1420ns$ (For Regular RU) <u>Verifications for Maximum Relative Packet Time Error (NOTE1)</u> - Perform time error measurements on Clock Tester at O-RU1 and O-RU2 plane independently. Collect the data ($maxTE1, minTE1$) and ($maxTE2, minTE2$) for both O-RU UNI at least for a period of 1000sec. - Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports. $max TE_{RL} (1,2) \leq 60ns$ (FR2) (For Enhanced RU) $max TE_{RL} (1,2) \leq 190ns$ (FR1) (For Enhanced RU) $max TE_{RL} (1,2) \leq 100ns$ (FR1) (For Regular RU) As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TERL(t)$, but to measure the maximum possible value it can take.

$$\max|TE_{RL}|(1,2) \text{ the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$$

Verifications for Maximum 1PPS Absolute Time Error

1. Verify that PTP and SyncE session is up and running.
2. Verify 1PPS time error measurements using Clock Tester at O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1465ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1420ns$ (For Regular RU)

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and SyncE session is up and running.
2. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

Verifications for Time Wander

1. Verify the Time Wander using Clock tester at O-RU UNI.
 - a. Frequency Limit is $\leq 36ppb$ (For O-DU Class-A)
 - b. Frequency Limit is $\leq 32ppb$ (For O-DU Class-B)
 - c. MTIE mask as per the limit specified.

Case-2: With background traffic

Detailed Test procedure:

1. Configure proper QoS on each of the TNEs so that PTP packets are not congested in the network.
2. Create traffic flows, especially the traffic types, rate and packet sizes should be as per FH requirements. (Refer the Section 7.2 in Xhaul Transport Requirements documents for details).
3. Start traffic flows on all the Traffic Tester ports and make sure that no traffic loss observed.

Verifications for Maximum Absolute Packet Time Error

1. Verify that the PTP traffic is not dropped with background traffic.
2. Verify time error measurements using the Clock Tester for the PTP flows from CSR at the O-RU UNI in presence of the Background traffic.
 - a. $\max|TE|$ measured on Clock Tester at the O-RU UNI $\leq 1465ns$ (For Enhanced RU)
 - b. $\max|TE|$ measured on Clock Tester at the O-RU UNI $\leq 1420ns$ (For Regular RU)

Verifications for Maximum Relative Packet Time Error (NOTE1)

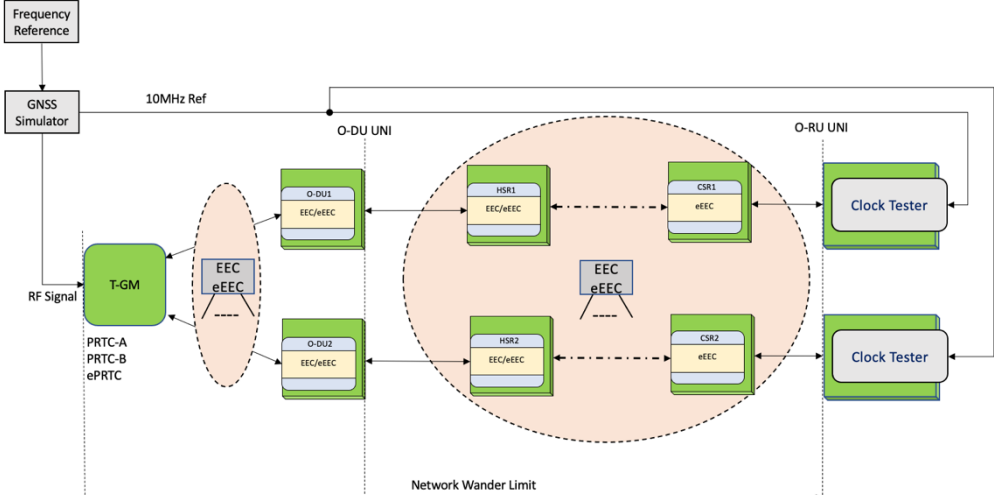
1. Perform time error measurements on Clock Tester at O-RU1 and O-RU2 plane independently. Collect the data $(\max TE1, \min TE1)$ and $(\max TE2, \min TE2)$ for both O-RU UNI at least for a period of 1000sec.
2. Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports in presence of the Background traffic.
 - a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)

	b. $\max TE_{RL} (1,2) \leq 190ns (FR1)$ (For Enhanced RU) c. $\max TE_{RL} (1,2) \leq 100ns (FR1)$ (For Regular RU) As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TERL(t)$, but to measure the maximum possible value it can take. $\max TE_{RL} (1,2) \text{the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$ Verifications for Maximum Absolute 1PPS Time Error 1. Verify that PTP session is up and running between T-GM and CSR. 2. Verify 1PPS time error measurements on Clock Tester at the O-RU UNI. a. $\max TE_L$ measured on Clock Tester at the O-RU UNI$\leq 1465ns$ (For Enhanced RU) b. $\max TE_L$ measured on Clock Tester at the O-RU UNI$\leq 1420ns$ (For Regular RU) Verifications for Maximum Relative 1PPS Time Error 1. Verify that PTP and SyncE session is up and running. 2. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RU UNI ports. a. $\max TE_{RL} (1,2) \leq 60ns (FR2)$ (For Enhanced RU) b. $\max TE_{RL} (1,2) \leq 190ns (FR1)$ (For Enhanced RU) c. $\max TE_{RL} (1,2) \leq 100ns (FR1)$ (For Regular RU) Verifications for Time Wander 1. Verify the Time Wander using Clock tester at the O-RU UNI. a. Frequency Limit is $\leq 36ppb$ (For O-DU Class-A) b. Frequency Limit is $\leq 32ppb$ (For O-DU Class-B) c. MTIE mask as per the limit specified. NOTE1 (As per G.8271.1): To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.																								
Pass/Fail Criteria:	Test shall pass if the Maximum absolute TE, Maximum Relative TE and Time Wander Limit measurement result is below the limit specified with and without background traffic.																								
Test Output	<table><tr><th rowspan="2">End-to-End Network Limit at R-RU UNI</th><th colspan="2">Expected Value</th><th rowspan="2">Observed Value</th></tr><tr><th>(For Enhanced RU)</th><th>(For Regular RU)</th></tr><tr><td>Packet $\max TE_L$</td><td>$\leq 1465ns$</td><td>$\leq 1420ns$</td><td></td></tr><tr><td rowspan="2">Packet $\max TE_{RL}$</td><td>$\leq 60ns(FR2)$</td><td rowspan="2">$\leq 100ns(FR1)$</td><td rowspan="2"></td></tr><tr><td>$\leq 190ns(FR1)$</td></tr><tr><td>1PPS $\max TE_L$</td><td>$\leq 1465ns$</td><td>$\leq 1420ns$</td><td></td></tr><tr><td rowspan="2">1PPS $\max TE_{RL}$</td><td>$\leq 60ns(FR2)$</td><td rowspan="2">$\leq 100ns(FR1)$</td><td rowspan="2"></td></tr><tr><td>$\leq 190ns(FR1)$</td></tr></table>	End-to-End Network Limit at R-RU UNI	Expected Value		Observed Value	(For Enhanced RU)	(For Regular RU)	Packet $\max TE_L $	$\leq 1465ns$	$\leq 1420ns$		Packet $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$		$\leq 190ns(FR1)$	1PPS $\max TE_L $	$\leq 1465ns$	$\leq 1420ns$		1PPS $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$		$\leq 190ns(FR1)$
End-to-End Network Limit at R-RU UNI	Expected Value		Observed Value																						
	(For Enhanced RU)	(For Regular RU)																							
Packet $\max TE_L $	$\leq 1465ns$	$\leq 1420ns$																							
Packet $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$																							
	$\leq 190ns(FR1)$																								
1PPS $\max TE_L $	$\leq 1465ns$	$\leq 1420ns$																							
1PPS $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$																							
	$\leq 190ns(FR1)$																								

	Frequency Limit (For O-DU Class-A)	$\leq 36ppb$	$\leq 36ppb$	
	Frequency Limit (For O-DU Class-B)	$\leq 32ppb$	$\leq 32ppb$	

1 9.1.12 Network Time Synchronization-LLS-C2 (Option-B) configuration

2 (SyncE network wander limit for chain of EEC and eEEC clock)

Test ID	SYNC.TE.12	Classification	Synchronization, Performance, Frequency Error
Test Title	Measurement of Time Wander at O-RU UNI. LLS-C2 (Option-B)		
Test Objective:	Verify the Time Wander at O-RU UNI.		
Pre-Req	TNEs in the network chain should support either ITU-T G.8262 EEC clock or ITU-T G.8262.1 eEEC clock. O-RU UNI is the measurement plane.		
Test Topology:	 The test topology above illustrates the placement of clock tester at measurement location with 10MHz reference connected to clock tester. The tester measurement ports at O-RU UNI connected to the CSRs.		
Test Configuration	1. Clock Tester at O-RU UNI receives timing input on its measurement port. 2. Provide 10MHz reference to Clock Tester from GNSS simulator.		
Procedure:	<u>Case1: Synchronous chain of ITU-T G.8262 EEC clock</u> <u>Detailed Test procedure:</u> 1. Configure Synchronous Ethernet with ESMC on all the nodes. 2. Connect Clock tester measurement ports to CSRs to monitor the output wander at the O-RU UNI. <u>Verifications needed:</u> 1. Verify that SyncE is up and ESMC-QL is traceable to that of T-GM. 2. Perform the Time Interval Error measurements on Clock Tester at the O-RU UNI.		

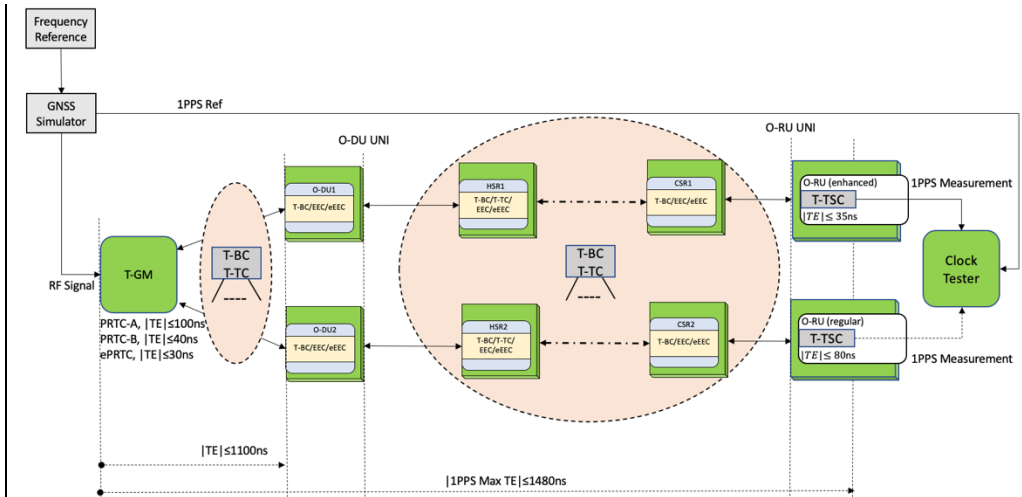
3. Verify the Time Wander at O-RU UNI.
 - a. Frequency Limit is $\leq 36ppb$ (For O-DU Class-A)
 - b. Frequency Limit is $\leq 32ppb$ (For O-DU Class-B)
 - c. MTIE mask as per the limit specified.
- Case2: Synchronous chain of ITU-T G.8262.1 eEEC clock**
- Detailed Test procedure:**
1. Configure Synchronous Ethernet with eESMC on all the nodes.
 2. Connect Clock tester measurement ports to CSRs to monitor the output wander at the O-RU UNI.
- Verifications needed:**
1. Verify that SyncE is up and eESMC-QL is traceable to that of T-GM.
 2. Perform the Time Interval Error measurements on Clock Tester at the O-RU UNI.
 3. Verify the Time Wander at O-RU UNI.
 - a. Frequency Limit is $\leq 36ppb$ (For O-DU Class-A)
 - b. Frequency Limit is $\leq 32ppb$ (For O-DU Class-B)
 - c. MTIE mask as per the limit specified.

Pass/Fail Criteria: Test shall pass if the measurement of Time Wander limits are below the specified limits.

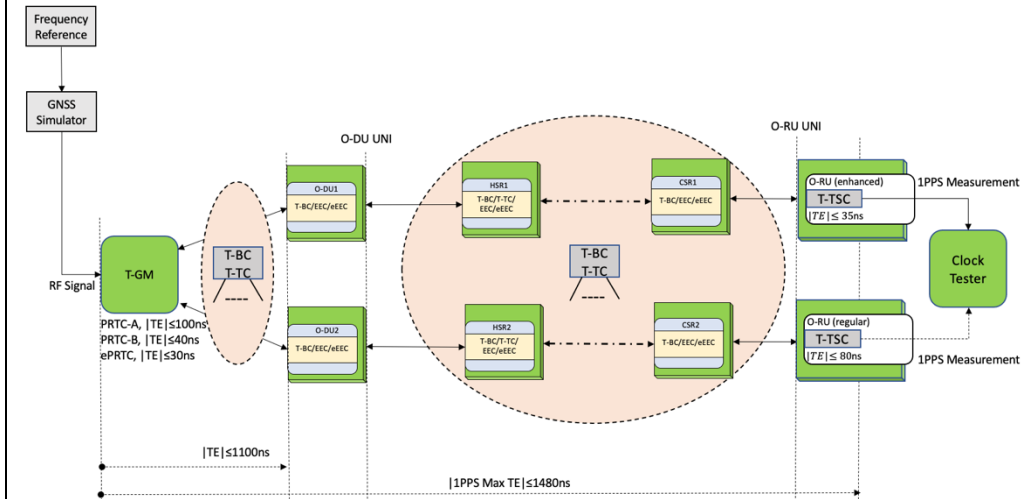
Test Output	End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value
		(Enhanced RU)	(Regular RU)	
	Frequency Limit (For O-DU Class-A)	$\leq 36ppb$	$\leq 36ppb$	
	Frequency Limit (For O-DU Class-B)	$\leq 32ppb$	$\leq 32ppb$	

9.1.13 Network Time Synchronization-LLS-C2 (Option-B) configuration (1PPS Time Error on O-RU)

Test ID	SYNC.TE.13	Classification	Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Measurement of 1PPS Absolute Time Error, 1PPS Relative Time Error at O-RU. LLS-C2 (Option-B)		
Test Objective:	1. Verify the 1PPS Maximum Absolute Time Error at O-RU. 2. Verify the 1PPS Maximum Relative Time Error between the O-RU UNI ports.		
Pre-Req	O-RU must support 1PPS out interface. T-GM/PRTC connected to O-DU via chain of T-BC/T-TCs. O-DU is locked.		
Test Topology:	1PPS Absolute Time Error		



1PPS Relative Time Error



Test Configuration

1. Clock Tester at O-RU UNI receives timing input on its measurement port.
2. Provide 1PPS reference to Clock Tester from GNSS simulator.

Procedure:

Case-1: 1PPS Absolute Time Error Measurement.

Detailed Test procedure:

1. Connect T-GM to O-DUs via chain of T-BC/T-TCs.
2. Configure O-DU and CSR as T-BC and HSR as either T-BC or T-TC.
3. Enable G.8275.1 profile on HSR (when acting as T-BC), CSR and T-GM.
4. Connect Clock tester 1PPS measurement ports to O-RU 1PPS output ports.
5. Establish PTP and SyncE session to the fronthaul network.

Verifications needed:

1. Verify that PTP session is up and running.
2. Verify 1PPS time error measurements using the Clock Tester at O-RU UNI.
 - a. $1PPS \max|TE_L|$ measured on Tester at the O-RU (Enhanced) output $\leq 1480ns$
 - b. $1PPS \max|TE_L|$ measured on Tester at the O-RU (Regular) output $\leq 1480ns$

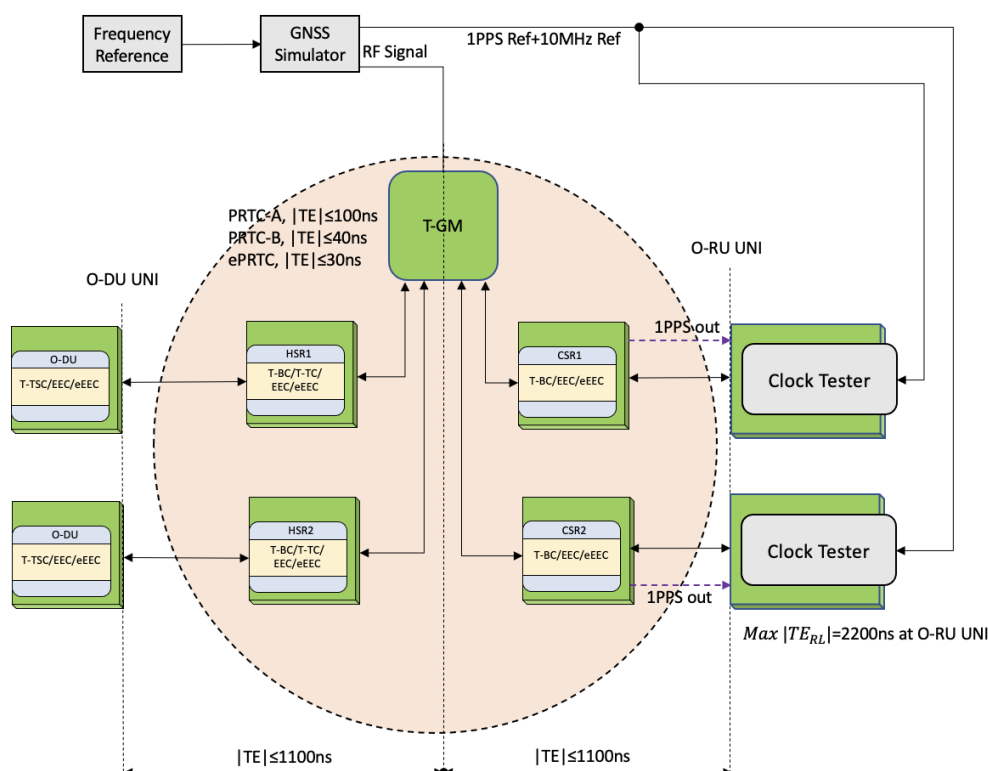
Case-1: 1PPS Relative Time Error Measurement.

	<u>Detailed Test procedure:</u> 1. Connect 1PPS output ports from both O-RU to tester 1PPS measurement Input ports simultaneously. <u>Verifications needed:</u> 1. Verify the 1PPS relative time error between O-RU1 and O-RU2 $\leq 2960ns$			
Pass/Fail Criteria:	Test shall pass if the 1PPS Maximum absolute TE and 1PPS Maximum Relative TE measurement result is below the limit specified.			
Test Output	End-to-End Network Limit	Expected Value		Observed Value
		(Enhanced RU)	(Regular RU)	
	1PPS max TE _L	$\leq 1480ns$	$\leq 1480ns$	
	1PPS max TE _{RL}	$\leq 2960ns$	$\leq 2960ns$	

1 9.1.14 Network Time Synchronization-LLS-C3 (Option-A) configuration

Test ID	SYNC.TE.14	Classification	Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Measurement of Frequency Error, Absolute Time Error, Relative Time Error at O-RU UNI. LLS-C3 (Option-A)		
Test Objective:	1. Verify the Maximum Absolute Packet Time Error ($\max TE_L $) at O-RU UNI. 2. Verify the Maximum Relative Packet Time Error ($\max TE_{RL} $) between the O-RU UNI ports. 3. Verify the Maximum Absolute 1PPS Time Error ($\max TE_L $) at O-RU UNI. 4. Verify the Maximum Relative 1PPS Time Error ($\max TE_{RL} $) between the O-RU UNI ports. 5. Verify the Time Wander at O-RU UNI.		
Pre-Req	T-GM/PRTC connected to GNSS simulator and ensure T-GM is locked. O-RU UNI is the measurement plane.		

Test Topology:



The test topology above illustrates the placement of clock tester at measurement plane with 1PPS and 10MHz reference connected to clock tester. The tester measurement ports at O-RU UNI connected to the CSR.

Test Configuration	1. Multiple Transport TNEs (HSR1, HSR2, CSR1 and CSR2) connected and configured as shown in the Test Topology. 2. Clock Tester at O-RU UNI receives timing input on its measurement port. 3. Provide 1PPS and 10MHz reference to Clock Tester from GNSS simulator. 4. G.8275.1 Full Timing Support profile.
Procedure:	<u>Detailed Test procedure:</u> 1. Connect T-GM to GNSS simulator and ensure that T-GM is locked. 2. Configure O-DU as T-TSC, CSR1 and CSR2 as T-BCs. HSR1 and HSR2 as either T-BCs or T-TCs. 3. Enable G.8275.1 profile on all transport TNEs (HSR and CSR). 4. Connect Clock tester measurement ports to CSRs TimeTransmitter port to monitor the PTP at O-RU UNI. 5. Establish PTP and SyncE session to the fronthaul network. <u>Verifications for Maximum Absolute Packet Time Error</u> 1. Verify that PTP session is up and running between T-GM and CSR. 2. Verify time error measurements using the Clock Tester for the PTP flows from CSR at the O-RU UNI. a. $\max TE_L$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU) b. $\max TE_L$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU) <u>Verifications for Maximum Relative Packet Time Error (NOTE1)</u>

1. Perform time error measurements on Clock Tester at O-RU1 and O-RU2 plane independently. Collect the data ($maxTE1, minTE1$) and ($maxTE2, minTE2$) for both O-RU UNI at least for a period of 1000sec.
2. Estimate the Maximum Relative Time Error measurements between the TimeTransmitter ports on CSR1 and CSR2 at O-RU UNI.

$$a. \max|TE_{RL}| (1,2) \leq 2200ns$$

As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TERL(t)$, but to measure the maximum possible value it can take.

$$\max|TE_{RL}| (1,2) \text{ the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$$

Verifications for Maximum Absolute 1PPS Time Error

1. Verify that PTP and SyncE session is up and running between T-GM and CSR.
2. Verify the 1PPS time error measurements on Clock Tester at the O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and SyncE session is up and running.
2. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}| (1,2) \leq 2200ns$

Verifications for Time Wander

1. Verify the Time Wander using Clock tester at the O-RU UNI.
 - a. Frequency Limit is $\leq 32ppb$ (For PRTC Class-A)
 - b. MTIE mask as per the limit specified.

NOTE1 (As per G.8271.1):

To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.

Pass/Fail Criteria:

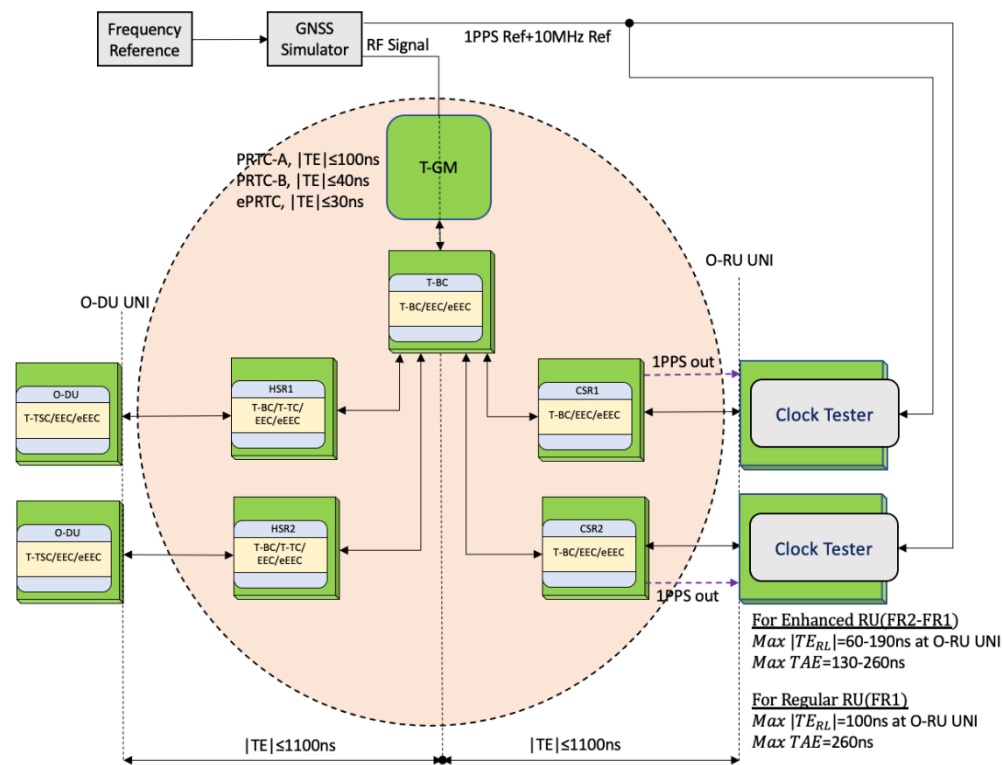
Test shall pass if the Maximum absolute TE, Maximum Relative TE and Time Wander Limit measurement result is below the limit specified with and without background traffic.

Test Output

End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value
	(For Enhanced RU)	(For Regular RU)	
Packet $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
Packet $\max TE_{RL} $	$\leq 2200ns$	$\leq 2200ns$	
1PPS $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
1PPS $\max TE_{RL} $	$\leq 2200ns$	$\leq 2200ns$	
Frequency Limit (For PRTC Class-A)	$\leq 32ppb$	$\leq 32ppb$	

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9.1.15 Network Time Synchronization-LLS-C3 (Option-B) configuration

Test ID	SYNC.TE.15	Classification	Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Measurement of Frequency Error, Absolute Time Error, Relative Time Error at O-RU UNI. LLS-C3 (Option-B)		
Test Objective:	1. Verify the Maximum Absolute Packet Time Error ($Max TE_L$) at O-RU UNI. 2. Verify the Maximum Relative Packet Time Error ($Max TE_{RL}$) between the O-RU UNI ports. 3. Verify the Maximum Absolute 1PPS Time Error ($Max TE_L$) at O-RU UNI. 4. Verify the Maximum Relative 1PPS Time Error ($Max TE_{RL}$) at O-RU UN. 5. Verify the Time Wander at O-RU UNI.		
Pre-Req	T-GM/PRTC connected to GNSS simulator and ensure T-GM is locked. O-RU UNI is the measurement plane.		
Test Topology:	 The test topology above illustrates the placement of clock tester at the measurement plane with 1PPS and 10MHz reference connected to clock tester. The tester measurement ports at O-RU UNI connected to the CSR. For Enhanced RU (FR2-FR1) $Max TE_{RL} = 60-190ns$ at O-RU UNI $Max TAE = 130-260ns$ For Regular RU (FR1) $Max TE_{RL} = 100ns$ at O-RU UNI $Max TAE = 260ns$		
Test Configuration	1. Multiple Transport TNEs (HSR1, HSR2, CSR1 and CSR2) connected and configured as shown in the Test Topology. 2. Clock Tester at O-RU UNI receives timing input on its measurement port. 3. Provide 1PPS and 10MHz reference to Clock Tester from GNSS simulator. 4. G.8275.1 Full Timing Support profile.		
Procedure:	Detailed Test procedure: 1. Connect T-GM to GNSS Simulator and ensure that T-GM is locked.		

2. Configure O-DU as T-TSC, CSR1 and CSR2 as T-BCs. HSR1 and HSR2 as either T-BCs or T-TCs.
3. Enable G.8275.1 profile on all transport TNEs (HSR, CSR and T-BC).
4. Connect Clock tester measurement ports to CSRs TimeTransmitter port to monitor the PTP at O-RU UNI.
5. Establish PTP and SyncE session to the fronthaul network.

Verifications for Maximum Absolute Packet Time Error

1. Verify that PTP session is up and running between T-GM and CSR.
2. Verify time error measurements using Clock Tester for the PTP flows from CSR at the O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative Packet Time Error (NOTE1)

1. Perform time error measurements on Clock Tester at O-RU1 and O-RU2 plane independently. Collect the data ($\max TE1, \min TE1$) and ($\max TE2, \min TE2$) for both O-RU UNI at least for a period of 1000sec.
2. Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TE_{RL}(t)$, but to measure the maximum possible value it can take.

$$\max|TE_{RL}|(1,2) \text{ the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$$

Verifications for Maximum Absolute 1PPS Time Error

1. Verify that PTP session is up and running between T-GM and CSR.
2. Verify the 1PPS time error measurements on Clock Tester at the O-RU UNI.
 - a. Max 1PPS $|TE|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU (Regular) UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and SyncE session is up and running.
2. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

Verifications for Time Wander

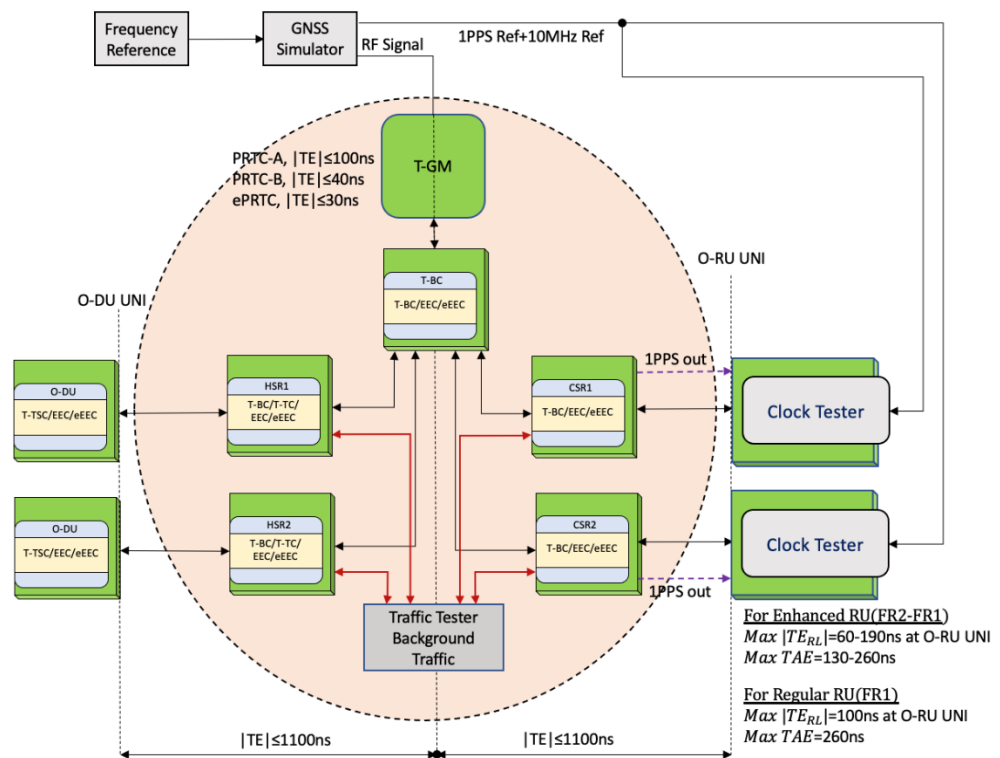
1. Verify the Time Wander using Clock tester at the O-RU UNI.

	a. Frequency Limit is $\leq 32ppb$ (For PRTC Class-A) b. MTIE mask as per the limit specified. NOTE1: To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.		
Pass/Fail Criteria:	Test shall pass if the Maximum absolute TE, Maximum Relative TE and Time Wander Limit measurement result is below the limit specified.		
Test Output	End-to-End Network Limit at O-RU UNI	Expected Value (For Enhanced RU) (For Regular RU)	Observed Value
	Packet $\max TE_L $	$\leq 1100ns$ $\leq 1100ns$	
	Packet $\max TE_{RL} $	$\leq 60ns(FR2)$ $\leq 100ns(FR1)$ $\leq 190ns(FR1)$	
	1PPS $\max TE_L $	$\leq 1100ns$ $\leq 1100ns$	
	1PPS $\max TE_{RL} $	$\leq 60ns(FR2)$ $\leq 100ns(FR1)$ $\leq 190ns(FR1)$	
	Frequency Limit (For PRTC Class-A)	$\leq 32ppb$ $\leq 32ppb$	

9.1.16 Network Time Synchronization-LLS-C3 (Option-B) configuration (Time Error with Background Traffic)

Test ID	SYNC.TE.16	Classification	Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Measurement of Frequency Error, Absolute Time Error, Relative Time Error at O-RU UNI with and without background traffic. LLS-C3 (Option-B)		
Test Objective:	1. Verify the Maximum Absolute Packet Time Error ($\max TE_L$) at O-RU UNI with and without background traffic. 2. Verify the Maximum Relative Packet Time Error ($\max TE_{RL}$) between the O-RU UNI ports. 3. Verify the Maximum Absolute 1PPS Time Error ($\max TE_L$) at O-RU UNI with and without background traffic. 4. Verify the Maximum Relative 1PPS Time Error ($\max TE_{RL}$) at O-RU UNI with and without background traffic. 5. Verify the Time Wander at O-RU UNI.		
Pre-Req	PRTC/T-GM connected to GNSS simulator and ensure that T-GM is locked. O-RU UNI is the measurement plane.		

Test Topology:



The test topology above illustrates the placement of clock tester at the measurement plane with 1PPS and 10MHz reference connected to clock tester. The tester measurement ports at O-RU UNI connected to the CSR.

Test Configuration

1. Multiple Transport TNEs (HSR1, HSR2, CSR1, CSR2 and T-BC) connected and configured as shown in the Test Topology.
2. Clock Tester at O-RU UNI receives timing input on its measurement port.
3. Provide 1PPS and 10MHz reference to Clock Tester from GNSS simulator.
4. Traffic Tester for emulating Background Traffic for the desired packet sizes and rate.
5. Proper QoS configuration on TNEs to make sure PTP packets not congested in the network.
6. G.8275.1 Full Timing Support profile.

Procedure:

Case-1: Normal Scenario (with no background traffic)

Detailed Test procedure:

1. Connect T-GM to GNSS Simulator and ensure that T-GM is locked.
2. Configure O-DU as T-TSC, CSR1 and CSR2 as T-BCs. HSR1 and HSR2 as either T-BCs or T-TCs.
3. Enable G.8275.1 profile on all transport TNEs (HSR, CSR and T-BC).
4. Connect Clock tester measurement ports to CSRs TimeTransmitter port to monitor the PTP at O-RU UNI.
5. Establish PTP and SyncE session to the fronthaul network.

Verifications for Maximum Absolute Packet Time Error

1. Verify that PTP session is up and running between T-GM and CSR.
2. Verify time error measurements using the Clock Tester for the PTP flows from CSR at the O-RU UNI.

- a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
- b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative Packet Time Error (NOTE1)

1. Perform time error measurements on Clock Tester at O-RU1 and O-RU2 plane independently. Collect the data ($\max TE1, \min TE1$) and ($\max TE2, \min TE2$) for both O-RU UNI at least for a period of 1000sec.
2. Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}| (1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}| (1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}| (1,2) \leq 100ns$ (FR1) (For Regular RU)

As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TE_{RL}(t)$, but to measure the maximum possible value it can take.

$$\max|TE_{RL}| (1,2) \text{ the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$$

Verifications for Maximum Absolute 1PPS Time Error

1. Verify that PTP session is up and running between T-GM and CSR.
2. Verify the 1PPS time error measurements on Clock Tester at the O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and SyncE session is up and running.
2. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}| (1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}| (1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}| (1,2) \leq 100ns$ (FR1) (For Regular RU)

Verifications for Time Wander

1. Verify the Time Wander using Clock tester at the O-RU UNI.
 - a. Frequency Limit is $\leq 32ppb$ (For PRTC Class-A)
 - b. MTIE mask as per the limit specified.

Case-2: With background traffic

Detailed Test procedure:

1. Configure proper QoS on each of the TNEs so that PTP packets are not congested in the network.
2. Create traffic flows, especially the traffic types, rate and packet sizes should be as per FH requirements. (Refer the Section 7.2 in Xhaul Transport Requirements documents for details).

3. Start traffic flows on all the Traffic Tester ports and make sure that no traffic loss observed.

Verifications for Maximum Absolute Packet Time Error

1. Verify that the PTP traffic is not dropped with background traffic.
2. Verify time error measurements using Clock Tester for the PTP flows from CSR at the O-RU UNI in presence of the Background traffic.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative Packet Time Error (NOTE1)

1. Perform time error measurements on Clock Tester at O-RU1 and O-RU2 plane independently. Collect the data $(\max TE1, \min TE1)$ and $(\max TE2, \min TE2)$ for both O-RU UNI at least for a period of 1000sec.
2. Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports in presence of the Background traffic.
 - a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TERL(t)$, but to measure the maximum possible value it can take.

$$\max|TE_{RL}|(1,2) \text{ the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$$

Verifications for Maximum Absolute 1PPS Time Error

1. Verify that PTP session is up and running between T-GM and CSR.
2. Verify 1PPS time error measurements on Clock Tester at the O-RU UNI in presence of the Background traffic.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and SyncE session is up and running.
2. Estimate the Maximum Relative 1PPS Time Error measurements between the TimeTransmitter ports of CSR1 and CSR2 at O-RU UNI in presence of Background Traffic.
 - a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

Verifications for Time Wander

1. Verify the Time Wander using Clock tester at the O-RU UNI.
 - a. Frequency Limit is $\leq 32ppb$ (For PRTC Class-A)
 - b. MTIE mask as per the limit specified.

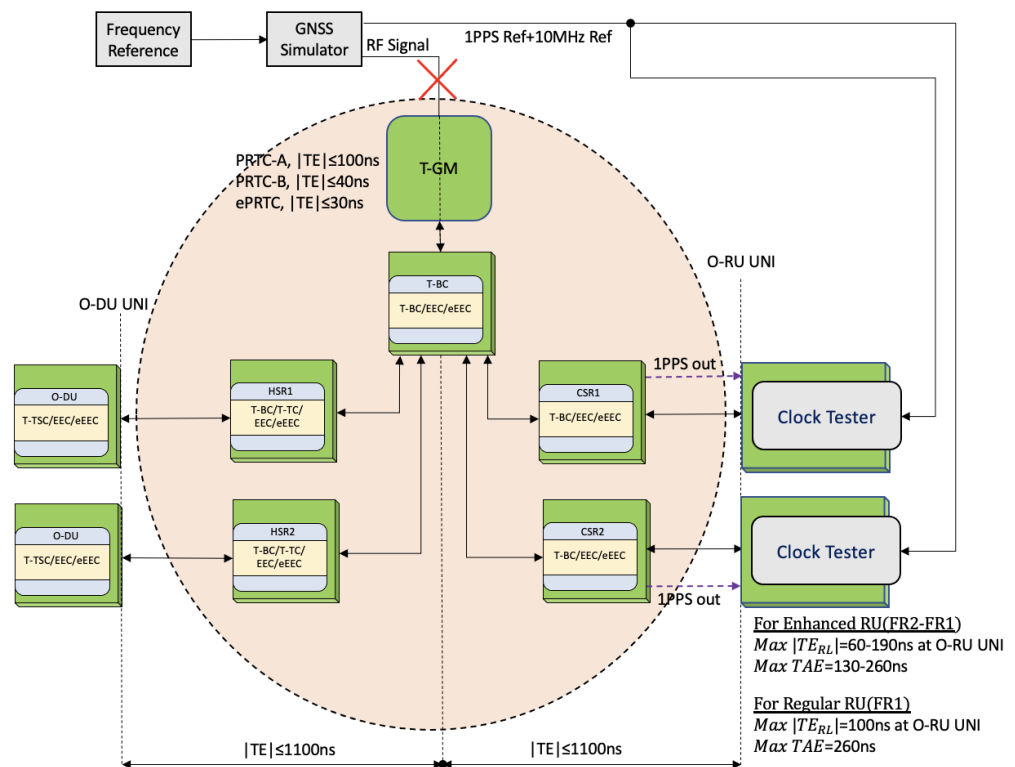
	NOTE1 (As per G.8271.1): To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.			
Pass/Fail Criteria:	Test shall pass if the Maximum absolute TE, Maximum Relative TE and Time Wander Limit measurement result is below the limit specified with and without background traffic.			
Test Output	End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value
		(For Enhanced RU)	(For Regular RU)	
	Packet $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
	Packet $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
		$\leq 190ns(FR1)$		
	1PPS $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
	1PPS $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
		$\leq 190ns(FR1)$		
	Frequency Limit (For PRTC Class-A)	$\leq 32ppb$	$\leq 32ppb$	

1 9.1.17 Network Time Synchronization-LLS-C3 (Option-B) configuration

2 (PTP Holdover test)

Test ID	SYNC.TE.17	Classification	Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Holdover performance test over LLS-C3 configuration		
Test Objective:	1. Verify the Maximum Absolute Packet Time Error ($\max TE_L$) at O-RU UNI. 2. Verify the Maximum Relative Packet Time Error ($\max TE_{RL}$) between the O-RU UNI ports. 3. Verify the Maximum Absolute 1PPS Time Error ($\max TE_L$) at O-RU UNI. 4. Verify the Maximum Relative 1PPS Time Error ($\max TE_{RL}$) at O-RU UNI. 5. Verify the Time Wander at O-RU UNI.		
Pre-Req	PRTC/T-GM connected to GNSS Simulator and locked. O-RU UNI is the measurement plane.		

Test Topology:



The test topology above illustrates the placement of clock tester at the measurement plane with 1PPS and 10MHz reference connected to clock tester. The tester measurement ports at O-RU UNI connected to the CSR.

Test Configuration

1. Multiple Transport TNEs (HSR, HSR and CSRs) connected and configured as shown in the Test Topology.
2. Clock Tester at O-RU UNI receives timing input on its measurement port.
3. Provide 1PPS and 10MHz reference to Clock Tester from GNSS simulator.
4. G8275.1 Full Timing Support profile

Procedure:

Case-1: Normal Scenario (GNSS connectivity to T-GM is Up)

Detailed Test procedure:

1. Connect T-GM to GNSS Simulator and ensure that T-GM is locked.
2. Configure O-DU as T-TSC, CSR1 and CSR2 as T-BCs. HSR1 and HSR2 as either T-BCs or T-TCs.
3. Enable G.8275.1 profile on all transport TNEs (HSR, CSR and T-BC).
4. Connect Clock tester measurement ports to CSRs TimeTransmitter port to monitor PTP at the O-RU UNI.
5. Establish PTP and SyncE session to the fronthaul network.

Verifications for Maximum Absolute Packet Time Error

1. Verify that PTP session is up and running between T-GM and CSR.
2. Verify time error measurements using Clock Tester for the PTP flows from CSR at the O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative Packet Time Error (NOTE1)

1. Perform time error measurements on Clock Tester at O-RU1 and O-RU2 plane independently. Collect the data ($maxTE1, minTE1$) and ($maxTE2, minTE2$) for both O-RU UNI at least for a period of 1000sec.
2. Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TE_{RL}(t)$, but to measure the maximum possible value it can take.

$$\max|TE_{RL}|(1,2) \text{ the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$$

Verifications for Maximum Absolute 1PPS Time Error

1. Verify that PTP session is up and running between T-GM and CSR.
2. Verify the 1PPS time error measurements using Clock Tester at the O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and SyncE session is up and running.
2. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

Verifications for Time Wander

1. Verify the Time Wander using Clock tester at the O-RU UNI.
 - a. Frequency Limit is $\leq 32ppb$ (For PRTC Class-A)
 - b. MTIE mask as per the limit specified.

Case-2: GNSS Connectivity to T-GM Fails (Short-interruption for 5 min)

Detailed Test procedure:

1. Simulate a Fault (Link down) between GNSS and T-GM for a short duration of 5 min.

Verifications for Maximum Absolute Packet Time Error

1. Verify that the T-GM moves to Phase Holdover and start advertising Clock Class=7.
2. Verify time error measurements on Clock Tester for the PTP flows from CSR at the O-RU UNI in while the T-GM is holdover.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU (Enhanced) UNI $\leq 1350ns$
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU (Regular) UNI $\leq 1350ns$

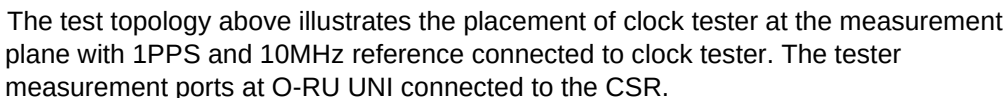
	<u>Verifications for Maximum Relative Packet Time Error (NOTE1)</u> 1. Perform time error measurements on Clock Tester at O-RU1 and O-RU2 plane independently. Collect the data ($maxTE1, minTE1$) and ($maxTE2, minTE2$) for both O-RU UNI at least for a period of 1000sec. 2. Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports. a. $\max TE_{RL} (1,2) \leq 60ns$ (FR2) (For Enhanced RU) b. $\max TE_{RL} (1,2) \leq 190ns$ (FR1) (For Enhanced RU) c. $\max TE_{RL} (1,2) \leq 100ns$ (FR1) (For Regular RU) As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TE_{RL}(t)$, but to measure the maximum possible value it can take. $\max TE_{RL} (1,2) \text{ the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$ <u>Verifications for Maximum Absolute 1PPS Time Error</u> 1. Verify that PTP session is up and running between T-GM and CSR. 2. Verify 1PPS time error measurements on Clock Tester at the O-RU UNI. a. $\max TE_L$ measured on Clock Tester at the O-RU UNI $\leq 1350ns$ (For Enhanced RU) b. $\max TE_L$ measured on Clock Tester at the O-RU UNI $\leq 1350ns$ (For Regular RU) <u>Verifications for Maximum Relative 1PPS Time Error</u> 1. Verify that PTP and SyncE session is up and running. 2. Estimate the Maximum Relative 1PPS Time Error measurements between the TimeTransmitter ports of CSR1 and CSR2 at O-RU UNI. a. $\max TE_{RL} (1,2) \leq 60ns$ (FR2) (For Enhanced RU) b. $\max TE_{RL} (1,2) \leq 190ns$ (FR1) (For Enhanced RU) c. $\max TE_{RL} (1,2) \leq 100ns$ (FR1) (For Regular RU) <u>Verifications for Time Wander</u> 1. Verify the Time Wander using Clock tester at the O-RU UNI. a. Frequency Limit is $\leq 32ppb$ (For PRTC Class-A) b. MTIE mask as per the limit specified.			
Pass/Fail Criteria:	Test shall pass if the Maximum absolute TE, Maximum Relative TE and Wander Limit measurement result remains under the limits for the entire holdover event, including the restoration.			
Test Output	End-to-End Network Limit at O-RU UNI	Expected Value (Normal Case) (For Enhanced RU) (For Regular RU)		Observed Value
	Packet $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
	Packet $\max TE_{RL} $	$\leq 60ns$ (FR2)	$\leq 100ns$ (FR1)	
		$\leq 190ns$ (FR1)		
	1PPS $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
	1PPS $\max TE_{RL} $	$\leq 60ns$ (FR2)	$\leq 100ns$ (FR1)	
		$\leq 190ns$ (FR1)		

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Test Topology:



Test Configuration

1. Multiple Transport TNEs (HSR1, HSR2, CSR1 and CSR2) connected and configured as shown in the Test Topology.

	2. Clock Tester at O-RU UNI receives timing input on its measurement port. 3. Provide 1PPS and 10MHz reference to Clock Tester from GNSS simulator. 4. G8275.1 profile configured for primary and backup path providing Full Timing Support (FTS)
Procedure:	<u>Case-1: Normal Scenario (GNSS connectivity to both GM is Up)</u> <u>Detailed Test procedure:</u> 1. Connect Edge PRTC/T-GM and Network T-GM to the GNSS Simulator and ensure both T-GMs locked. 2. Enable network connectivity among T-GMs, CSRs and HSRs. 3. Enable both Edge T-GM and Network T-GM with G.8275.1 profile. 4. Establish PTP connection (G.8275.1) sessions at each of the TNEs. 5. Configure Clock Tester for measurement profiles. 6. Perform the Time error measurement. <u>Verifications for Maximum Absolute Packet Time Error</u> 1. Verify that PTP session is up and running between Edge T-GM and CSR. 2. Verify time error measurements using Clock Tester for the PTP flows from CSR at the O-RU UNI. a. $\max TE_L$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU) b. $\max TE_L$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU) <u>Verifications for Maximum Relative Packet Time Error (NOTE1)</u> 1. Perform time error measurements on Clock Tester at O-RU1 and O-RU2 plane independently. Collect the data ($\max TE1, \min TE1$) and ($\max TE2, \min TE2$) for both O-RU UNI at least for a period of 1000sec. 2. Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports. a. $\max TE_{RL} (1,2) \leq 60ns$ (FR2) (For Enhanced RU) b. $\max TE_{RL} (1,2) \leq 190ns$ (FR1) (For Enhanced RU) c. $\max TE_{RL} (1,2) \leq 100ns$ (FR1) (For Regular RU) As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TE_{RL}(t)$, but to measure the maximum possible value it can take. $\max TE_{RL} (1,2) \text{ the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$ <u>Verifications for Maximum Absolute 1PPS Time Error</u> 1. Verify that PTP session is up and running between T-GM and CSR. 2. Verify the 1PPS time error measurements on Clock Tester at the O-RU UNI. a. $\max TE_L$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU) b. $\max TE_L$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU) <u>Verifications for Maximum Relative 1PPS Time Error</u>

1. Verify that PTP and SyncE session is up and running.
2. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

Verifications for Time Wander

1. Verify the Time Wander using Clock tester at the O-RU UNI.
 - a. Frequency Limit is $\leq 32ppb$ (For PRTC Class-A)
 - b. MTIE mask as per the limit specified.

Case-2: GNSS Connectivity to Edge T-GM Fails

Detailed Test procedure:

1. Disconnect GNSS Simulator connection to Edge PRTC/T-GM and switch the Timing source to Network T-GM.
2. Monitor and log the Time Error at specified time interval.

Verifications for Maximum Absolute Packet Time Error

1. Verify that the Edge T-GM moves to Phase Holdover and start advertising Clock Class=7 and CSR recover clock from Network T-GM via FTS network based on A-BTCA.

Verify time error measurements using Clock Tester for the PTP flows from CSR at the O-RU UNI in while the Edge T-GM is in holdover.

- a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
- b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative Packet Time Error (NOTE1)

1. Perform time error measurements on Clock Tester at O-RU1 and O-RU2 plane independently. Collect the data ($\max TE1, \min TE1$) and ($\max TE2, \min TE2$) for both O-RU UNI at least for a period of 1000sec.
2. Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TE_{RL}(t)$, but to measure the maximum possible value it can take.

$$\max|TE_{RL}|(1,2) \text{ the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$$

Verifications for Maximum Absolute 1PPS Time Error

1. Verify that PTP session is up and running between T-GM and CSR.
2. Verify 1PPS time error measurements on Clock Tester at the O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)

	b. $\max TE_L$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU) Verifications for Maximum Relative 1PPS Time Error - Verify that PTP and SyncE session is up and running. - Estimate the Maximum Relative 1PPS Time Error measurements between the O-RU UNI ports. $\max TE_{RL} (1,2) \leq 60ns$ (FR2) (For Enhanced RU) $\max TE_{RL} (1,2) \leq 190ns$ (FR1) (For Enhanced RU) $\max TE_{RL} (1,2) \leq 100ns$ (FR1) (For Regular RU) Verifications for Time Wander - Verify the Time Wander using Clock tester at the O-RU UNI. - Frequency Limit is $\leq 32ppb$ (For PRTC Class-A) - MTIE mask as per the limit specified.		
Pass/Fail Criteria:	Test shall pass if the Maximum absolute TE, Maximum Relative TE and Wander Limit measurement result is below the limit specified.		
Test Output	End-to-End Network Limit at O-RU UNI	Expected Value (For Enhanced RU) (For Regular RU)	Observed Value
	Packet $\max TE_L $	$\leq 1100ns$ $\leq 1100ns$	
	Packet $\max TE_{RL} $	$\leq 60ns$ (FR2) $\leq 100ns$ (FR1) $\leq 190ns$ (FR1)	
	1PPS $\max TE_L $	$\leq 1100ns$ $\leq 1100ns$	
	1PPS $\max TE_{RL} $	$\leq 60ns$ (FR2) $\leq 100ns$ (FR1) $\leq 190ns$ (FR1)	
	Frequency Limit (PRTC Class-A)	$\leq 32ppb$ $\leq 32ppb$	

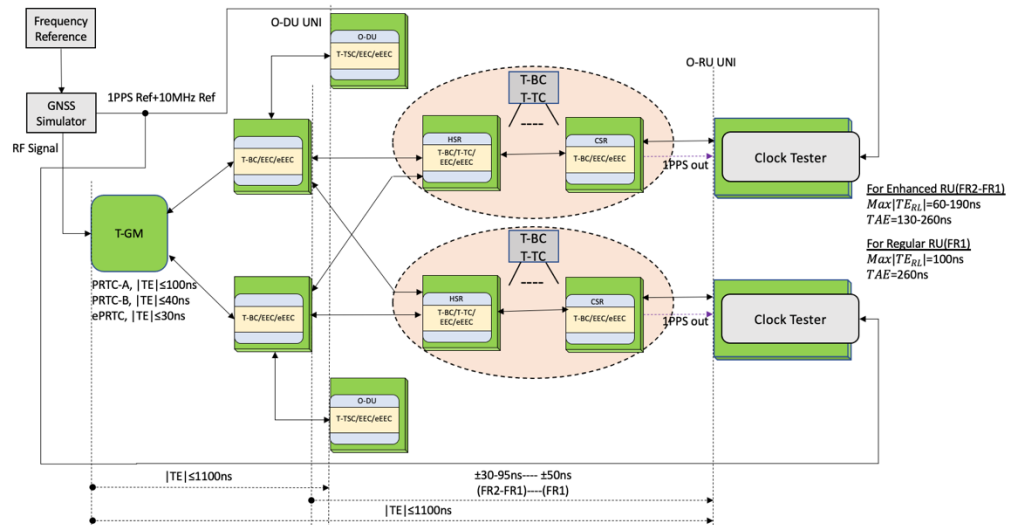
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9.1.19 Network Time Synchronization-LLS-C3 (Option-C) configuration

Test ID	SYNC.TE.19	Classification	Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Measurement of Frequency Error, Absolute Time Error, Relative Time Error at O-RU UNI. LLS-C3 (Option-C)		
Test Objective:	- Verify the Maximum Absolute Packet Time Error ($\max TE_L$) at O-RU UNI. - Verify the Maximum Relative Packet Time Error ($\max TE_{RL}$) between the O-RU UNI ports. - Verify the Maximum Absolute 1PPS Time Error ($\max TE_L$) at O-RU UNI. - Verify the Maximum Relative 1PPS Time Error ($\max TE_{RL}$) at O-RU UNI. - Verify the Time Wander at O-RU UNI.		
Pre-Req	T-GM is placed either in the Midhaul or Backhaul. T-GM/PRTC connected to GNSS Simulator, and T-GM is locked. O-RU UNI is the measurement plane.		

Test Topology:



The test topology above illustrates the placement of clock tester at the measurement plane with 1PPS and 10MHz reference connected to clock tester. The tester measurement ports at O-RU UNI connected to the CSR.

Test Configuration

1. Multiple Transport TNEs (HSR, HSR-B and CSRs) connected and configured as shown in the Test Topology.
2. Clock Tester at O-RU UNI receives timing input on its measurement port.
3. Provide 1PPS and 10MHz reference to Clock Tester from GNSS simulator.
4. G8275.1 profile for primary and G8275.2 profile for backup and provides partial timing support.

Procedure:

Detailed Test procedure:

1. Connect T-GM to GNSS Simulator and ensure that T-GM is locked.
2. Configure O-DU as T-TSC, CSR1 and CSR2 as T-BCs. HSR1 and HSR2 as either T-BCs or T-TCs.
3. Enable G.8275.1 profile on all transport TNEs (HSR, CSR and T-BC).
4. Connect Clock tester measurement ports to CSRs TimeTransmitter port to monitor PTP at the O-RU UNI.
5. Establish PTP and SyncE session to the fronthaul network.

Verifications for Maximum Absolute Packet Time Error

1. Verify that PTP session is up and running between T-GM and CSR.
2. Verify time error measurements using Clock Tester for the PTP flows from CSR at the O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative Packet Time Error (NOTE1)

1. Perform time error measurements on Clock Tester at O-RU1 and O-RU2 plane independently. Collect the data ($\max TE1, \min TE1$) and ($\max TE2, \min TE2$) for both O-RU UNI at least for a period of 1000sec.
2. Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}| (1,2) \leq 60ns$ (FR2) (For Enhanced RU)

- b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
- c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TE_{RL}(t)$, but to measure the maximum possible value it can take.

$$\max|TE_{RL}|(1,2) \text{ the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$$

Verifications for Maximum Absolute 1PPS Time Error

1. Verify that PTP session is up and running between T-GM and CSR.
2. Verify the 1PPS time error measurements on Clock Tester at the O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and SyncE session is up and running.
2. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

Verifications for Time Wander

1. Verify the Time Wander using Clock tester at the O-RU UNI.
 - a. Frequency Limit is $\leq 32ppb$ (For PRTC Class-A)
 - b. MTIE mask as per the limit specified.

NOTE1:

To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.

Pass/Fail Criteria:

Test shall pass if the Maximum absolute TE, Maximum Relative TE and Wander Limit measurement result is below the limit specified.

Test Output

End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value
	(For Enhanced RU)	(For Regular RU)	
Packet $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
Packet $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
	$\leq 190ns(FR1)$		
1PPS $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
1PPS $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
	$\leq 190ns(FR1)$		
Frequency Limit (If PRTC is Class-A device)	$\leq 32ppb$	$\leq 32ppb$	

9.2 Timing Solution & Resiliency in Ring Topology (LLS-C3)

This chapter delivers the testing methodologies, parameters and performance limits for PTP-based network technologies in normal and fail-over or rearrangement scenarios of T-GM and TNEs across various network segments in the LLS-C3 configurations. The chapter is organized in different sections each representing a typical synchronization configuration mode. For each mode, all related metrics are identified, and limits are provided to deliver pass/fail verdicts.

Various synchronization resiliency options have been defined in the O-RAN WG9 Synchronization Architecture and Solution Specification. Depending on the specific O-RAN deployment being considered, not all of them might be relevant. When testing the S-Plane, the System Integrator shall identify which of the test cases are relevant depending on the specific deployment scenarios addressed.

Following high level options have been covered in this chapter and each of these options, clock advertisements and performance limits are discussed for normal and failover scenarios for T-GMs and TNEs in Backhaul, Midhaul and Fronthaul network segments in LLS-C3 configuration. Also scenarios like Asymmetry and Passive Port Monitoring are covered.

1. Ring topology in Midhaul and Backhaul (LLS-C3 Configuration)
 - a. O-DU connected to HSR and SYNC from Backhaul.
 - b. O-DU connected to HSR and SYNC from Backhaul. (Aggregated Link)
 - c. O-DU connected to CSR and SYNC from Backhaul.
2. Ring topology in Midhaul and Backhaul (LLS-C3 Configuration)
 - a. O-DU connected to HSR and SYNC from Midhaul.
 - b. O-DU connected to HSR and SYNC from Midhaul. (Aggregated Link)
 - c. O-DU connected to CSR and SYNC from Midhaul.

It is important to note that, though Passive Port Monitoring is an OPTIONAL feature in ITU-T G.8275.1 ANNEX-G, if implemented, it is necessary for the equipment to conform to the requirement as per the specifications.

For PTP rearrangement scenarios like member-link failover in Aggregated Ethernet Bundle and for other failover scenarios, operator can make choice based on the vendor TNE capability to estimate the end-to-end clock performance limit. However, the performance limits specified in this chapter applicable for both normal and rearrangement scenarios.

The test cases apply to both lab and field deployment cases. For lab cases, the type and number of inserted Transport Node Elements (TNE) will be determined by operators (or any other applicable end customer). The tests are performed at Xhaul UNI interfaces, and not between individual TNEs. The Xhaul UNI interfaces include O-RU Fronthaul UNI, O-DU Fronthaul or Midhaul UNI.

1PPS test at O-RU UNI is typically used for debugging and monitoring the packet layer performance of the nearest TNE in the fronthaul. The intention of such test is that if the TNE vendor implemented the 1PPS interface, the 1PPS error measured on such interface at the O-RU UNI should be a reflection of the output time error of TNE. Operator shall pay attention to an unacceptably large deviation of 1PPS error, beyond the limits specified, which may indicate an incorrect behaviour of TNEs.

The Test Topologies in all the test cases uses GNSS simulator to generate the GNSS signal to the network PRTC/T-GM. This high-quality reference from the simulator is used as the external reference for clock testers as well. The resultant accuracy of this method is a few nanoseconds. However, one can also use a GNSS signal feeding both the initial PRTC/T-GM in the network and the reference clock generating a 1pps/10MHz used to synchronize the clock testers for the measurements. Operator can make an allowance in this case, for all of the measurements for the accuracy of the reference. The magnitude of this allowance will depend on the specific GNSS reference in use.

For relative time error measurements at UNI (e.g., O-RU Fronthaul UNI) this document uses the metric $\max|TE_{RL}|$ (1,2) as defined in [88], Appendix XI. Please note this metric represents an estimation and the worst-case scenario. Tests that meet (are below) the limits are seen as PASS. Tests not meeting this metric may still be acceptable. The decision is for operators to make. However, if the measurement equipment support relative time error measurement, operator can use such equipment. Refer ITU-T G.8271.1 Appendix XI for more details.

For Frequency Limit at UNI (e.g., O-RU Fronthaul UNI) this document uses the metric defined as per WG4 section H.2 based on PRTC class for LLS-C3 configuration. Applicable MTIE limit shall apply for Frequency as per WG4 specification.

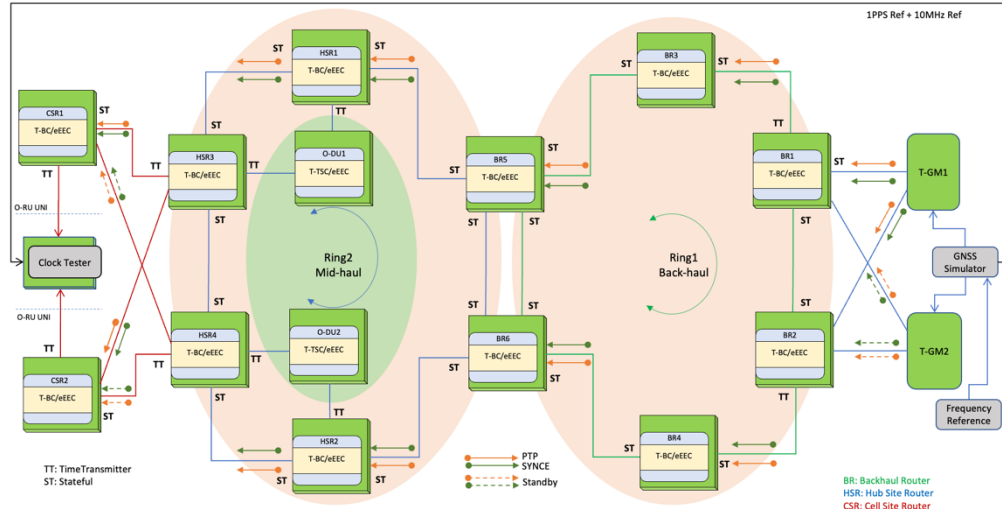
The acceptable limits in all the below tests shall meet CUS specification worst case limits based on the maximum network size, but operator “should” apply the limits more aligned with the real tested configuration based on the network size.

9.2.1 Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC from Backhaul (LLS-C3)-Normal State (Case-1: T-GM1 is the preferred GM)

Test ID	SYNC.RE.01	Classification	Resiliency, Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC from Backhaul-Normal State without any Fail-over conditions exist (Case-1: T-GM1 is the preferred GM)		
Test Objective:	1. Verify the Clock Lock states on BRs, HSRs and CSRs. 2. Verify the steady-state clock advertisements on BRs, HSRs and CSRs. 3. Verify the Maximum Absolute Packet Time Error ($\max TE_L$) at O-RU UNI. 4. Verify the Maximum Relative Packet Time Error ($\max TE_{RL}$) between two O-RU UNI ports. 5. Verify the Maximum Absolute 1PPS Time Error ($\max TE_L$) at O-RU UNI. 6. Verify the Time Wander at the O-RU UNI.		
Pre-Req	✓ T-GM1 is the Primary GM and T-GM2 is the Secondary GM. ✓ Clock parameters of both T-GM1 and T-GM2 are same except Priority-2 & GMCID.		

✓ Priority-2 on T-GM1=1 and T-GM2=2.

Test Topology:



The test topology above illustrates the placement of clock tester at the measurement plane with 1PPS and 10MHz reference from GNSS simulator connected to clock tester. The tester measurement ports at O-RU UNI connected to CSR.

Test Configuration

1. Clock Tester at O-RU UNI receives timing input on its measurement port.
2. Provide 1PPS and 10MHz reference to Clock Tester from GNSS simulator.
3. G.8275.1 Full Timing Support profile.

Procedure:

Detailed Test procedure:

1. Connect both T-GMs to GNSS Simulator.
2. Configure T-GMs, BRs, HSRs and CSRs with G.8275.1 profile.
3. Connect Clock Tester measurement ports to CSRs TimeTransmitter ports to monitor PTP at the O-RU UNI.

Verifications for Clock Advertisements

1. Verify the clock lock state on BRs, HSRs, CSRs. The lock state should be Phase-aligned.
2. Verify the steady-state clock advertisements on BRs, HSRs and CSRs as per ITU-T G.8275.1 Standard-Appendix V, Table V.3.

Verifications for Maximum Absolute Packet Time Error

1. Verify that PTP session is up and running between T-GM and CSR.
2. Verify time error measurements using Clock Tester for the PTP flows from CSR at the O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative Packet Time Error (NOTE1)

1. Perform time error measurements on Clock Tester at O-RU1 and O-RU2 plane independently. Collect the data ($\max TE1, \min TE1$) and ($\max TE2, \min TE2$) for both O-RU UNI at least for a period of 1000sec.
2. Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)

- b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
- c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TE_{RL}(t)$, but to measure the maximum possible value it can take.

$$\max|TE_{RL}|(1,2) \text{ the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$$

Verifications for Maximum Absolute 1PPS Time Error

1. Verify that PTP session is up and running between T-GM and CSR.
2. Verify the 1PPS time error measurements using Clock Tester at the O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and SyncE session is up and running.
2. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

Verifications for Time Wander

1. Verify the Time Wander using Clock tester at the O-RU UNI.
 - a. Frequency Limit is $\leq 32ppb$ (For PRTC Class-A)
 - b. MTIE mask as per the limit specified.

NOTE1 (As per G.8271.1):

To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.

Pass/Fail Criteria:

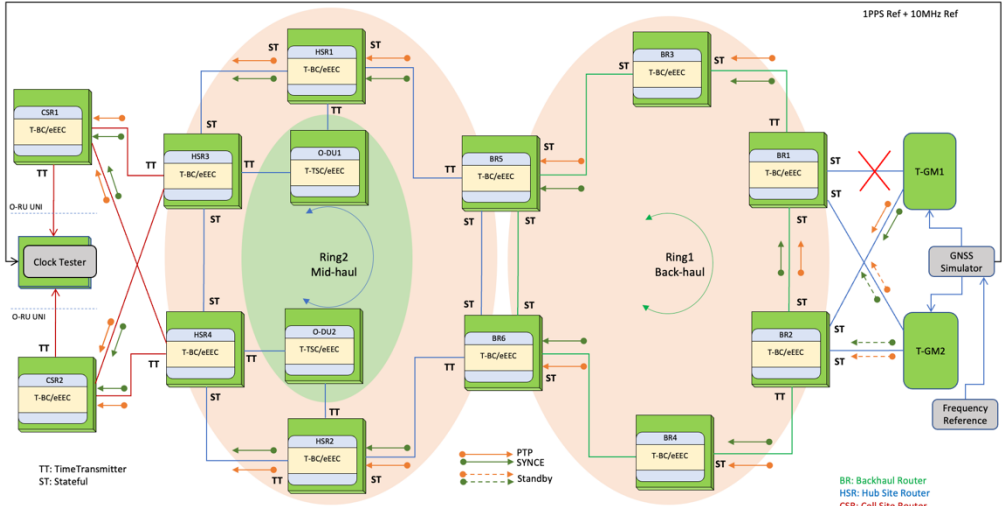
Test shall pass if the Maximum Absolute TE, Relative TE and Time Wander Limit result is within the specified limits.

Test Output

End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value
	(For Enhanced RU)	(For Regular RU)	
Packet $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
Packet $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
	$\leq 190ns(FR1)$		
1PPS $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
1PPS $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
	$\leq 190ns(FR1)$		
Frequency Limit (PRTC Class-A)	$\leq 32ppb$	$\leq 32ppb$	

The above limits applicable for both Normal and Re-arrangement scenarios.

9.2.2 Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC from Backhaul (LLS-C3)-T-GM1 to BR1 Link Down. (Case-1: T-GM1 is the preferred GM)

Test ID	SYNC.RE.02	Classification	Resiliency, Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC from Backhaul-T-GM1 to BR1 Link Down. (Case-1: T-GM1 is the preferred GM)		
Test Objective:	1. Verify the Clock Lock states on BRs, HSRs and CSRs. 2. Verify the clock advertisements on BRs, HSRs and CSRs. 3. Verify the Maximum Absolute Packet Time Error ($Max TE_L$) at O-RU UNI. 4. Verify the Maximum Relative Packet Time Error ($Max TE_{RL}$) between two O-RU UNI ports. 5. Verify the Maximum Absolute 1PPS Time Error ($Max TE_L$) at O-RU UNI. 6. Verify the Time Wander at the O-RU UNI.		
Pre-Req	✓ T-GM1 is the Primary GM and T-GM2 is the Secondary GM. ✓ Clock parameters of both T-GM1 and T-GM2 are same except Priority-2 & GMCID. ✓ Priority-2 on T-GM1=1 and T-GM2=2. ✓ Test 9.2.1 MUST PASS.		
Test Topology:	 The test topology above illustrates the placement of clock tester at the measurement plane with 1PPS and 10MHz reference from GNSS simulator connected to clock tester. The tester measurement ports at O-RU UNI connected to CSR. Stateful (ST) refers to the port configuration of neither explicit timeTransmitter nor explicit timeReceiver. When configured, it refers to the dynamic state of the port based on the received Announce message.		
Test Configuration	1. As per Test 9.2.1. 2. Fail the link between T-GM1 and BR1.		

Procedure:
Detailed Test procedure:

1. Connect both T-GMs to GNSS Simulator.
2. Configure T-GMs, BRs, HSRs and CSRs with G.8275.1 profile.
3. Connect Clock Tester measurement ports to CSRs TimeTransmitter ports to monitor PTP at the O-RU UNI.

Verifications for Clock Advertisements

1. Verify the clock lock state on BRs, HSRs, CSRs. The lock state should be Phase-aligned.
2. Verify the steady-state clock advertisements on BRs, HSRs and CSRs as per ITU-T G.8275.1 Standard-Appendix V, Table V.3.

Verifications for Maximum Absolute Packet Time Error

1. Verify that PTP session is up and running between T-GM and CSR.
2. Verify time error measurements using Clock Tester for the PTP flows from CSR at the O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative Packet Time Error (NOTE1)

1. Perform time error measurements on Clock Tester at O-RU1 and O-RU2 plane independently. Collect the data ($\max TE1, \min TE1$) and ($\max TE2, \min TE2$) for both O-RU UNI at least for a period of 1000sec.
2. Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}| (1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}| (1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}| (1,2) \leq 100ns$ (FR1) (For Regular RU)

As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TE_{RL}(t)$, but to measure the maximum possible value it can take.

$$\max|TE_{RL}| (1,2) \text{ the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$$

Verifications for Maximum Absolute 1PPS Time Error

1. Verify that PTP session is up and running between T-GM and CSR.
2. Verify the 1PPS time error measurements using Clock Tester at the O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and SyncE session is up and running.
2. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}| (1,2) \leq 60ns$ (FR2) (For Enhanced RU)

- b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
- c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

Verifications for Time Wander

1. Verify the Time Wander using Clock tester at the O-RU UNI.
 - a. Frequency Limit is $\leq 32ppb$ (For PRTC Class-A)
 - b. MTIE mask as per the limit specified.

NOTE1 (As per G.8271.1):

To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.

Pass/Fail Criteria:

Test shall pass if the Maximum Absolute TE, Relative TE and Time Wander Limit result is within the specified limits.

Test Output

End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value
	(For Enhanced RU)	(For Regular RU)	
Packet $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
Packet $\max TE_{RL} $	$\leq 60ns$ (FR2)	$\leq 100ns$ (FR1)	
	$\leq 190ns$ (FR1)		
1PPS $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
1PPS $\max TE_{RL} $	$\leq 60ns$ (FR2)	$\leq 100ns$ (FR1)	
	$\leq 190ns$ (FR1)		
Frequency Limit (PRTC Class-A)	$\leq 32ppb$	$\leq 32ppb$	

The above limits applicable for both Normal and Re-arrangement scenarios.

1

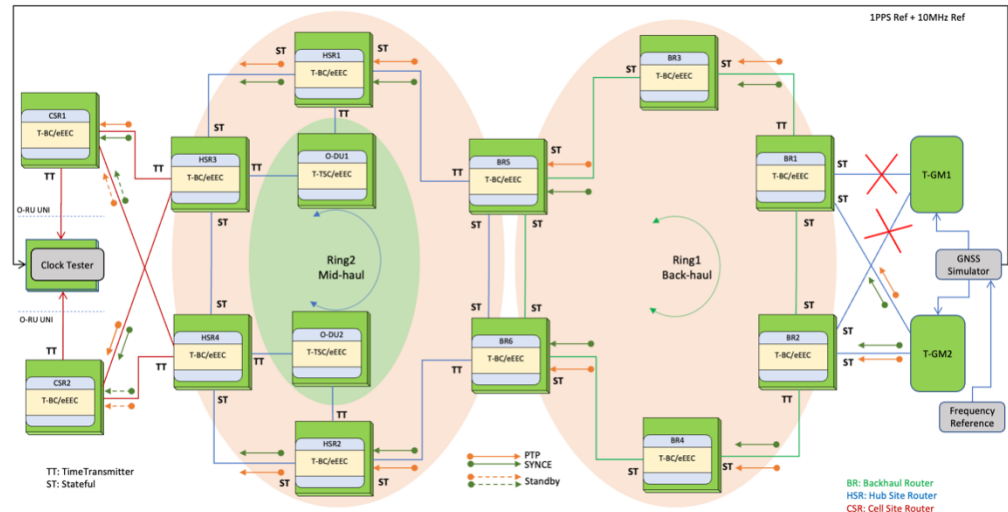
9.2.3 Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC from Backhaul (LLS-C3)-T-GM1 Fail-over. (Case-1: T-GM1 is the preferred GM)

3

Test ID	SYNC.RE.03	Classification	Resiliency, Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC from Backhaul-T-GM1 Fail-over. (Case-1: T-GM1 is the preferred GM)		
Test Objective:	1. Verify the Clock Lock states on BRs, HSRs and CSRs. 2. Verify the clock advertisements on BRs, HSRs and CSRs. 3. Verify the Maximum Absolute Packet Time Error ($\max TE_L$) at O-RU UNI. 4. Verify the Maximum Relative Packet Time Error ($\max TE_{RL}$) between two O-RU UNI ports. 5. Verify the Maximum Absolute 1PPS Time Error ($\max TE_L$) at O-RU UNI. 6. Verify the Time Wander at the O-RU UNI.		
Pre-Req	✓ T-GM1 is the Primary GM and T-GM2 is the Secondary GM. ✓ Clock parameters of both T-GM1 and T-GM2 are same except Priority-2 & GMCID.		

- ✓ Priority-2 on T-GM1=1 and T-GM2=2.
- ✓ Test 9.2.1 MUST PASS.

Test Topology:



The test topology above illustrates the placement of clock tester at the measurement plane with 1PPS and 10MHz reference from GNSS simulator connected to clock tester. The tester measurement ports at O-RU UNI connected to CSR.

Stateful (ST) refers to the port configuration of neither explicit timeTransmitter nor explicit timeReceiver. When configured, it refers to the dynamic state of the port based on the received Announce message.

Test Configuration

1. As per Test 9.2.1.
2. Fail the link between T-GM1 to BR1 and BR2 OR Power OFF the T-GM1.

Procedure:

Detailed Test procedure:

1. Connect both T-GMs to GNSS Simulator.
2. Configure T-GMs, BRs, HSRs and CSRs with G.8275.1 profile.
3. Connect Clock Tester measurement ports to CSRs TimeTransmitter ports to monitor PTP at the O-RU UNI.

Verifications for Clock Advertisements

1. Verify that BR1 shall select clock from T-GM1 via BR2.
2. Verify the clock lock state on BRs, HSRs, CSRs. The lock state should be Phase-aligned.
3. Verify the steady-state clock advertisements on BRs, HSRs and CSRs as per ITU-T G.8275.1 Standard-Appendix V, Table V.3.

Verifications for Maximum Absolute Packet Time Error

1. Verify that PTP session is up and running between T-GM and CSR.
2. Verify time error measurements using Clock Tester for the PTP flows from CSR at the O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative Packet Time Error (NOTE1)

1. Perform time error measurements on Clock Tester at O-RU1 and O-RU2 plane independently. Collect the data ($maxTE1, minTE1$) and ($maxTE2, minTE2$) for both O-RU UNI at least for a period of 1000sec.
2. Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TE_{RL}(t)$, but to measure the maximum possible value it can take.

$$\max|TE_{RL}|(1,2) \text{ the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$$

Verifications for Maximum Absolute 1PPS Time Error

1. Verify that PTP session is up and running between T-GM and CSR.
2. Verify the 1PPS time error measurements using Clock Tester at the O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and SyncE session is up and running.
2. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

Verifications for Time Wander

1. Verify the Time Wander using Clock tester at the O-RU UNI.
 - a. Frequency Limit is $\leq 32ppb$ (For PRTC Class-A)
 - b. MTIE mask as per the limit specified.

NOTE1 (As per G.8271.1):

To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.

Pass/Fail Criteria:

Test shall pass if the Maximum Absolute TE, Relative TE and Time Wander Limit result is within the specified limits.

Test Output

End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value
	(For Enhanced RU)	(For Regular RU)	
Packet $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
Packet $\max TE_{RL} $	$\leq 60ns$ (FR2)	$\leq 100ns$ (FR1)	
	$\leq 190ns$ (FR1)		
1PPS $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
1PPS $\max TE_{RL} $	$\leq 60ns$ (FR2)	$\leq 100ns$ (FR1)	

		$\leq 190ns(FR1)$		
	Frequency Limit (PRTC Class-A)	$\leq 32ppb$	$\leq 32ppb$	
The above limits applicable for both Normal and Re-arrangement scenarios.				

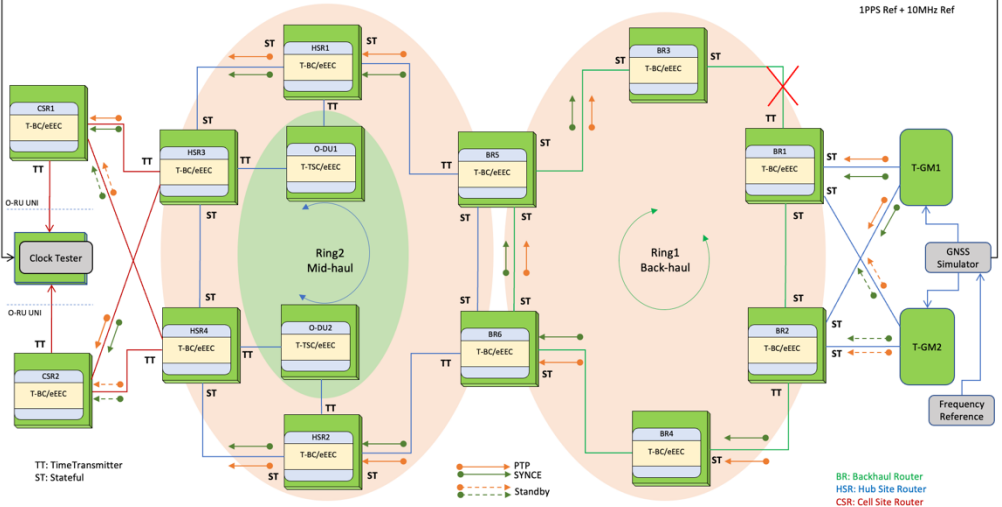
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9.2.4 Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC from Backhaul (LLS-C3)-Ring Fail-over in Backhaul. (Case-1: T-GM1 is the preferred GM)

Test ID	SYNC.RE.04	Classification	Resiliency, Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC from Backhaul-Ring Fail-over in Backhaul. (Case-1: T-GM1 is the preferred GM)		
Test Objective:	1. Verify the Clock Lock states on BRs, HSRs and CSRs. 2. Verify the clock advertisements on BRs, HSRs and CSRs. 3. Verify the Maximum Absolute Packet Time Error ($Max TE_L$) at O-RU UNI. 4. Verify the Maximum Relative Packet Time Error ($Max TE_{RL}$) between two O-RU UNI ports. 5. Verify the Maximum Absolute 1PPS Time Error ($Max TE_L$) at O-RU UNI. 6. Verify the Time Wander at the O-RU UNI.		
Pre-Req	✓ T-GM1 is the Primary GM and T-GM2 is the Secondary GM. ✓ Clock parameters of both T-GM1 and T-GM2 are same except Priority-2 & GMCID. ✓ Priority-2 on T-GM1=1 and T-GM2=2. ✓ Test 9.2.1 MUST PASS.		
Test Topology:			

	The test topology above illustrates the placement of clock tester at the measurement plane with 1PPS and 10MHz reference from GNSS simulator connected to clock tester. The tester measurement ports at O-RU UNI connected to CSR. Stateful (ST) refers to the port configuration of neither explicit timeTransmitter nor explicit timeReceiver. When configured, it refers to the dynamic state of the port based on the received Announce message.
Test Configuration	- As per Test 9.2.1. - Fail the link between BR1 and BR3.
Procedure:	<u>Detailed Test procedure:</u> - Connect both T-GMs to GNSS Simulator. - Configure T-GMs, BRs, HSRs and CSRs with G.8275.1 profile. - Connect Clock Tester measurement ports to CSRs TimeTransmitter ports to monitor PTP at the O-RU UNI. <u>Verifications for Clock Advertisements</u> - Verify that BR1 shall select clock from T-GM1 via BR2. - Verify the clock lock state on BRs, HSRs, CSRs. The lock state should be Phase-aligned. - Verify the steady-state clock advertisements on BRs, HSRs and CSRs as per ITU-T G.8275.1 Standard-Appendix V, Table V.3. <u>Verifications for Maximum Absolute Packet Time Error</u> - Verify that PTP session is up and running between T-GM and CSR. - Verify time error measurements using Clock Tester for the PTP flows from CSR at the O-RU UNI. $\max TE_L$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU) $\max TE_L$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU) <u>Verifications for Maximum Relative Packet Time Error (NOTE1)</u> - Perform time error measurements on Clock Tester at O-RU1 and O-RU2 plane independently. Collect the data ($\max TE1, \min TE1$) and ($\max TE2, \min TE2$) for both O-RU UNI at least for a period of 1000sec. - Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports. $\max TE_{RL} (1,2) \leq 60ns$ (FR2) (For Enhanced RU) $\max TE_{RL} (1,2) \leq 190ns$ (FR1) (For Enhanced RU) $\max TE_{RL} (1,2) \leq 100ns$ (FR1) (For Regular RU) As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TE_{RL}(t)$, but to measure the maximum possible value it can take. $\max TE_{RL} (1,2) \text{ the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$ <u>Verifications for Maximum Absolute 1PPS Time Error</u> - Verify that PTP session is up and running between T-GM and CSR. - Verify the 1PPS time error measurements using Clock Tester at the O-RU UNI.

- a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
- b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and SyncE session is up and running.
2. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}| (1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}| (1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}| (1,2) \leq 100ns$ (FR1) (For Regular RU)

Verifications for Time Wander

1. Verify the Time Wander using Clock tester at the O-RU UNI.
 - a. Frequency Limit is $\leq 32ppb$ (For PRTC Class-A)
 - b. MTIE mask as per the limit specified.

NOTE1 (As per G.8271.1):

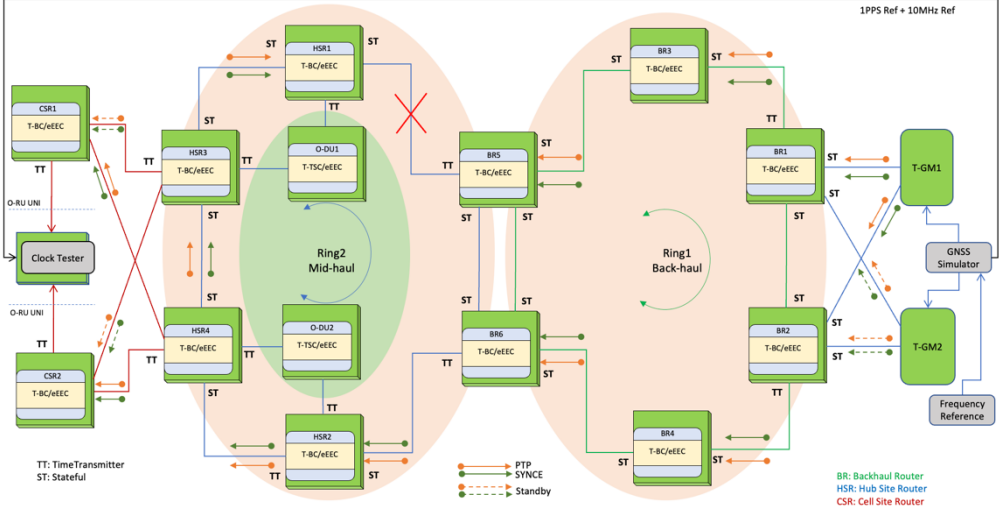
To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.

Pass/Fail Criteria:	Test shall pass if the Maximum Absolute TE, Relative TE and Time Wander Limit result is within the specified limits.			
Test Output	End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value
		(For Enhanced RU)	(For Regular RU)	
	Packet max TE _L	≤ 1100ns	≤ 1100ns	
	Packet max TE _{RL}	≤ 60ns(FR2)	≤ 100ns(FR1)	
		≤ 190ns(FR1)		
	1PPS max TE _L	≤ 1100ns	≤ 1100ns	
	1PPS max TE _{RL}	≤ 60ns(FR2)	≤ 100ns(FR1)	
		≤ 190ns(FR1)		
	Frequency Limit (PRTC Class-A)	≤ 32ppb	≤ 32ppb	
The above limits applicable for both Normal and Re-arrangement scenarios.				

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2 9.2.5 Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC from
 3 Backhaul (LLS-C3)-Ring Fail-over in Midhaul. (Case-1: T-GM1 is the
 4 preferred GM)

Test ID	SYNC.RE.05	Classification	Resiliency, Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC from Backhaul-Ring Fail-over in Midhaul. (Case-1: T-GM1 is the preferred GM)		

Test Objective:	1. Verify the Clock Lock states on BRs, HSRs and CSRs. 2. Verify the clock advertisements on BRs, HSRs and CSRs. 3. Verify the Maximum Absolute Packet Time Error ($Max TE_L$) at O-RU UNI. 4. Verify the Maximum Relative Packet Time Error ($Max TE_{RL}$) between two O-RU UNI ports. 5. Verify the Maximum Absolute 1PPS Time Error ($Max TE_L$) at O-RU UNI. 6. Verify the Time Wander at the O-RU UNI.
Pre-Req	✓ T-GM1 is the Primary GM and T-GM2 is the Secondary GM. ✓ Clock parameters of both T-GM1 and T-GM2 are same except Priority-2 & GMCID. ✓ Priority-2 on T-GM1=1 and T-GM2=2. ✓ Test 9.2.1 MUST PASS.
Test Topology:	 The test topology above illustrates the placement of clock tester at the measurement plane with 1PPS and 10MHz reference from GNSS simulator connected to clock tester. The tester measurement ports at O-RU UNI connected to CSR. Stateful (ST) refers to the port configuration of neither explicit timeTransmitter nor explicit timeReceiver. When configured, it refers to the dynamic state of the port based on the received Announce message.
Test Configuration	1. As per Test 9.2.1. 2. Fail the link between BR1 and BR3.
Procedure:	Detailed Test procedure: 1. Connect both T-GMs to GNSS Simulator. 2. Configure T-GMs, BRs, HSRs and CSRs with G.8275.1 profile. 3. Connect Clock Tester measurement ports to CSRs TimeTransmitter ports to monitor PTP at the O-RU UNI. Verifications for Clock Advertisements 1. Verify that BR1 shall select clock from T-GM1 via BR2. 2. Verify the clock lock state on BRs, HSRs, CSRs. The lock state should be Phase-aligned. 3. Verify the steady-state clock advertisements on BRs, HSRs and CSRs as per ITU-T G.8275.1 Standard-Appendix V, Table V.3.

Verifications for Maximum Absolute Packet Time Error

1. Verify that PTP session is up and running between T-GM and CSR.
2. Verify time error measurements using Clock Tester for the PTP flows from CSR at the O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative Packet Time Error (NOTE1)

1. Perform time error measurements on Clock Tester at O-RU1 and O-RU2 plane independently. Collect the data $(\max TE1, \min TE1)$ and $(\max TE2, \min TE2)$ for both O-RU UNI at least for a period of 1000sec.
2. Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}| (1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}| (1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}| (1,2) \leq 100ns$ (FR1) (For Regular RU)

As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TE_{RL}(t)$, but to measure the maximum possible value it can take.

$$\max|TE_{RL}| (1,2) \text{ the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$$

Verifications for Maximum Absolute 1PPS Time Error

1. Verify that PTP session is up and running between T-GM and CSR.
2. Verify the 1PPS time error measurements using Clock Tester at the O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and SyncE session is up and running.
2. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}| (1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}| (1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}| (1,2) \leq 100ns$ (FR1) (For Regular RU)

Verifications for Time Wander

1. Verify the Time Wander using Clock tester at the O-RU UNI.
 - a. Frequency Limit is $\leq 32ppb$ (For PRTC Class-A)
 - b. MTIE mask as per the limit specified.

NOTE1 (As per G.8271.1):

To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.

Pass/Fail Criteria:	Test shall pass if the Maximum Absolute TE, Relative TE and Time Wander Limit result is within the specified limits.			
Test Output	End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value
		(For Enhanced RU)	(For Regular RU)	
	Packet max TE _L	≤ 1100ns	≤ 1100ns	
	Packet max TE _{RL}	≤ 60ns(FR2)	≤ 100ns(FR1)	
		≤ 190ns(FR1)		
	1PPS max TE _L	≤ 1100ns	≤ 1100ns	
	1PPS max TE _{RL}	≤ 60ns(FR2)	≤ 100ns(FR1)	
		≤ 190ns(FR1)		
Frequency Limit (PRTC Class-A)	≤ 32ppb	≤ 32ppb		
The above limits applicable for both Normal and Re-arrangement scenarios.				

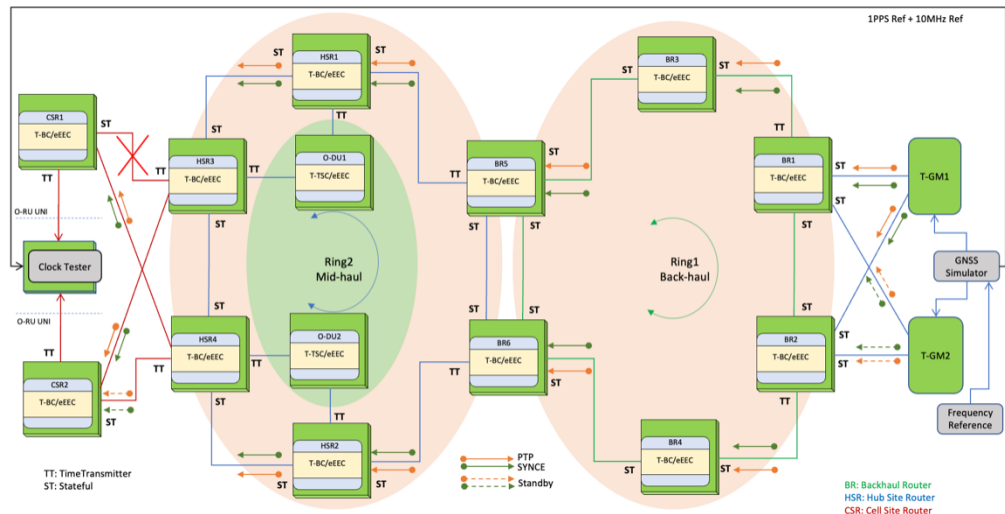
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9.2.6 Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC from Backhaul (LLS-C3)-Ring Fail-over in Fronthaul. (Case-1: T-GM1 is the preferred GM)

4

Test ID	SYNC.RE.06	Classification	Resiliency, Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC from Backhaul-Ring Fail-over in Fronthaul. (Case-1: T-GM1 is the preferred GM)		
Test Objective:	1. Verify the Clock Lock states on BRs, HSRs and CSRs. 2. Verify the clock advertisements on BRs, HSRs and CSRs. 3. Verify the Maximum Absolute Packet Time Error ($Max TE_L$) at O-RU UNI. 4. Verify the Maximum Relative Packet Time Error ($Max TE_{RL}$) between two O-RU UNI ports. 5. Verify the Maximum Absolute 1PPS Time Error ($Max TE_L$) at O-RU UNI. 6. Verify the Time Wander at the O-RU UNI.		
Pre-Req	✓ T-GM1 is the Primary GM and T-GM2 is the Secondary GM. ✓ Clock parameters of both T-GM1 and T-GM2 are same except Priority-2 & GMCID. ✓ Priority-2 on T-GM1=1 and T-GM2=2. ✓ Test 9.2.1 MUST PASS.		

Test Topology:



The test topology above illustrates the placement of clock tester at the measurement plane with 1PPS and 10MHz reference from GNSS simulator connected to clock tester. The tester measurement ports at O-RU UNI connected to CSR.

Stateful (ST) refers to the port configuration of neither explicit timeTransmitter nor explicit timeReceiver. When configured, it refers to the dynamic state of the port based on the received Announce message.

Test Configuration

- As per Test 9.2.1.
- Fail the link between HSR3 and CSR1.

Procedure:

Detailed Test procedure:

- Connect both T-GMs to GNSS Simulator.
- Configure T-GMs, BRs, HSRs and CSRs with G.8275.1 profile.
- Connect Clock Tester measurement ports to CSRs TimeTransmitter ports to monitor PTP at the O-RU UNI.

Verifications for Clock Advertisements

- Verify that BR1 shall select clock from T-GM1 via BR2.
- Verify the clock lock state on BRs, HSRs, CSRs. The lock state should be Phase-aligned.
- Verify the steady-state clock advertisements on BRs, HSRs and CSRs as per ITU-T G.8275.1 Standard-Appendix V, Table V.3.

Verifications for Maximum Absolute Packet Time Error

- Verify that PTP session is up and running between T-GM and CSR.
- Verify time error measurements using Clock Tester for the PTP flows from CSR at the O-RU UNI.
 - $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative Packet Time Error (NOTE1)

- Perform time error measurements on Clock Tester at O-RU1 and O-RU2 plane independently. Collect the data ($\max TE1, \min TE1$) and ($\max TE2, \min TE2$) for both O-RU UNI at least for a period of 1000sec.
- Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports.

- a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
- b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
- c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TE_{RL}(t)$, but to measure the maximum possible value it can take.

$$\max|TE_{RL}|(1,2) \text{ the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$$

Verifications for Maximum Absolute 1PPS Time Error

1. Verify that PTP session is up and running between T-GM and CSR.
2. Verify the 1PPS time error measurements using Clock Tester at the O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and SyncE session is up and running.
2. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

Verifications for Time Wander

1. Verify the Time Wander using Clock tester at the O-RU UNI.
 - a. Frequency Limit is $\leq 32ppb$ (For PRTC Class-A)
 - b. MTIE mask as per the limit specified.

NOTE1 (As per G.8271.1):

To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.

Pass/Fail Criteria:

Test shall pass if the Maximum Absolute TE, Relative TE and Time Wander Limit result is within the specified limits.

Test Output

End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value
	(For Enhanced RU)	(For Regular RU)	
Packet $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
Packet $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
	$\leq 190ns(FR1)$		
1PPS $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
1PPS $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
	$\leq 190ns(FR1)$		
Frequency Limit (PRTC Class-A)	$\leq 32ppb$	$\leq 32ppb$	

The above limits applicable for both Normal and Re-arrangement scenarios.

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9.2.7 Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC from

3

Backhaul (LLS-C3)-Asymmetry Test (Case-1: T-GM1 is the preferred GM)

Test ID	SYNC.RE.07	Classification	Resiliency, Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC from Backhaul-Asymmetry Test. (Case-1: T-GM1 is the preferred GM)		
Test Objective:	1. Verify the Clock Lock states on BRs, HSRs and CSRs. 2. Verify the steady-state clock advertisements on BRs, HSRs and CSRs. 3. Verify the Maximum Absolute Packet Time Error ($Max TE_L$) at O-RU UNI. 4. Verify the Maximum Relative Packet Time Error ($Max TE_{RL}$) between two O-RU UNI ports. 5. Verify the Maximum Absolute 1PPS Time Error ($Max TE_L$) at O-RU UNI. 6. Verify the Time Wander at the O-RU UNI.		
Pre-Req	✓ T-GM1 is the Primary GM and T-GM2 is the Secondary GM. ✓ Clock parameters of both T-GM1 and T-GM2 are same except Priority-2 & GMCID. ✓ Priority-2 on T-GM1=1 and T-GM2=2.		
Test Topology:	The test topology above illustrates the placement of clock tester at the measurement plane with 1PPS and 10MHz reference from GNSS simulator connected to clock tester. The tester measurement ports at O-RU UNI connected to CSR. Stateful (ST) refers to the port configuration of neither explicit timeTransmitter nor explicit timeReceiver. When configured, it refers to the dynamic state of the port based on the received Announce message.		
Test Configuration	1. As per Test 9.2.1. 2. Introduce Asymmetry of +1000ns at the O/p of T-GM1 and No Asymmetry at the O/p of T-GM2.		

	3. Configure Asymmetry Compensation of +1000ns on BR1 and BR2
Procedure:	<u>Detailed Test procedure:</u> 1. Connect both T-GMs to GNSS Simulator. 2. Configure T-GMs, BRs, HSRs and CSRs with G.8275.1 profile. 3. Connect Clock Tester measurement ports to CSRs TimeTransmitter ports to monitor PTP at the O-RU UNI. <u>Verifications for Clock Advertisements</u> 1. Verify the clock lock state on BRs, HSRs, CSRs. The lock state should be Phase-aligned. 2. Verify the steady-state clock advertisements on BRs, HSRs and CSRs as per ITU-T G.8275.1 Standard-Appendix V, Table V.3. <u>After Step-2 of Test Configurations</u> <u>Verifications for Maximum Absolute Packet Time Error and 1PPS Time Error</u> 1. Verify Performance Measurement during steady state. (1PPS and Packet Time Error should show an additional Error of +/-500ns due to Asymmetry) <u>Verifications for Maximum Relative Packet Time Error (NOTE1)</u> 1. Perform time error measurements on Clock Tester at O-RU1 and O-RU2 plane independently. Collect the data ($maxTE1, minTE1$) and ($maxTE2, minTE2$) for both O-RU UNI at least for a period of 1000sec. 2. Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports. a. $\max TE_{RL} (1,2) \leq 60ns$ (FR2) (For Enhanced RU) b. $\max TE_{RL} (1,2) \leq 190ns$ (FR1) (For Enhanced RU) c. $\max TE_{RL} (1,2) \leq 100ns$ (FR1) (For Regular RU) As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TE_{RL}(t)$, but to measure the maximum possible value it can take. $\max TE_{RL} (1,2) \text{ the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$ <u>Verifications for Maximum Relative 1PPS Time Error</u> 1. Verify that PTP and SyncE session is up and running. 2. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RU UNI ports. a. $\max TE_{RL} (1,2) \leq 60ns$ (FR2) (For Enhanced RU) b. $\max TE_{RL} (1,2) \leq 190ns$ (FR1) (For Enhanced RU) c. $\max TE_{RL} (1,2) \leq 100ns$ (FR1) (For Regular RU) <u>Verifications for Time Wander</u> 1. Verify the Time Wander using Clock tester at the O-RU UNI. a. Frequency Limit is $\leq 32ppb$ (For PRTC Class-A) b. MTIE mask as per the limit specified. NOTE1 (As per G.8271.1): To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.

After Step-3 of Test Configurations Verifications for Maximum Absolute Packet Time Error and 1PPS Time Error 1. Verify Performance Measurement during steady state. (1PPS and Packet Time Error shouldn't show any additional Error of +/-500ns due to Asymmetry) Verifications for Maximum Relative Packet Time Error and 1PPS Time Error 1. Verify the Maximum Relative 1PPS Time Error measurements between the O-RU UNI ports. The results should be as per Test 9.2.1.				
Pass/Fail Criteria:	Test shall pass if the Maximum Absolute TE, Relative TE and Time Wander Limit result is within the specified limits.			
Test Output	End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value
		(For Enhanced RU)	(For Regular RU)	
	Packet max TE _L	≤ 1100ns	≤ 1100ns	
	Packet max TE _{RL}	≤ 60ns(FR2)	≤ 100ns(FR1)	
		≤ 190ns(FR1)		
	1PPS max TE _L	≤ 1100ns	≤ 1100ns	
	1PPS max TE _{RL}	≤ 60ns(FR2)	≤ 100ns(FR1)	
		≤ 190ns(FR1)		
	Frequency Limit (PRTC Class-A)	≤ 32ppb	≤ 32ppb	
The above limits for max TE _L doesn't include the error due to Manual Asymmetry.				

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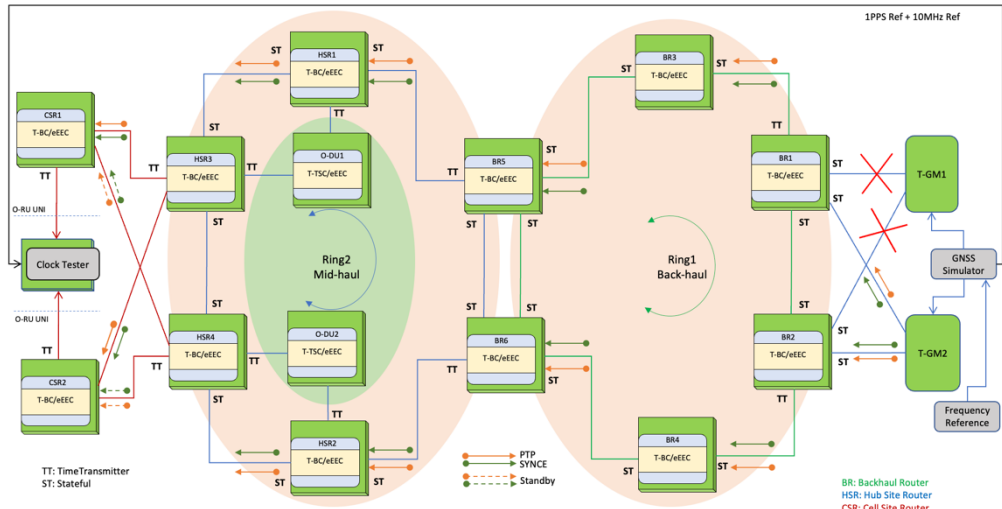
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9.2.8 Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC from Backhaul (LLS-C3)-Asymmetry Test with Link-Fail. (Case-1: T-GM1 is the preferred GM)

Test ID	SYNC.RE.08	Classification	Resiliency, Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC from Backhaul-Asymmetry Test with Link-Fail. (Case-1: T-GM1 is the preferred GM)		
Test Objective:	1. Verify the Clock Lock states on BRs, HSRs and CSRs. 2. Verify the steady-state clock advertisements on BRs, HSRs and CSRs. 3. Verify the Maximum Absolute Packet Time Error ($\max TE_L $) at O-RU UNI. 4. Verify the Maximum Relative Packet Time Error ($\max TE_{RL} $) between two O-RU UNI ports. 5. Verify the Maximum Absolute 1PPS Time Error ($\max TE_L $) at O-RU UNI. 6. Verify the Time Wander at the O-RU UNI.		
Pre-Req	✓ T-GM1 is the Primary GM and T-GM2 is the Secondary GM. ✓ Clock parameters of both T-GM1 and T-GM2 are same except Priority-2 & GMCID. ✓ Priority-2 on T-GM1=1 and T-GM2=2.		

Test Topology:



The test topology above illustrates the placement of clock tester at the measurement plane with 1PPS and 10MHz reference from GNSS simulator connected to clock tester. The tester measurement ports at O-RU UNI connected to CSR.

Stateful (ST) refers to the port configuration of neither explicit timeTransmitter nor explicit timeReceiver. When configured, it refers to the dynamic state of the port based on the received Announce message.

Test Configuration

1. Pre-condition-Test 9.2.7 Configuration.
2. Remove the configured Asymmetry Compensation of +1000ns on BR1 and BR2 in TEST-7.
3. Fail the link between BR1 and T-GM1 and BR2 and T-GM1.

Procedure:

Detailed Test procedure:

1. Connect both T-GMs to GNSS Simulator.
2. Configure T-GMs, BRs, HSRs and CSRs with G.8275.1 profile.
3. Connect Clock Tester measurement ports to CSRs TimeTransmitter ports to monitor PTP at the O-RU UNI.

Verifications for Clock Advertisements

1. Verify the clock lock state on BRs, HSRs, CSRs. The lock state should be Phase-aligned.
2. Verify the steady-state clock advertisements on BRs, HSRs and CSRs as per ITU-T G.8275.1 Standard-Appendix V, Table V.3.

After Step-3 of Test Configurations

Verifications for Maximum Absolute Packet Time Error and 1PPS Time Error

1. Verify Performance Measurement during steady state. (1PPS and Packet Time Error should show an additional Error of +/-500ns due to Asymmetry)

Verifications for Maximum Relative Packet Time Error (NOTE1)

1. Perform time error measurements on Clock Tester at O-RU1 and O-RU2 plane independently. Collect the data ($maxTE1, minTE1$) and ($maxTE2, minTE2$) for both O-RU UNI at least for a period of 1000sec.
2. Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports.
 - a. $max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)

- c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TE_{RL}(t)$, but to measure the maximum possible value it can take.

$$\max|TE_{RL}|(1,2) \text{ the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$$

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and SyncE session is up and running.
2. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

Verifications for Time Wander

1. Verify the Time Wander using Clock tester at the O-RU UNI.
 - a. Frequency Limit is $\leq 32ppb$ (For PRTC Class-A)
 - b. MTIE mask as per the limit specified.

After Step-3 of Test Configurations

Verifications for Maximum Absolute Packet Time Error and 1PPS Time Error

1. Verify Performance Measurement during steady state. (1PPS and Packet Time Error shouldn't show any additional Error of +/-500ns due to Asymmetry)

Verifications for Maximum Relative Packet Time Error and 1PPS Time Error

1. Verify the Maximum Relative 1PPS Time Error measurements between the O-RU UNI ports. The results should be as per Test 9.2.1.

NOTE1 (As per G.8271.1):

To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.

Pass/Fail Criteria:

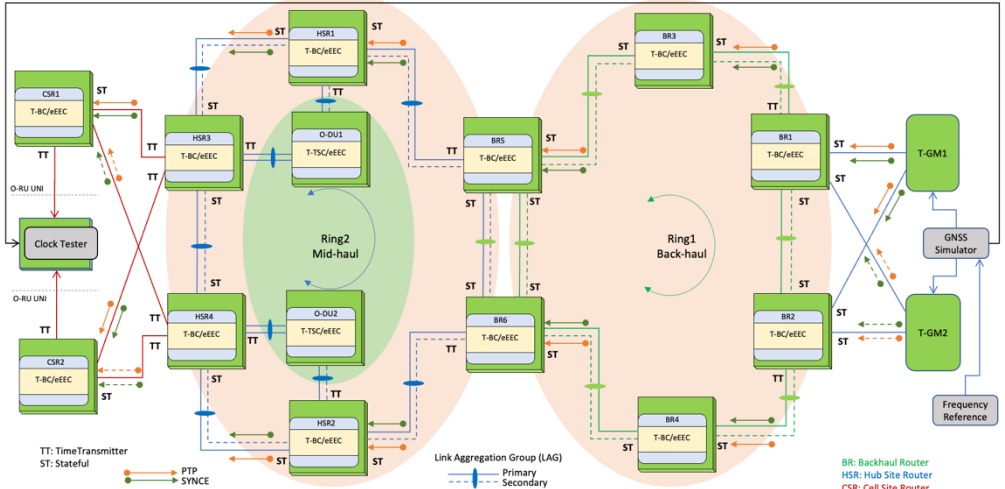
Test shall pass if the Maximum Absolute TE, Relative TE and Time Wander Limit result is within the specified limits.

Test Output

End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value
	(For Enhanced RU)	(For Regular RU)	
Packet $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
Packet $\max TE_{RL} $	$\leq 60ns$ (FR2)	$\leq 100ns$ (FR1)	
	$\leq 190ns$ (FR1)		
1PPS $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
1PPS $\max TE_{RL} $	$\leq 60ns$ (FR2)	$\leq 100ns$ (FR1)	
	$\leq 190ns$ (FR1)		
Frequency Limit (PRTC Class-A)	$\leq 32ppb$	$\leq 32ppb$	

The above limits for $\max|TE_L|$ doesn't include the error due to Manual Asymmetry.

9.2.9 Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC from Backhaul (LLS-C3)-Aggregated Link (Case-1: T-GM1 is the preferred GM)

Test ID	SYNC.RE.09	Classification	Resiliency, Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC from Backhaul-Aggregated Link. (Case-1: T-GM1 is the preferred GM)		
Test Objective:	1. Verify the Clock Lock states on BRs, HSRs and CSRs. 2. Verify the steady-state clock advertisements on BRs, HSRs and CSRs. 3. Verify the Maximum Absolute Packet Time Error ($Max TE_L$) at O-RU UNI. 4. Verify the Maximum Relative Packet Time Error ($Max TE_{RL}$) between two O-RU UNI ports. 5. Verify the Maximum Absolute 1PPS Time Error ($Max TE_L$) at O-RU UNI. 6. Verify the Time Wander at the O-RU UNI.		
Pre-Req	✓ T-GM1 is the Primary GM and T-GM2 is the Secondary GM. ✓ Clock parameters of both T-GM1 and T-GM2 are same except Priority-2 & GMCID. ✓ Priority-2 on T-GM1=1 and T-GM2=2.		
Test Topology:	 The test topology above illustrates the placement of clock tester at the measurement plane with 1PPS and 10MHz reference from GNSS simulator connected to clock tester. The tester measurement ports at O-RU UNI connected to CSR. Stateful (ST) refers to the port configuration of neither explicit timeTransmitter nor explicit timeReceiver. When configured, it refers to the dynamic state of the port based on the received Announce message.		
Test Configuration	1. Clock Tester at O-RU UNI receives timing input on its measurement port. 2. Provide 1PPS and 10MHz reference to Clock Tester from GNSS simulator. 3. G.8275.1 Full Timing Support profile.		
Procedure:	Detailed Test procedure: 1. Connect both T-GMs to GNSS Simulator.		

2. Configure T-GMs, BRs, HSRs and CSRs with G.8275.1 profile over Aggregated Link in backhaul and midhaul.
3. Connect Clock Tester measurement ports to CSRs TimeTransmitter ports to monitor PTP at the O-RU UNI.

Verifications for Clock Advertisements

1. Verify the clock lock state on BRs, HSRs, CSRs. The lock state should be Phase-aligned.
2. Verify the steady-state clock advertisements on BRs, HSRs and CSRs as per ITU-T G.8275.1 Standard-Appendix V, Table V.3.

Verifications for Maximum Absolute Packet Time Error

1. Verify that PTP session is up and running between T-GM and CSR.
2. Verify time error measurements using Clock Tester for the PTP flows from CSR at the O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative Packet Time Error (NOTE1)

1. Perform time error measurements on Clock Tester at O-RU1 and O-RU2 plane independently. Collect the data ($\max TE1, \min TE1$) and ($\max TE2, \min TE2$) for both O-RU UNI at least for a period of 1000sec.
2. Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TE_{RL}(t)$, but to measure the maximum possible value it can take.

$$\max|TE_{RL}|(1,2) \text{ the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$$

Verifications for Maximum Absolute 1PPS Time Error

1. Verify that PTP session is up and running between T-GM and CSR.
2. Verify the 1PPS time error measurements using Clock Tester at the O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and SyncE session is up and running.
2. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

	<u>Verifications for Time Wander</u> 1. Verify the Time Wander using Clock tester at the O-RU UNI. a. Frequency Limit is $\leq 32ppb$ (For PRTC Class-A) b. MTIE mask as per the limit specified. NOTE1 (As per G.8271.1): To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.			
Pass/Fail Criteria:	Test shall pass if the Maximum Absolute TE, Relative TE and Time Wander Limit result is within the specified limits.			
Test Output	End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value
		(For Enhanced RU)	(For Regular RU)	
	Packet $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
	Packet $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
		$\leq 190ns(FR1)$		
	1PPS $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
	1PPS $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
		$\leq 190ns(FR1)$		
	Frequency Limit (PRTC Class-A)	$\leq 32ppb$	$\leq 32ppb$	
The above limits applicable for both Normal and Re-arrangement scenarios.				

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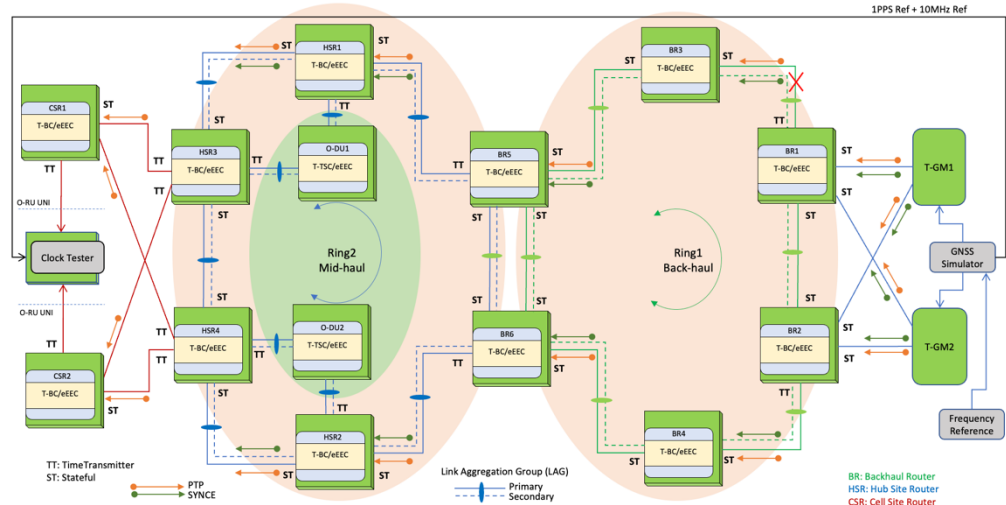
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9.2.10 Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC from Backhaul (LLS-C3)-Aggregated Link-BR1 to BR3 primary member link down. (Case-1: T-GM1 is the preferred GM)

Test ID	SYNC.RE.10	Classification	Resiliency, Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC from Backhaul-Aggregated Link-BR1 to BR3 primary member link down. (Case-1: T-GM1 is the preferred GM)		
Test Objective:	1. Verify the Clock Lock states on BRs, HSRs and CSRs. 2. Verify the clock advertisements on BRs, HSRs and CSRs. 3. Verify the Maximum Absolute Packet Time Error ($\max TE_L $) at O-RU UNI. 4. Verify the Maximum Relative Packet Time Error ($\max TE_{RL} $) between two O-RU UNI ports. 5. Verify the Maximum Absolute 1PPS Time Error ($\max TE_L $) at O-RU UNI. 6. Verify the Time Wander at the O-RU UNI.		
Pre-Req	✓ T-GM1 is the Primary GM and T-GM2 is the Secondary GM. ✓ Clock parameters of both T-GM1 and T-GM2 are same except Priority-2 & GMCID.		

✓ Priority-2 on T-GM1=1 and T-GM2=2.

Test Topology:



The test topology above illustrates the placement of clock tester at the measurement plane with 1PPS and 10MHz reference from GNSS simulator connected to clock tester. The tester measurement ports at O-RU UNI connected to CSR.

Stateful (ST) refers to the port configuration of neither explicit timeTransmitter nor explicit timeReceiver. When configured, it refers to the dynamic state of the port based on the received Announce message.

Test Configuration

1. Pre-condition-Test 9.2.9 Configuration.
2. Fail the Primary member link between BR1 and BR3 in the Aggregated Ethernet Bundle.

Procedure:

Detailed Test procedure:

1. Connect both T-GMs to GNSS Simulator.
2. Configure T-GMs, BRs, HSRs and CSRs with G.8275.1 profile over Aggregated Link in backhaul and midhaul.
3. Connect Clock Tester measurement ports to CSRs TimeTransmitter ports to monitor PTP at the O-RU UNI.

Verifications for Clock Advertisements

1. Verify the clock lock state on BRs, HSRs, CSRs. The lock state should be Phase-aligned.
2. Verify the steady-state clock advertisements on BRs, HSRs and CSRs as per ITU-T G.8275.1 Standard-Appendix V, Table V.3.

Verifications for Maximum Absolute Packet Time Error

1. Verify that PTP session is up and running between T-GM and CSR.
2. Verify time error measurements using Clock Tester for the PTP flows from CSR at the O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative Packet Time Error (NOTE1)

1. Perform time error measurements on Clock Tester at O-RU1 and O-RU2 plane independently. Collect the data ($\max TE1, \min TE1$) and ($\max TE2, \min TE2$) for both O-RU UNI at least for a period of 1000sec.

2. Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports.

- a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
- b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
- c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TE_{RL}(t)$, but to measure the maximum possible value it can take.

$$\max|TE_{RL}|(1,2) \text{ the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$$

Verifications for Maximum Absolute 1PPS Time Error

1. Verify that PTP session is up and running between T-GM and CSR.
2. Verify the 1PPS time error measurements using Clock Tester at the O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and SyncE session is up and running.
2. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

Verifications for Time Wander

1. Verify the Time Wander using Clock tester at the O-RU UNI.
 - a. Frequency Limit is $\leq 32ppb$ (For PRTC Class-A)
 - b. MTIE mask as per the limit specified.

NOTE1 (As per G.8271.1):

To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.

Pass/Fail Criteria:

Test shall pass if the Maximum Absolute TE, Relative TE and Time Wander Limit result is within the specified limits.

Test Output

End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value
	(For Enhanced RU)	(For Regular RU)	
Packet $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
Packet $\max TE_{RL} $	$\leq 60ns$ (FR2)	$\leq 100ns$ (FR1)	
	$\leq 190ns$ (FR1)		
1PPS $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
1PPS $\max TE_{RL} $	$\leq 60ns$ (FR2)	$\leq 100ns$ (FR1)	
	$\leq 190ns$ (FR1)		
Frequency Limit (PRTC Class-A)	$\leq 32ppb$	$\leq 32ppb$	

The above limits applicable for both Normal and Re-arrangement scenarios.

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9.2.11 Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC

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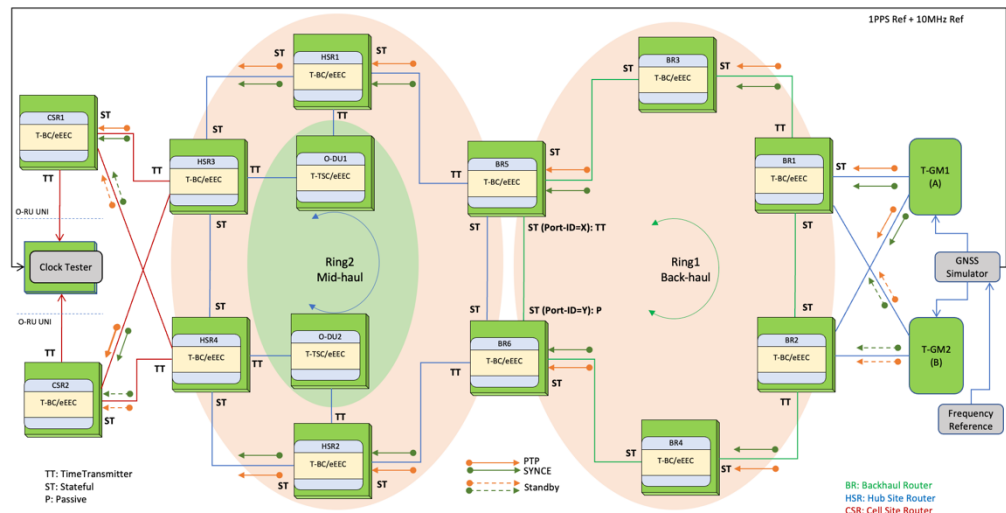
from Backhaul (LLS-C3)-Normal State (Case-2: Both T-GMs are of equal

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preference)

Test ID	SYNC.RE.011	Classification	Resiliency, Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC from Backhaul-Normal State without any Fail-over conditions exist. (Case-2: Both T-GMs are of equal preference)		
Test Objective:	1. Verify the Clock Lock states on BRs, HSRs and CSRs. 2. Verify the steady-state clock advertisements on BRs, HSRs and CSRs. 3. Verify the Maximum Absolute Packet Time Error ($Max TE_L$) at O-RU UNI. 4. Verify the Maximum Relative Packet Time Error ($Max TE_{RL}$) between two O-RU UNI ports. 5. Verify the Maximum Absolute 1PPS Time Error ($Max TE_L$) at O-RU UNI. 6. Verify the Time Wander at the O-RU UNI.		
Pre-Req	✓ Both T-GM1 and T-GM2 are available to the network. ✓ Clock parameters of both T-GM1 and T-GM2 are same except GMCID. ✓ Port-Identity of T-GM1 < Port-Identity of T-GM2.		

Test Topology:



The test topology above illustrates the placement of clock tester at the measurement plane with 1PPS and 10MHz reference from GNSS simulator connected to clock tester. The tester measurement ports at O-RU UNI connected to CSR.

Stateful (ST) refers to the port configuration of neither explicit timeTransmitter nor explicit timeReceiver. When configured, it refers to the dynamic state of the port based on the received Announce message.

Test Configuration	1. Clock Tester at O-RU UNI receives timing input on its measurement port. 2. Provide 1PPS and 10MHz reference to Clock Tester from GNSS simulator. 3. G.8275.1 Full Timing Support profile.
Procedure:	<u>Detailed Test procedure:</u> 1. Connect both T-GMs to GNSS Simulator. 2. Configure T-GMs, BRs, HSRs and CSRs with G.8275.1 profile. 3. Connect Clock Tester measurement ports to CSRs TimeTransmitter ports to monitor PTP at the O-RU UNI. <u>Verifications for Clock Advertisements</u> 1. BR1 shall select clock from T-GM1 (T-GM1 is better by topology) based on Port-id of Sender of A and sender of B). 2. BR2 shall select clock from T-GM1 (T-GM1 is better by topology) based on Port-id of Sender of A and sender of B). 3. Verify the clock lock state on BRs, HSRs, CSRs. The lock state should be Phase-aligned. 4. Verify the steady-state clock advertisements on BRs, HSRs and CSRs as per ITU-T G.8275.1 Standard-Appendix V, Table V.3. <u>Verifications for Maximum Absolute Packet Time Error</u> 1. Verify that PTP session is up and running between T-GM and CSR. 2. Verify time error measurements using Clock Tester for the PTP flows from CSR at the O-RU UNI. a. $\max TE_L$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU) b. $\max TE_L$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU) <u>Verifications for Maximum Relative Packet Time Error (NOTE1)</u> 1. Perform time error measurements on Clock Tester at O-RU1 and O-RU2 plane independently. Collect the data ($\max TE1, \min TE1$) and ($\max TE2, \min TE2$) for both O-RU UNI at least for a period of 1000sec. 2. Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports. a. $\max TE_{RL} (1,2) \leq 60ns$ (FR2) (For Enhanced RU) b. $\max TE_{RL} (1,2) \leq 190ns$ (FR1) (For Enhanced RU) c. $\max TE_{RL} (1,2) \leq 100ns$ (FR1) (For Regular RU) As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TE_{RL}(t)$, but to measure the maximum possible value it can take. $\max TE_{RL} (1,2) \text{ the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$ <u>Verifications for Maximum Absolute 1PPS Time Error</u> 1. Verify that PTP session is up and running between T-GM and CSR. 2. Verify the 1PPS time error measurements using Clock Tester at the O-RU UNI. a. $\max TE_L$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)

	b. $\max TE_L$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU) <u>Verifications for Maximum Relative 1PPS Time Error</u> 1. Verify that PTP and SyncE session is up and running. 2. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RU UNI ports. a. $\max TE_{RL} (1,2) \leq 60ns$ (FR2) (For Enhanced RU) b. $\max TE_{RL} (1,2) \leq 190ns$ (FR1) (For Enhanced RU) c. $\max TE_{RL} (1,2) \leq 100ns$ (FR1) (For Regular RU) <u>Verifications for Time Wander</u> 1. Verify the Time Wander using Clock tester at the O-RU UNI. a. Frequency Limit is$\leq 32ppb$ (For PRTC Class-A) b. MTIE mask as per the limit specified. NOTE1 (As per G.8271.1): To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.																															
Pass/Fail Criteria:	Test shall pass if the Maximum Absolute TE, Relative TE and Time Wander Limit result is within the specified limits.																															
Test Output	<table><tr><td rowspan="2">End-to-End Network Limit at O-RU UNI</td><td colspan="2">Expected Value</td><td rowspan="2">Observed Value</td></tr><tr><td>(For Enhanced RU)</td><td>(For Regular RU)</td></tr><tr><td>Packet $\max TE_L$</td><td>$\leq 1100ns$</td><td>$\leq 1100ns$</td><td></td></tr><tr><td rowspan="2">Packet $\max TE_{RL}$</td><td>$\leq 60ns(FR2)$</td><td rowspan="2">$\leq 100ns(FR1)$</td><td rowspan="2"></td></tr><tr><td>$\leq 190ns(FR1)$</td></tr><tr><td>1PPS $\max TE_L$</td><td>$\leq 1100ns$</td><td>$\leq 1100ns$</td><td></td></tr><tr><td rowspan="2">1PPS $\max TE_{RL}$</td><td>$\leq 60ns(FR2)$</td><td rowspan="2">$\leq 100ns(FR1)$</td><td rowspan="2"></td></tr><tr><td>$\leq 190ns(FR1)$</td></tr><tr><td>Frequency Limit (PRTC Class-A)</td><td>$\leq 32ppb$</td><td>$\leq 32ppb$</td><td></td></tr></table> The above limits applicable for both Normal and Re-arrangement scenarios.				End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value	(For Enhanced RU)	(For Regular RU)	Packet $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$		Packet $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$		$\leq 190ns(FR1)$	1PPS $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$		1PPS $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$		$\leq 190ns(FR1)$	Frequency Limit (PRTC Class-A)	$\leq 32ppb$	$\leq 32ppb$	
End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value																													
	(For Enhanced RU)	(For Regular RU)																														
Packet $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$																														
Packet $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$																														
	$\leq 190ns(FR1)$																															
1PPS $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$																														
1PPS $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$																														
	$\leq 190ns(FR1)$																															
Frequency Limit (PRTC Class-A)	$\leq 32ppb$	$\leq 32ppb$																														

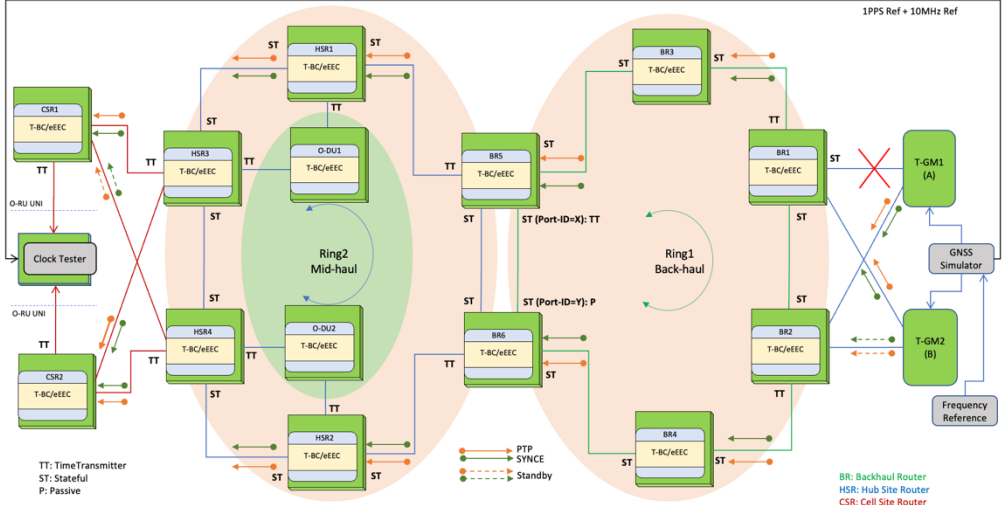
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2 9.2.12 Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC

3 from Backhaul (LLS-C3)-T-GM1 to BR1 Link Down. (Case-2: Both T-GMs

4 are of equal preference)

Test ID	SYNC.RE.12	Classification	Resiliency, Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC from Backhaul-T-GM1 to BR1 Link Down. (Case-2: Both T-GMs are of equal preference)		
Test Objective:	1. Verify the Clock Lock states on BRs, HSRs and CSRs. 2. Verify the clock advertisements on BRs, HSRs and CSRs.		

	- Verify the Maximum Absolute Packet Time Error ($Max TE_L$) at O-RU UNI. - Verify the Maximum Relative Packet Time Error ($Max TE_{RL}$) between two O-RU UNI ports. - Verify the Maximum Absolute 1PPS Time Error ($Max TE_L$) at O-RU UNI. - Verify the Time Wander at the O-RU UNI.
Pre-Req	- Both T-GM1 and T-GM2 are available to the network. - Clock parameters of both T-GM1 and T-GM2 are same except GMCID. - Port-Identity of T-GM1 < Port-Identity of T-GM2. - Test 9.2.11 MUST PASS.
Test Topology:	 The test topology above illustrates the placement of clock tester at the measurement plane with 1PPS and 10MHz reference from GNSS simulator connected to clock tester. The tester measurement ports at O-RU UNI connected to CSR. Stateful (ST) refers to the port configuration of neither explicit timeTransmitter nor explicit timeReceiver. When configured, it refers to the dynamic state of the port based on the received Announce message.
Test Configuration	- As per Test 9.2.11. - Fail the link between T-GM1 and BR1.
Procedure:	Detailed Test procedure: - Connect both T-GMs to GNSS Simulator. - Configure T-GMs, BRs, HSRs and CSRs with G.8275.1 profile. - Connect Clock Tester measurement ports to CSRs TimeTransmitter ports to monitor PTP at the O-RU UNI. Verifications for Clock Advertisements - BR1 shall select clock from T-GM2 (T-GM2 is better (or better by topology) based on Port-id of receiver of A and sender of A) on the shortest path with Step-removal=0. - BR2 shall select clock from T-GM1 (T-GM1 is better by topology) based on Port-id of sender of A and sender of B). - Verify the clock lock state on BRs, HSRs, CSRs. The lock state should be Phase-aligned.

4. Verify the steady-state clock advertisements on BRs, HSRs and CSRs as per ITU-T G.8275.1 Standard-Appendix V, Table V.3.

Verifications for Maximum Absolute Packet Time Error

1. Verify that PTP session is up and running between T-GM and CSR.
2. Verify time error measurements using Clock Tester for the PTP flows from CSR at the O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative Packet Time Error (NOTE1)

1. Perform time error measurements on Clock Tester at O-RU1 and O-RU2 plane independently. Collect the data ($\max TE1, \min TE1$) and ($\max TE2, \min TE2$) for both O-RU UNI at least for a period of 1000sec.
2. Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}| (1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}| (1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}| (1,2) \leq 100ns$ (FR1) (For Regular RU)

As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TE_{RL}(t)$, but to measure the maximum possible value it can take.

$$\max|TE_{RL}| (1,2) \text{ the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$$

Verifications for Maximum Absolute 1PPS Time Error

1. Verify that PTP session is up and running between T-GM and CSR.
2. Verify the 1PPS time error measurements using Clock Tester at the O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and SyncE session is up and running.
2. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}| (1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}| (1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}| (1,2) \leq 100ns$ (FR1) (For Regular RU)

Verifications for Time Wander

1. Verify the Time Wander using Clock tester at the O-RU UNI.
 - a. Frequency Limit is $\leq 32ppb$ (For PRTC Class-A)
 - b. MTIE mask as per the limit specified.

NOTE1 (As per G.8271.1):

	To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.			
Pass/Fail Criteria:	Test shall pass if the Maximum Absolute TE, Relative TE and Time Wander Limit result is within the specified limits.			
Test Output	End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value
		(For Enhanced RU)	(For Regular RU)	
	Packet max TE _L	≤ 1100ns	≤ 1100ns	
	Packet max TE _{RL}	≤ 60ns(FR2)	≤ 100ns(FR1)	
		≤ 190ns(FR1)		
	1PPS max TE _L	≤ 1100ns	≤ 1100ns	
	1PPS max TE _{RL}	≤ 60ns(FR2)	≤ 100ns(FR1)	
		≤ 190ns(FR1)		
	Frequency Limit (PRTC Class-A)	≤ 32ppb	≤ 32ppb	
The above limits applicable for both Normal and Re-arrangement scenarios.				

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9.2.13 Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC

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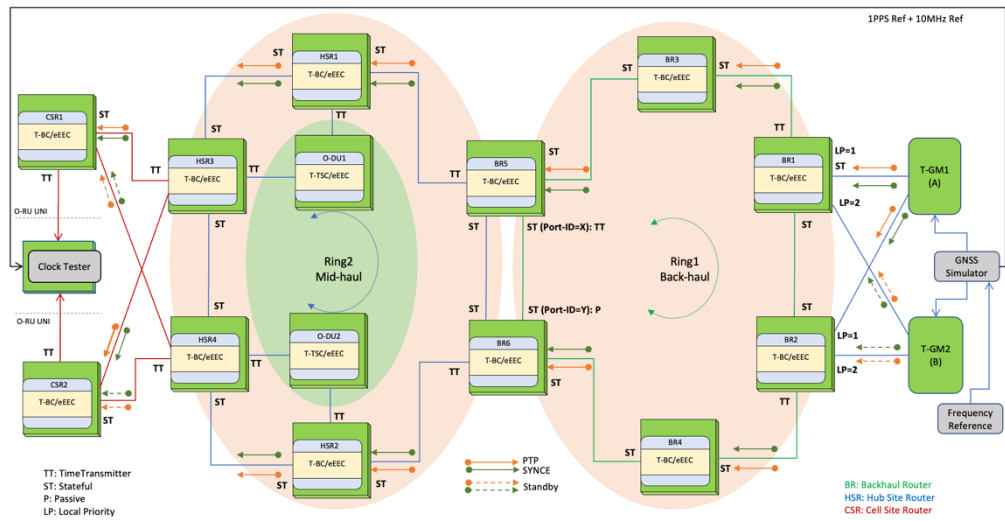
from Backhaul (LLS-C3)-Normal State (Case-3: T-GM1 is preferred through

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explicit configuration via Local-Priority on TNEs)

Test ID	SYNC.RE.13	Classification	Resiliency, Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC from Backhaul-Normal State without any Fail-over conditions exist. (Case-3: T-GM1 is preferred through explicit configuration via Local-Priority on TNEs)		
Test Objective:	1. Verify the Clock Lock states on BRs, HSRs and CSRs. 2. Verify the steady-state clock advertisements on BRs, HSRs and CSRs. 3. Verify the Maximum Absolute Packet Time Error ($\max TE_L$) at O-RU UNI. 4. Verify the Maximum Relative Packet Time Error ($\max TE_{RL}$) between two O-RU UNI ports. 5. Verify the Maximum Absolute 1PPS Time Error ($\max TE_L$) at O-RU UNI. 6. Verify the Time Wander at the O-RU UNI.		
Pre-Req	✓ Both T-GM1 and T-GM2 are available to the network. ✓ Clock parameters of both T-GM1 and T-GM2 are same except GMCID. ✓ Local Priority Configuration on BR1 and BR2 for T-GM1 is 1 and that for T-GM2 is 2.		

Test Topology:



The test topology above illustrates the placement of clock tester at the measurement plane with 1PPS and 10MHz reference from GNSS simulator connected to clock tester. The tester measurement ports at O-RU UNI connected to CSR.

Stateful (ST) refers to the port configuration of neither explicit timeTransmitter nor explicit timeReceiver. When configured, it refers to the dynamic state of the port based on the received Announce message.

Test Configuration

1. Clock Tester at O-RU UNI receives timing input on its measurement port.
2. Provide 1PPS and 10MHz reference to Clock Tester from GNSS simulator.
3. G.8275.1 Full Timing Support profile.

Procedure:

Detailed Test procedure:

1. Connect both T-GMs to GNSS Simulator.
2. Configure T-GMs, BRs, HSRs and CSRs with G.8275.1 profile.
3. Connect Clock Tester measurement ports to CSRs TimeTransmitter ports to monitor PTP at the O-RU UNI.

Verifications for Clock Advertisements

1. BR1 and BR2 shall select clock from T-GM1.
2. Verify the clock lock state on BRs, HSRs, CSRs. The lock state should be Phase-aligned.
3. Verify the steady-state clock advertisements on BRs, HSRs and CSRs as per ITU-T G.8275.1 Standard-Appendix V, Table V.3.

Verifications for Maximum Absolute Packet Time Error

1. Verify that PTP session is up and running between T-GM and CSR.
2. Verify time error measurements using Clock Tester for the PTP flows from CSR at the O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative Packet Time Error (NOTE1)

1. Perform time error measurements on Clock Tester at O-RU1 and O-RU2 plane independently. Collect the data ($\max TE1, \min TE1$) and ($\max TE2, \min TE2$) for both O-RU UNI at least for a period of 1000sec.

2. Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports.

- a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
- b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
- c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TE_{RL}(t)$, but to measure the maximum possible value it can take.

$$\max|TE_{RL}|(1,2) \text{ the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$$

Verifications for Maximum Absolute 1PPS Time Error

1. Verify that PTP session is up and running between T-GM and CSR.
2. Verify the 1PPS time error measurements using Clock Tester at the O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and SyncE session is up and running.
2. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

Verifications for Time Wander

1. Verify the Time Wander using Clock tester at the O-RU UNI.
 - a. Frequency Limit is $\leq 32ppb$ (For PRTC Class-A)
 - b. MTIE mask as per the limit specified.

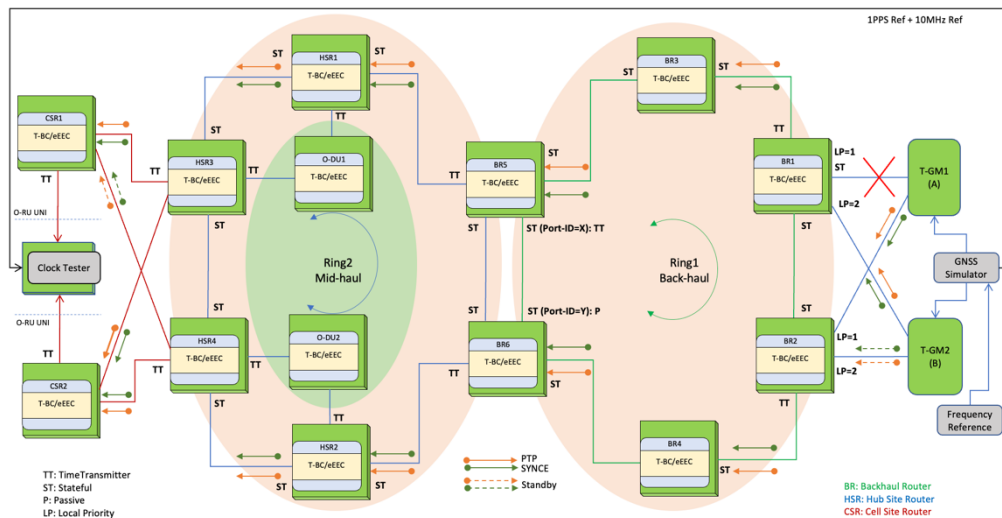
NOTE1 (As per G.8271.1):

To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.

Pass/Fail Criteria:	Test shall pass if the Maximum Absolute TE, Relative TE and Time Wander Limit result is within the specified limits.			
Test Output	End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value
		(For Enhanced RU)	(For Regular RU)	
	Packet max TE _L	≤ 1100ns	≤ 1100ns	
	Packet max TE _{RL}	≤ 60ns(FR2)	≤ 100ns(FR1)	
		≤ 190ns(FR1)		
	1PPS max TE _L	≤ 1100ns	≤ 1100ns	
	1PPS max TE _{RL}	≤ 60ns(FR2)	≤ 100ns(FR1)	
		≤ 190ns(FR1)		
	Frequency Limit (PRTC Class-A)	≤ 32ppb	≤ 32ppb	

The above limits applicable for both Normal and Re-arrangement scenarios.

9.2.14 Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC from Backhaul (LLS-C3)-T-GM1 to BR1 Link Down. (Case-3: T-GM1 is preferred through explicit configuration via Local-Priority on TNEs)

Test ID	SYNC.RE.14	Classification	Resiliency, Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC from Backhaul-T-GM1 to BR1 Link Down. (Case-3: T-GM1 is preferred through explicit configuration via Local-Priority on TNEs)		
Test Objective:	1. Verify the Clock Lock states on BRs, HSRs and CSRs. 2. Verify the clock advertisements on BRs, HSRs and CSRs. 3. Verify the Maximum Absolute Packet Time Error ($Max TE_L$) at O-RU UNI. 4. Verify the Maximum Relative Packet Time Error ($Max TE_{RL}$) between two O-RU UNI ports. 5. Verify the Maximum Absolute 1PPS Time Error ($Max TE_L$) at O-RU UNI. 6. Verify the Time Wander at the O-RU UNI.		
Pre-Req	✓ Both T-GM1 and T-GM2 are available to the network. ✓ Clock parameters of both T-GM1 and T-GM2 are same except GMCID. ✓ Local Priority Configuration on BR1 and BR2 for T-GM1 is 1 and that for T-GM2 is 2. ✓ Test 9.2.13 MUST PASS.		
Test Topology:			

	The test topology above illustrates the placement of clock tester at the measurement plane with 1PPS and 10MHz reference from GNSS simulator connected to clock tester. The tester measurement ports at O-RU UNI connected to CSR. Stateful (ST) refers to the port configuration of neither explicit timeTransmitter nor explicit timeReceiver. When configured, it refers to the dynamic state of the port based on the received Announce message.
Test Configuration	- As per Test 9.2.13. - Fail the link between T-GM1 and BR1.
Procedure:	<u>Detailed Test procedure:</u> - Connect both T-GMs to GNSS Simulator. - Configure T-GMs, BRs, HSRs and CSRs with G.8275.1 profile. - Connect Clock Tester measurement ports to CSRs TimeTransmitter ports to monitor PTP at the O-RU UNI. <u>Verifications for Clock Advertisements</u> - BR1 shall select clock from T-GM2. - BR2 shall select clock from T-GM1. - Verify the clock lock state on BRs, HSRs, CSRs. The lock state should be Phase-aligned. - Verify the steady-state clock advertisements on BRs, HSRs and CSRs as per ITU-T G.8275.1 Standard-Appendix V, Table V.3. <u>Verifications for Maximum Absolute Packet Time Error</u> - Verify that PTP session is up and running between T-GM and CSR. - Verify time error measurements using Clock Tester for the PTP flows from CSR at the O-RU UNI. $\max TE_L$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU) $\max TE_L$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU) <u>Verifications for Maximum Relative Packet Time Error (NOTE1)</u> - Perform time error measurements on Clock Tester at O-RU1 and O-RU2 plane independently. Collect the data ($\max TE1, \min TE1$) and ($\max TE2, \min TE2$) for both O-RU UNI at least for a period of 1000sec. - Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports. $\max TE_{RL} (1,2) \leq 60ns$ (FR2) (For Enhanced RU) $\max TE_{RL} (1,2) \leq 190ns$ (FR1) (For Enhanced RU) $\max TE_{RL} (1,2) \leq 100ns$ (FR1) (For Regular RU) As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TE_{RL}(t)$, but to measure the maximum possible value it can take. $\max TE_{RL} (1,2) \text{ the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$ <u>Verifications for Maximum Absolute 1PPS Time Error</u> - Verify that PTP session is up and running between T-GM and CSR.

2. Verify the 1PPS time error measurements using Clock Tester at the O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and SyncE session is up and running.
2. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

Verifications for Time Wander

1. Verify the Time Wander using Clock tester at the O-RU UNI.
 - a. Frequency Limit is $\leq 32ppb$ (For PRTC Class-A)
 - b. MTIE mask as per the limit specified.

NOTE1 (As per G.8271.1):

To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.

Pass/Fail Criteria:	Test shall pass if the Maximum Absolute TE, Relative TE and Time Wander Limit result is within the specified limits.			
Test Output	End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value
		(For Enhanced RU)	(For Regular RU)	
	Packet $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
	Packet $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
		$\leq 190ns(FR1)$		
	1PPS $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
	1PPS $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
		$\leq 190ns(FR1)$		
	Frequency Limit (PRTC Class-A)	$\leq 32ppb$	$\leq 32ppb$	
The above limits applicable for both Normal and Re-arrangement scenarios.				

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9.2.15 Timing Resiliency in Ring Topology-O-DU connected to HSR and

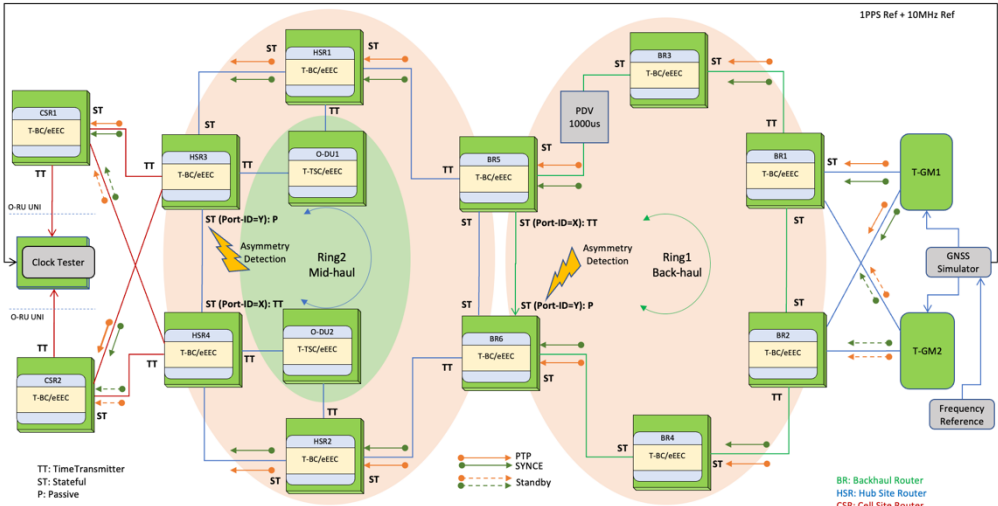
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SYNC from Backhaul (LLS-C3)-Normal State (Passive Port

4

Monitoring)

Test ID	SYNC.RE.15	Classification	Resiliency, Synchronization, Performance, Absolute Time Error, Relative Time Error
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Test Title	Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC from Backhaul-Normal State without any Fail-over conditions exist. (Passive Port Monitoring)
Test Objective:	1. Verify the Clock Lock states on BRs, HSRs and CSRs. 2. Verify the steady-state clock advertisements on BRs, HSRs and CSRs. 3. Verify the Maximum Absolute Packet Time Error ($Max TE_L$) at O-RU UNI. 4. Verify the Maximum Relative Packet Time Error ($Max TE_{RL}$) between two O-RU UNI ports. 5. Verify the Maximum Absolute 1PPS Time Error ($Max TE_L$) at O-RU UNI. 6. Verify the Time Wander at the O-RU UNI.
Pre-Req	✓ Both T-GM1 and T-GM2 are available to the network. ✓ Clock parameters of both T-GM1 and T-GM2 are same except GMCID. ✓ Port-Identity of T-GM1 < Port-Identity of T-GM2.
Test Topology:	 The test topology above illustrates the placement of clock tester at the measurement plane with 1PPS and 10MHz reference from GNSS simulator connected to clock tester. The tester measurement ports at O-RU UNI connected to CSR. Stateful (ST) refers to the port configuration of neither explicit timeTransmitter nor explicit timeReceiver. When configured, it refers to the dynamic state of the port based on the received Announce message.
Test Configuration	1. Clock Tester at O-RU UNI receives timing input on its measurement port. 2. Provide 1PPS and 10MHz reference to Clock Tester from GNSS simulator. 3. G.8275.1 Full Timing Support profile. 4. Introduce Asymmetry of 1000ns between BR3 and BR5.
Procedure:	Detailed Test procedure: 1. Connect both T-GMs to GNSS Simulator. 2. Configure T-GMs, BRs, HSRs and CSRs with G.8275.1 profile. 3. Connect Clock Tester measurement ports to CSRs TimeTransmitter ports to monitor PTP at the O-RU UNI. Verifications for Clock Advertisements

1. BR1 shall select clock from T-GM1 (T-GM1 is better by topology) based on Port-id of Sender of A and sender of B).
2. BR2 shall select clock from T-GM1 (T-GM1 is better by topology) based on Port-id of Sender of A and sender of B).
3. Verify the clock lock state on BRs, HSRs, CSRs. The lock state should be Phase-aligned.
4. After step-4 in Test Configuration, Verify Performance Measurement during steady state. (1PPS and Packet Time Error should show an Error of +/-500ns)
5. BR6 should raise an alarm based on the configured Alarm Threshold.
6. BR6 should estimate the path asymmetry between the two paths.
7. Verify the steady-state clock advertisements on BRs, HSRs and CSRs as per ITU-T G.8275.1 Standard-Appendix V, Table V.3.

Verifications for Maximum Absolute Packet Time Error

1. Verify that PTP session is up and running between T-GM and CSR.
2. Verify time error measurements using Clock Tester for the PTP flows from CSR at the O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative Packet Time Error (NOTE1)

1. Perform time error measurements on Clock Tester at O-RU1 and O-RU2 plane independently. Collect the data ($\max TE1, \min TE1$) and ($\max TE2, \min TE2$) for both O-RU UNI at least for a period of 1000sec.
2. Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}| (1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}| (1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}| (1,2) \leq 100ns$ (FR1) (For Regular RU)

As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TE_{RL}(t)$, but to measure the maximum possible value it can take.

$$\max|TE_{RL}| (1,2) \text{ the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$$

Verifications for Maximum Absolute 1PPS Time Error

1. Verify that PTP session is up and running between T-GM and CSR.
2. Verify the 1PPS time error measurements using Clock Tester at the O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and SyncE session is up and running.
2. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RU UNI ports.

	a. $\max TE_{RL} (1,2) \leq 60ns$ (FR2) (For Enhanced RU) b. $\max TE_{RL} (1,2) \leq 190ns$ (FR1) (For Enhanced RU) c. $\max TE_{RL} (1,2) \leq 100ns$ (FR1) (For Regular RU) <u>Verifications for Time Wander</u> 1. Verify the Time Wander using Clock tester at the O-RU UNI. a. Frequency Limit is$\leq 32ppb$ (For PRTC Class-A) b. MTIE mask as per the limit specified. NOTE1 (As per G.8271.1): To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.																												
Pass/Fail Criteria:	Test shall pass if the Maximum Absolute TE, Relative TE and Time Wander Limit result is within the specified limits.																												
Test Output	<table><tr><td rowspan="2">End-to-End Network Limit at O-RU UNI</td><td colspan="2">Expected Value</td><td rowspan="2">Observed Value</td></tr><tr><td>(For Enhanced RU)</td><td>(For Regular RU)</td></tr><tr><td>Packet $\max TE_L$</td><td>$\leq 1100ns$</td><td>$\leq 1100ns$</td><td></td></tr><tr><td rowspan="2">Packet $\max TE_{RL}$</td><td>$\leq 60ns$(FR2)</td><td rowspan="2">$\leq 100ns$(FR1)</td><td rowspan="2"></td></tr><tr><td>$\leq 190ns$(FR1)</td></tr><tr><td>1PPS $\max TE_L$</td><td>$\leq 1100ns$</td><td>$\leq 1100ns$</td><td></td></tr><tr><td rowspan="2">1PPS $\max TE_{RL}$</td><td>$\leq 60ns$(FR2)</td><td rowspan="2">$\leq 100ns$(FR1)</td><td rowspan="2"></td></tr><tr><td>$\leq 190ns$(FR1)</td></tr><tr><td>Frequency Limit (PRTC Class-A)</td><td>$\leq 32ppb$</td><td>$\leq 32ppb$</td><td></td></tr></table>	End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value	(For Enhanced RU)	(For Regular RU)	Packet $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$		Packet $\max TE_{RL} $	$\leq 60ns$ (FR2)	$\leq 100ns$ (FR1)		$\leq 190ns$ (FR1)	1PPS $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$		1PPS $\max TE_{RL} $	$\leq 60ns$ (FR2)	$\leq 100ns$ (FR1)		$\leq 190ns$ (FR1)	Frequency Limit (PRTC Class-A)	$\leq 32ppb$	$\leq 32ppb$	
End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value																										
	(For Enhanced RU)	(For Regular RU)																											
Packet $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$																											
Packet $\max TE_{RL} $	$\leq 60ns$ (FR2)	$\leq 100ns$ (FR1)																											
	$\leq 190ns$ (FR1)																												
1PPS $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$																											
1PPS $\max TE_{RL} $	$\leq 60ns$ (FR2)	$\leq 100ns$ (FR1)																											
	$\leq 190ns$ (FR1)																												
Frequency Limit (PRTC Class-A)	$\leq 32ppb$	$\leq 32ppb$																											
The above limits for $\max TE_L $ doesn't include the error due to Manual Asymmetry.																													

1 9.2.16 Timing Resiliency in Ring Topology-O-DU connected to HSR and

2 SYNC from Midhaul (LLS-C3)-Normal State (Case-1: T-GM1 is the

3 preferred GM)

Test ID	SYNC.RE.16	Classification	Resiliency, Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC from Midhaul-Normal State without any Fail-over conditions exist. (Case-1: T-GM1 is the preferred GM)		
Test Objective:	- Verify the Clock Lock states on BRs, HSRs and CSRs. - Verify the steady-state clock advertisements on BRs, HSRs and CSRs. - Verify the Maximum Absolute Packet Time Error ($\max TE_L$) at O-RU UNI. - Verify the Maximum Relative Packet Time Error ($\max TE_{RL}$) between two O-RU UNI ports. - Verify the Maximum Absolute 1PPS Time Error ($\max TE_L$) at O-RU UNI. - Verify the Time Wander at the O-RU UNI.		
Pre-Req	✓ T-GM1 is the Primary GM and T-GM2 is the Secondary GM in the Midhaul.		

- b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative Packet Time Error (NOTE1)

1. Perform time error measurements on Clock Tester at O-RU1 and O-RU2 plane independently. Collect the data $(\max TE1, \min TE1)$ and $(\max TE2, \min TE2)$ for both O-RU UNI at least for a period of 1000sec.
2. Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TE_{RL}(t)$, but to measure the maximum possible value it can take.

$$\max|TE_{RL}|(1,2) \text{ the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$$

Verifications for Maximum Absolute 1PPS Time Error

1. Verify that PTP session is up and running between T-GM and CSR.
2. Verify the 1PPS time error measurements using Clock Tester at the O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and SyncE session is up and running.
2. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

Verifications for Time Wander

1. Verify the Time Wander using Clock tester at the O-RU UNI.
 - a. Frequency Limit is $\leq 32ppb$ (For PRTC Class-A)
 - b. MTIE mask as per the limit specified.

NOTE1 (As per G.8271.1):

To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.

Pass/Fail Criteria:	Test shall pass if the Maximum Absolute TE, Relative TE and Time Wander Limit result is within the specified limits.			
Test Output	End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value
		(For Enhanced RU)	(For Regular RU)	
	Packet $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
	Packet $\max TE_{RL} $	$\leq 60ns$ (FR2)	$\leq 100ns$ (FR1)	

		$\leq 190ns(FR1)$		
	1PPS $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
	1PPS $\max TE_{RL} $	$\leq 60ns(FR2)$ $\leq 190ns(FR1)$	$\leq 100ns(FR1)$	
	Frequency Limit (PRTC Class-A)	$\leq 32ppb$	$\leq 32ppb$	
The above limits applicable for both Normal and Re-arrangement scenarios.				

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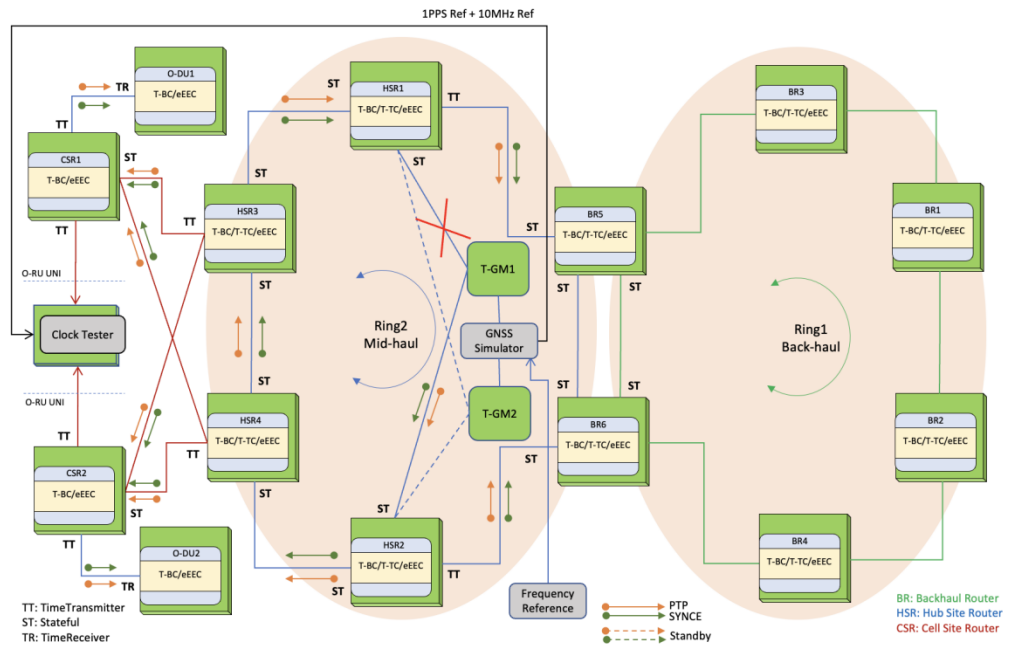
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9.2.17 Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC from Midhaul (LLS-C3)-T-GM1 to HSR1 Link Down. (Case-1: T-GM1 is the preferred GM)

Test ID	SYNC.RE.17	Classification	Resiliency, Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC from Midhaul-T-GM to HSR1 Link Down. (Case-1: T-GM1 is the preferred GM)		
Test Objective:	1. Verify the Clock Lock states on BRs, HSRs and CSRs. 2. Verify the steady-state clock advertisements on BRs, HSRs and CSRs. 3. Verify the Maximum Absolute Packet Time Error ($\max TE_L$) at O-RU UNI. 4. Verify the Maximum Relative Packet Time Error ($\max TE_{RL}$) between two O-RU UNI ports. 5. Verify the Maximum Absolute 1PPS Time Error ($\max TE_L$) at O-RU UNI. 6. Verify the Time Wander at the O-RU UNI.		
Pre-Req	✓ T-GM1 is the Primary GM and T-GM2 is the Secondary GM in the Midhaul. ✓ Clock parameters of both T-GM1 and T-GM2 are same except Priority-2 & GMCID. ✓ Priority-2 on T-GM1=1 and T-GM2=2.		

Test Topology:



The test topology above illustrates the placement of clock tester at the measurement plane with 1PPS and 10MHz reference from GNSS simulator connected to clock tester. The tester measurement ports at O-RU UNI connected to CSR.

Stateful (ST) refers to the port configuration of neither explicit timeTransmitter nor explicit timeReceiver. When configured, it refers to the dynamic state of the port based on the received Announce message.

Test Configuration

1. Pre-condition-Test 9.2.11 Configuration.
2. Fail the link between T-GM1 and HSR1.

Procedure:

Detailed Test procedure:

1. Connect both T-GMs to GNSS Simulator.
2. Configure T-GMs, BRs, HSRs and CSRs with G.8275.1 profile.
3. Connect Clock Tester measurement ports to CSRs TimeTransmitter ports to monitor PTP at the O-RU UNI.

Verifications for Clock Advertisements

1. Verify the clock lock state on BRs, HSRs, CSRs. The lock state should be Phase-aligned.
2. Verify the clock advertisements on BRs, HSRs and CSRs as per ITU-T G.8275.1 Standard-Appendix V, Table V.3.

Verifications for Maximum Absolute Packet Time Error

1. Verify that PTP session is up and running between T-GM and CSR.
2. Verify time error measurements using Clock Tester for the PTP flows from CSR at the O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative Packet Time Error (NOTE1)

1. Perform time error measurements on Clock Tester at O-RU1 and O-RU2 plane independently. Collect the data ($\max TE1, \min TE1$) and ($\max TE2, \min TE2$) for both O-RU UNI at least for a period of 1000sec.

2. Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports.

- a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
- b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
- c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TE_{RL}(t)$, but to measure the maximum possible value it can take.

$$\max|TE_{RL}|(1,2) \text{ the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$$

Verifications for Maximum Absolute 1PPS Time Error

1. Verify that PTP session is up and running between T-GM and CSR.
2. Verify the 1PPS time error measurements using Clock Tester at the O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and SyncE session is up and running.
2. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

Verifications for Time Wander

1. Verify the Time Wander using Clock tester at the O-RU UNI.
 - a. Frequency Limit is $\leq 32ppb$ (For PRTC Class-A)
 - b. MTIE mask as per the limit specified.

NOTE1 (As per G.8271.1):

To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.

Pass/Fail Criteria:

Test shall pass if the Maximum Absolute TE, Relative TE and Time Wander Limit result is within the specified limits.

Test Output

End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value
	(For Enhanced RU)	(For Regular RU)	
Packet $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
Packet $\max TE_{RL} $	$\leq 60ns$ (FR2)	$\leq 100ns$ (FR1)	
	$\leq 190ns$ (FR1)		
1PPS $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
1PPS $\max TE_{RL} $	$\leq 60ns$ (FR2)	$\leq 100ns$ (FR1)	
	$\leq 190ns$ (FR1)		
Frequency Limit (PRTC Class-A)	$\leq 32ppb$	$\leq 32ppb$	

The above limits applicable for both Normal and Re-arrangement scenarios.

9.2.18 Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC from Midhaul (LLS-C3)-T-GM1 Fail-over. (Case-1: T-GM1 is the preferred GM)

Test ID	SYNC.RE.18	Classification	Resiliency, Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC from Midhaul-T-GM1 Fail-over. (Case-1: T-GM1 is the preferred GM)		
Test Objective:	1. Verify the Clock Lock states on BRs, HSRs and CSRs. 2. Verify the steady-state clock advertisements on BRs, HSRs and CSRs. 3. Verify the Maximum Absolute Packet Time Error ($Max TE_L$) at O-RU UNI. 4. Verify the Maximum Relative Packet Time Error ($Max TE_{RL}$) between two O-RU UNI ports. 5. Verify the Maximum Absolute 1PPS Time Error ($Max TE_L$) at O-RU UNI. 6. Verify the Time Wander at the O-RU UNI.		
Pre-Req	✓ T-GM1 is the Primary GM and T-GM2 is the Secondary GM in the Midhaul. ✓ Clock parameters of both T-GM1 and T-GM2 are same except Priority-2 & GMCID. ✓ Priority-2 on T-GM1=1 and T-GM2=2.		
Test Topology:	The test topology above illustrates the placement of clock tester at the measurement plane with 1PPS and 10MHz reference from GNSS simulator connected to clock tester. The tester measurement ports at O-RU UNI connected to CSR.		

	Stateful (ST) refers to the port configuration of neither explicit timeTransmitter nor explicit timeReceiver. When configured, it refers to the dynamic state of the port based on the received Announce message.
Test Configuration	1. Pre-condition-Test 9.2.11 Configuration. 2. Fail the links between T-GM1 to HSR1 and HSR2.
Procedure:	<u>Detailed Test procedure:</u> 1. Connect both T-GMs to GNSS Simulator. 2. Configure T-GMs, BRs, HSRs and CSRs with G.8275.1 profile. 3. Connect Clock Tester measurement ports to CSRs TimeTransmitter ports to monitor PTP at the O-RU UNI. <u>Verifications for Clock Advertisements</u> 1. Verify the clock lock state on BRs, HSRs, CSRs. The lock state should be Phase-aligned. 2. Verify the clock advertisements on BRs, HSRs and CSRs as per ITU-T G.8275.1 Standard-Appendix V, Table V.3. <u>Verifications for Maximum Absolute Packet Time Error</u> 1. Verify that PTP session is up and running between T-GM and CSR. 2. Verify time error measurements using Clock Tester for the PTP flows from CSR at the O-RU UNI. a. $\max TE_L$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU) b. $\max TE_L$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU) <u>Verifications for Maximum Relative Packet Time Error (NOTE1)</u> 3. Perform time error measurements on Clock Tester at O-RU1 and O-RU2 plane independently. Collect the data ($\max TE1, \min TE1$) and ($\max TE2, \min TE2$) for both O-RU UNI at least for a period of 1000sec. 4. Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports. a. $\max TE_{RL} (1,2) \leq 60ns$ (FR2) (For Enhanced RU) b. $\max TE_{RL} (1,2) \leq 190ns$ (FR1) (For Enhanced RU) c. $\max TE_{RL} (1,2) \leq 100ns$ (FR1) (For Regular RU) As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TE_{RL}(t)$, but to measure the maximum possible value it can take. $\max TE_{RL} (1,2) \text{ the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$ <u>Verifications for Maximum Absolute 1PPS Time Error</u> 1. Verify that PTP session is up and running between T-GM and CSR. 2. Verify the 1PPS time error measurements using Clock Tester at the O-RU UNI. a. $\max TE_L$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU) b. $\max TE_L$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and SyncE session is up and running.
2. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

Verifications for Time Wander

1. Verify the Time Wander using Clock tester at the O-RU UNI.
 - a. Frequency Limit is $\leq 32ppb$ (For PRTC Class-A)
 - b. MTIE mask as per the limit specified.

NOTE1 (As per G.8271.1):

To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.

Pass/Fail Criteria:

Test shall pass if the Maximum Absolute TE, Relative TE and Time Wander Limit result is within the specified limits.

Test Output

End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value
	(For Enhanced RU)	(For Regular RU)	
Packet $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
Packet $\max TE_{RL} $	$\leq 60ns$ (FR2)	$\leq 100ns$ (FR1)	
	$\leq 190ns$ (FR1)		
1PPS $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
1PPS $\max TE_{RL} $	$\leq 60ns$ (FR2)	$\leq 100ns$ (FR1)	
	$\leq 190ns$ (FR1)		
Frequency Limit (PRTC Class-A)	$\leq 32ppb$	$\leq 32ppb$	

The above limits applicable for both Normal and Re-arrangement scenarios.

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9.2.19 Timing Resiliency in Ring Topology-O-DU connected to HSR and

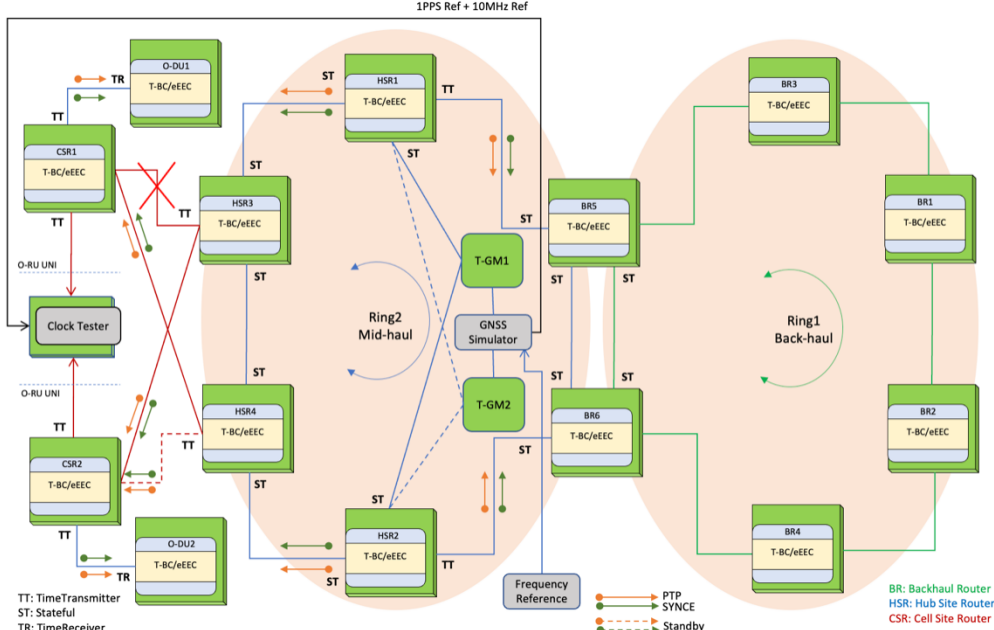
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SYNC from Midhaul (LLS-C3)-Link Failure in Fronthaul. (Case-1: T-

4

GM1 is the preferred GM)

Test ID	SYNC.RE.19	Classification	Resiliency, Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC from Midhaul-Link Failure in Fronthaul. (Case-1: T-GM1 is the preferred GM)		
Test Objective:	1. Verify the Clock Lock states on BRs, HSRs and CSRs. 2. Verify the steady-state clock advertisements on BRs, HSRs and CSRs. 3. Verify the Maximum Absolute Packet Time Error ($\max TE_L$) at O-RU UNI.		

	- Verify the Maximum Relative Packet Time Error ($Max TE_{RL}$) between two O-RU UNI ports. - Verify the Maximum Absolute 1PPS Time Error ($Max TE_L$) at O-RU UNI. - Verify the Time Wander at the O-RU UNI.
Pre-Req	✓ T-GM1 is the Primary GM and T-GM2 is the Secondary GM in the Midhaul. ✓ Clock parameters of both T-GM1 and T-GM2 are same except Priority-2 & GMCID. ✓ Priority-2 on T-GM1=1 and T-GM2=2.
Test Topology:	 The test topology above illustrates the placement of clock tester at the measurement plane with 1PPS and 10MHz reference from GNSS simulator connected to clock tester. The tester measurement ports at O-RU UNI connected to CSR. Stateful (ST) refers to the port configuration of neither explicit timeTransmitter nor explicit timeReceiver. When configured, it refers to the dynamic state of the port based on the received Announce message.
Test Configuration	- Pre-condition-Test 9.2.11 Configuration. - Fail the link between HSR3 and CSR1.
Procedure:	Detailed Test procedure: - Connect both T-GMs to GNSS Simulator. - Configure T-GMs, BRs, HSRs and CSRs with G.8275.1 profile. - Connect Clock Tester measurement ports to CSRs TimeTransmitter ports to monitor PTP at the O-RU UNI. Verifications for Clock Advertisements - Verify the clock lock state on BRs, HSRs, CSRs. The lock state should be Phase-aligned. - Verify the clock advertisements on BRs, HSRs and CSRs as per ITU-T G.8275.1 Standard-Appendix V, Table V.3. Verifications for Maximum Absolute Packet Time Error - Verify that PTP session is up and running between T-GM and CSR.

2. Verify time error measurements using Clock Tester for the PTP flows from CSR at the O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative Packet Time Error (NOTE1)

1. Perform time error measurements on Clock Tester at O-RU1 and O-RU2 plane independently. Collect the data ($\max TE1, \min TE1$) and ($\max TE2, \min TE2$) for both O-RU UNI at least for a period of 1000sec.
2. Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}| (1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}| (1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}| (1,2) \leq 100ns$ (FR1) (For Regular RU)

As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TE_{RL}(t)$, but to measure the maximum possible value it can take.

$$\max|TE_{RL}| (1,2) \text{ the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$$

Verifications for Maximum Absolute 1PPS Time Error

1. Verify that PTP session is up and running between T-GM and CSR.
2. Verify the 1PPS time error measurements using Clock Tester at the O-RU UNI.
 - a. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $\max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and SyncE session is up and running.
2. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}| (1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}| (1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}| (1,2) \leq 100ns$ (FR1) (For Regular RU)

Verifications for Time Wander

1. Verify the Time Wander using Clock tester at the O-RU UNI.
 - a. Frequency Limit is $\leq 32ppb$ (For PRTC Class-A)
 - b. MTIE mask as per the limit specified.

NOTE1 (As per G.8271.1):

To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.

Pass/Fail Criteria:	Test shall pass if the Maximum Absolute TE, Relative TE and Time Wander Limit result is within the specified limits.
----------------------------	--

Test Output	End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value
		(For Enhanced RU)	(For Regular RU)	
	Packet $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
	Packet $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
		$\leq 190ns(FR1)$		
	1PPS $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
	1PPS $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
		$\leq 190ns(FR1)$		
	Frequency Limit (PRTC Class-A)	$\leq 32ppb$	$\leq 32ppb$	
The above limits applicable for both Normal and Re-arrangement scenarios.				

1

2

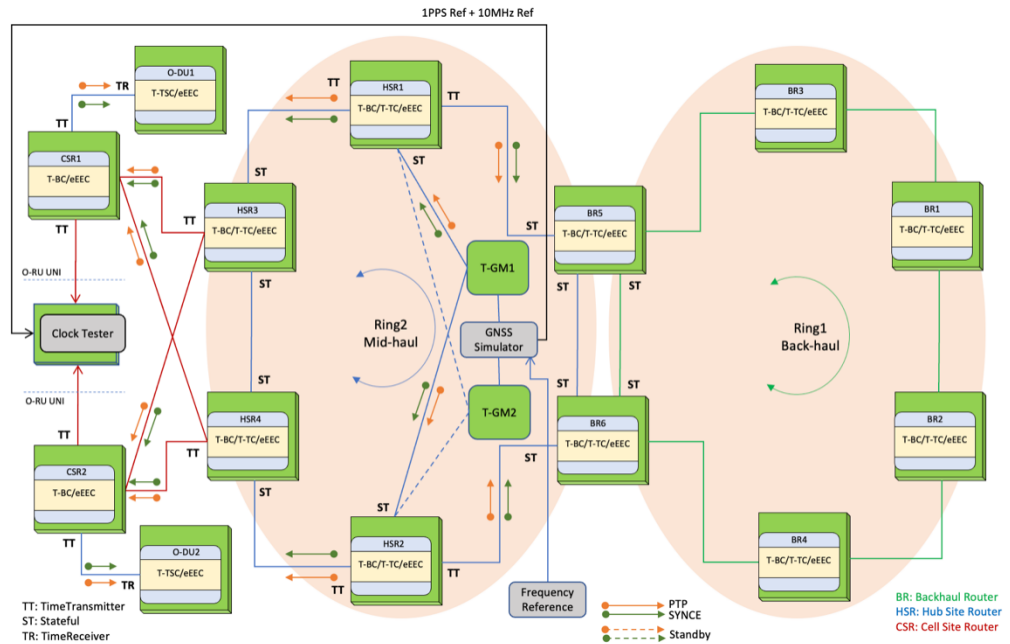
3

4

9.2.20 Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC from Midhaul (LLS-C3)-Asymmetry Test. (Case-1: T-GM1 is the preferred GM)

Test ID	SYNC.RE.20	Classification	Resiliency, Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Timing Resiliency in Ring Topology-O-DU connected to HSR and SYNC from Midhaul-Asymmetry Test. (Case-1: T-GM1 is the preferred GM)		
Test Objective:	1. Verify the Clock Lock states on BRs, HSRs and CSRs. 2. Verify the steady-state clock advertisements on BRs, HSRs and CSRs. 3. Verify the Maximum Absolute Packet Time Error ($\max TE_L$) at O-RU UNI. 4. Verify the Maximum Relative Packet Time Error ($\max TE_{RL}$) between two O-RU UNI ports. 5. Verify the Maximum Absolute 1PPS Time Error ($\max TE_L$) at O-RU UNI. 6. Verify the Time Wander at the O-RU UNI.		
Pre-Req	✓ T-GM1 is the Primary GM and T-GM2 is the Secondary GM. ✓ Clock parameters of both T-GM1 and T-GM2 are same except Priority-2 & GMCID. ✓ Priority-2 on T-GM1=1 and T-GM2=2.		

Test Topology:



The test topology above illustrates the placement of clock tester at the measurement plane with 1PPS and 10MHz reference from GNSS simulator connected to clock tester. The tester measurement ports at O-RU UNI connected to CSR.

Stateful (ST) refers to the port configuration of neither explicit timeTransmitter nor explicit timeReceiver. When configured, it refers to the dynamic state of the port based on the received Announce message.

Test Configuration

1. As per Test 9.2.11.
2. Introduce Asymmetry of +1000ns at the O/p of T-GM1 and No Asymmetry at the O/p of T-GM2.
3. Configure Asymmetry Compensation of +1000ns on HSR1 and HSR2.

Procedure:

Detailed Test procedure:

1. Connect both T-GMs to GNSS Simulator.
2. Configure T-GMs, BRs, HSRs and CSRs with G.8275.1 profile.
3. Connect Clock Tester measurement ports to CSRs TimeTransmitter ports to monitor PTP at the O-RU UNI.

Verifications for Clock Advertisements

1. Verify the clock lock state on BRs, HSRs, CSRs. The lock state should be Phase-aligned.
2. Verify the steady-state clock advertisements on BRs, HSRs and CSRs as per ITU-T G.8275.1 Standard-Appendix V, Table V.3.

After Step-2 of Test Configurations

Verifications for Maximum Absolute Packet Time Error and 1PPS Time Error

1. Verify Performance Measurement during steady state. (1PPS and Packet Time Error should show an additional Error of +/-500ns due to Asymmetry)

Verifications for Maximum Relative Packet Time Error (NOTE1)

1. Perform time error measurements on Clock Tester at O-RU1 and O-RU2 plane independently. Collect the data ($maxTE1$, $minTE1$) and ($maxTE2$, $minTE2$) for both O-RU UNI at least for a period of 1000sec.
2. Estimate the Maximum Relative Time Error measurements between the O-RU UNI ports.

- a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
- b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
- c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

As per ITU-T G.8271.1 Appendix-V1, the requirement is not to measure the continuous relative time error, $TE_{RL}(t)$, but to measure the maximum possible value it can take.

$$\max|TE_{RL}|(1,2) \text{ the greatest of } \begin{cases} \max TE_L(1) - \min TE_L(2) \\ \max TE_L(2) - \min TE_L(1) \end{cases}$$

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and SyncE session is up and running.
2. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

Verifications for Time Wander

1. Verify the Time Wander using Clock tester at the O-RU UNI.
 - a. Frequency Limit is $\leq 32ppb$ (For PRTC Class-A)
 - b. MTIE mask as per the limit specified.

NOTE1 (As per G.8271.1):

To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.

After Step-3 of Test Configurations

Verifications for Maximum Absolute Packet Time Error and 1PPS Time Error

1. Verify Performance Measurement during steady state. (1PPS and Packet Time Error shouldn't show any additional Error of +/-500ns due to Asymmetry)

Verifications for Maximum Relative Packet Time Error and 1PPS Time Error

1. Verify the Maximum Relative 1PPS Time Error measurements between the O-RU UNI ports. The results should be as per Test 9.2.11.

Pass/Fail Criteria:

Test shall pass if the Maximum Absolute TE, Relative TE and Time Wander Limit result is within the specified limits.

Test Output

End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value
	(For Enhanced RU)	(For Regular RU)	
Packet $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
Packet $\max TE_{RL} $	$\leq 60ns$ (FR2)	$\leq 100ns$ (FR1)	
	$\leq 190ns$ (FR1)		
1PPS $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
1PPS $\max TE_{RL} $	$\leq 60ns$ (FR2)	$\leq 100ns$ (FR1)	
	$\leq 190ns$ (FR1)		
Frequency Limit (PRTC Class-A)	$\leq 32ppb$	$\leq 32ppb$	

The above limits for $\max|TE_L|$ doesn't include the error due to Manual Asymmetry.

9.3 Timing Solution Additional use cases in LLS-C3 and LLS-C4 config

This chapter specifies a few additional use cases in LLS-C3 and LLS-C4 config models. Also LLS-C3 and LLS-C4 mixed configuration with and without GNSS assist and redundant T-GMs for co-located DUs and RUs are discussed. Each of these use cases are specified with performance limits and Pass/Fail verdict in normal and fail-over scenarios of GNSS and T-GM.

These use cases in high level are already mentioned in the O-RAN WG9 Synchronization Architecture and Solution Specification. Depending on the specific O-RAN deployment being considered, not all of them might be relevant. When testing the S-Plane, the System Integrator shall identify which of the test cases are relevant depending on the specific deployment scenarios addressed.

Following high level use cases have been covered in this chapter and each of these use cases, clock performance limits are discussed for normal and failover scenarios.

1. LLS-C3 and LLS-C4 Configuration and Use case.

- a. LLS-C4 with PTP frequency backup from T-GM
- b. LLS-C4 with PTP time backup from T-GM
- c. LLS-C4 and LLS-C3 mixed configurations for co-located DUs and RUs
- d. LLS-C4 and LLS-C3 mixed configurations for co-located DUs and RUs (GNSS Failure at O-RU site)
- e. LLS-C4 and LLS-C3 mixed configurations with backup from T-GM for co-located DUs and RUs
- f. LLS-C4 and LLS-C3 mixed configurations with backup from T-GM for co-located DUs and RUs (GNSS Failure at O-RU site)
- g. LLS-C4 and LLS-C3 mixed configurations with backup from redundant T-GM for co-located DUs and RUs (GNSS Failure at O-RU site and T-GM-A Failure)
- h. LLS-C3 configurations with GNSS assisted CSR for co-located DUs and RUs
- i. LLS-C3 configurations with (GNSS+PRTC) assisted CSR for co-located DUs and RUs
- j. LLS-C3 configurations with GNSS assisted CSR with backup from redundant T-GM for co-located DUs and RUs
- k. LLS-C3 configurations with GNSS assisted CSR with backup from redundant T-GM for co-located DUs and RUs (GNSS Failure at CSR site)
- l. LLS-C3 configurations with GNSS assisted CSR with backup from redundant T-GM for co-located DUs and RUs (GNSS Failure at CSR site and T-GM-A Failure)

O-RU UNI specified in the topology is for the network measurement. Two measurement planes have been specified in the document to cover all measurements. One at O-RU UNI for measuring X-haul network limits (Packet + 1PPS) and other at 1PPS measurement plane of O-RU to measure the 1PPS performance limits of O-RU in LLS-C4 configuration. Operator to position the Tester on these measurement planes for conducting the respective test.

The performance limits of O-RU in LLS-C4 when it is locked to local GNSS receiver has been specified in O-RAN.WG4.CUS documents. As per this specifications, the budget shall be based on ITU-T G.8272 (PRTC class A, with max |TE|=100ns, or PRTC class B, with max |TE|=40ns on each GNSS receiver and max |TE|=30ns on O-RU. Max|TE| O-RU includes the O-RU clock recovery error from Local Time Source and any O-RU internal error. The LLS-C4 test cases in this section assumes the O-RU T-TSC Error of 10ns and O-RU internal Error of 20ns to arrive at the possible limits.

The test cases apply to both lab and field deployment cases. The test topologies in this test specification specify only few Transport Node Elements (TNEs) for illustration. However, for lab cases, the type and number of inserted Transport Node Elements (TNE) will be determined by operators (or any other applicable end customer) subject to the O-RU UNI limits. Some of the tests are performed at Xhaul UNI interfaces and some are at the O-RU 1PPS interfaces. The Xhaul UNI interfaces include O-RU Fronthaul UNI, O-DU Fronthaul or Midhaul UNI.

1PPS test at O-RU UNI is typically used for debugging and monitoring the packet layer performance of the nearest TNE in the fronthaul. The intention of such test is that if the TNE vendor implemented the 1PPS interface, the 1PPS error measured on such interface at the O-RU UNI should be a reflection of the output time error of TNE. Operator shall pay attention to an unacceptably large deviation of 1PPS error, beyond the limits specified, which may indicate an incorrect behaviour of TNEs.

The Test Topologies in all the test cases uses GNSS simulator to generate the GNSS signal to the network PRTC/T-GM. This high-quality reference from the simulator is used as the external reference for clock testers as well. The resultant accuracy of this method is a few nanoseconds. However, one can also use a GNSS signal feeding both the initial PRTC/T-GM in the network and the reference clock generating a 1pps/10MHz used to synchronize the clock testers for the measurements. Operator can make an allowance in this case, for all of the measurements for the accuracy of the reference. The magnitude of this allowance will depend on the specific GNSS reference in use.

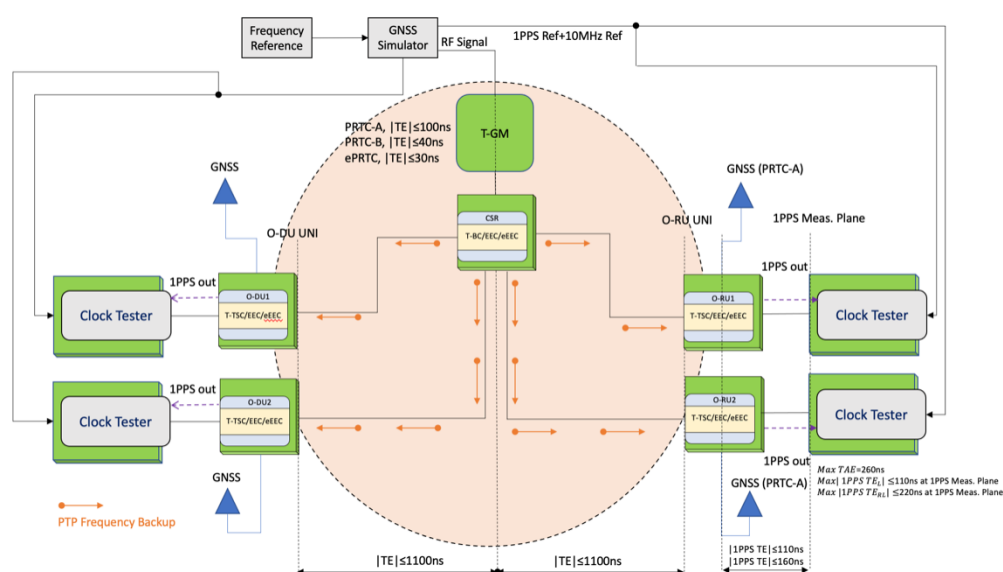
For relative time error measurements at UNI (e.g., O-RU Fronthaul UNI) this document uses the metric $\max|TE_{RL}|$ (1,2) as defined in [88], Appendix XI. Please note this metric represents an estimation and the worst-case scenario. Tests that meet (are below) the limits are seen as PASS. Tests not meeting this metric may still be acceptable. The decision is for operators to make. However, if the measurement equipment support relative time error measurement, operator can use such equipment. Refer ITU-T G.8271.1 Appendix XI for more details.

For Frequency Limit at UNI (e.g., O-RU Fronthaul UNI) this document uses the metric defined as per WG4 section H.2 based on PRTC class for LLS-C3 configuration. Applicable MTIE limit shall apply for Frequency as per WG4 specification.

9.3.1 Network Time Synchronization-LLS-C4 with PTP frequency backup from T-GM

Test ID	SYNC.TE.3	Classification	Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Measurement of Absolute Time Error, Relative Time Error at the 1PPS measurement plane of O-RU. LLS-C4 with PTP frequency backup from T-GM.		
Test Objective:	1. Verify the Maximum Absolute 1PPS Time Error ($\max TE_L$) at 1PPS Measurement Plane of O-RUs. 2. Verify the Maximum Relative 1PPS Time Error ($\max TE_{RL}$) at 1PPS Measurement Plane between O-RUs.		
Pre-Req	T-GM/PRTC connected to GNSS simulator and ensure T-GM is locked. Tester 1PPS measurement port connected to O-RU 1PPS out is the measurement plane.		

Test Topology:



The test topology above illustrates the placement of clock tester at the 1PPS measurement plane with 1PPS and 10MHz reference connected to clock tester.

Test Configuration

1. T-GM, CSR, O-RUs and O-DUs are connected and configured as shown in the Test Topology.
2. Clock Tester at O-RU 1PPS measurement plane receives timing input on its measurement port.
3. Provide 1PPS and 10MHz reference to Clock Tester from GNSS simulator.
4. GNSS (PRTC-A) is connected to O-RU and O-DUs.
5. G.8275.1 Full Timing Support profile.

Procedure:

Detailed Test procedure:

1. GNSS (PRTC-A) is connected directly to O-RUs and O-DUs.
2. O-RU and O-DU use the GNSS (PRTC-A) as the primary clock source and use Frequency as backup from the T-GM at the central location.
3. Connect T-GM to GNSS Simulator and ensure that T-GM is locked.
4. Configure O-RU and O-DU as T-TSC and CSR as T-BC.
5. Enable G.8275.1 profile on all transport TNEs (CSR).
6. Connect Clock tester measurement ports to O-RU 1PPS monitor port to measure the 1PPS at measurement plane.
7. Establish PTP and Sync-E session to the fronthaul network.

Verifications for Maximum Absolute 1PPS Time Error

1. Verify that PTP session is up and running between T-GM and CSR.
2. Verify that both O-RU and O-DU locked to GNSS (PRTC-A) as the primary clock source.
3. Verify the 1PPS time error measurements on Clock Tester at the 1PPS measurement plane.
 - a. $\text{Max}|1PPS TE_L|$ measured on Clock Tester at the 1PPS measurement plane of O-RU $\leq 110ns$

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and Sync-E session is up and running.

2. Verify that both O-RU and O-DU locked to GNSS (PRTC-A) as the primary clock source.
3. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RUs at 1PPS measurement plane.
 - a. $\max|TE_{RL}|(1,2) \leq 220ns$

NOTE1:

To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.

Pass/Fail Criteria:

Test shall pass if the Maximum absolute TE, Maximum Relative TE, and Time Wander Limit measurement result is below the limit specified.

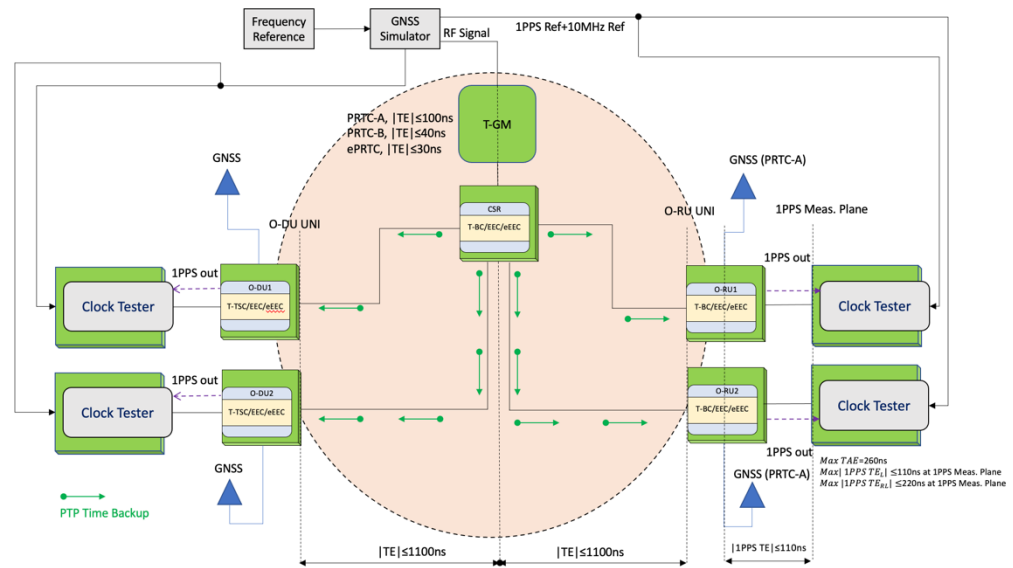
Test Output

End-to-End Network Limit	Expected Value	Observed Value
1PPS $\max TE_L $ at 1PPS measurement plane	$\leq 110ns$	
1PPS $\max TE_{RL} $ at 1PPS measurement plane	$\leq 220ns$	

9.3.2 Network Time Synchronization-LLS-C4 with PTP Time backup from T-GM

Test ID	SYNC.TE.4	Classification	Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Measurement of Absolute Time Error, Relative Time Error at 1PPS measurement plane of O-RU. LLS-C4 with PTP Time backup from T-GM.		
Test Objective:	1. Verify the Maximum Absolute 1PPS Time Error ($\max TE_L$) at 1PPS Measurement Plane of O-RUs. 2. Verify the Maximum Relative 1PPS Time Error ($\max TE_{RL}$) at 1PPS Measurement Plane between O-RUs.		
Pre-Req	T-GM/PRTC connected to GNSS simulator and ensure T-GM is locked. Tester 1PPS measurement port connected to O-RU 1PPS out is the measurement plane.		

Test Topology:



The test topology above illustrates the placement of clock tester at the 1PPS measurement plane with 1PPS and 10MHz reference connected to clock tester.

Test Configuration

1. T-GM, CSR, O-RUs and O-DUs are connected and configured as shown in the Test Topology.
2. Clock Tester at O-RU 1PPS measurement plane receives timing input on its measurement port.
3. Provide 1PPS and 10MHz reference to Clock Tester from GNSS simulator.
4. GNSS (PRTC-A) is connected to O-RU and O-DUs.
5. G.8275.1 Full Timing Support profile.

Procedure:

Detailed Test procedure:

1. GNSS (PRTC-A) is connected directly to O-RUs and O-DUs.
2. O-RU and O-DU use the GNSS (PRTC-A) as the primary clock source and use Time as backup from the T-GM at the central location.
3. Connect T-GM to GNSS Simulator and ensure that T-GM is locked.
4. Configure O-RU and O-DU as T-TSC and CSR as T-BC.
5. Enable G.8275.1 profile on all transport TNEs (CSR).
6. Connect Clock tester measurement ports to O-RU 1PPS monitor port to measure the 1PPS at measurement plane.
7. Establish PTP and Sync-E session to the fronthaul network.

Verifications for Maximum Absolute 1PPS Time Error

1. Verify that PTP session is up and running between T-GM and CSR.
2. Verify that both O-RU and O-DU locked to GNSS (PRTC-A) as the primary clock source.
3. Verify the 1PPS time error measurements on Clock Tester at the 1PPS measurement plane.
 - a. Max 1PPS $|TE|$ measured on Clock Tester at 1PPS measurement plane of O-RU $\leq 110ns$

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and Sync-E session is up and running.

2. Verify that both O-RU and O-DU locked to GNSS (PRTC-A) as the primary clock source.
3. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RUs at 1PPS measurement plane.
 - a. $\max|TE_{RL}| (1,2) \leq 220ns$

NOTE1:

To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.

Pass/Fail Criteria:

Test shall pass if the Maximum absolute TE, Maximum Relative TE and Time Wander Limit measurement result is below the limit specified.

Test Output

End-to-End Network Limit	Expected Value	Observed Value
1PPS $\max TE_L $ at 1PPS measurement plane	$\leq 110ns$	
1PPS $\max TE_{RL} $ at 1PPS measurement plane	$\leq 220ns$	

9.3.3 Network Time Synchronization-LLS-C4 and LLS-C3 mixed configurations for co-located DUs and RUs

Test ID

SYNC.TE.5

Classification

Synchronization, Performance, Absolute Time Error, Relative Time Error

Test Title

Measurement of Absolute Time Error, Relative Time Error at 1PPS measurement plane of O-RU. LLS-C4 and LLS-C3 mixed configurations for co-located DUs and RUs.

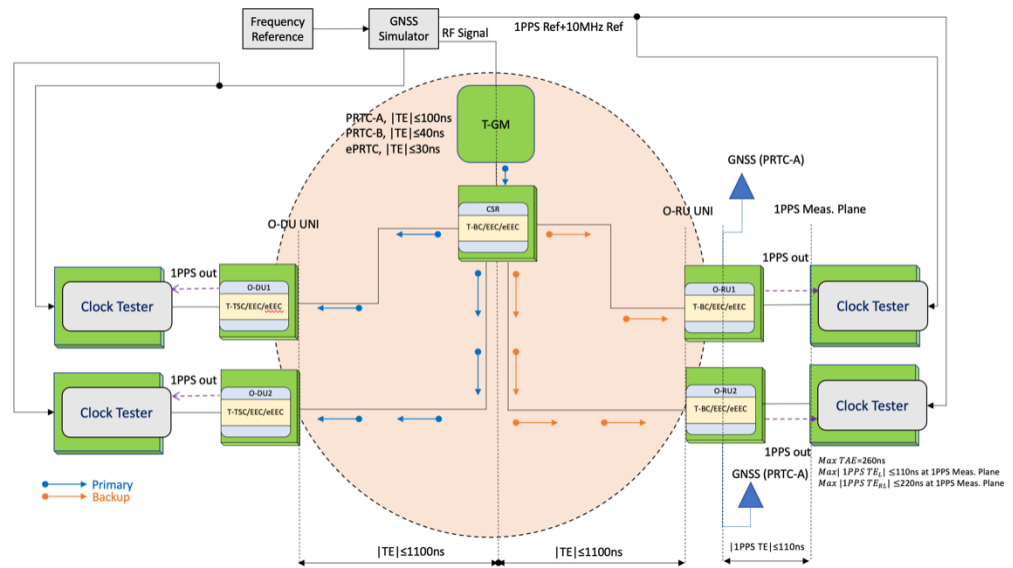
Test Objective:

1. Verify the Maximum Absolute 1PPS Time Error ($\max|TE_L|$) at 1PPS Measurement Plane of O-RUs.
2. Verify the Maximum Relative 1PPS Time Error ($\max|TE_{RL}|$) at 1PPS Measurement Plane between O-RUs.

Pre-Req

T-GM/PRTC connected to GNSS simulator and ensure T-GM is locked.
Tester 1PPS measurement port connected to O-RU/O-DU 1PPS out is the measurement plane.

Test Topology:



The test topology above illustrates the placement of clock tester at the 1PPS measurement plane with 1PPS and 10MHz reference connected to clock tester.

Test Configuration

1. T-GM, CSR, O-RUs and O-DUs are connected and configured as shown in the Test Topology.
2. Clock Tester at O-RU 1PPS measurement plane receives timing input on its measurement port.
3. Provide 1PPS and 10MHz reference to Clock Tester from GNSS simulator.
4. GNSS (PRTC-A) is connected to O-RUs.
5. G.8275.1 Full Timing Support profile.

Procedure:

Detailed Test procedure:

1. GNSS (PRTC-A) is connected directly to O-RUs.
2. O-RU use the GNSS (PRTC-A) as the primary clock source and use Time as backup from the T-GM at the central location.
3. O-DU lock to PTP clock from T-GM at the central location.
4. Connect T-GM to GNSS Simulator and ensure that T-GM is locked.
5. Configure O-RU and O-DU as T-TSC and CSR as T-BC.
6. Enable G.8275.1 profile on all transport TNEs (CSR).
7. Connect Clock tester measurement ports to O-RU 1PPS monitor port to measure the 1PPS at measurement plane.
8. Establish PTP and Sync-E session to the fronthaul network.

Verifications for Maximum Absolute 1PPS Time Error

1. Verify that PTP session is up between GNSS (PRTC-A) and O-RU (T-TSC).
2. Verify that PTP session is up between T-GM, CSR (T-BC) and O-DU (T-TSC).
3. Verify that O-RU locked to GNSS (PRTC-A) as the primary clock source.
4. Verify the 1PPS time error measurements on Clock Tester at the 1PPS measurement plane.

- a. Max 1PPS $|TE_L|$ measured on Clock Tester at the 1PPS measurement plane of O-RU $\leq 110ns$

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and Sync-E session is up.
2. Verify that both O-RU locked to GNSS (PRTC-A) as the primary clock source.
3. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RUs at 1PPS measurement plane.
 - a. $\max|TE_{RL}| (1,2) \leq 220ns$

NOTE1:

To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.

Pass/Fail Criteria:

Test shall pass if the Maximum absolute TE, Maximum Relative TE and Time Wander Limit measurement result is below the limit specified.

Test Output

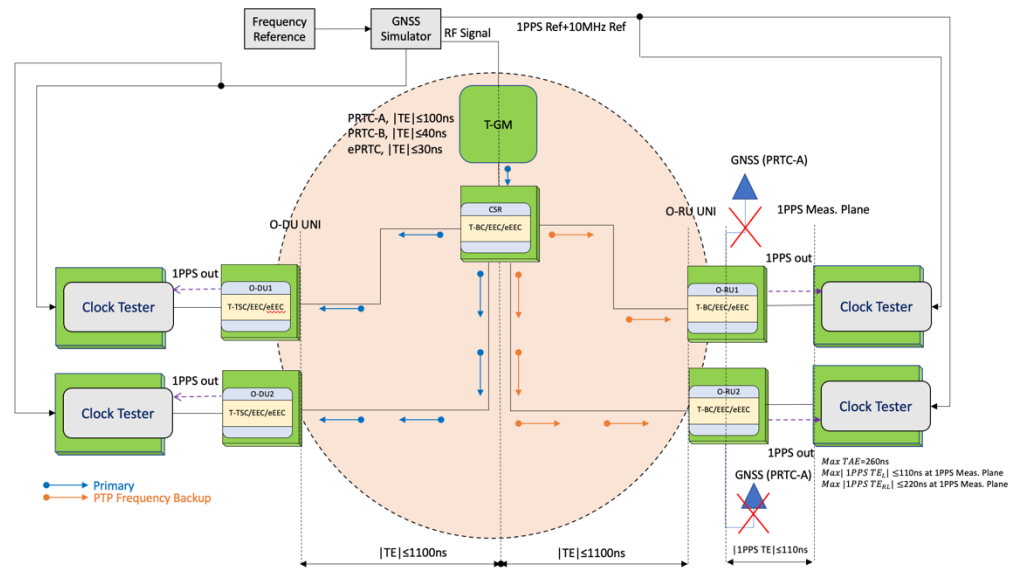
End-to-End Network Limit	Expected Value	Observed Value
1PPS $\max TE_L $ at 1PPS measurement plane	$\leq 110ns$	
1PPS $\max TE_{RL} $ at 1PPS measurement plane	$\leq 220ns$	

9.3.4 Network Time Synchronization-LLS-C4 and LLS-C3 mixed

configurations for co-located DUs and RUs-GNSS Failure at O-RU site.

Test ID	SYNC.TE.6	Classification	Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Measurement of Absolute Time Error, Relative Time Error at 1PPS measurement plane of O-RU-LLS-C4 and LLS-C3 mixed configurations for co-located DUs and RUs. GNSS Failure at O-RU site.		
Test Objective:	1. Verify the Maximum Absolute 1PPS Time Error ($\max TE_L$) at 1PPS Measurement Plane of O-RUs. 2. Verify the Maximum Relative 1PPS Time Error ($\max TE_{RL}$) at 1PPS Measurement Plane between O-RUs.		
Pre-Req	Test Case 9.3.5 MUST be a PASS.		

Test Topology:



The test topology above illustrates the placement of clock tester at the 1PPS measurement plane with 1PPS and 10MHz reference connected to clock tester.

Test Configuration

1. FAIL the GNSS connected to O-RU1 and O-RU2.

Procedure:

Verifications for Maximum Absolute 1PPS Time Error

1. Verify that PTP session is down between GNSS (PRTC-A) and O-RU (T-TSC).
2. Verify that PTP session is up between T-GM, CSR (T-BC) and O-RU (T-TSC).
3. Verify that O-RU is locked to PTP Frequency from T-GM.
4. Verify the 1PPS time error measurements on Clock Tester at the 1PPS measurement plane. (O-RU should be in Phase-holdover using frequency from Backup T-GM)
 - a. Max 1PPS $|TE_L|$ measured on Clock Tester at the 1PPS measurement plane of O-RU $\leq 110ns$ (NOTE2)

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and Sync-E session is up.
2. Verify that both O-RU locked to PTP Frequency from T-GM.
3. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RUs at 1PPS measurement plane.
 - a. $\max|TE_{RL}|(1,2) \leq 220ns$

NOTE1:

To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.

NOTE2:

O-RU in this topology is LLS-C4 O-RU as defined in the CUS specification [O-RAN.WG4.CUS.0-R003-v12.00, Section 11.3.2.3] and as such Note 3 and Note 4 apply. The Limit 110ns of phase error when GNSS fails with frequency assist from backup PTP is an assumption and the allowable limits are not yet defined for LLS-C4.

Pass/Fail Criteria:

Test shall pass if the Maximum absolute TE, Maximum Relative TE and Time Wander Limit measurement result is below the limit specified.

Test Output

	Expected Value	
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1 9.3.5 Network Time Synchronization-LLS-C4 and LLS-C3 mixed
2 configurations with backup from redundant T-GM for co-located DUs
3 and RUs

Test Topology:

Frequency Reference

GNSS Simulator

1PPS Ref+10MHz Ref

RF Signal

PRTC-A, $|TE| \leq 100\text{ns}$
PRTC-B, $|TE| \leq 40\text{ns}$
ePRTC, $|TE| \leq 30\text{ns}$

T-GM-A

T-GM-B

CSR

CSR

O-DU UNI

O-RU UNI

O-DU1

O-DU2

O-RU1

O-RU2

Clock Tester

Clock Tester

Clock Tester

Clock Tester

1PPS out

1PPS out

1PPS out

1PPS out

GNSS

GNSS

1PPS Meas. Plane

1PPS TE $\leq 110\text{ns}$

1PPS TE $\leq 110\text{ns}$

1PPS TE $\leq 110\text{ns}$

1PPS TE $\leq 110\text{ns}$

Max TAE=260ns
Max $|1PPS TE_{\text{R1}}| \leq 110\text{ns}$ at 1PPS Meas. Plane
Max $|1PPS TE_{\text{R2}}| \leq 220\text{ns}$ at 1PPS Meas. Plane

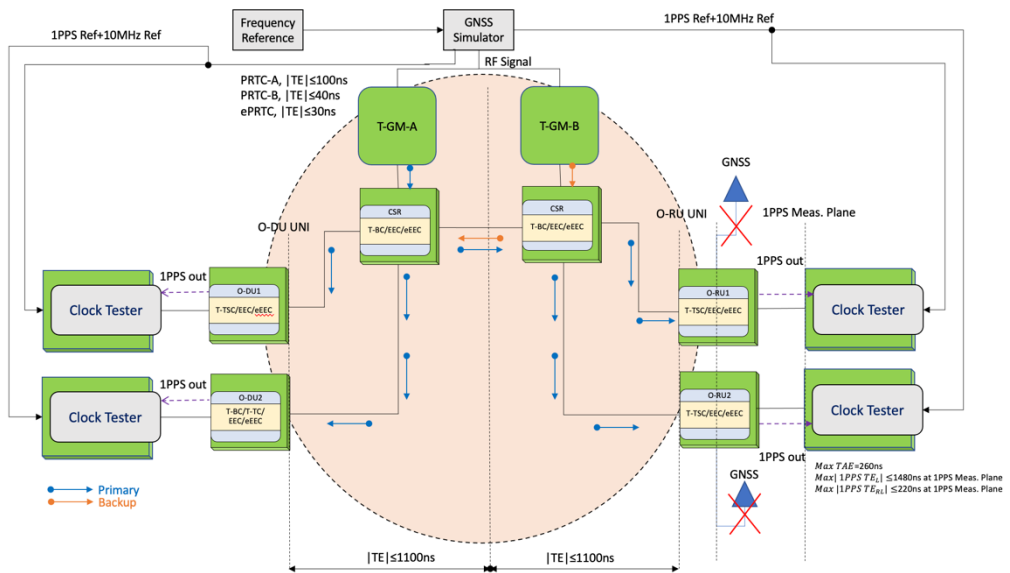
Legend: Blue arrow = Primary, Orange arrow = Backup

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Test Configuration	1. T-GM-A, T-GM-B, CSR, O-RUs and O-DUs are connected and configured as shown in the Test Topology. 2. Clock Tester at O-RU 1PPS measurement plane receives timing input on its measurement port. 3. Provide 1PPS and 10MHz reference to Clock Tester from GNSS simulator. 4. GNSS (PRTC-A) is connected to O-RU and O-DUs. 5. G.8275.1 Full Timing Support profile.		
Procedure:	<u>Detailed Test procedure:</u> 1. GNSS (PRTC-A) is connected directly to O-RUs. 2. O-RU use the GNSS (PRTC-A) as the primary clock source and use PTP Time as backup from the T-GM-B at the central location. 3. O-DU use the T-GM-A as the Primary Clock Source. 4. Connect T-GM-A and T-GM-B to GNSS Simulator and ensure that they are locked. 5. Configure O-RU and O-DU as T-TSC and CSR as T-BC. 6. Enable G.8275.1 profile on all transport TNEs (CSR). 7. Connect Clock tester measurement ports to O-RU 1PPS monitor port to measure the 1PPS at measurement plane. 8. Establish PTP and Sync-E session to the fronthaul network. <u>Verifications for Maximum Absolute 1PPS Time Error</u> 1. Verify that PTP session is up between GNSS (PRTC-A) and O-RU (T-TSC). 2. Verify that PTP session is up between T-GM-A, T-GM-B, CSR (T-BC) and O-DU (T-TSC). 3. Verify that O-RU locked to GNSS (PRTC-A) as the primary clock source. 4. Verify the 1PPS time error measurements on Clock Tester at the 1PPS measurement plane. a. Max 1PPS TE_L measured on Clock Tester at the 1PPS measurement plane of O-RU $\leq 110ns$ <u>Verifications for Maximum Relative 1PPS Time Error</u> 1. Verify that PTP and Sync-E session is up. 2. Verify that both O-RU locked to GNSS (PRTC-A) as the primary clock source. 3. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RUs at 1PPS measurement plane. a. $\max TE_{RL} (1,2) \leq 220ns$ NOTE1: To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.		
Pass/Fail Criteria:	Test shall pass if the Maximum absolute TE, Maximum Relative TE and Time Wander Limit measurement result is below the limit specified.		
Test Output	End-to-End Network Limit 1PPS $\max TE_L$ at 1PPS	Expected Value $\leq 110ns$	Observed Value

	measurement plane			
	1PPS $\max TE_{RL} $ at 1PPS measurement plane	$\leq 220ns$		

- 1 9.3.6 Network Time Synchronization-LLS-C4 and LLS-C3 mixed
- 2 configurations with backup from redundant T-GM for co-located DUs
- 3 and RUs (GNSS Failure at O-RU site).

Test ID	SYNC.TE.8	Classification	Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Measurement of Absolute Time Error, Relative Time Error at 1PPS measurement plane of O-RU. LLS-C4 and LLS-C3 mixed configurations with backup from redundant T-GM for co-located DUs and RUs (GNSS Failure at O-RU site).		
Test Objective:	1. Verify the Maximum Absolute 1PPS Time Error ($\max TE_L$) at 1PPS Measurement Plane of O-RUs. 2. Verify the Maximum Relative 1PPS Time Error ($\max TE_{RL}$) at 1PPS Measurement Plane between O-RUs.		
Pre-Req	Test Case 9.3.7 MUST be a PASS. T-GM-A Priority-2 better than that of T-GM-B		
Test Topology:	 The test topology above illustrates the placement of clock tester at the 1PPS measurement plane with 1PPS and 10MHz reference connected to clock tester.		
Test Configuration	1. FAIL the GNSS connection to O-RU1 and O-RU2.		
Procedure:	Detailed Test procedure: 1. O-RU use Time as backup from the T-GM-A at the central location. Verifications for Maximum Absolute 1PPS Time Error 1. Verify that PTP session is up between T-GM-A, T-GM-B, CSR (T-BC) and O-DU (T-TSC).		

2. Verify that O-RU locked to T-GM-A as the clock source.
3. Verify the 1PPS time error measurements on Clock Tester at the 1PPS measurement plane.
 - a. Max 1PPS $|TE_L|$ measured on Clock Tester at the 1PPS measurement plane of O-RU $\leq 1480ns$

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and Sync-E session is up.
2. Verify that both O-RU locked to T-GM-A as the primary clock source.
3. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RUs at 1PPS measurement plane.
 - a. $\max|TE_{RL}| (1,2) \leq 220ns$

NOTE1:

To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.

Pass/Fail Criteria: Test shall pass if the Maximum absolute TE, Maximum Relative TE and Time Wander Limit measurement result is below the limit specified.

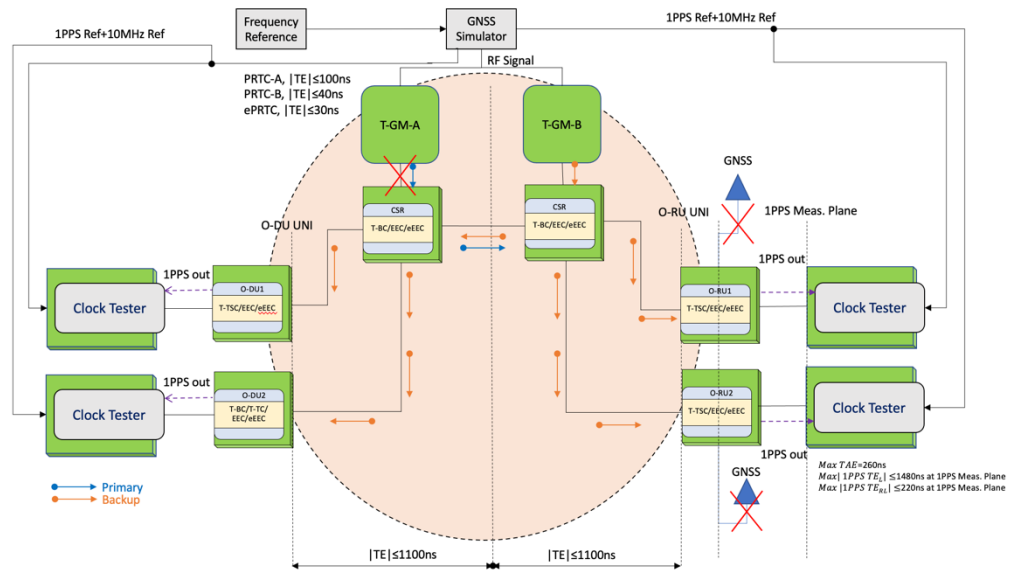
Test Output

End-to-End Network Limit	Expected Value	Observed Value
1PPS $\max TE_L $ at 1PPS measurement plane	$\leq 1480ns$	
1PPS $\max TE_{RL} $ at 1PPS measurement plane	$\leq 220ns$	

- 1 9.3.7 Network Time Synchronization-LLS-C4 and LLS-C3 mixed
- 2 configurations with backup from redundant T-GM for co-located DUs
- 3 and RUs (GNSS & T-GM-A Failure).

Test ID	SYNC.TE.9	Classification	Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Measurement of Absolute Time Error, Relative Time Error at 1PPS measurement plane of O-RU. LLS-C4 and LLS-C3 mixed configurations with backup from redundant T-GM for co-located DUs and RUs (GNSS & T-GM-A Failure).		
Test Objective:	1. Verify the Maximum Absolute 1PPS Time Error ($Max TE_L$) at 1PPS Measurement Plane of O-RUs. 2. Verify the Maximum Relative 1PPS Time Error ($Max TE_{RL}$) at 1PPS Measurement Plane between O-RUs.		
Pre-Req	Test Case 9.3.8 MUST be a PASS. T-GM-A Priority-2 better than that of T-GM-B		

Test Topology:



The test topology above illustrates the placement of clock tester at the 1PPS measurement plane with 1PPS and 10MHz reference connected to clock tester.

Test Configuration

1. FAIL the GNSS connection to T-GM-A.

Procedure:

Detailed Test procedure:

1. O-RU and O-DU use T-GM-B as backup.

Verifications for Maximum Absolute 1PPS Time Error

1. Verify that PTP session is up between T-GM-B, CSR (T-BC) and O-DU (T-TSC).
2. Verify that O-RU locked to T-GM-B as the primary clock source.
3. Verify the 1PPS time error measurements on Clock Tester at the 1PPS measurement plane.
 - a. Max 1PPS $|TE_L|$ measured on Clock Tester at the 1PPS measurement plane of O-RU $\leq 1480ns$

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and Sync-E session is up.
2. Verify that both O-RU locked to T-GM-A as the primary clock source.
3. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RUs at 1PPS measurement plane.
 - a. $\max|1PPS TE_{RL}|(1,2) \leq 220ns$

NOTE1:

To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.

Pass/Fail Criteria:

Test shall pass if the Maximum absolute TE, Maximum Relative TE and Time Wander Limit measurement result is below the limit specified.

Test Output

End-to-End Network Limit	Expected Value	Observed Value
1PPS $\max TE_L $ at 1PPS measurement plane	$\leq 1480ns$	

1PPS max $ TE_{RL} $ at 1PPS measurement plane	$\leq 220ns$		
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9.3.8 Network Time Synchronization-LLS-C3 configurations with GNSS assisted CSR for co-located DUs and RUs

Test ID	SYNC.TE.10	Classification	Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Measurement of Absolute Time Error, Relative Time Error at measurement plane of O-RU. LLS-C3 configurations with GNSS assisted CSR for co-located DUs and RUs.		
Test Objective:	1. Verify the Maximum Absolute Packet Time Error ($Max TE_L$) at O-RU UNI. 2. Verify the Maximum Relative Packet Time Error ($Max TE_{RL}$) between the O-RU UNI ports. 3. Verify the Maximum Absolute 1PPS Time Error ($Max TE_L$) at O-RU UNI. 4. Verify the Maximum Relative 1PPS Time Error ($Max TE_{RL}$) at O-RU UN. 5. Verify the Time Wander at O-RU UNI.		
Pre-Req	T-GM/PRTC connected to GNSS simulator and ensure T-GM is locked. O-RU UNI is the measurement plane.		
Test Topology:	The test topology above illustrates the placement of clock tester at the O-RU UNI measurement plane with 1PPS and 10MHz reference connected to clock tester.		
Test Configuration	1. T-GM, CSR, O-RUs and O-DUs are connected and configured as shown in the Test Topology. 2. Clock Tester at O-RU UNI measurement plane receives timing input on its measurement port. 3. Provide 1PPS and 10MHz reference to Clock Tester from GNSS simulator. 4. GNSS is connected to O-RU and O-DUs. 5. G.8275.1 Full Timing Support profile.		

Procedure:
Detailed Test procedure:

1. GNSS (PRTC-A) is connected directly to O-RUs and O-DUs.
2. Both O-RU and O-DU use the GNSS (PRTC-A) as the primary clock source and use Time as backup from the T-GM at the central location.
3. Connect T-GM to GNSS Simulator and ensure that they are locked.
4. Configure O-DU as T-TSC and CSR as T-BC.
5. Enable G.8275.1 profile on all transport TNEs (CSR).
6. Connect Clock tester measurement ports to O-RU 1PPS monitor port to measure the 1PPS at O-RU UNI.
7. Establish PTP and Sync-E session to the fronthaul network.

Verifications for Maximum Absolute Packet Time Error

1. Verify that PTP session is up and running between T-GM, CSR (T-BC) and O-DU (T-TSC).
2. Verify that O-DU is locked to GNSS.
3. Verify time error measurements using Clock Tester for the PTP flows from CSR at the O-RU UNI.
 - a. $\text{Max}|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 100ns$ (For Enhanced RU)
 - b. $\text{Max}|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 100ns$ (For Regular RU)

Verifications for Maximum Relative Packet Time Error

1. Verify that PTP and Sync-E session is up.
2. Verify that CSR locked to T-GM as the clock source.
3. Verify the Maximum Relative Time Error measurements between the O-RU UNI ports.
 - a. $\text{max}|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\text{max}|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\text{max}|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

Verifications for Maximum Absolute 1PPS Time Error

1. Verify that PTP session is up between GNSS (PRTC-A) and O-RU (T-TSC).
2. Verify that PTP session is up between T-GM, CSR (T-BC) and O-DU (T-TSC).
3. Verify that O-RU locked to GNSS (PRTC-A) as the primary clock source.
4. Verify the 1PPS time error measurements on Clock Tester at O-RU UNI.
 - a. Max 1PPS $|TE_L|$ measured on Clock Tester at O-RU UNI $\leq 100ns$ (For Enhanced RU)
 - b. Max 1PPS $|TE_L|$ measured on Clock Tester at O-RU UNI $\leq 100ns$ (For Regular RU)

Verifications for Maximum Relative 1PPS Time Error

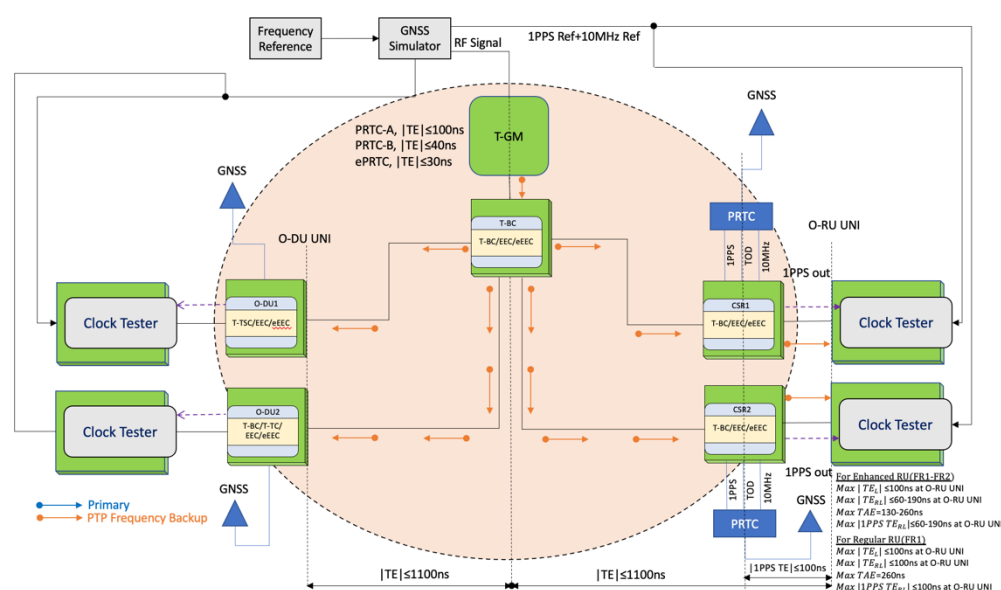
1. Verify that PTP and Sync-E session is up.
2. Verify that both O-RU locked to GNSS (PRTC-A) as the primary clock source.
3. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RUs UNI.
 - a. $\text{max}|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\text{max}|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)

	c. $\max TE_{RL} (1,2) \leq 100ns$ (FR1) (For Regular RU)			
	NOTE1: To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.			
Pass/Fail Criteria:	Test shall pass if the Maximum absolute TE, Maximum Relative TE and Time Wander Limit measurement result is below the limit specified.			
Test Output	End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value
		(For Enhanced RU)	(For Regular RU)	
	Packet $\max TE_L $	$\leq 100ns$	$\leq 100ns$	
	Packet $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
		$\leq 190ns(FR1)$		
	1PPS $\max TE_L $	$\leq 100ns$	$\leq 100ns$	
	1PPS $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
		$\leq 190ns(FR1)$		
Frequency Limit (For PRTC Class-A)	$\leq 32ppb$	$\leq 32ppb$		

9.3.9 Network Time Synchronization-LLS-C3 configurations with (GNSS+PRTC) assisted CSR for co-located DUs and RUs

Test ID	SYNC.TE.11	Classification	Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Measurement of Absolute Time Error, Relative Time Error at measurement plane of O-RU UNI. LLS-C3 configurations with (GNSS+PRTC) assisted CSR for co-located DUs and RUs.		
Test Objective:	1. Verify the Maximum Absolute Packet Time Error ($\max TE_L$) at O-RU UNI. 2. Verify the Maximum Relative Packet Time Error ($\max TE_{RL}$) between the O-RU UNI ports. 3. Verify the Maximum Absolute 1PPS Time Error ($\max TE_L$) at O-RU UNI. 4. Verify the Maximum Relative 1PPS Time Error ($\max TE_{RL}$) at O-RU UN. 5. Verify the Time Wander at O-RU UNI.		
Pre-Req	T-GM/PRTC connected to GNSS simulator and ensure T-GM is locked. Tester at O-RU UNI is the measurement plane.		

Test Topology:



The test topology above illustrates the placement of clock tester at the O-RU UNI measurement plane with 1PPS and 10MHz reference connected to clock tester.

Test Configuration	1. T-GM, CSR, O-RUs and O-DUs are connected and configured as shown in the Test Topology. 2. Clock Tester at O-RU UNI receives timing input on its measurement port. 3. Provide 1PPS and 10MHz reference to Clock Tester from GNSS simulator. 4. GNSS is connected to O-RU and O-DUs. 5. G.8275.1 Full Timing Support profile.
Procedure:	<u>Detailed Test procedure:</u> 1. GNSS via PRTC is connected directly to CSRs. 2. Connect 1PPS, TOD, 10MHz cable from PRTC to CSR. 3. CSR use the (GNSS + PRTC) as the primary clock source and backup from the T-GM at the central location. 4. O-DU use the GNSS (PRTC-A) as the Primary Clock Source. 5. Connect T-GM to GNSS Simulator and ensure that they are locked. 6. Configure O-DU as T-TSC and CSR as T-BC. 7. Enable G.8275.1 profile on all transport TNEs (CSR). 8. Connect Clock tester measurement ports to CSR 1PPS monitor port to measure the 1PPS at O-RU UNI. 9. Establish PTP and Sync-E session to the fronthaul network. <u>Verifications for Maximum Absolute Packet Time Error</u> 1. Verify that PTP session is up and running between T-GM, CSR (T-BC) and O-DU (T-TSC). 2. Verify time error measurements using Clock Tester for the PTP flows from CSR at the O-RU UNI. a. $\text{Max} TE_L$ measured on Clock Tester at the O-RU UNI $\leq 100ns$ (For Enhanced RU) b. $\text{Max} TE_L$ measured on Clock Tester at the O-RU UNI $\leq 100ns$ (For Regular RU)

Verifications for Maximum Relative Packet Time Error

1. Verify that PTP and Sync-E session is up.
2. Verify that CSR locked to T-GM as the clock source.
3. Verify the Maximum Relative Time Error measurements between the O-RU UNI ports.
 - a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

Verifications for Maximum Absolute 1PPS Time Error

1. Verify that PTP session is up between (GNSS + PRTC) and CSR (T-BC).
2. Verify that CSR locked to GNSS (PRTC-A) as the primary clock source.
3. Verify the 1PPS time error measurements on Clock Tester at the O-RU UNI.
 - a. Max 1PPS $|TE_L|$ measured on Clock Tester at O-RU UNI $\leq 100ns$ (For Enhanced RU)
 - b. Max 1PPS $|TE_L|$ measured on Clock Tester at O-RU UNI $\leq 100ns$ (For Regular RU)

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and Sync-E session is up.
2. Verify that CSR locked to (GNSS+PRTC) as the primary clock source.
3. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RUs UNI.
 - a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

NOTE1:

To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.

Pass/Fail Criteria:

Test shall pass if the Maximum absolute TE, Maximum Relative TE and Time Wander Limit measurement result is below the limit specified.

Test Output

End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value
	(For Enhanced RU)	(For Regular RU)	
Packet $\max TE_L $	$\leq 100ns$	$\leq 100ns$	
Packet $\max TE_{RL} $	$\leq 60ns$ (FR2)	$\leq 100ns$ (FR1)	
	$\leq 190ns$ (FR1)		
1PPS $\max TE_L $	$\leq 100ns$	$\leq 100ns$	
1PPS $\max TE_{RL} $	$\leq 60ns$ (FR2)	$\leq 100ns$ (FR1)	
	$\leq 190ns$ (FR1)		
Frequency Limit (For PRTC Class-A)	$\leq 32ppb$	$\leq 32ppb$	

assisted CSR with backup from redundant T-GM for co-located DUs

and RUs

[illegible]

Procedure:
Detailed Test procedure:

1. GNSS (PRTC-A) is connected directly to CSRs.
2. CSR use the GNSS (PRTC-A) as the primary clock source and use Time as backup from the T-GM-B at the central location.
3. O-DU use the T-GM-A as the Primary Clock Source.
4. Configure O-RU and O-DU as T-TSC and CSR as T-BC.
5. Enable G.8275.1 profile on all transport TNEs (CSR).
6. Connect Clock tester measurement ports to CSR 1PPS monitor port to measure the 1PPS at measurement plane.
7. Establish PTP and Sync-E session to the fronthaul network.

Verifications for Maximum Absolute Packet Time Error

1. Verify that PTP session is up and running between T-GM-A, T-GM-B, CSR (T-BC) and O-DU (T-TSC).
2. Verify time error measurements using Clock Tester for the PTP flows from CSR at the O-RU UNI.
 - a. $\text{Max}|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 100ns$ (For Enhanced RU)
 - b. $\text{Max}|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 100ns$ (For Regular RU)

Verifications for Maximum Relative Packet Time Error

1. Verify that PTP and Sync-E session is up.
2. Verify that CSR locked to T-GM-A as the clock source.
3. Verify the Maximum Relative Time Error measurements between the O-RU UNI ports.
 - a. $\text{max}|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\text{max}|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\text{max}|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

Verifications for Maximum Absolute 1PPS Time Error

1. Verify that PTP session is up between GNSS (PRTC-A) and CSR (T-BC).
2. Verify that CSR locked to T-GM-A as the primary clock source.
3. Verify the 1PPS time error measurements on Clock Tester at the O-RU UNI.
 - a. Max 1PPS $|TE_L|$ measured on Clock Tester O-RU UNI $\leq 100ns$ (For Enhanced RU)
 - b. Max 1PPS $|TE_L|$ measured on Clock Tester at O-RU UNI $\leq 100ns$ (For Regular RU)

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and Sync-E session is up.
2. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RUs UNI.
 - a. $\text{max}|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\text{max}|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\text{max}|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

NOTE1:

	To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.			
Pass/Fail Criteria:	Test shall pass if the Maximum absolute TE, Maximum Relative TE and Time Wander Limit measurement result is below the limit specified.			
Test Output	End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value
		(For Enhanced RU)	(For Regular RU)	
	Packet $\max TE_L $	$\leq 100ns$	$\leq 100ns$	
	Packet $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
		$\leq 190ns(FR1)$		
	1PPS $\max TE_L $	$\leq 100ns$	$\leq 100ns$	
	1PPS $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
		$\leq 190ns(FR1)$		
	Frequency Limit (For PRTC Class-A)	$\leq 32ppb$	$\leq 32ppb$	

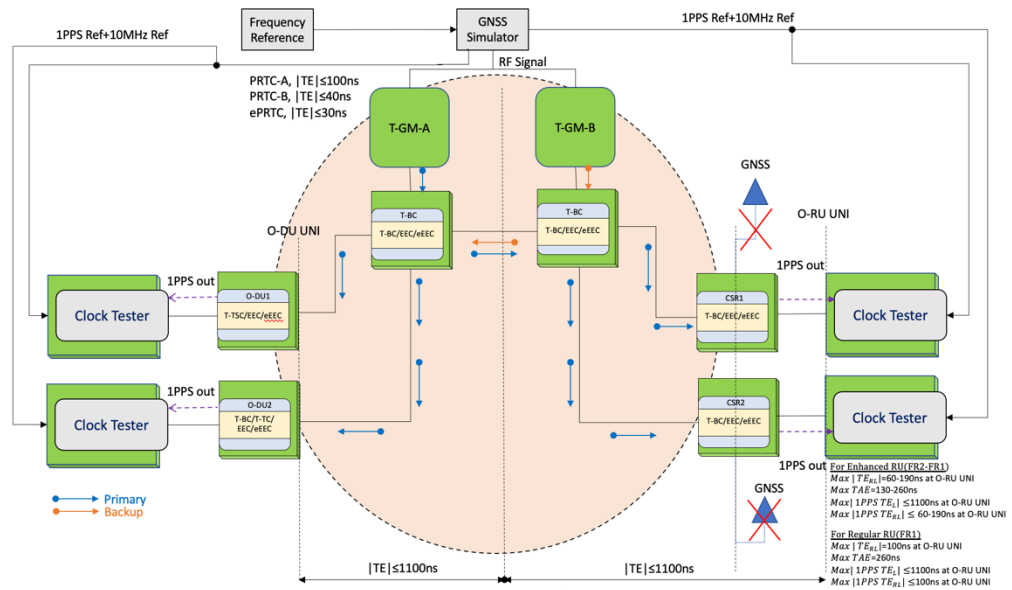
9.3.11 Network Time Synchronization-LLS-C3 configurations with GNSS

assisted CSR with backup from redundant T-GM for co-located DUs

and RUs (GNSS Failure at CSR site)

Test ID	SYNC.TE.13	Classification	Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Measurement of Absolute Time Error, Relative Time Error at measurement plane of O-RU. LLS-C3 configurations with GNSS assisted CSR with backup from redundant T-GM for co-located DUs and Rus (GNSS Failure at RU site).		
Test Objective:	1. Verify the Maximum Absolute Packet Time Error ($\max TE_L$) at O-RU UNI. 2. Verify the Maximum Relative Packet Time Error ($\max TE_{RL}$) between the O-RU UNI ports. 3. Verify the Maximum Absolute 1PPS Time Error ($\max TE_L$) at O-RU UNI. 4. Verify the Maximum Relative 1PPS Time Error ($\max TE_{RL}$) at O-RU UN. 5. Verify the Time Wander at O-RU UNI.		
Pre-Req	O-RU UNI is the measurement plane. Testcase 9.3.12 MUST be a PASS. T-GM-A Priority-2 better than that of T-GM-B		

Test Topology:



The test topology above illustrates the placement of clock tester at the O-RU UNI with 1PPS and 10MHz reference connected to clock tester.

Test Configuration

1. Fail the GNSS connection to CSRs.

Procedure:

Detailed Test procedure:

1. CSR and O-DU use the T-GM-A as the clock source.
2. Connect Clock tester measurement ports to CSR 1PPS monitor port to measure the 1PPS at O-RU UNI.
3. Establish PTP and Sync-E session to the fronthaul network.

Verifications for Maximum Absolute Packet Time Error

1. Verify that PTP session is up and running between T-GM-A, T-GM-B, CSR (T-BC) and O-DU (T-TSC).
2. Verify that O-DU is locked to T-GM-A via CSR.
3. Verify time error measurements using Clock Tester for the PTP flows from CSR at the O-RU UNI.
 - a. Max $|TE_L|$ measured on Clock Tester at the O-RU UNI ≤ 1100ns (For Enhanced RU)
 - b. Max $|TE_L|$ measured on Clock Tester at the O-RU UNI ≤ 1100ns (For Regular RU)

Verifications for Maximum Relative Packet Time Error

1. Verify that PTP and Sync-E session is up.
2. Verify that CSR locked to T-GM-A as the clock source.
3. Verify the Maximum Relative Time Error measurements between the O-RU UNI ports.
 - a. max $|TE_{RL}|$ (1,2) ≤ 60ns (FR2) (For Enhanced RU)
 - b. max $|TE_{RL}|$ (1,2) ≤ 190ns (FR1) (For Enhanced RU)
 - c. max $|TE_{RL}|$ (1,2) ≤ 100ns (FR1) (For Regular RU)

Verifications for Maximum Absolute 1PPS Time Error

1. Verify that PTP session is up between T-GM-A, T-GM-B, CSR (T-BC) and O-DU (T-TSC).
2. Verify that CSR locked to T-GM-A as the clock source.

3. Verify the 1PPS time error measurements on Clock Tester at the 1PPS measurement plane.
 - a. Max 1PPS $|TE_L|$ measured on Clock Tester at the 1PPS measurement plane of O-RU $\leq 1100ns$ (For Enhanced RU)
 - b. Max 1PPS $|TE_L|$ measured on Clock Tester at the 1PPS measurement plane of O-RU $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and Sync-E session is up.
2. Verify that CSR locked to T-GM-A as the clock source.
3. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RUs UNI.
 - a. $\max|1PPS TE_{RL}| (1,2) \leq 60ns (FR2)$ (For Enhanced RU)
 - b. $\max|1PPS TE_{RL}| (1,2) \leq 190ns (FR1)$ (For Enhanced RU)
 - c. $\max|1PPS TE_{RL}| (1,2) \leq 100ns (FR1)$ (For Regular RU)

NOTE1:

To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.

Pass/Fail Criteria:

Test shall pass if the Maximum absolute TE, Maximum Relative TE and Time Wander Limit measurement result is below the limit specified.

Test Output

End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value
	(For Enhanced RU)	(For Regular RU)	
Packet $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
Packet $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
	$\leq 190ns(FR1)$		
1PPS $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
1PPS $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
	$\leq 190ns(FR1)$		
Frequency Limit (For PRTC Class-A)	$\leq 32ppb$	$\leq 32ppb$	

9.3.12 Network Time Synchronization-LLS-C3 configurations with GNSS assisted CSR with backup from redundant T-GM for co-located DUs and RUs (GNSS Failure at CSR site and T-GM-A Failure)

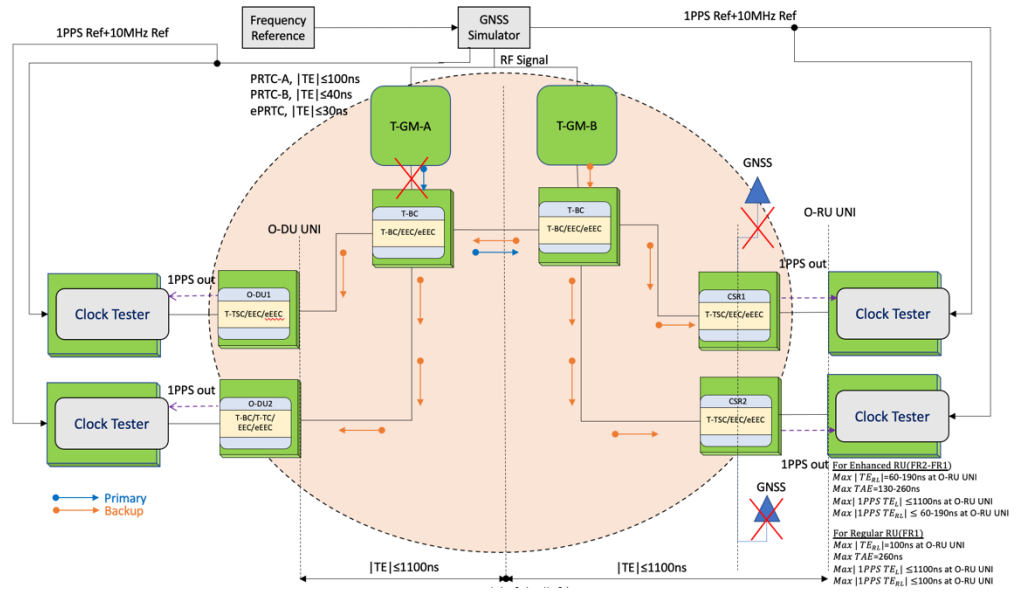
Test ID	SYNC.TE.14	Classification	Synchronization, Performance, Absolute Time Error, Relative Time Error
Test Title	Measurement of Absolute Time Error, Relative Time Error at measurement plane of O-RU UNI. LLS-C3 configurations with GNSS assisted CSR for co-located DUs and RUs (GNSS Failure).		
Test Objective:	1. Verify the Maximum Absolute Packet Time Error ($\max TE_L$) at O-RU UNI. 2. Verify the Maximum Relative Packet Time Error ($\max TE_{RL}$) between the O-RU UNI ports.		

3. Verify the Maximum Absolute 1PPS Time Error ($Max|TE_L|$) at O-RU UNI.
4. Verify the Maximum Relative 1PPS Time Error ($Max|TE_{RL}|$) at O-RU UN.
5. Verify the Time Wander at O-RU UNI.

Pre-Req

O-RU UNI is the measurement plane.
Test Case 9.3.13 MUST be a PASS.
T-GM-A Priority-2 better than that of T-GM-B

Test Topology:



The test topology above illustrates the placement of clock tester at the O-RU UNI with 1PPS and 10MHz reference connected to clock tester.

Test Configuration

FAIL the GNSS connection to T-GM-A.

Procedure:

Detailed Test procedure:

1. O-DU and CSR use the T-GM-B as the Primary Clock Source.

Verifications for Maximum Absolute Packet Time Error

1. Verify that CSR is locked to T-GM-B.
2. Verify time error measurements using Clock Tester for the PTP flows from CSR at the O-RU UNI.
 - a. $Max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $Max|TE_L|$ measured on Clock Tester at the O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative Packet Time Error

1. Verify that PTP and Sync-E session is up.
2. Verify that CSR locked to T-GM-B as the primary clock source.
3. Verify the Maximum Relative Time Error measurements between the O-RU UNI ports.
 - a. $max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

Verifications for Maximum Absolute 1PPS Time Error

1. Verify that PTP session is down between GNSS (PRTC-A) and CSR (T-BC).
2. Verify that PTP session is up between T-GM-B, CSR (T-BC) and O-DU (T-TSC).
3. Verify that CSR locked to T-GM-B as the primary clock source.
4. Verify the 1PPS time error measurements on Clock Tester at O-RU UNI.
 - a. Max 1PPS $|TE_L|$ measured on Clock Tester at O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. Max 1PPS $|TE_L|$ measured on Clock Tester at O-RU UNI $\leq 11000ns$ (For Regular RU)

Verifications for Maximum Relative 1PPS Time Error

1. Verify that PTP and Sync-E session is up.
2. Verify that CSR locked to T-GM-B as the primary clock source.
3. Estimate the Maximum Relative 1PPS Time Error measurements between the O-RUs UNI.
 - a. $\max|1PPS TE_{RL}| (1,2) \leq 60ns (FR2)$ (For Enhanced RU)
 - b. $\max|1PPS TE_{RL}| (1,2) \leq 190ns (FR1)$ (For Enhanced RU)
 - c. $\max|1PPS TE_{RL}| (1,2) \leq 100ns (FR1)$ (For Regular RU)

NOTE1:

To measure the relative time error, measure the time error from each point with respect to the internal time reference of measurement equipment, and then subtracting time error of one point from the other.

Pass/Fail Criteria:

Test shall pass if the Maximum absolute TE, Maximum Relative TE and Time Wander Limit measurement result is below the limit specified.

Test Output

End-to-End Network Limit at O-RU UNI	Expected Value		Observed Value
	(For Enhanced RU)	(For Regular RU)	
Packet $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
Packet $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
	$\leq 190ns(FR1)$		
1PPS $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
1PPS $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
	$\leq 190ns(FR1)$		
Frequency Limit (For PRTC Class-A)	$\leq 32ppb$	$\leq 32ppb$	

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9.4 Timing Solution use cases in LLS-C3 configuration for Shared O-RU

This chapter specifies the test specification for a few use cases in LLS-C3 configurations for Shared O-RU. The test specification will address cases where two O-DUs connected to O-RUs (Shared O-RUs), via an Access Transport Networks (TN). On resiliency note, we are dealing with different elements (such as T-GM and HSR associated with an 'ACTIVE' O-DU) that have either partially or completely failed or when there are failures in the connected interfaces on the Transport Networks.

A Shared O-RU maintains connectivity to more than one O-DUs, for example, two O-DUs, with one of them is considered as the 'ACTIVE' O-DU and other may be considered as the 'STANDBY' O-DU.

It is also expected that an O-DU shall support multiple Shared O-RUs. For example, O-DU1 may be connected to O-RU1 and O-RU2; O-DU2 may also be connected to O-RU1 and O-RU2. If O-DU1 fails, both O-RUs switch to O-DU2.

For the case of Shared O-RU with common transport, O-RU and Access Transport Network (TN) are all shared with a single Operator, however one or more O-DUs are managed by more than one SROs. For the Shared O-RU with separate transport, the O-RU is configured to operate with more than one Shared Resource Operators O-DUs and the associated access TN.

For Shared O-RU using a common transport network managed by Shared O-RU host (SOH), the resiliency solution will not differ from the non-Shared O-RU. Hence the resiliency scenario is not discussed in this section. For Shared O-RU supported by separate transport networks that are managed by different SROs, each of SROs takes its own responsibility to the resiliency design to its transport network.

The test specification assume that there are no failures on O-DUs and Shared O-RUs. The resiliency is based on the premise that there will always be redundant path operational for S-plane communications in the Access Transport Network between O-DUs and Shared O-RUs.

The Shared O-RU use cases in high level are already mentioned in the O-RAN WG9 Synchronization Architecture and Solution Specification. Following use cases have been covered in this chapter and for each of these use cases, clock performance limits are discussed for normal and failover scenarios.

LLS-C3 Configuration for Shared O-RU Use case.

- a. A Shared O-RU using common transport with both O-DUs managed by different SROs.
- b. Shared O-RUs using separate transport with both O-DUs managed by different SROs.
- c. Shared O-RUs using separate transport with both O-DUs managed by different SROs (Link Failure on T-GM1).
- d. Shared O-RUs using separate transport with both O-DUs managed by different SROs (Redundant TN).
- e. Shared O-RUs using separate transport with both O-DUs managed by different SROs (Link Failure on HSR1).
- f. Shared O-RUs using separate transport with both O-DUs managed by different SROs (Full Redundancy).
- g. Shared O-RUs using separate transport with both O-DUs managed by different SROs (GM & HSR Failure).

The performance limits at the input of Shared O-RU UNI for various Shared O-RU use cases mentioned above for LLS-C3 configuration remains same as that of non-Shared O-RU. Refer clause 14.3.1 in O-RAN.WG4.CUS.0-R003-v14.00 for details. However, it is important for each operator to design and manage the network to ensure LLS-C3 network limits are met under various network failover conditions.

The test cases apply to both lab and field deployment cases. The test topologies in this section specify only few Transport Node Elements (TNEs) for illustration. However, for lab cases, the type and number of inserted Transport Node Elements (TNE) will be determined by operators (or any other applicable end customer) subject to the Shared O-RU UNI limits. The test in this section is performed at Shared O-RU UNI interfaces.

The Test Topologies in all the test cases uses GNSS simulator to generate the GNSS signal to the network PRTC/T-GM. This high-quality reference from the simulator is used as the external reference for clock testers as well. The resultant accuracy of this method is a few nanoseconds. However, one can also use a GNSS signal feeding both the initial PRTC/T-GM in the network and the reference clock generating a 1pps/10MHz used to synchronize the clock testers for the measurements. Operator can make an allowance in this case, for all of the measurements for the accuracy of the reference. The magnitude of this allowance will depend on the specific GNSS reference in use.

For relative time error measurements at UNI (e.g., Shared O-RU Fronthaul UNI) this document uses the metric $\max|TE_{RL}|$ (1,2) as defined in [88], Appendix XI. Please note this metric represents an estimation and the worst-case scenario. Tests that meet (are below) the limits are seen as PASS. Tests not meeting this metric may still be acceptable. The decision is for operators to make. However, if the measurement equipment support relative time error measurement, operator can use such equipment. Refer ITU-T G.8271.1 Appendix XI for more details.

For Frequency Limit at UNI (e.g., Shared O-RU Fronthaul UNI) this document uses the metric defined as per WG4 section H.2 based on PRTC class for LLS-C3 configuration. Applicable MTIE limit shall apply for Frequency as per WG4 specification.

9.4.1 Timing Solution for Shared O-RU using common transport

Test ID	SYNC.TE.01	Classification	Synchronization, Performance, Absolute Time Error, Relative Time Error, Shared O-RU
Test Title	Measurement of Absolute Time Error, Relative Time Error at measurement plane of Shared O-RU UNI.		
Test Description	LLS-C3 configurations for Shared O-RU with common transport. A single Shared O-RU connected to two O-DUs via Access TN and O-DU1 and O-DU2 are managed by SRO-1 and SRO-2 respectively. No redundancy in the Access TN. Only SRO-1 has the host privilege. This means SRO-1 additionally acts as Shared O-RU host (SOH).		
Test Objective:	- Verify the Clock Lock states on O-DUs, HSRs and CSRs. - Verify the steady-state clock attributes on O-DUs, HSRs and CSRs. - Verify the Maximum Absolute Packet Time Error ($\max TE_L$) at Shared O-RU UNI. - Verify the Maximum Relative Packet Time Error ($\max TE_{RL}$) at Shared O-RU UNI.		

5. Verify the Maximum Absolute 1PPS Time Error ($Max|TE_L|$) at Shared O-RU UNI.
6. Verify the Maximum Relative 1PPS Time Error ($Max|TE_{RL}|$) at Shared O-RU UNI.
7. Verify the Time Wander at Shared O-RU UNI.

Pre-Req

O-RU with Shared O-RU Feature (Not shown in the diagram).

Shared O-RU UNI is the measurement plane.

T-GM, HSR, CSR1, CSR2 are part of common TN managed by SRO-1.

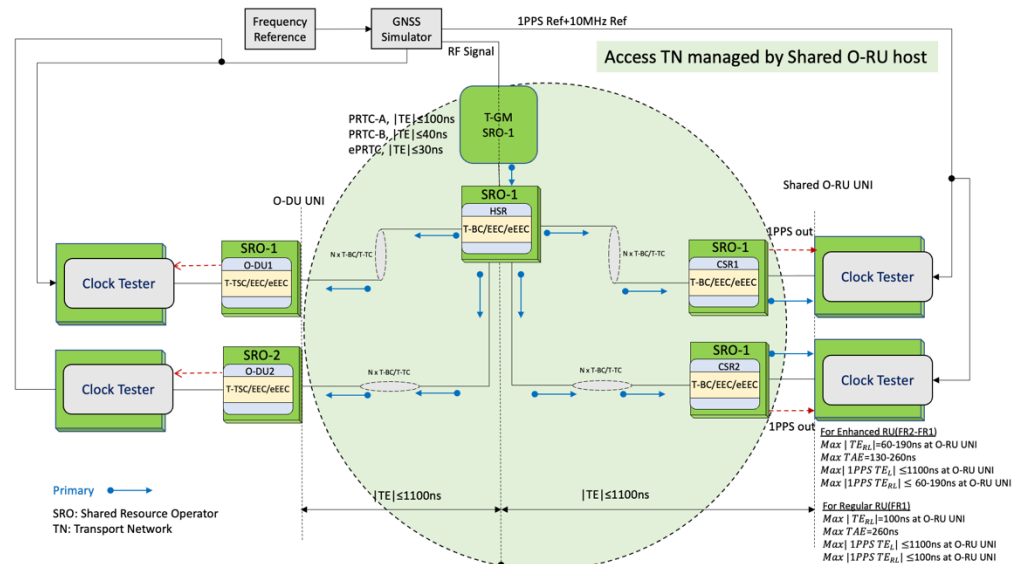
O-DU1 and O-DU2 are managed by SRO-1 and SRO-2 respectively.

Only SRO-1 has the host privilege.

O-DU1 is the ACTIVE O-DU managed by SRO-1 and O-DU2 is the STANDBY O-DU managed by SRO-2.

When the ACTIVE O-DU fails, the Shared O-RU helps identify the possible connectivity with STANDBY O-DU managed by SRO-2.

Test Topology:



The test topology above illustrates the placement of clock testers at the Shared O-RUs UNI connected to CSRs. 1PPS and 10MHz reference from GNSS Simulator connected to the clock reference inputs of the clock tester.

Test Configuration

Configure G.8275.1 profile on all devices.

Procedure:

Detailed Test procedure:

1. HSR uses T-GM as the Primary Clock Source.
2. CSR1, CSR2, O-DU1 and O-DU2 are locked to HSR.

Verifications for Clock Attributes

1. Verify the clock lock states on O-DUs, HSRs, CSRs. The lock state should be Phase-aligned.
2. Verify the clock attributes on O-DUs, HSRs and CSRs as per ITU-T G.8275.1 Standard-Appendix V, Table V.3.

Verifications for Maximum Absolute Packet Time Error

1. Verify time error measurements using Clock Tester for the PTP flows from CSRs at the Shared O-RUs UNI.
 - a. $Max|TE_L|$ measured on Clock Tester at Shared O-RUs UNI $\leq 1100ns$ (For Enhanced RU)

	b. $\text{Max} TE_L $ measured on Clock Tester at Shared O-RU UNI $\leq 1100ns$ (For Regular RU)			
	<u>Verifications for Maximum Relative Packet Time Error</u>			
	1. Verify the Maximum Relative Time Error measurements between the Shared O-RUs UNI.			
	a. $\text{max} TE_{RL} (1,2) \leq 60ns$ (FR2) (For Enhanced RU)			
	b. $\text{max} TE_{RL} (1,2) \leq 190ns$ (FR1) (For Enhanced RU)			
	c. $\text{max} TE_{RL} (1,2) \leq 100ns$ (FR1) (For Regular RU)			
	<u>Verifications for Maximum Absolute 1PPS Time Error</u>			
	1. Verify the 1PPS time error measurements on Clock Tester at Shared O-RUs UNI.			
	a. Max 1PPS $ TE_L $ measured on Clock Tester at Shared O-RUs UNI $\leq 1100ns$ (For Enhanced RU)			
	b. Max 1PPS $ TE_L $ measured on Clock Tester at Shared O-RUs UNI $\leq 1100ns$ (For Regular RU)			
	<u>Verifications for Maximum Relative 1PPS Time Error</u>			
	1. Estimate the Maximum Relative 1PPS Time Error measurements between the Shared O-RUs UNI.			
	a. $\text{max} 1PPS TE_{RL} (1,2) \leq 60ns$ (FR2) (For Enhanced RU)			
	b. $\text{max} 1PPS TE_{RL} (1,2) \leq 190ns$ (FR1) (For Enhanced RU)			
	c. $\text{max} 1PPS TE_{RL} (1,2) \leq 100ns$ (FR1) (For Regular RU)			
Pass/Fail Criteria:	Test shall pass if the Maximum absolute TE, Maximum Relative TE and Time Wander Limit measurement result is below the limit specified.			
Test Output	End-to-End Network Limit at Shared O-RU UNI	Expected Value		Observed Value
		(For Enhanced RU)	(For Regular RU)	
	Packet $\text{max} TE_L $	$\leq 1100ns$	$\leq 1100ns$	
	Packet $\text{max} TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
		$\leq 190ns(FR1)$		
	1PPS $\text{max} TE_L $	$\leq 1100ns$	$\leq 1100ns$	
	1PPS $\text{max} TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
		$\leq 190ns(FR1)$		
Frequency Limit (For PRTC Class-A)	$\leq 32ppb$	$\leq 32ppb$		

1

9.4.2 Timing Solution for Shared O-RU with separate transport.

Test ID	SYNC.TE.02	Classification	Synchronization, Performance, Absolute Time Error, Relative Time Error, Shared O-RU
Test Title	Measurement of Absolute Time Error, Relative Time Error at measurement plane of Shared O-RU UNI.		
Test Description	LLS-C3 configurations for Shared O-RU with common transport. Two Shared O-RUs are connected to two O-DUs via Access TN and O-DU1 and O-DU2 are managed by SRO-1 and SRO-2 respectively.		

	Only SRO-1 has the host privilege. This means SRO-1 additionally act as Shared O-RU host (SOH).										
Test Objective:	1. Verify the Clock Lock states on O-DUs, HSRs and CSRs. 2. Verify the steady-state clock attributes on O-DUs, HSRs and CSRs. 3. Verify the Maximum Absolute Packet Time Error ($Max TE_L$) at Shared O-RU UNI. 4. Verify the Maximum Relative Packet Time Error ($Max TE_{RL}$) between the Shared O-RUs UNI. 5. Verify the Maximum Absolute 1PPS Time Error ($Max TE_L$) at Shared O-RU UNI. 6. Verify the Maximum Relative 1PPS Time Error ($Max TE_{RL}$) between the Shared O-RUs UNI. 7. Verify the Time Wander at Shared O-RU UNI.										
Pre-Req	O-RU with Shared O-RU Feature (Not shown in the diagram). Shared O-RU UNI is the measurement plane. Both T-GM1 and T-GM2 share a common Timing Source (e.g. GNSS). T-GM1, HSR1, CSR1 and O-DU1 are managed by SRO-1. T-GM2, HSR2 and O-DU2 are managed by SRO-2. SRO-1 has host privilege. Link between HSR1 and HSR2 doesn't use for S-Plane traffic. It is meant to carry only C/U plane traffic should the need arise. O-DU1 is the ACTIVE O-DU managed by SRO-1 and O-DU2 is the STANDBY O-DU managed by SRO-2. When the ACTIVE O-DU fails, the Shared O-RU helps identify the possible connectivity with STANDBY O-DU managed by SRO-2.										
Test Topology:	The test topology above illustrates the placement of two clock testers at the Shared O-RUs UNI connected to a CSR. 1PPS and 10MHz reference from GNSS Simulator connected to the clock reference inputs of the clock tester. <table><tr><th>For Enhanced RU(FB2-FR1)</th></tr><tr><td>$Max TE_{RL} = 60-190ns$ at O-RU UNI</td></tr><tr><td>$Max TAE = 130-260ns$</td></tr><tr><td>$Max 1PPS TE_L \leq 1100ns$ at O-RU UNI</td></tr><tr><td>$Max 1PPS TE_{RL} \leq 60-190ns$ at O-RU UNI</td></tr><tr><th>For Regular RU(FB1)</th></tr><tr><td>$Max TE_{RL} = 100ns$ at O-RU UNI</td></tr><tr><td>$Max TAE = 260ns$</td></tr><tr><td>$Max 1PPS TE_L \leq 1100ns$ at O-RU UNI</td></tr><tr><td>$Max 1PPS TE_{RL} \leq 100ns$ at O-RU UNI</td></tr></table>	For Enhanced RU(FB2-FR1)	$Max TE_{RL} = 60-190ns$ at O-RU UNI	$Max TAE = 130-260ns$	$Max 1PPS TE_L \leq 1100ns$ at O-RU UNI	$Max 1PPS TE_{RL} \leq 60-190ns$ at O-RU UNI	For Regular RU(FB1)	$Max TE_{RL} = 100ns$ at O-RU UNI	$Max TAE = 260ns$	$Max 1PPS TE_L \leq 1100ns$ at O-RU UNI	$Max 1PPS TE_{RL} \leq 100ns$ at O-RU UNI
For Enhanced RU(FB2-FR1)											
$Max TE_{RL} = 60-190ns$ at O-RU UNI											
$Max TAE = 130-260ns$											
$Max 1PPS TE_L \leq 1100ns$ at O-RU UNI											
$Max 1PPS TE_{RL} \leq 60-190ns$ at O-RU UNI											
For Regular RU(FB1)											
$Max TE_{RL} = 100ns$ at O-RU UNI											
$Max TAE = 260ns$											
$Max 1PPS TE_L \leq 1100ns$ at O-RU UNI											
$Max 1PPS TE_{RL} \leq 100ns$ at O-RU UNI											
Test Configuration	Configure G.8275.1 profile on all devices. Priority2 of T-GM1 < Priority2 of T-GM2.										
Procedure:	Detailed Test procedure:										

1. HSR1 uses the T-GM1 as the Primary Clock Source.
2. HSR2 uses the T-GM2 as the Primary Clock Source.
3. CSR1 and O-DU1 are locked to HSR1.
4. O-DU2 is locked to HSR2.

Verifications for Clock States and Attributes

1. Verify the clock lock state on O-DUs, HSRs, CSRs. The lock state should be Phase-aligned.
2. Verify the clock attributes on O-DUs, HSRs and CSRs as per ITU-T G.8275.1 Standard-Appendix V, Table V.3.

Verifications for Maximum Absolute Packet Time Error

1. Verify time error measurements using Clock Tester for the PTP flows from CSR at the Shared O-RU UNI.
 - a. $\text{Max}|TE_L|$ measured on Clock Tester at the Shared O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $\text{Max}|TE_L|$ measured on Clock Tester at the Shared O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative Packet Time Error

1. Verify the Maximum Relative Time Error measurements between the Shared O-RUs UNI.
 - a. $\text{max}|TE_{RL}| (1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\text{max}|TE_{RL}| (1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\text{max}|TE_{RL}| (1,2) \leq 100ns$ (FR1) (For Regular RU)

Verifications for Maximum Absolute 1PPS Time Error

1. Verify the 1PPS time error measurements on Clock Tester at the Shared O-RU UNI.
 - a. Max 1PPS $|TE_L|$ measured on Clock Tester at Shared O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. Max 1PPS $|TE_L|$ measured on Clock Tester at Shared O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative 1PPS Time Error

1. Estimate the Maximum Relative 1PPS Time Error measurements between the Shared O-RUs UNI.
 - a. $\text{max}|1PPS TE_{RL}| (1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\text{max}|1PPS TE_{RL}| (1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\text{max}|1PPS TE_{RL}| (1,2) \leq 100ns$ (FR1) (For Regular RU)

Pass/Fail Criteria:

Test shall pass if the Maximum absolute TE, Maximum Relative TE and Time Wander Limit measurement result is below the limit specified.

Test Output

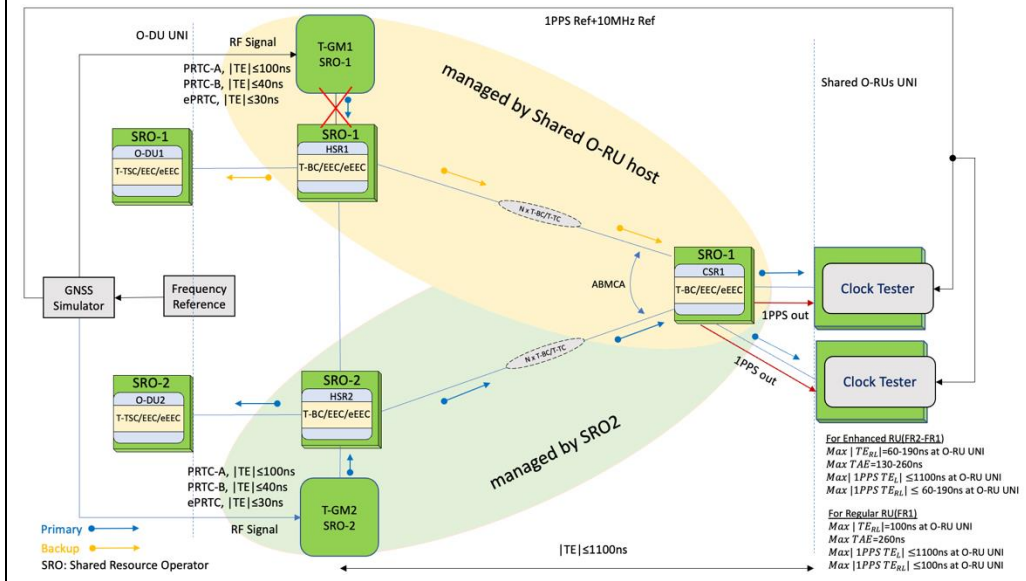
End-to-End Network Limit at Shared O-RU UNI	Expected Value		Observed Value
	(For Enhanced RU)	(For Regular RU)	
Packet $\text{max} TE_L $	$\leq 1100ns$	$\leq 1100ns$	
Packet $\text{max} TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
	$\leq 190ns(FR1)$		
1PPS $\text{max} TE_L $	$\leq 1100ns$	$\leq 1100ns$	
1PPS $\text{max} TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
	$\leq 190ns(FR1)$		

	Frequency Limit (For PRTC Class-A)	$\leq 32ppb$	$\leq 32ppb$	
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9.4.3 Timing Solution Resiliency for Shared O-RU with separate transport (Link Failure on T-GM1)

Test ID	SYNC.TE.03	Classification	Synchronization, Performance, Absolute Time Error, Relative Time Error, Shared O-RU
Test Title	Measurement of Absolute Time Error, Relative Time Error at measurement plane of Shared O-RU UNI.		
Test Description	LLS-C3 configurations for Shared O-RU with separate transport. Two Shared O-RUs are connected to two O-DUs via Access TN and O-DU1 and O-DU2 are managed by SRO-1 and SRO-2 respectively. Only SRO-1 has the host-privilege. This means SRO-1 additionally act as Shared O-RU host (SOH). Link Failure on T-GM1.		
Test Objective	1. Verify the Clock Lock states on O-DUs, HSRs and CSRs. 2. Verify the steady-state clock attributes on O-DUs, HSRs and CSRs. 3. Verify the Maximum Absolute Packet Time Error ($Max TE_L$) at Shared O-RU UNI. 4. Verify the Maximum Relative Packet Time Error ($Max TE_{RL}$) between the Shared O-RUs UNI. 5. Verify the Maximum Absolute 1PPS Time Error ($Max TE_L$) at O-RU UNI. 6. Verify the Maximum Relative 1PPS Time Error ($Max TE_{RL}$) between the Shared O-RUs UNI. 7. Verify the Time Wander at Shared O-RU UNI.		
Pre-Req	Testcase 9.4.2 MUST be a PASS.		

Test Topology:



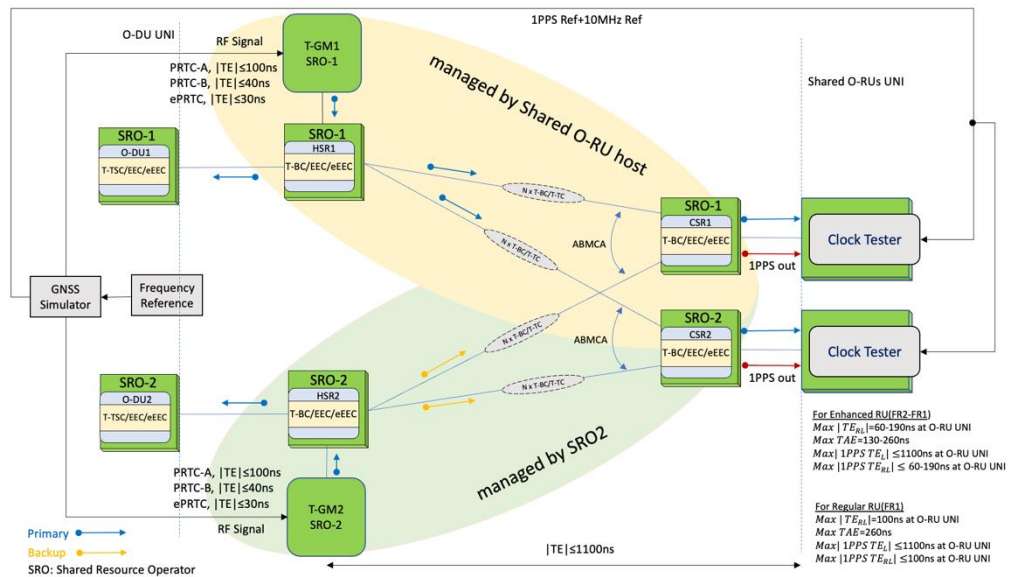
	The test topology above illustrates the placement of clock tester at the Shared O-RUs UNI connected to a CSR. 1PPS and 10MHz reference from GNSS Simulator connected to the clock reference inputs of the clock tester.			
Test Configuration	Fail the Link between T-GM1 and HSR1.			
Procedure:	<u>Detailed Test procedure:</u> - HSR1 moves to phase-holdover. - HSR2 uses the T-GM2 as the Primary Clock Source. - CSR1 and O-DU1 are locked to HSR2. - O-DU2 is locked to HSR2. <u>Verifications for Clock State and Attributes</u> - Verify the clock lock state on O-DUs, HSRs, CSRs. The lock state should be Phase-aligned. - Verify the clock attributes on O-DUs, HSRs and CSRs as per ITU-T G.8275.1 Standard-Appendix V, Table V.3. <u>Verifications for Maximum Absolute Packet Time Error</u> - Verify time error measurements using Clock Tester for the PTP flows from CSR at the Shared O-RU UNI. $\text{Max} TE_L$ measured on Clock Tester at the Shared O-RU UNI $\leq 1100ns$ (For Enhanced RU) $\text{Max} TE_L$ measured on Clock Tester at the Shared O-RU UNI $\leq 1100ns$ (For Regular RU) <u>Verifications for Maximum Relative Packet Time Error</u> - Verify the Maximum Relative Time Error measurements between the Shared O-RUs UNI. $\text{max} TE_{RL} (1,2) \leq 60ns$ (FR2) (For Enhanced RU) $\text{max} TE_{RL} (1,2) \leq 190ns$ (FR1) (For Enhanced RU) $\text{max} TE_{RL} (1,2) \leq 100ns$ (FR1) (For Regular RU) <u>Verifications for Maximum Absolute 1PPS Time Error</u> - Verify the 1PPS time error measurements on Clock Tester at Shared O-RU UNI. - Max 1PPS TE_L measured on Clock Tester at Shared O-RU UNI $\leq 1100ns$ (For Enhanced RU) - Max 1PPS TE_L measured on Clock Tester at Shared O-RU UNI $\leq 1100ns$ (For Regular RU) <u>Verifications for Maximum Relative 1PPS Time Error</u> - Estimate the Maximum Relative 1PPS Time Error measurements between the Shared O-RUs UNI. $\text{max} 1PPS TE_{RL} (1,2) \leq 60ns$ (FR2) (For Enhanced RU) $\text{max} 1PPS TE_{RL} (1,2) \leq 190ns$ (FR1) (For Enhanced RU) $\text{max} 1PPS TE_{RL} (1,2) \leq 100ns$ (FR1) (For Regular RU)			
Pass/Fail Criteria:	Test shall pass if the Maximum absolute TE, Maximum Relative TE and Time Wander Limit measurement result is below the limit specified.			
Test Output	End-to-End Network Limit at Shared O-RU UNI	Expected Value		Observed Value
		(For Enhanced RU)	(For Regular RU)	
	Packet $\text{max} TE_L $	$\leq 1100ns$	$\leq 1100ns$	

	Packet max $ TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
		$\leq 190ns(FR1)$		
	1PPS max $ TE_L $	$\leq 1100ns$	$\leq 1100ns$	
	1PPS max $ TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
		$\leq 190ns(FR1)$		
	Frequency Limit (For PRTC Class-A)	$\leq 32ppb$	$\leq 32ppb$	

9.4.4 Timing Solution for Shared O-RU with separate transport (Redundant T-GM and HSR).

Test ID	SYNC.TE.04	Classification	Synchronization, Performance, Absolute Time Error, Relative Time Error, Shared O-RU
Test Title	Measurement of Absolute Time Error, Relative Time Error at measurement plane of Shared O-RU UNI.		
Test Descriptions	LLS-C3 configurations for Shared O-RU with separate transport. Two Shared O-RUs are connected to two O-DUs via Access TN and O-DU1 and O-DU2 are managed by SRO-1 and SRO-2 respectively. Redundancy at T-GM, HSR and Transport connectivity. Both SRO-1 and SRO-2 has the host-privileges. This means both SRO-1 and SRO-2 additionally act as Shared O-RU host (SOH).		
Test Objective	1. Verify the Clock Lock states on O-DUs, HSRs and CSRs. 2. Verify the steady-state clock attributes on O-DUs, HSRs and CSRs. 3. Verify the Maximum Absolute Packet Time Error ($Max TE_L$) at Shared O-RU UNI. 4. Verify the Maximum Relative Packet Time Error ($Max TE_{RL}$) between the Shared O-RUs UNI. 5. Verify the Maximum Absolute 1PPS Time Error ($Max TE_L$) at Shared O-RU UNI. 6. Verify the Maximum Relative 1PPS Time Error ($Max TE_{RL}$) between the Shared O-RUs UNI. 7. Verify the Time Wander at Shared O-RU UNI.		
Pre-Req	O-RU with Shared O-RU Feature (Not shown in the diagram). Shared O-RU UNI is the measurement plane. Both T-GM1 and T-GM2 share a common Timing Source (e.g. GNSS). T-GM1, HSR1, CSR1 and O-DU1 are managed by SRO-1. T-GM2, HSR2, CSR2 and O-DU2 are managed by SRO-2. Both SRO-1 and SRO-2 MUST have host privileges. O-DU1 is the ACTIVE O-DU managed by SRO-1. When the ACTIVE O-DU Fails, the Shared O-RU helps identify the possible connectivity with STANDBY O-DU managed by SRO-2.		

Test Topology



The test topology above illustrates the placement of two clock testers at the O-RUs UNI connected to CSRs. 1PPS and 10MHz reference from GNSS Simulator connected to the clock reference inputs of the clock tester.

Test Configuration

Configure G.8275.1 profile on all devices.
 Priority2 of T-GM1 < Priority2 of T-GM2.

Procedure

Detailed Test procedure:

- HSR1 uses T-GM1 as the Primary Clock Source.
- HSR2 uses T-GM2 as the Primary Clock Source.
- CSR1 & CSR2 run A-BTCA, and both select clock from HSR1.
- O-DU1 locked to HSR1.
- O-DU2 locked to HSR2.

Verifications for Clock States and Attributes

- Verify the clock lock state on O-DUs, HSRs, CSRs. The lock state should be Phase-aligned.
- Verify the clock attributes on O-DUs, HSRs and CSRs as per ITU-T G.8275.1 Standard-Appendix V, Table V.3.

Verifications for Maximum Absolute Packet Time Error

- Verify time error measurements using Clock Tester for the PTP flows from CSR at the Shared O-RU UNI.
 - $\text{Max } |TE_L|$ measured on Clock Tester at the Shared O-RU UNI $\leq 1100\text{ns}$ (For Enhanced RU)
 - $\text{Max } |TE_L|$ measured on Clock Tester at the Shared O-RU UNI $\leq 1100\text{ns}$ (For Regular RU)

Verifications for Maximum Relative Packet Time Error

- Verify the Maximum Relative Time Error measurements between the Shared O-RUs UNI.
 - $\text{max } |TE_{RL}| (1,2) \leq 60\text{ns}$ (FR2) (For Enhanced RU)
 - $\text{max } |TE_{RL}| (1,2) \leq 190\text{ns}$ (FR1) (For Enhanced RU)
 - $\text{max } |TE_{RL}| (1,2) \leq 100\text{ns}$ (FR1) (For Regular RU)

Verifications for Maximum Absolute 1PPS Time Error

	1. Verify the 1PPS time error measurements on Clock Tester at Shared O-RU UNI. a. Max 1PPS TE_L measured on Clock Tester at Shared O-RU UNI $\leq 1100ns$ (For Enhanced RU) b. Max 1PPS TE_L measured on Clock Tester at Shared O-RU UNI $\leq 1100ns$ (For Regular RU) Verifications for Maximum Relative 1PPS Time Error 1. Estimate the Maximum Relative 1PPS Time Error measurements between the Shared O-RUs UNI. a. $\max 1PPS TE_{RL} (1,2) \leq 60ns (FR2)$ (For Enhanced RU) b. $\max 1PPS TE_{RL} (1,2) \leq 190ns (FR1)$ (For Enhanced RU) c. $\max 1PPS TE_{RL} (1,2) \leq 100ns (FR1)$ (For Regular RU)			
Pass/Fail Criteria	Test shall pass if the Maximum absolute TE, Maximum Relative TE and Time Wander Limit measurement result is below the limit specified.			
Test Output	End-to-End Network Limit at Shared O-RU UNI	Expected Value		Observed Value
		(For Enhanced RU)	(For Regular RU)	
	Packet $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
	Packet $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
		$\leq 190ns(FR1)$		
	1PPS $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
	1PPS $\max TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
		$\leq 190ns(FR1)$		
	Frequency Limit (For PRTC Class-A)	$\leq 32ppb$	$\leq 32ppb$	

9.4.5 Timing Solution Resiliency for Shared O-RU with separate transport (Link Failure on HSR1)

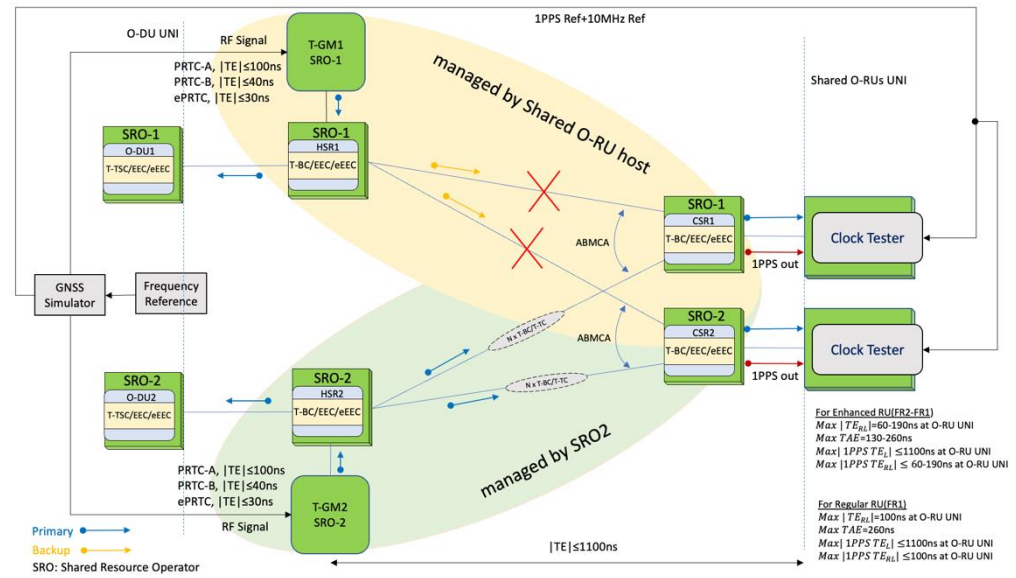
Test ID	SYNC.TE.05	Classification	Synchronization, Performance, Absolute Time Error, Relative Time Error, Shared O-RU
Test Title	Measurement of Absolute Time Error, Relative Time Error at measurement plane of O-RU UNI.		
Test Descriptions	LLS-C3 configurations for Shared O-RU with separate transport. Two Shared O-RUs are connected to two O-DUs via Access TN and O-DU1 and O-DU2 are managed by SRO-1 and SRO-2 respectively. Redundancy at T-GM, HSR and Transport connectivity. Both SRO-1 and SRO-2 have the host privileges. This means both SRO-1 and SRO-2 additionally act as Shared O-RU host (SOH). Link Failure on HSR1.		
Test Objective	1. Verify the Clock Lock states on O-DUs, HSRs and CSRs. 2. Verify the steady-state clock attributes on O-DUs, HSRs and CSRs. 3. Verify the Maximum Absolute Packet Time Error ($\max TE_L$) at Shared O-RU UNI.		

4. Verify the Maximum Relative Packet Time Error ($Max|TE_{RL}|$) at the Shared O-RUs UNI ports.
5. Verify the Maximum Absolute 1PPS Time Error ($Max|TE_L|$) at Shared O-RU UNI.
6. Verify the Maximum Relative 1PPS Time Error ($Max|TE_{RL}|$) at Shared O-RU UNI.
7. Verify the Time Wander at Shared O-RU UNI.

Pre-Req

Testcase 9.4.4 MUST be a PASS.

Test Topology



The test topology above illustrates the placement of two clock testers at the Shared O-RUs UNI connected to CSRs. 1PPS and 10MHz reference from GNSS Simulator connected to the clock reference inputs of the clock tester.

Test Configuration

Fail the links between HSR1 and CSRs.

Procedure:

Detailed Test procedure:

1. HSR1 uses the T-GM1 as the Primary Clock Source.
2. HSR2 uses the T-GM2 as the Primary Clock Source.
3. CSR1 & CSR2 both select clock from HSR2.
4. O-DU1 locked to HSR1.
5. O-DU2 locked to HSR2.

Verifications for Clock States and Attributes

1. Verify the clock lock state on O-DUs, HSRs, CSRs. The lock state should be Phase-aligned.
2. Verify the clock attributes on O-DUs, HSRs and CSRs as per ITU-T G.8275.1 Standard-Appendix V, Table V.3.

Verifications for Maximum Absolute Packet Time Error

1. Verify time error measurements using Clock Tester for the PTP flows from CSR at the Shared O-RU UNI.
 - a. $Max|TE_L|$ measured on Clock Tester at Shared O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $Max|TE_L|$ measured on Clock Tester at Shared O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative Packet Time Error

1. Verify the Maximum Relative Time Error measurements between the Shared O-RUs UNI.

- a. $\max|TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
- b. $\max|TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
- c. $\max|TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

Verifications for Maximum Absolute 1PPS Time Error

1. Verify the 1PPS time error measurements on Clock Tester at Shared O-RU UNI.

- a. Max 1PPS $|TE_L|$ measured on Clock Tester at Shared O-RU UNI $\leq 1100ns$ (For Enhanced RU)
- b. Max 1PPS $|TE_L|$ measured on Clock Tester at Shared O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative 1PPS Time Error

1. Estimate the Maximum Relative 1PPS Time Error measurements between the Shared O-RUs UNI.

- a. $\max|1PPS TE_{RL}|(1,2) \leq 60ns$ (FR2) (For Enhanced RU)
- b. $\max|1PPS TE_{RL}|(1,2) \leq 190ns$ (FR1) (For Enhanced RU)
- c. $\max|1PPS TE_{RL}|(1,2) \leq 100ns$ (FR1) (For Regular RU)

Pass/Fail Criteria:

Test shall pass if the Maximum absolute TE, Maximum Relative TE and Time Wander Limit measurement result is below the limit specified.

Test Output

End-to-End Network Limit at Shared O-RU UNI	Expected Value		Observed Value
	(For Enhanced RU)	(For Regular RU)	
Packet $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
Packet $\max TE_{RL} $	$\leq 60ns$ (FR2)	$\leq 100ns$ (FR1)	
	$\leq 190ns$ (FR1)		
1PPS $\max TE_L $	$\leq 1100ns$	$\leq 1100ns$	
1PPS $\max TE_{RL} $	$\leq 60ns$ (FR2)	$\leq 100ns$ (FR1)	
	$\leq 190ns$ (FR1)		
Frequency Limit (For PRTC Class-A)	$\leq 32ppb$	$\leq 32ppb$	

1 9.4.6 Timing Solution for Shared O-RU with separate transport (Full 2 Redundancy)

Test ID	SYNC.TE.06	Classification	Synchronization, Performance, Absolute Time Error, Relative Time Error, Shared O-RU
Test Title	Measurement of Absolute Time Error, Relative Time Error at measurement plane of O-RU UNI.		
Test Description	LLS-C3 configurations for Shared O-RU with separate transport. Two Shared O-RUs are connected to two O-DUs via Access TN and O-DU1 and O-DU2 are managed by SRO-1 and SRO-2 respectively. Full Redundancy in Access Transport Network. Both SRO-1 and SRO-2 have the host privileges. This means both SRO-1 and SRO-2 additionally act as Shared O-RU host (SOH).		

Test Objective

1. Verify the Clock Lock states on O-DUs, HSRs and CSRs.
2. Verify the steady-state clock attributes on O-DUs, HSRs and CSRs.
3. Verify the Maximum Absolute Packet Time Error ($Max|TE_L|$) at Shared O-RU UNI.
4. Verify the Maximum Relative Packet Time Error ($Max|TE_{RL}|$) between the Shared O-RUs UNI.
5. Verify the Maximum Absolute 1PPS Time Error ($Max|TE_L|$) at Shared O-RU UNI.
6. Verify the Maximum Relative 1PPS Time Error ($Max|TE_{RL}|$) between the Shared O-RUs UNI.
7. Verify the Time Wander at Shared O-RU UNI.

Pre-Req

O-RU with Shared O-RU Feature (Not shown in the diagram).

Shared O-RU UNI is the measurement plane.

Both T-GM1 and T-GM2 share a common Timing Source (e.g. GNSS).

T-GM1A, T-GM1B, HSR1A, HSR1B, CSR1 and O-DU1 are managed by SRO-1.

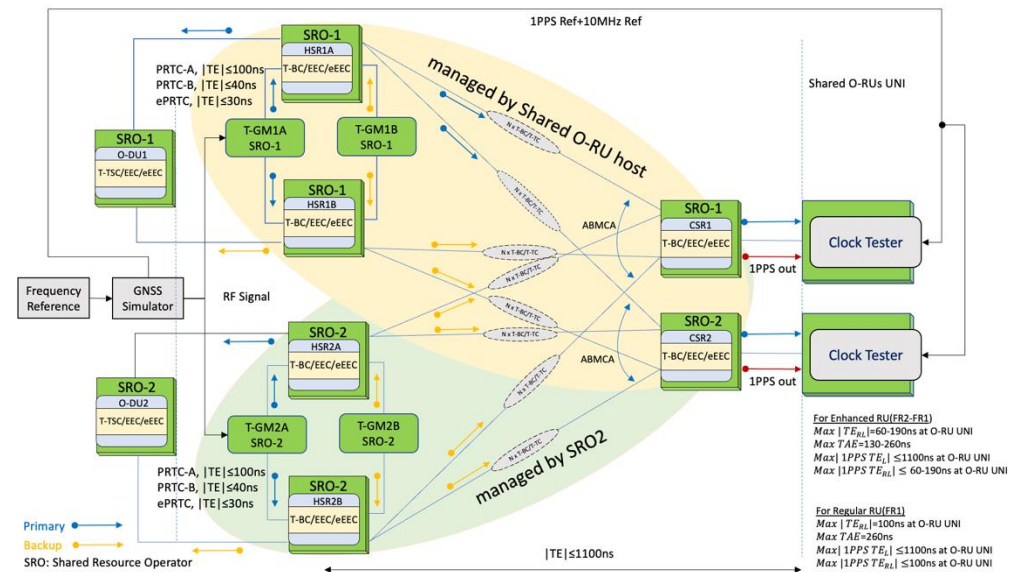
T-GM2A, T-GM2B, HSR2A, HSR2B, CSR2 and O-DU2 are managed by SRO-2.

Both SRO-1 and SRO-2 MUST have host privileges.

O-DU1 is the ACTIVE O-DU managed by SRO-1.

When the ACTIVE O-DU Fails, the Shared O-RU helps identify the possible connectivity with STANDBY O-DU managed by SRO-2.

Test Topology



The test topology above illustrates the placement of two clock tester at the Shared O-RUs UNI connected to CSRs. 1PPS and 10MHz reference from GNSS Simulator connected to the clock reference inputs of the clock tester.

Test Configuration

Configure G.8275.1 profile on all devices.

Configure Priority2 of T-GM1 < Priority2 of T-GM2.

Procedure

Detailed Test procedure:

1. HSR1A and HSR1B uses the T-GM1A as the Primary Clock Source.
2. HSR2A and HSR2B uses the T-GM2A as the Primary Clock Source.
3. CSR1 & CSR2 run A-BTCA, and both select clock from HSR1.
4. O-DU1 locked to HSR1A.

5. O-DU2 locked to HSR2A.

Verifications for Clock State and Attributes

1. Verify the clock lock state on O-DUs, HSRs, CSRs. The lock state should be Phase-aligned.
2. Verify the clock attributes on O-DUs, HSRs and CSRs as per ITU-T G.8275.1 Standard-Appendix V, Table V.3.

Verifications for Maximum Absolute Packet Time Error

1. Verify time error measurements using Clock Tester for the PTP flows from CSR at the Shared O-RU UNI.
 - a. $\text{Max}|TE_L|$ measured on Clock Tester at the Shared O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. $\text{Max}|TE_L|$ measured on Clock Tester at the Shared O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative Packet Time Error

1. Verify the Maximum Relative Time Error measurements between the Shared O-RUs UNI.
 - a. $\text{max}|TE_{RL}| (1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\text{max}|TE_{RL}| (1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\text{max}|TE_{RL}| (1,2) \leq 100ns$ (FR1) (For Regular RU)

Verifications for Maximum Absolute 1PPS Time Error

1. Verify the 1PPS time error measurements on Clock Tester at Shared O-RU UNI.
 - a. Max 1PPS $|TE_L|$ measured on Clock Tester at Shared O-RU UNI $\leq 1100ns$ (For Enhanced RU)
 - b. Max 1PPS $|TE_L|$ measured on Clock Tester at Shared O-RU UNI $\leq 1100ns$ (For Regular RU)

Verifications for Maximum Relative 1PPS Time Error

1. Estimate the Maximum Relative 1PPS Time Error measurements between the Shared O-RUs UNI.
 - a. $\text{max}|1PPS TE_{RL}| (1,2) \leq 60ns$ (FR2) (For Enhanced RU)
 - b. $\text{max}|1PPS TE_{RL}| (1,2) \leq 190ns$ (FR1) (For Enhanced RU)
 - c. $\text{max}|1PPS TE_{RL}| (1,2) \leq 100ns$ (FR1) (For Regular RU)

Pass/Fail Criteria:

Test shall pass if the Maximum absolute TE, Maximum Relative TE and Time Wander Limit measurement result is below the limit specified.

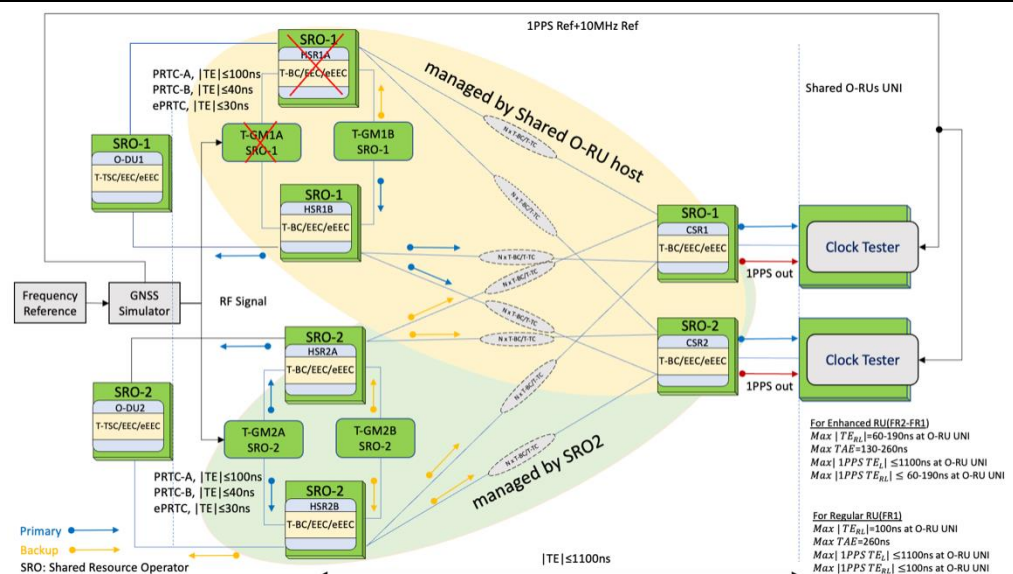
Test Output

End-to-End Network Limit at Shared O-RU UNI	Expected Value		Observed Value
	(For Enhanced RU)	(For Regular RU)	
Packet $\text{max} TE_L $	$\leq 1100ns$	$\leq 1100ns$	
Packet $\text{max} TE_{RL} $	$\leq 60ns$ (FR2)	$\leq 100ns$ (FR1)	
	$\leq 190ns$ (FR1)		
1PPS $\text{max} TE_L $	$\leq 1100ns$	$\leq 1100ns$	
1PPS $\text{max} TE_{RL} $	$\leq 60ns$ (FR2)	$\leq 100ns$ (FR1)	
	$\leq 190ns$ (FR1)		
Frequency Limit (For PRTC Class-A)	$\leq 32ppb$	$\leq 32ppb$	

9.4.7 Timing Solution Resiliency for Shared O-RU with separate transport (GM1A and HSR1A Failure)

Test ID	SYNC.TE.07	Classification	Synchronization, Performance, Absolute Time Error, Relative Time Error, Shared O-RU
Test Title	Measurement of Absolute Time Error, Relative Time Error at measurement plane of O-RU UNI.		
Test Description	LLS-C3 configurations for Shared O-RU with separate transport. Two Shared O-RUs are connected to two O-DUs via Access TN and O-DU1 and O-DU2 are managed by SRO-1 and SRO-2 respectively. Full Redundancy in Access Transport Network. Both SRO-1 and SRO-2 have the host privileges. This means both SRO-1 and SRO-2 additionally act as Shared O-RU host (SOH). Link Failure on T-GM1A and HSR1A.		
Test Objective	1. Verify the Clock Lock states on O-DUs, HSRs and CSRs. 2. Verify the steady-state clock attributes on O-DUs, HSRs and CSRs. 3. Verify the Maximum Absolute Packet Time Error ($Max TE_L$) at Shared O-RU UNI. 4. Verify the Maximum Relative Packet Time Error ($Max TE_{RL}$) between the Shared O-RUs UNI. 5. Verify the Maximum Absolute 1PPS Time Error ($Max TE_L$) at O-RU UNI. 6. Verify the Maximum Relative 1PPS Time Error ($Max TE_{RL}$) between the Shared O-RUs UNI. 7. Verify the Time Wander at Shared O-RU UNI.		
Pre-Req	Testcase 9.4.6 MUST be a PASS.		

Test Topology



The test topology above illustrates the placement of two clock testers at the Shared O-RUs UNI connected to CSRs. 1PPS and 10MHz reference from GNSS Simulator connected to the clock reference inputs of the clock tester.

Test Configuration	Fail the HSR1A and T-GM1A.			
Procedure	<u>Detailed Test procedure:</u>			
	1. HSR1B uses the T-GM1B as the Primary Clock Source.			
	2. HSR2A and HSR2B uses the T-GM2A as the Primary Clock Source.			
	3. CSR1 & CSR2 run A-BTCA, and both select clock from HSR1B.			
	4. O-DU1 is locked to HSR1B.			
	5. O-DU2 is locked to HSR2A.			
	<u>Verifications for Clock States and Attributes</u>			
	1. Verify the clock lock state on O-DUs, HSRs, CSRs. The lock state should be Phase-aligned.			
	2. Verify the clock attributes on O-DUs, HSRs and CSRs as per ITU-T G.8275.1 Standard-Appendix V, Table V.3.			
	<u>Verifications for Maximum Absolute Packet Time Error</u>			
1. Verify time error measurements using Clock Tester for the PTP flows from CSR at the Shared O-RU UNI.				
a. $\text{Max} TE_L $ measured on Clock Tester at the Shared O-RU UNI $\leq 1100ns$ (For Enhanced RU)				
b. $\text{Max} TE_L $ measured on Clock Tester at the Shared O-RU UNI $\leq 1100ns$ (For Regular RU)				
<u>Verifications for Maximum Relative Packet Time Error</u>				
1. Verify the Maximum Relative Time Error measurements between the Shared O-RUs UNI.				
a. $\text{max} TE_{RL} (1,2) \leq 60ns (FR2)$ (For Enhanced RU)				
b. $\text{max} TE_{RL} (1,2) \leq 190ns (FR1)$ (For Enhanced RU)				
c. $\text{max} TE_{RL} (1,2) \leq 100ns (FR1)$ (For Regular RU)				
<u>Verifications for Maximum Absolute 1PPS Time Error</u>				
1. Verify the 1PPS time error measurements on Clock Tester at Shared O-RU UNI.				
a. Max 1PPS $ TE_L $ measured on Clock Tester at Shared O-RU UNI $\leq 1100ns$ (For Enhanced RU)				
b. Max 1PPS $ TE_L $ measured on Clock Tester at Shared O-RU UNI $\leq 1100ns$ (For Regular RU)				
<u>Verifications for Maximum Relative 1PPS Time Error</u>				
1. Estimate the Maximum Relative 1PPS Time Error measurements between the Shared O-RUs UNI.				
a. $\text{max} 1PPS TE_{RL} (1,2) \leq 60ns (FR2)$ (For Enhanced RU)				
b. $\text{max} 1PPS TE_{RL} (1,2) \leq 190ns (FR1)$ (For Enhanced RU)				
c. $\text{max} 1PPS TE_{RL} (1,2) \leq 100ns (FR1)$ (For Regular RU)				
Pass/Fail Criteria:	Test shall pass if the Maximum absolute TE, Maximum Relative TE and Time Wander Limit measurement result is below the limit specified.			
Test Output	End-to-End Network Limit at Shared O-RU UNI	Expected Value		Observed Value
		(For Enhanced RU)	(For Regular RU)	
	Packet $\text{max} TE_L $	$\leq 1100ns$	$\leq 1100ns$	
	Packet $\text{max} TE_{RL} $	$\leq 60ns(FR2)$	$\leq 100ns(FR1)$	
		$\leq 190ns(FR1)$		

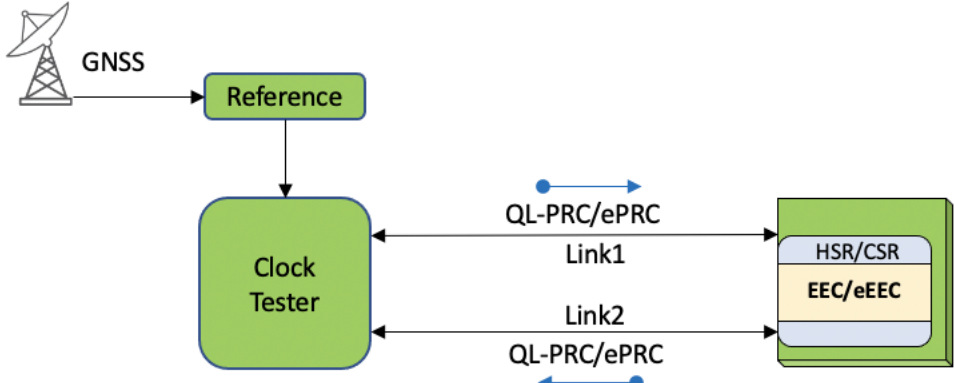
	1PPS max TE _L	≤ 1100ns	≤ 1100ns	
	1PPS max TE _{RL}	≤ 60ns(FR2)	≤ 100ns(FR1)	
		≤ 190ns(FR1)		
	Frequency Limit (For PRTC Class-A)	≤ 32ppb	≤ 32ppb	

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Annex A: ITU G.8262/G.8262.1 Clock specification test cases for TNEs

SyncE performance as per G.8262/G.8262.1 clock specifications

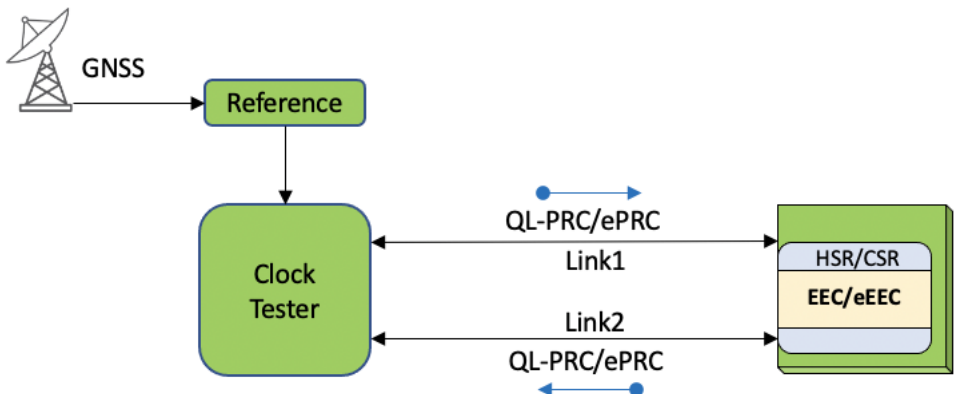
A.1 Pull-in/Hold-in/Pull-out Limits

Test ID	SYNC.SE.01	Classification	Synchronization, Performance
Test Title	Pull-in/hold-in/pull-out limits on HSR/CSR		
Test Objective:	Verify the Pull-in/hold-in/pull-out range meets ITU-T G.8262/G.8262.1		
Pre-Req	HSR/CSR connected to a tester which uses a stable reference clock		
Test Topology:			
Test Configuration	To measure the 'pull-in' range, apply a high frequency offset to the selected clock input using clock tester so that the device will enter into holdover mode. Now reduce the offset in arbitrary small step until EEC/eEEC locks to the input source. EEC/eEEC should lock to the input source before the offset reaches to ± 4.6PPM. To measure the 'hold-in' range, vary the frequency offset arbitrarily, slowly, and measure the highest frequency offset within which the device maintains lock. 'Pull-out' range is the offset within which the device stays in locked state and outside of which the device loses the lock irrespective of the rate of the frequency change.		
Procedure:	Detailed Test procedure: 1. Connect the tester to HSR/CSR fronthaul interface. 2. Configure tester to provide the clock to DUT configured interface. 3. Establish Frequency Synchronization on DUT. 4. Inject PPM offset from the tester as per the ITU-T G.8262/G.8262.1 Verifications needed: 1. Verify that SyncE and ESMC/eESMC are up. 2. Perform the Pull-in/hold-in/Pull-out test on the device under test. 3. Compare the test result as per ITU-T G.8262/G.8262.1		
Pass/Fail Criteria:	Test shall pass if the measurement result is below the limit specified by ITU-T G.8262/ G.8262.1 clock specifications.		
Test Output	ITU-T G.8262/G.8262.1 defined the limit of ± 4.6 PPM		

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A.2 Wander Generation Limit for EEC/eEEC Clock

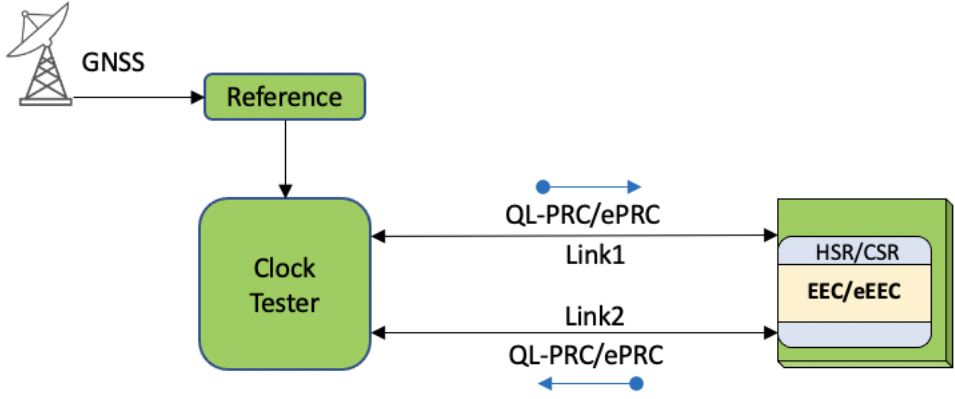
Test ID	SYNC.SE.02	Classification	Synchronization, Performance,
Test Title	Frequency Error/TIE/MTIE/TDEV on HSR/CSR		
Test Objective:	Verify the device maximum output wander or TIE/MTIE/TDEV		
Pre-Req	HSR/CSR connected to a tester which uses a stable reference clock. TNE should support ESMC or eESMC as applicable.		
Test Topology:			
Test Configuration	This section to be filled out when putting the test report (may include device and/or tester configuration)		
Procedure:	Detailed Test procedure: 1. Connect the tester to HSR/CSR interface. 2. Configure tester to provide the clock to DUT configured interface. 3. Establish Frequency Synchronization on DUT. 4. Perform the wander generation test. Verifications needed: 1. Verify that SyncE and ESMC/eESMC are up. 2. Verify the Frequency Error/TIE/MTIE/TDEV a. All the results should be within ITU-T G.8262/G.8262.1 clock specification limits		
Pass/Fail Criteria:	Test shall pass if the measurement result is below the limit specified by ITU-T G.8262/G.8262.1 clock specifications.		
Test Output	This section is the placeholder that will capture the output/configurations after test is executed, typically done after test execution		

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A.3 Wander Tolerance/Transfer Test for EEC/eEEC Clock

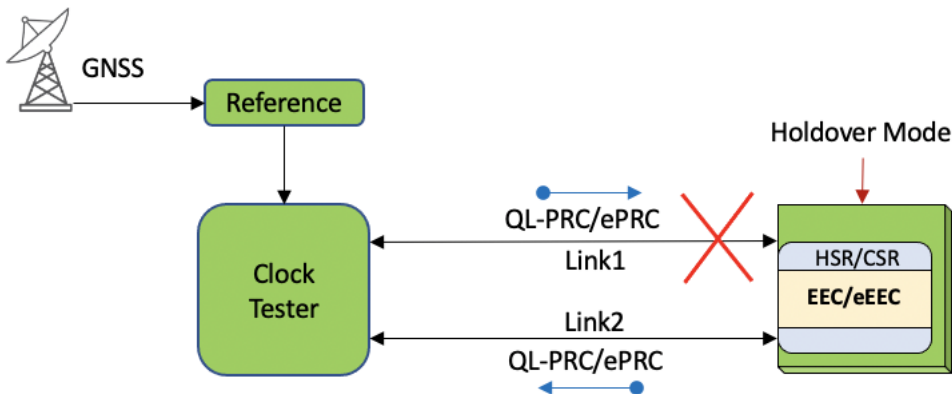
Test ID	SYNC.SE.03	Classification	Synchronization, Performance
Test Title	Wander Tolerance/Wander Transfer @ HSR/CSR		

Test Objective:	Verify the device wander tolerance and wander transfer performance
Pre-Req	HSR/CSR connected to a tester which uses a stable reference clock. TNE should support ESMC or eESMC as applicable. Stimulus for input wander noise to be enabled in the tester.
Test Topology:	
Test Configuration	This section to be filled out when putting the test report (may include device and/or tester configuration)
Procedure:	<u>Detailed Test procedure:</u> 1. Connect the tester to HSR/CSR fronthaul interface. 2. Configure tester to provide the clock to DUT configured interface. 3. Establish Frequency Synchronization on DUT. 4. Perform the wander tolerance/wander transfer test by adding wander noises as input stimulus to the device under test. <u>Verifications needed:</u> 1. Verify that SyncE and ESMC/eESMC are up. 2. Device shouldn't lose lock/no alarm/no reference switch-over with the input wander. 3. Criteria for wander tolerance test as per G.8262/G.8262.1 EEC/eEEC specification. a. Device should maintain the performance within limits. b. Device shouldn't lose lock/no alarm/no reference switch-over with the input wander. c. Device shouldn't move to holdover state. 4. Verify the wander transfer characteristics. a. Wander transfer characteristics should conform ITU G.8262/G.8262.1 EEC/eEEC limits. b. In the passband (1Hz-3Hz), the phase gain of the eEEC should be smaller than 0.2dB. 5. Compare the test result as per ITU-T G.8262/G.8262.1 clock specifications.
Pass/Fail Criteria:	Test shall pass if the measurement result is below the limit specified by ITU-T G.8262/G.8262.1 clock specifications.
Test Output:	This section is the placeholder that will capture the output/configurations after test is executed, typically done after test execution

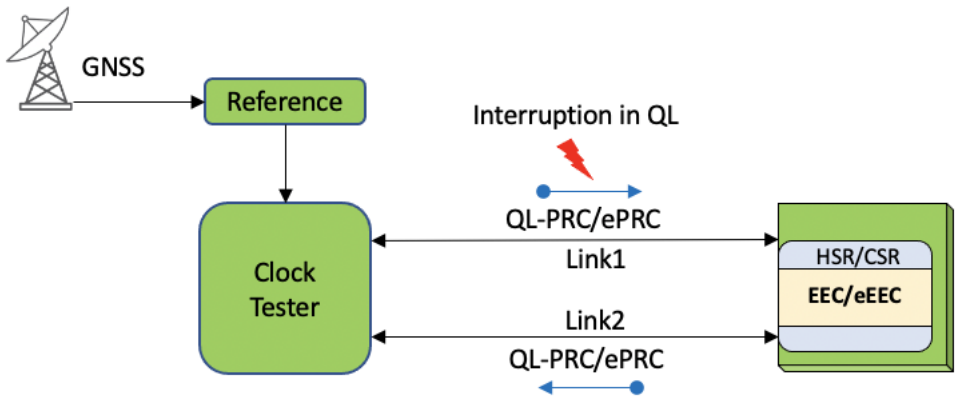
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A.4 Frequency Holdover Limits for EEC/eEEC Clock

Test ID	SYNC.SE.04	Classification	Synchronization, Performance,
Test Title	Frequency holdover limits on HSR/CSR		
Test Objective:	Verify the device maximum wander or TIE/MTIE/TDEV in the Frequency holdover state.		
Pre-Req	HSR/CSR connected to a tester which uses a stable reference clock. TNE should support ESMC or eESMC as applicable.		
Test Topology:			
Test Configuration	This section to be filled out when putting the test report (may include device and/or tester configuration)		
Procedure:	<u>Detailed Test procedure:</u> 1. Connect the tester to HSR/CSR fronthaul interface. 2. Configure tester to provide the clock to DUT configured interface. 3. Establish Frequency Synchronization on DUT. 4. Perform the wander generation test for 15 min. 5. Break the Link1 or stops ESMC/eESMC transmission on Link1. 6. Perform the wander generation test for 24 hrs. <u>Verifications needed:</u> 1. Verify that SyncE and ESMC/eESMC are up. 2. Device should be Frequency synchronized to tester via Link1. 3. Verify the Frequency Error/TIE/MTIE/TDEV before breaking the link. a. All the results should be within ITU-T G.8262/G.8262.1 specification limits. 4. Continue verify the Frequency Error/TIE/MTIE/TDEV after breaking the link. a. All the results should be within ITU-T G.8262/G.8262.1 holdover specification limits		
Pass/Fail Criteria:	Test shall pass if the measurement result is below the limit specified by ITU-T G.8262/G.8262.1 clock specifications.		
Test Output	This section is the placeholder that will capture the output/configurations after test is executed, typically done after test execution		

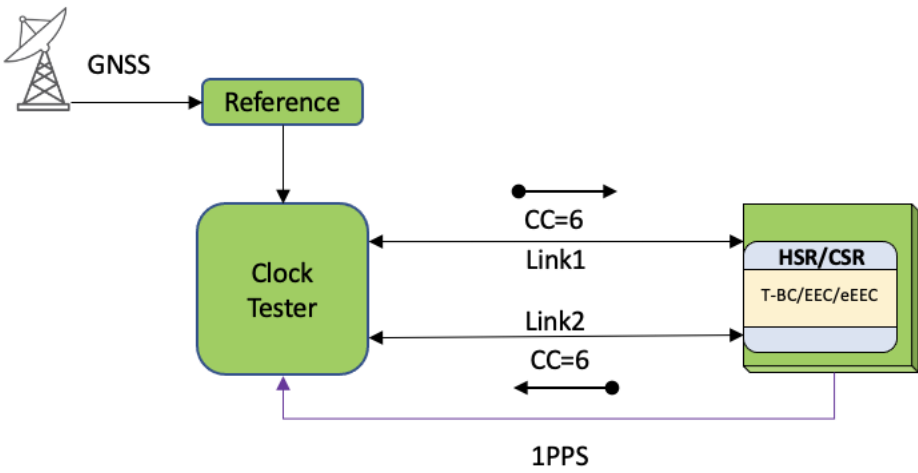
A.5 Short-term phase-transients as per ITU-T G.8262/G.8262.1

Test ID	SYNC.SE.05	Classification	Synchronization, Performance, TIE/MTIE
Test Title	Failover performance or Short-term phase-transients		
Test Objective:	Verify the TIE/MTIE during the failure of primary reference for a short duration (<15 sec)		
Pre-Req	HSR/CSR connected to a tester which uses a stable reference clock		
Test Topology:			
Test Configuration	This section to be filled out when putting the test report (may include device and/or tester configuration)		
Procedure:	<u>Detailed Test procedure:</u> 1. Connect the tester to HSR/CSR fronthaul interface as per the Topology. 2. Configure tester to provide the clock to DUT configured primary interface. 3. Establish Frequency Synchronization on DUT. 4. Perform the wander generation test. 5. Stop the ESMC/eESMC from tester for up to 15 sec. 6. Second reference or the previously selected clock source traceable to primary to be available after 15 sec. <u>Verifications needed:</u> 1. Verify that SyncE and ESMC/eESMC are up. 2. Device should be Frequency synchronized to tester via Link1. 3. Verify the maximum phase-transient at the output of the device under test. a. Result should meet ITU-T G.8262/G.8262.1 mask. b. The overall output phase error should remain constant and smaller than 170ns for G.8262.1 clock and 1000ns for G.8262 clock.		
Pass/Fail Criteria:	Test shall pass if the measurement result is below the limit specified by ITU-T G.8262/G.8262.1 clock specifications.		
Test Output	This section is the placeholder that will capture the output/configurations after test is executed, typically done after test execution		

Annex B: ITU G.8273.2 Clock specification test cases for TNEs

PTP performance as per G.8273.2 clock specifications

B.1 Time Noise Generation Performance

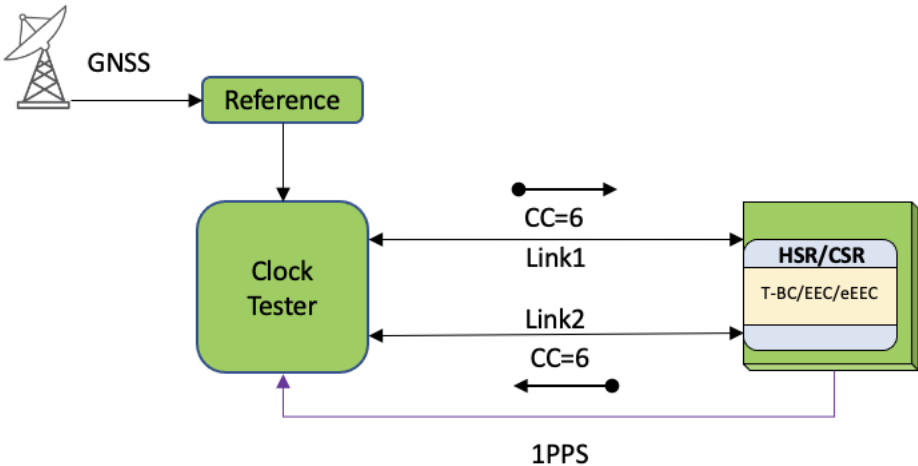
Test ID	SYNC.SE.06	Classification	Synchronization, Performance, TIE/MTIE
Test Title	Time Noise Generation Performance		
Test Objective:	Verify the Max TE , cTE, dTE of the HSR/CSR		
Pre-Req	HSR/CSR connected to a Clock tester which uses a high-quality stable reference clock.		
Test Topology:			
Test Configuration	This section to be filled out when putting the test report (may include device and/or tester configuration)		
Procedure:	Detailed Test procedure: 1. Connect the tester to HSR/CSR fronthaul interface as per the Topology. 2. Configure tester as TimeTransmitter on the link1 and TimeReceiver on the link2. 3. Establish Frequency and Phase Synchronization on DUT. 4. Perform the noise-generation test for duration of 1000 sec or more. Verifications needed: 1. Verify both frequency and phase lock on device under test. 2. Verify the time error performance as per ITU-T G.8273.2 a. Frequency measurement should be as per ITU-T G.8262/G.8262.1 EEC/eEEEC. b. For enhanced TE specifications, the device conforms following Class-C specifications. 1. Maximum 2way absolute Time Error of 30 nsec. 2. Constant Time Error of +/-10 nsec. 3. Dynamic Time Error (Filtered) for low and high frequency 10 nsec and 30 nsec respectively.		

	4. Dynamic MTIE (LF) should be as per G.8273.2 mask. NOTE: For TNEs doesn't support enhanced Class-C TE specifications shall support either Class-A or Class-B specifications as per ITU-T G.8273.2.																							
Pass/Fail Criteria:	Test shall pass if the measurement result is below the limit specified by ITU-T G.8273.2 clock specifications.																							
Test Output	This section is the placeholder that will capture the output/configurations after test is executed, typically done after test execution <table><tr><th rowspan="2">Time Error Limit</th><th colspan="3">Expected Value</th></tr><tr><th>Class-A</th><th>Class-B</th><th>Class-C</th></tr><tr><td>Max Time Error-2WAY (Max TE)</td><td>≤ 100ns</td><td>≤ 70ns</td><td>≤ 30ns</td></tr><tr><td>Constant Time Error-2WAY (cTE-2WAY)</td><td>≤ +50ns</td><td>≤ +20ns</td><td>≤ +10ns</td></tr><tr><td>Dynamic Time Error-HF (dTE-HF)</td><td>≤ 70ns</td><td>≤ 70ns</td><td>≤ 70ns</td></tr><tr><td>Dynamic Time Error-LF (dTE-LF)</td><td>≤ 40ns</td><td>≤ 40ns</td><td>≤ 10ns</td></tr></table>	Time Error Limit	Expected Value			Class-A	Class-B	Class-C	Max Time Error-2WAY (Max TE)	≤ 100ns	≤ 70ns	≤ 30ns	Constant Time Error-2WAY (cTE-2WAY)	≤ +50ns	≤ +20ns	≤ +10ns	Dynamic Time Error-HF (dTE-HF)	≤ 70ns	≤ 70ns	≤ 70ns	Dynamic Time Error-LF (dTE-LF)	≤ 40ns	≤ 40ns	≤ 10ns
Time Error Limit	Expected Value																							
	Class-A	Class-B	Class-C																					
Max Time Error-2WAY (Max TE)	≤ 100ns	≤ 70ns	≤ 30ns																					
Constant Time Error-2WAY (cTE-2WAY)	≤ +50ns	≤ +20ns	≤ +10ns																					
Dynamic Time Error-HF (dTE-HF)	≤ 70ns	≤ 70ns	≤ 70ns																					
Dynamic Time Error-LF (dTE-LF)	≤ 40ns	≤ 40ns	≤ 10ns																					

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B.2 Time Noise Tolerance Performance Test

Test ID	SYNC.SE.07	Classification	Synchronization, Performance, TIE/MTIE
Test Title	Time Noise tolerance Performance		
Test Objective:	Verify the device Time Noise tolerance performance		
Pre-Req	HSR/CSR connected to a tester which uses a high-quality stable reference clock. Stimulus for input time noise to be enabled in the tester.		
Test Topology:			
Test Configuration	This section to be filled out when putting the test report (may include device and/or tester configuration)		
Procedure:	Detailed Test procedure: 1. Connect the tester to HSR/CSR fronthaul interface as per the Topology. 2. Configure tester as TimeTransmitter on the link1 and TimeReceiver on the link2. 3. Establish Frequency and Phase Synchronization on DUT. 4. Perform the time noise tolerance test for duration of 1000 sec or more.		

	Verifications needed: 1. Verify both frequency and phase lock on device under test. 2. Criteria for noise tolerance test as per G.8273.2 specification a. Device should maintain the performance within limits. b. Device shouldn't lose lock/no alarm/no reference switch-over with the input wander. c. Device shouldn't move to holdover state. 3. Verify the time error performance as per ITU-T G.8273.2 a. Frequency measurement should be as per ITU-T G.8262.1 eEEEC. b. For enhanced TE specifications, the device conforms following Class-C specifications. 1. Maximum 2way absolute Time Error of 30 nsec. 2. Constant Time Error of +/-10 nsec. 3. Dynamic Time Error (Filtered) for low and high frequency 10 nsec and 30 nsec respectively. 4. Dynamic MTIE (LF) should be as per G.8273.2 mask. NOTE: For TNEs doesn't support enhanced Class-C TE specifications shall support either Class-A or Class-B specifications as per ITU-T G.8273.2.
Pass/Fail Criteria:	Test shall pass if the measurement result is below the limit specified by ITU-T G.8273.2 clock specifications.
Test Output	This section is the placeholder that will capture the output/configurations after test is executed, typically done after test execution

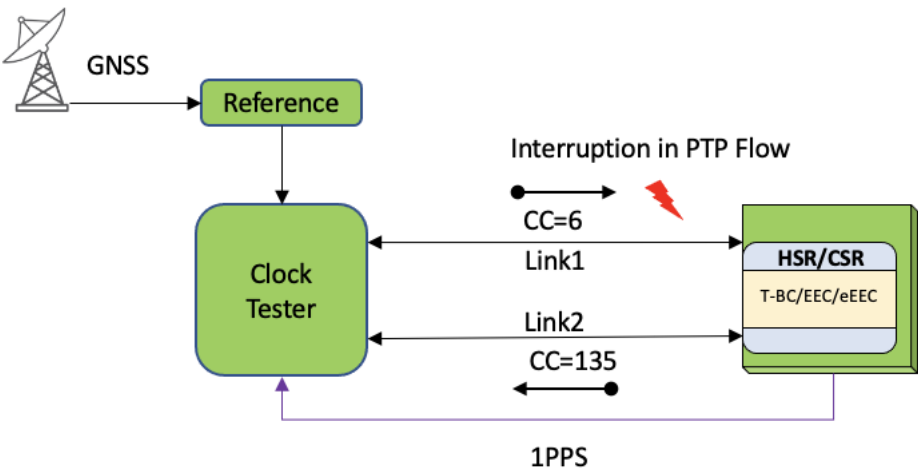
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B.3 Time Holdover Performance Test

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Test ID	SYNC.SE.08	Classification	Synchronization, Performance, TIE/MTIE
Test Title	Time holdover Performance		
Test Objective:	Verify the device Time holdover performance (Physical Layer Assisted)		
Pre-Req	HSR/CSR connected to a tester which uses a high-quality stable reference clock		

Test Topology:	
Test Configuration	This section to be filled out when putting the test report (may include device and/or tester configuration)
Procedure:	<u>Detailed Test procedure:</u> 1. Connect the tester to HSR/CSR fronthaul interface as per the Topology. 2. Configure tester as TimeTransmitter on the link1 and TimeReceiver on the link2. 3. Apply a stable time and frequency from the Tester to DUT input. 4. Establish Frequency and Phase Synchronization on DUT. 5. Perform the holdover test for duration of 1000 sec or more (Stopping the PTP flows) <u>Verifications needed:</u> 1. Verify both frequency and phase lock on device under test (HSR/CSR) after stopping the PTP Flow, a. PTP should be in holdover state. b. Frequency should be in Locked state. 2. Verify the time error performance as per ITU-T G.8273.2 a. Frequency measurement should be as per ITU-T G.8262.1 eEEEC. b. For enhanced TE specifications, the device conforms following Class-C specifications. 1. Maximum 2way absolute Time Error of 30 nsec. 2. Constant Time Error of +/-10 nsec. 3. Dynamic Time Error (Filtered) for low and high frequency 10 nsec and 30 nsec respectively. 4. Dynamic MTIE (LF) should be as per G.8273.2 mask. 3. Verify the MTIE mask defined as per ITU-T G.8273.2. NOTE: For TNEs doesn't support enhanced Class-C TE specifications shall support either Class-A or Class-B specifications as per ITU-T G.8273.2.
Pass/Fail Criteria:	Test shall pass if the measurement result is below the limit specified by ITU-T G.8273.2 clock specifications.
Test Output	This section is the placeholder that will capture the output/configurations after test is executed, typically done after test execution

